

Danish Technical University
41684 Materials Technology summary
Comprehensive Study Notes from spring 2026



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1 Ceramics

Materials typically considered ceramics

- Polycrystalline ceramics fabricated by sintering
 - Al_2O_3
 - TiO_2
 - ZrO_2
 - BaTiO_3
 - SiC
 - Si_3N_4
- Glass
 - SiO_2
 - Na_2O , Na_2CO_3 , CaO
 - B_2O_3
 - PbO
- Glass ceramics
 - $\text{Li}_2\text{O} \times \text{Al}_2\text{O}_3 \times n\text{SiO}_2$
- Single crystals
 - Quartz (SiO_2)
 - Diamonds
 - NaCl
- Room temperature chemical reaction
 - Cement
 - Plaster
- Natural ceramics
 - Stone
 - Bone
 - Teeth
 - Nacre
 - Crustations
 - Diatoms

1.1 History

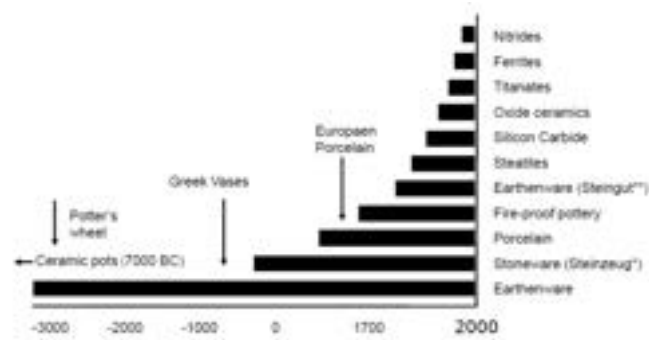


Figure 1: Ceramics History

Caves and clay

- Cave pictures - Chauvet Cave, 35000 years old, containing pigment and oil/water.
- Clay figures - Tuc d'Audobert Cave, France 14000 years old, containing unfired but preserved in cave.
- Clay tablets with inscriptions - cuneiform - 3000 BC

Modern ceramics

- From the 18th:
 - Higher furnace temperature (1200 – 2300 °C)
 - Chemistry knowledge
 - Microscopes
 - Machines, assembly lines, printing techniques
- Automobile industry & development of spark plugs
 - porcelain from natural clay, flint, and feldspar made on a potter's wheel. Impurities, breakdown
 - Complex shaping (isostatic press, injection molding, 3D printing)
- Applications:
 - 5 million tonnes Al_2O_3 products every year (1995)
 - High temperature stable products
 - Microelectronics and space applications - Functional materials
 - Bulk materials, porous materials, coatings
 - Energy conversion and storage devices - batteries, solid oxide cells, membranes

1.2 Learning objectives

Evaluating the differences between the properties resulting from the amorphous and crystalline state of a solid material.

- Amorphous materials (non-crystalline solids) have a short range of ordering.
- Melts over a range of temperatures

- Glass ceramics and ceramic glasses

Relating the internal structure of ceramics to their properties.

Identifying principles and mechanisms, which allow tailoring the materials specific properties and performance:

- Composition
- Crystal structures
- Defects
- Microstructures

1.2.1 Imperfections in ceramics

Electroneutrality (charge balance) must be maintained when impurities are present.

Example: NaCl

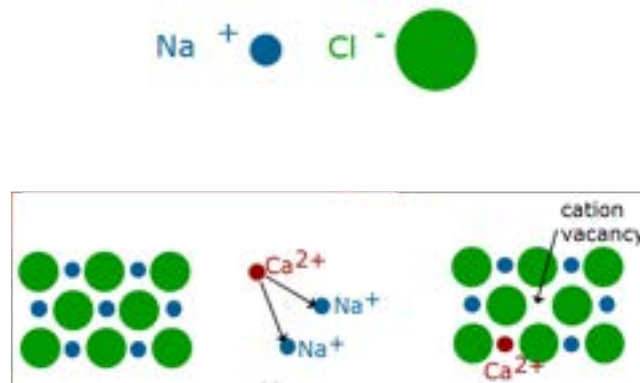


Figure 2: Substitutional cation impurity, with and without the Ca^{2+} impurity

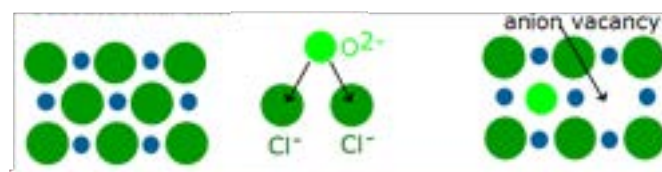


Figure 3: Substitutional anion impurity, with and without the O^{2-} impurity

1.3 Properties

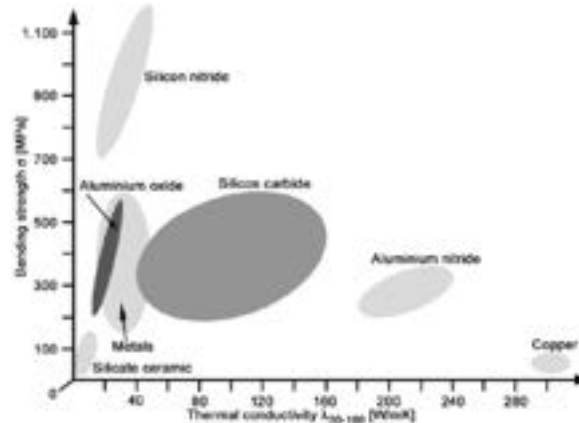
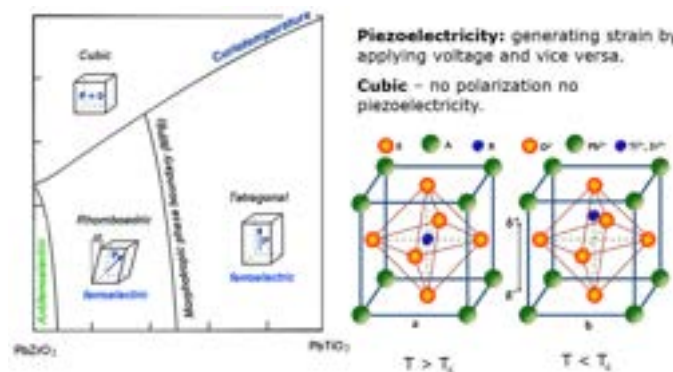


Figure 4: Thermal and mechanical properties in relation to metals

1.3.1 Mechanical properties

- Impact of microstructure on properties
 - Dense ceramics
 - Porous ceramics
- Fracture behavior in ceramics
 - Characterized commonly as brittle fracture
 - Can be transgranular or intergranular
 - Flaws acts as stress raisers
- Fractography
 - Analysis of the fracture
 - Focus on location, type and source of the defect/flaw
 - Useful in taking measures to reduce similar fractures

Property modification through structure



1.4 Ceramic component fabrication and classification

Powder synthesis → Powder treatment → Powder conditioning → Shaping/forming

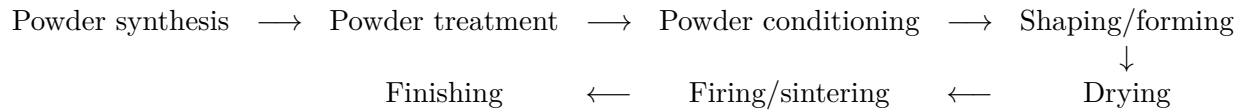


Figure 5: Ceramic classification

1.5 Composites

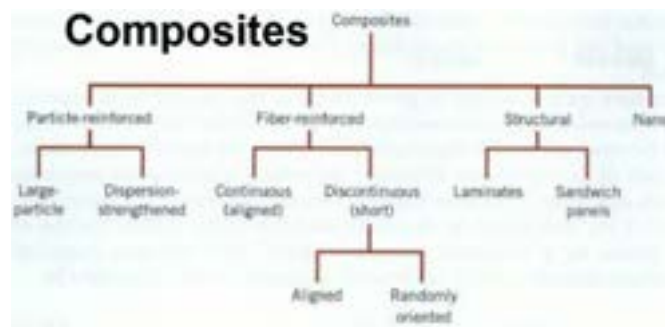


Figure 6: Composites

- Improved the mechanical properties
- Chemical and corrosion resistance
- Longer lifetime
- Ease of processing and designing
- High temperature performance
- Selected functionality

1.6 Amorphous Ceramics

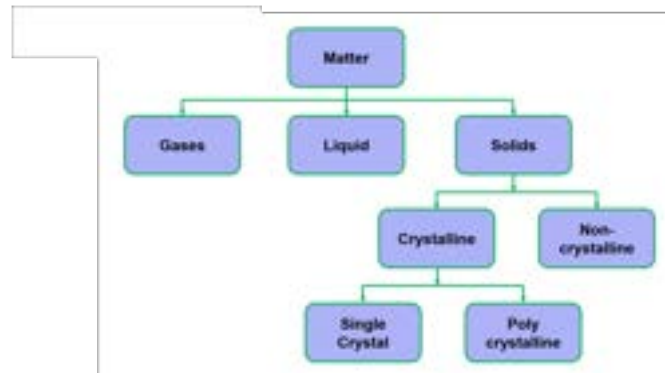


Figure 7: Different states of matter

Non-crystalline Solids (Glasses)

- A solid that is lacking of long-range ordering
- A short-range ordering may exist
- Have an internal structure of basic units (lattices), without long range order,
- Melts over a range of temperatures
- Also known as amorphous solids

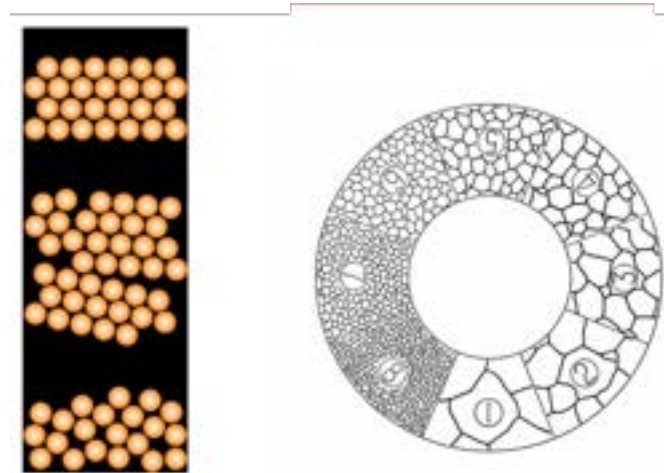


Figure 8: a) crystalline to Non-Crystalline, b) structural point of view

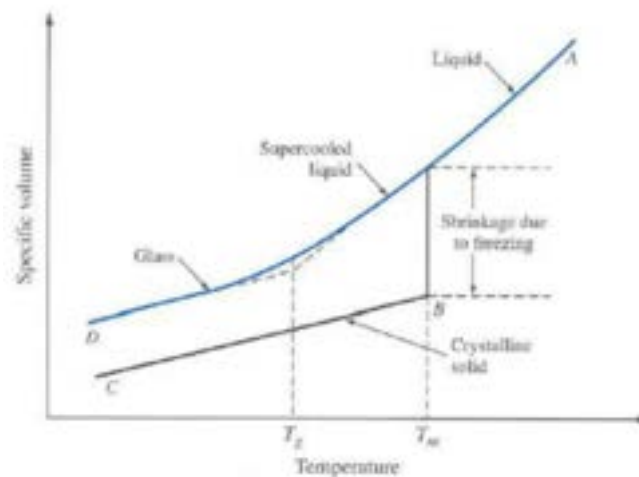
Natural crystalline and non-crystalline materials

- Crystalline
 - Galena (PbS)
 - Quartz (SiO_2)
 - Pyrite (FeS_2)
- Non-crystalline
 - Obsidian (similar to granite KAlSi_3O_8)

Non-Crystalline ceramics are mainly based on SiO_2 (glasses), some other materials possible, but not common.

Glasses

- Mainly based on SiO_2
- Can be pure SiO_2 , fused silica
- With impurities (alkaline metals, soda glass)
- Wide range of properties based on composition
- Glass is an *inorganic* product of fusion that has cooled to a solid without crystallization
- Glass does not crystallize upon cooling
- Upon cooling it transforms to a rigid solid.



Glasses with impurities

- Glass ceramics, huge family of non-crystalline ceramics
- SiO_2 without impurities is fused silica
- The addition of impurities $\rightarrow$ breaks the regular structure $\rightarrow$ non-crystalline $\text{SiO}_2 \rightarrow$ other types of glasses

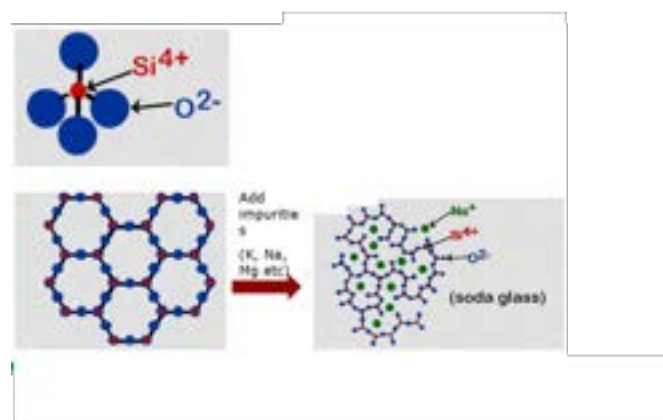


Figure 9: left: Crystalline SiO_2 , Right: Non-crystalline glass

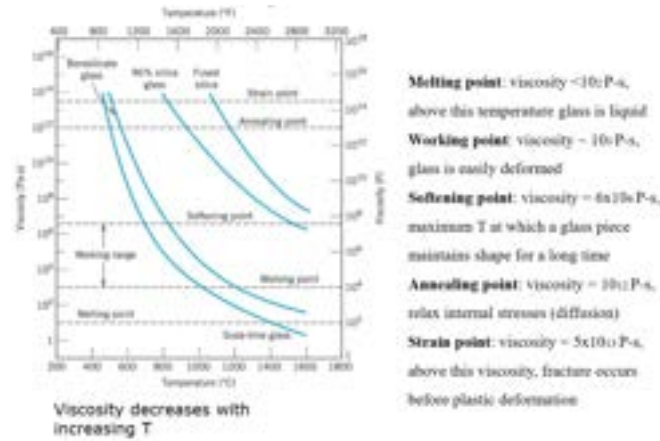


Figure 10: Glasses - Viscosity

Glass, fused Silica

- Amorphous form of crystalline silica
- No or very low impurities
- Posses excellent optical transparency
- Low electrical conductivity
- Stable at high temperatures
- Low thermal expansion
- Because of above $\rightarrow$ wide variety of applications

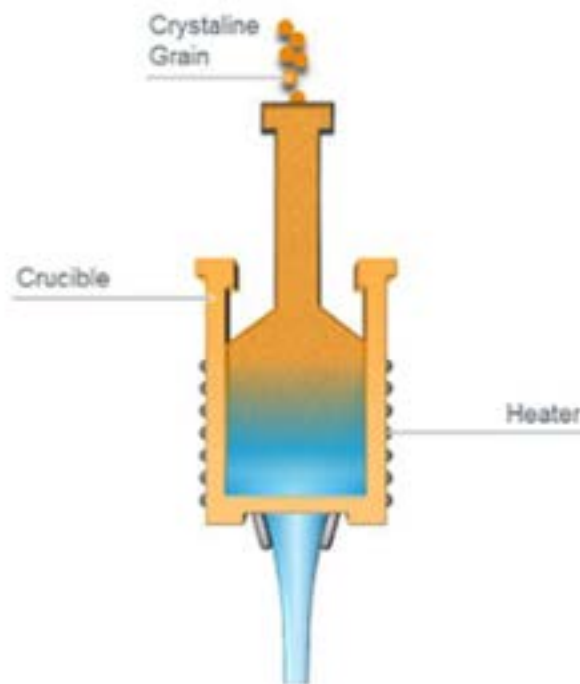
Applications of fused silica

- Refractory
 - High temperature lamp envelopes
 - Spacecraft windows
 - Crucibles and furnace refractories
- High optical transparency
 - UV optics
 - Optical fiber for telecommunications
- Low thermal expansion
 - High temperature lamp envelopes
 - Spacecraft windows
 - Thermally stable mirrors (astronomical telescopes, IX microlithography stepper camera optics)
- High purity
 - IC microlithography stepper camera optics

Fused silica, Manufacturing

- Crystalline forms of silica

- Cristobalite 1470 – 1723 °C
 - Tridymite 870 – 1470 °C
 - Quartz below 870 °C
- $T > 1730\text{ °C}$ required to form an all liquid melt
- Simple melting methods, difficult to apply
 - Electric arc melting
 - Flame heating
- Reactions with gases and crucibles → introduce impurities
- very challenging to get high pure fused silica



Above can be seen the most used method in Fused Silica manufacturing,

- can be used as a batch or continuous process
- Uses electrical heating to fuse the SiO_2
- Where the melt from the exit is then used to make the products

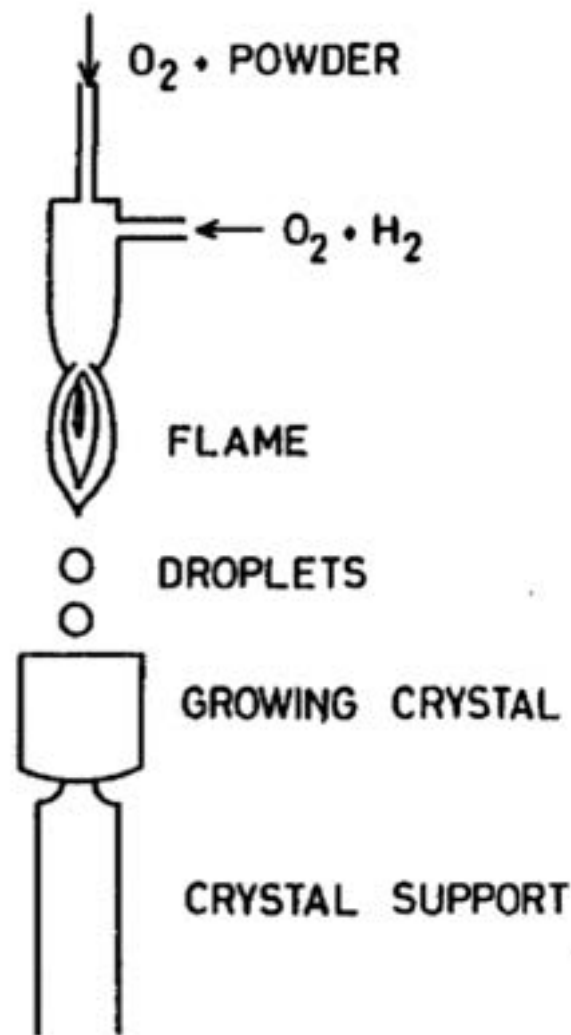


Figure 11: Vernueil process in Silica manufacturing

- Silica is an excellent glass former
- When molten quartz droplets settle on a substrate above 1600°C , they fuse together into a molten mass
- This mass does not crystallize at those temperatures nor when being cooled to room temperature.

Glass modifying oxides and modified glasses

Glass modifying oxides

- Network modifiers: Oxides that break up the glass network.
 - Added to glass to increase workability
 - Oxygen atom enters networks and other ion stay in interstices
 - Examples: Na_2O , K_2O , MgO , CaO
- Intermediate oxides: cannot form glass network by themselves but can join into an existing network
 - Added to obtain special properties
 - Examples: PbO , Al_2O_3

Modified glasses

- Soda lime glass: Very common glass (90%)
 - $\text{SiO}_2\text{--Na}_2\text{O--CaO}$
 - Easy to shape
 - Used in flat glass containers
- Borosilicate glass: Alkali oxides are replaced by boric oxide in silica glass network
 - Used for lab equipment and piping (pyrex)
- Lead glass: Lead oxide acts as network modifier and network former
 - Low melting point, used for solder sealing
 - Used in radiation shields, optical glass and TV bulbs

Glass Ceramics

Glass ceramics are,

- Controlled crystallization of glasses $\rightarrow$ Glass ceramics
- materials that have both glass and ceramic properties
- made to obtain specific properties, particularly thermomechanical properties.
- Example of glass ceramics are: Li_2O_3 , Al_2O_3 , SiO_2

Properties of some glass ceramics:

- Al_2O_3 , MgO , CaO , SiO_2
 - Property: Translucent, chemically resistant
 - Use: Window glass, bottles
- PbO , SiO_2
 - Property: High refractive index
 - Use: Lead crystal
- B_2O_3 , SiO_2 , Na_2O
 - Property: Acid resistant, low expansion on heating
 - Use: Pyrex
- MgO , Al_2O_3 , SiO_2
 - Property: Insulator with high mechanical strength at high temperatures
 - Use: Spark plug insulators
- CaSiO_3 , $\text{CaMgSi}_2\text{O}_6$, $\text{CaAl}_2\text{Si}_2\text{O}_8$, $\text{CaAl}_2\text{Si}_2\text{O}_8$
 - Property: Wear resistant
 - Use: Building materials
- $\text{Li}_2\text{Si}_2\text{O}_5$
 - Property: Resistant to thermal shock
 - Use: Nose cones on rockets, cookware

Other non-crystals ceramics

- Efforts of being made to make other amorphous ceramics
- Examples are:
 - $\text{Al}_2\text{O}_3 - \text{ZrO}_2$ system
 - Si_3N_4
 - $\text{Si}-\text{C}-\text{N}$
 - $\text{Si}-\text{B}-\text{C}-\text{N}$
- No component level fabrication, more work to be done.

1.7 Internal Structure

Parameters that influence the internal structure of ceramics are, Bonding, Crystal structure and Imperfections (chemical and physical).

Bonding

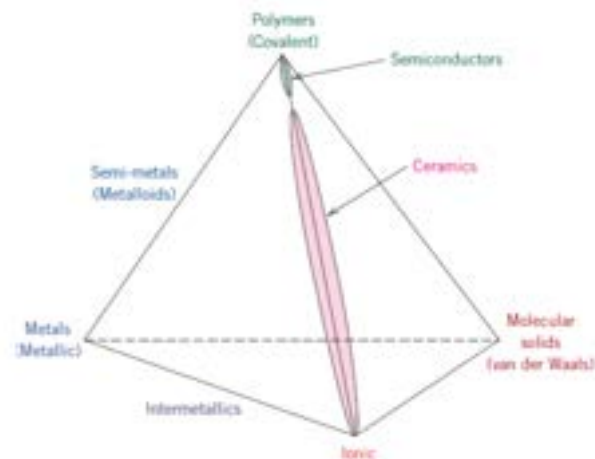


Figure 12: Mixed bonding

To find the percent ionic character of a bond %IC the following equation is used,

$$\%IC = \left(1 - \exp(-(0.25)(X_A - X_B)^2)\right) \times 100$$

Where X_A and X_B are the electronegativities of the atoms, found using the following table,

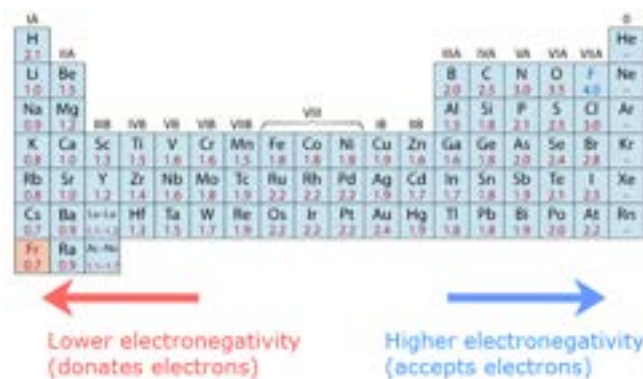


Figure 13: Electronegativity table

Bond strength

- Structure is determined by size and charge
- Electrical neutrality is required at the unit cell level as well as throughout the crystal structure
- The charge of each cation (positive charge) and anion (negative charge) must be mutually balance by the combined charge of the surrounding ions of the opposite charge, this places a limitation on the positions of the ions in the structure
- Portion of the charge of each cation depends on valence (V) and Coordination Number (CN, number of nearest neighbors)
- The following is called the bond strength

$$|V_{\text{cation}}|/(CN)_{\text{cation}} \text{ or } |V_{\text{anion}}|/(CN)_{\text{anion}}$$

$$|V_{\text{cation}}|/(CN)_{\text{cation}} = |V_{\text{anion}}|/(CN)_{\text{anion}}$$

Crystal Structures

The governing factors in the crystal structures are the Relative size of the atoms and charge neutrality.

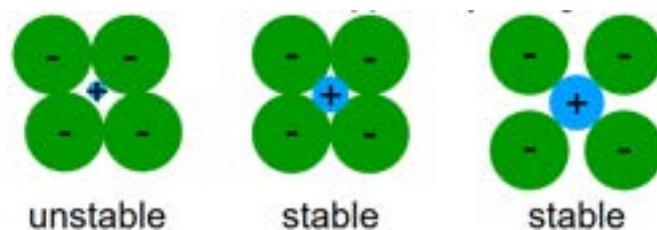
Some characteristics,

- Coordination number (CN), The number of atoms immediatly surrounding to the atom in consideration
- Atomic packing fctor (APF), Ration of the volume of the atoms in a unit cell to the volume of the unit cell
- Ionic radii, Radius of an ion in an ionic crystal

Factors that determine structure

Relative sizes of ions:

Formation of stable structures, maximizing the number of oppositely charged ion neighbors.



Maintenance of charge neutrality:

Net charge in ceramic should be zero.

Reflected in chemical formula, A_mX_p , where m, p values to achieve charge neutrality,



Interstitial sites

The most common are Tetrahedral (4 atoms as the nearest neighbors) and Octahedral (6 atoms as the nearest neighbors) sites, for each one of the anion spheres, two tetrahedral and one octahedral site exists.

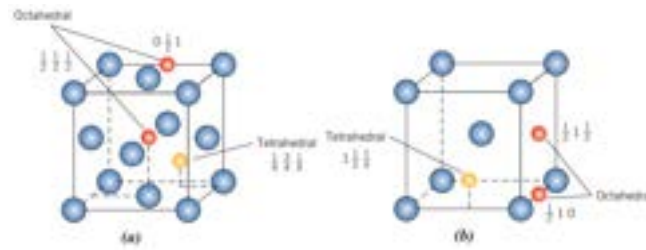


Figure 14: Locations of tetrahedral and octahedral interstitial sites within (a) FCC and (b) BCC unit cells

Coordination Number (CN) and Ionic Radii

Coordination number increases with

$$\frac{r_{\text{cation}}}{r_{\text{anion}}}$$

To form a stable structure, how many anions can surround around a cation?,

Coordination Number	Cation-Anion Radius Ratio	Coordination Geometry
2	<0.155	
3	$0.155-0.225$	
4	$0.225-0.414$	
6	$0.414-0.732$	
8	$0.732-1.0$	

Figure 15: Coordination numbers and geometries for various Cation-Anion radius ratios

The name of the geometries are Linear (2), Triangular (3), Tetrahedral (4), Octahedral (6) and Cubic (8).

ZnS (zinc blende) is known to have tetrahedral geometry

NaCl (sodium chloride) is known to have octahedral geometry

CsCl (cesium chloride) is known to have cubic geometry

Example problem: Predicting the crystal structure of FeO

On the basis of ionic radii,

Fe^{2+} has a ionic radius of 0.077 nm

O^{2-} has a ionic radius of 0.140 nm

Therefore,

$$\frac{r_{\text{cation}}}{r_{\text{anion}}} = \frac{0.077}{0.140} = 0.550$$

Based on this ratio, coordination number of 6 because, $0.414 < 0.550 < 0.732$, therefore the crystal structure is NaCl

How to determine crystal structures

In practice X-ray diffraction is mainly used

$$N\lambda = 2d \sin \theta$$

But other methods such as spectroscopy and microscopy are also used.

Typical ceramic crystal structures

- Ax-type compounds, rock salt structure, CsCl, NaCl, and zinc blend structure

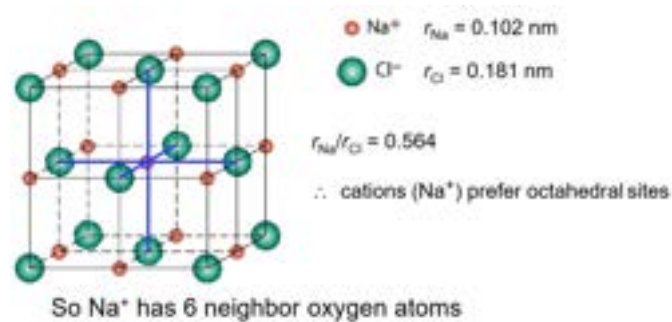


Figure 16: Rock Salt structure

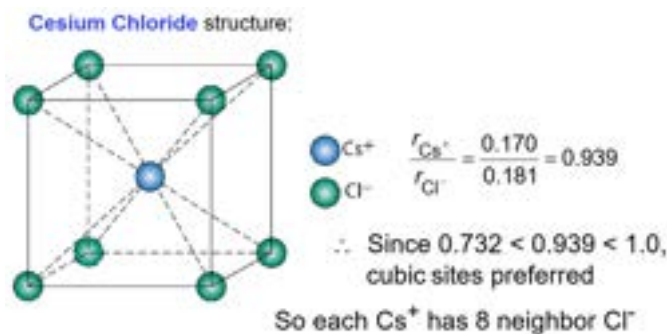


Figure 17: Cesium Chloride structure

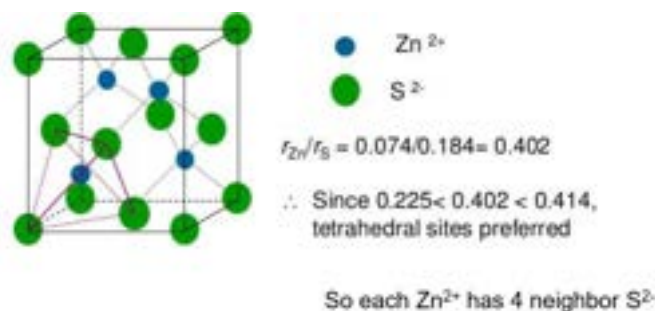


Figure 18: Zinc blend structure, Most often seen in covalent compounds

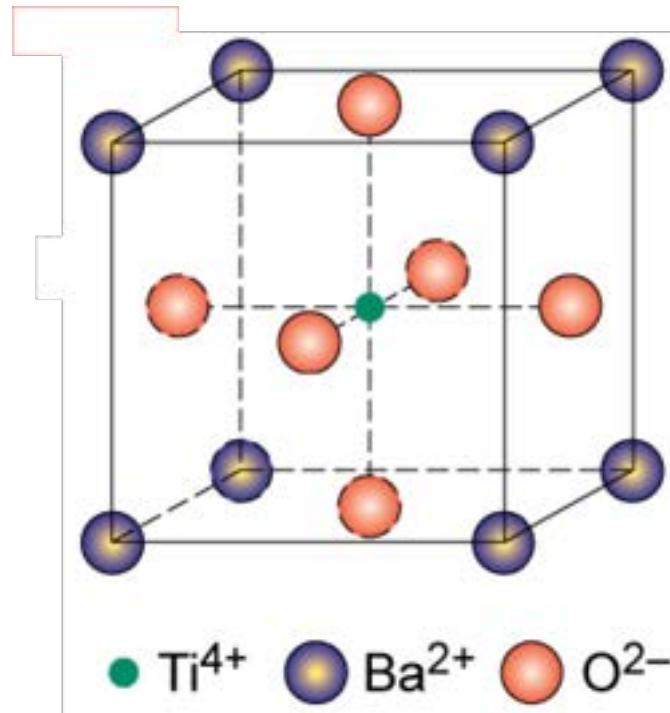


Figure 20: Perovskite structure, ABO_3 , seen above is the complex oxide $BaTiO_3$

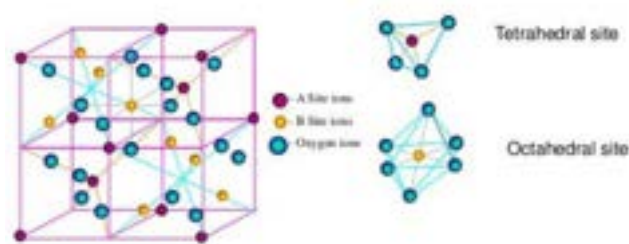


Figure 21: Spinel structure AB_2X_4 in this case $MgAl_2O_4$

- A_mX_p compounds, Fluorite structure

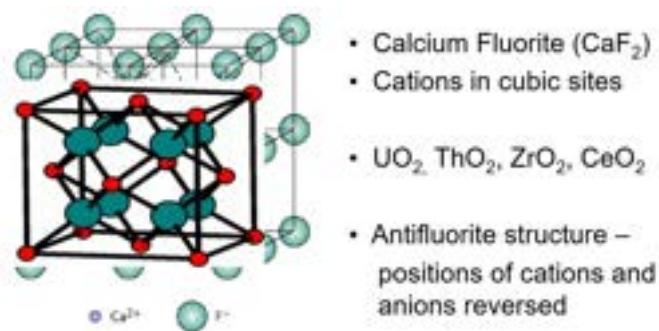


Figure 19: Fluorite structure

- $A_mB_nX_p$ compounds, Perovskite structure
 - Structural distortion of this crystal structure can lead to interesting properties
 - These distortions can be caused by different sized cations
 - The degree of distortion can be predicted by Goldschmidt tolerance factor

$$t = \frac{r_A + r_X}{\sqrt{2}(r_B + r_X)}$$

- Ideal Perovskite has the $t = 1$. The distortions can cause $0.89 < t < 1$
- $t < 1$, if size of A is smaller than ideal and/or size of B ion is larger than ideal. Result in for example, orthorhombic structures.
- $t > 1$, can result in tetragonal or hexagonal structures
- Interesting electromechanical properties

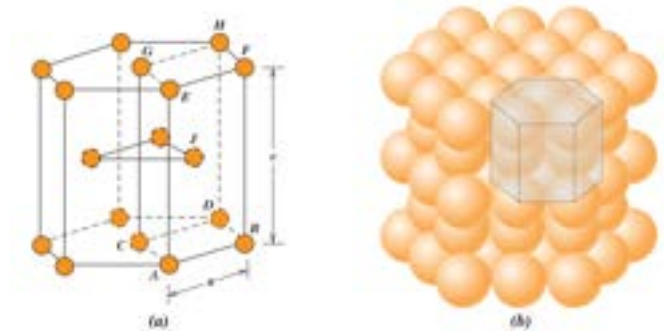
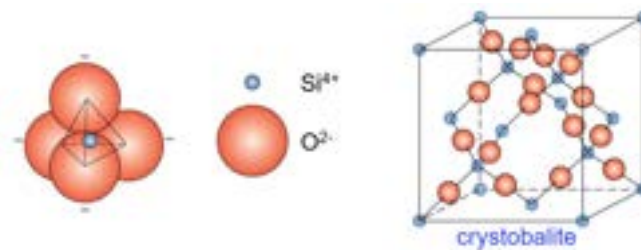


Figure 22: HCP unit cell

HCP unitcells contain 6 atoms, 12 tetrahedral interstitial sites and 6 octohedral interstitial sites.

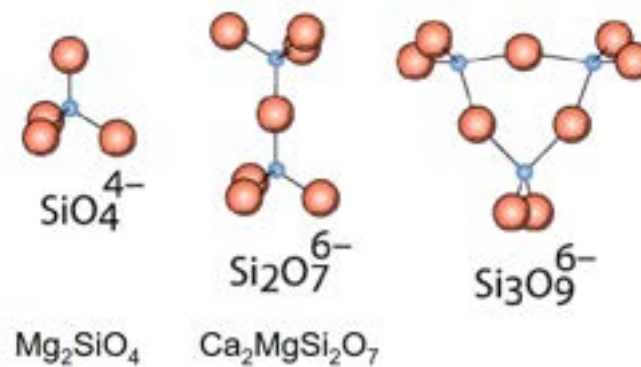
Silicate Ceramics

Most common elements on earth are Si and O



- SiO_2 (silica) polymorphic forms are quartz, crystobalite and tridymite
- The strong Si–O bonds lead to a high melting temperature (1710°C) for this material

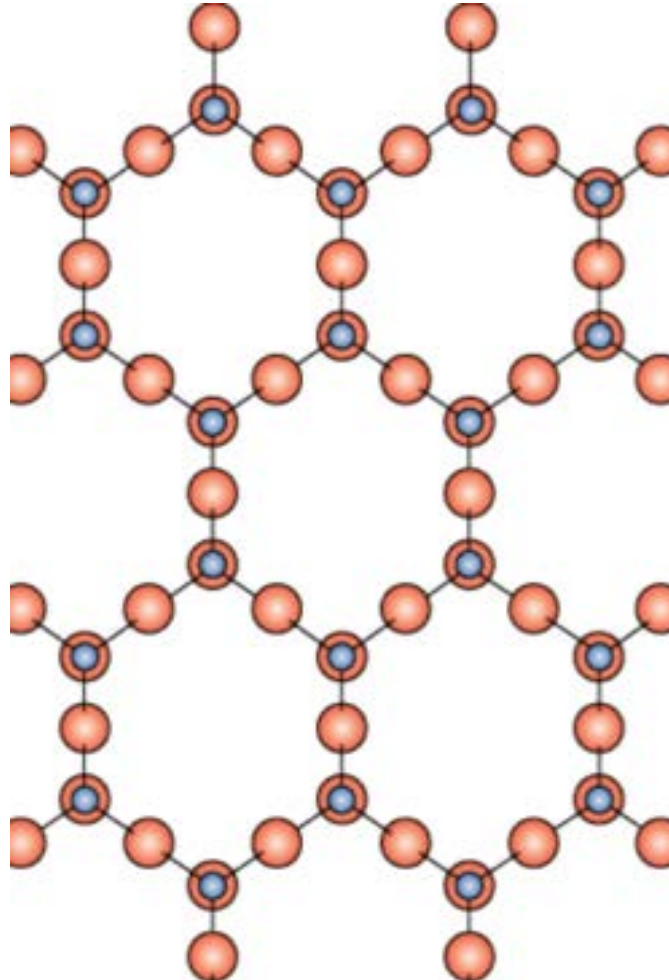
Bonding of adjacent SiO_4^{4-} accomplished by the sharing of common corners, edges, or faces



Presence of cations such as Ca^{2+} , Mg^{2+} , and Al^{3+} maintain charge neutrality and ionically bond SiO_4^{4-} to one another.

Layered Silicates

- Layered silicates (e.g., clays, mica, talc)
 - SiO_4 tetrahedra connected together to form 2D plane
- A net negative charge is associated with each $(\text{Si}_2\text{O}_5)^{2-}$ unit
- Negative charge balanced by adjacent plane rich in positively charged cations



Kaolinite clay alternates $(\text{Si}_2\text{O}_5)^{2-}$ layer with $\text{Al}_2(\text{OH})_4^{2+}$ layer

Property relationships

The properties of the ceramics can be modified.

- Crystal structure changes
- Introducing defects/imperfections
 - Composition changes
 - Microstructure changes

Polymorphic Forms of Carbon Diamond

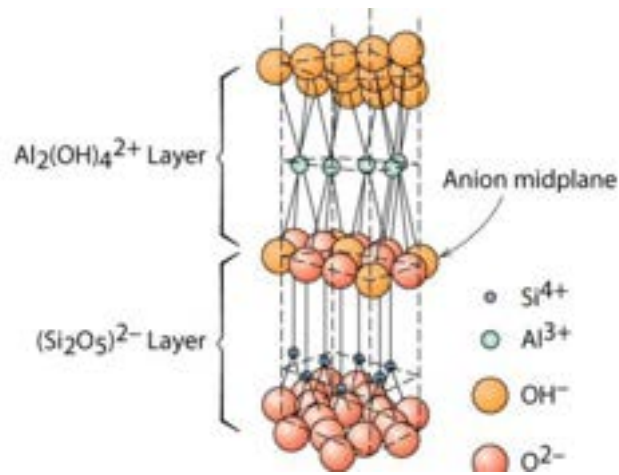
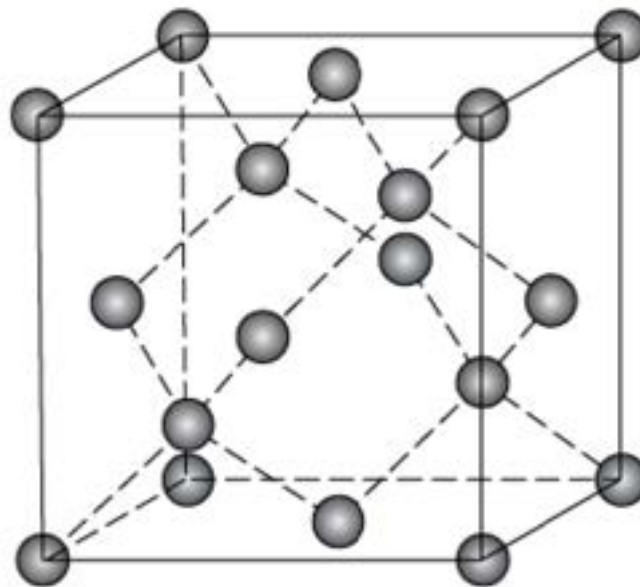
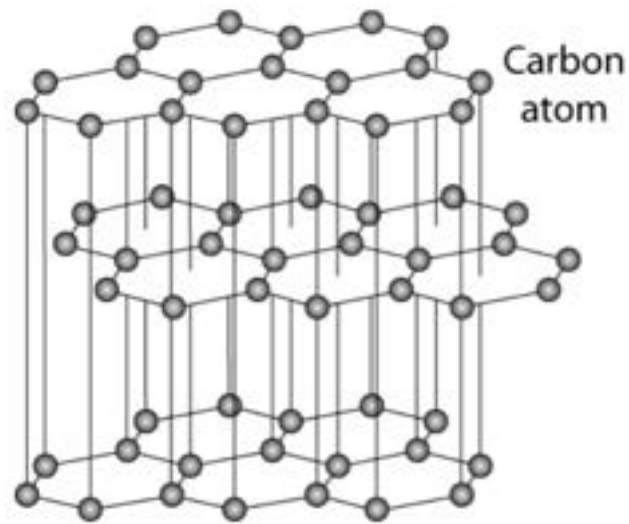


Figure 23: Note that adjacent sheets of this type are loosely bound to one another by Van der Waal's forces



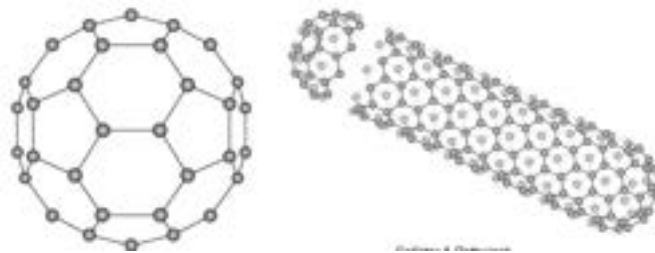
- Tetrahedral bonding of carbon
 - Hardest material known
 - Very high thermal conductivity
- Large single crystals, gemstones
- Small crystals, are used to grind/cut other materials
- Diamond thin films
 - Hard surface coatings, used for cutting tools, medical devices, etc.

Polymorphic Forms of Carbon Graphite



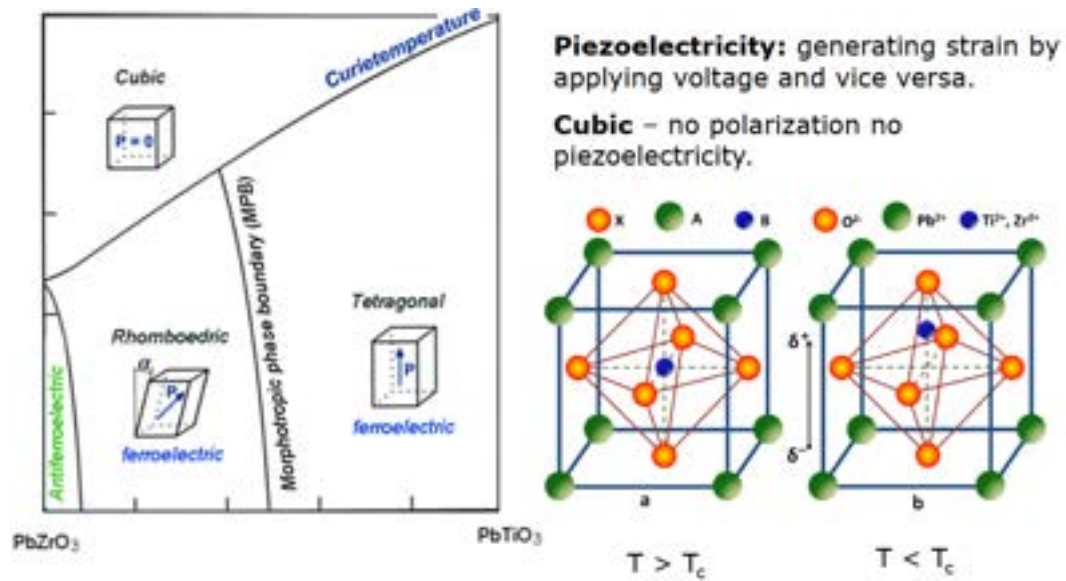
- Layered structure, parallel hexagonal arrays of carbon atoms
- weak Van der Waal's forces between layers
- Planes slide easily over one another, good lubricant

Polymorphic Forms of Carbon Fullerenes and Nanotubes

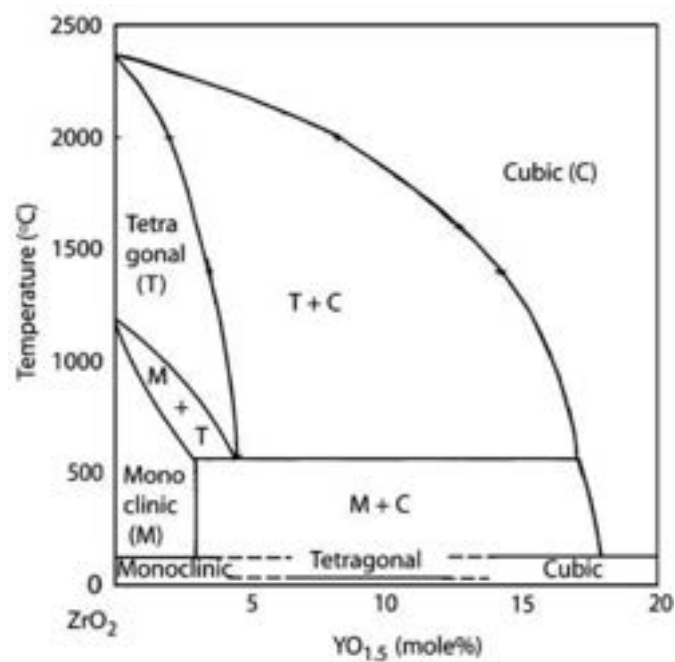


- Fullerenes, spherical cluster of 60 carbon atoms, C₆₀
 - Like a soccer ball
- Carbon nanotubes, sheet of graphite rolled into a tube
 - ends capped with fullerene hemispheres

Property modification through composition/structure $\text{Pb}(\text{Zr}, \text{Ti})\text{O}_3$ – PZT



Another example is ZrO_2



- Substitutional impurity of Y
- Increase of Y, leads to structural changes
- Structural changes then lead to lead to property changes

Property	ZrO ₂ (monoclinic)	Y-TZP (Tetragonal)	Ce-TZP (Tetragonal)	Mg-PSZ (Monoclinic)	FSZ (Cubic)
Density (g/cm ³)	6.15	6.05	6.15	5.75	5.5-5.8
Hardness (HV _{0.05})	1900	1350	900	1020	2447
Bend Stc (MPa)	1000	1000	350	800	1400
Compressive Str (MPa)	5200	2000	-	2000	5000
Young's Modulus (GPa)	250	205	215	205	205
Poisson's Ratio	0.32	0.3	-	0.23	-
Fracture Toughness (MPa m ^{1/2})	8	9.5	15-20	8-15	1.7
Thermal Exp. Co-EF (x10 ⁻⁶ °C ⁻¹)	12.2	10	8	10	8
Thermal Conductivity (W m ⁻¹ K ⁻¹)	2.7	2	2	1.8	3.2

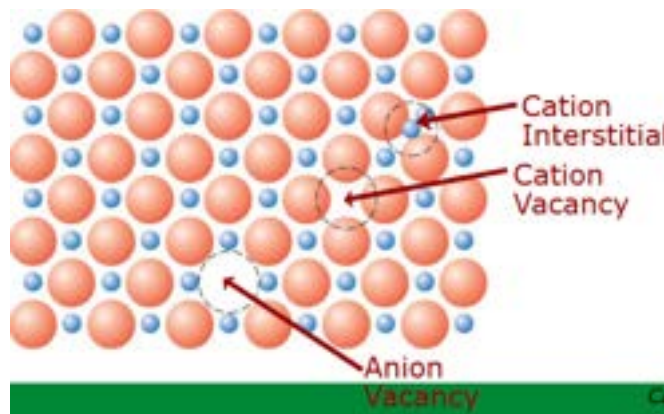
Figure 24: Typical properties of various types of zirconia

1.8 Imperfections in Ceramics

Point Defects

Vacancies, They exist in ceramics for both cations and anions

Interstitials, Exist mostly for cations, they are not normally observed for anions because anions are large relative to the interstitial sites



Frenkel Defect, , a cation vacancy-cation interstitial pair occur

$$N_{fr} = \sqrt{NN^*} \exp\left(\frac{\Delta H}{2RT}\right)$$

Where N is the number of normally occupied positions, N^* is the number of available positions for the moving ion, ΔH of formation is the enthalpy formation of one Frenkel defect, $R = 8.63 \cdot 10^{-5} \text{ eV} \cdot \text{K}^{-1} \cdot \text{atom}^{-1}$ is the gas constant, and T is the temperature in Kelvin.

Shottky Defect, a paired set of cation and anion vacancies

$$N_s = \sqrt{NN^*} \exp\left(\frac{\Delta H}{2RT}\right)$$

Where N is the number of normally occupied positions, N^* is the number of available positions for the moving ion, ΔH of formation is the enthalpy formation of one Schottky defect, $R = 8.63 \cdot 10^{-5} \text{ eV} \cdot \text{K}^{-1} \cdot \text{atom}^{-1}$ is the gas constant, and T is the temperature in Kelvin.

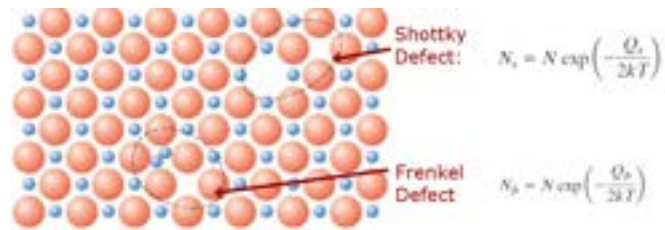


Figure 25: Equilibrium concentration of defects is maintained

Calculating defect concentrations from composition

When a ceramic is non-stoichiometric, the type and concentration of defects can be determined from the chemical composition.

Example: Spinel (MgAl_2O_4)

1. Determine formula units per fixed anion count

- Spinel structure has 4 O^{2-} per formula unit
- From oxide mol%, calculate total cations and total O atoms
- Find cations per 4 O^{2-}

2. Compare to stoichiometric formula

- Stoichiometric MgAl_2O_4 : 3 cations per 4 O^{2-}
- If cations < 3: cation vacancies
- If cations > 3: interstitial cations or anion vacancies

3. Calculate vacancy percentage

$$\% \text{vacancies} = \frac{3 - (\text{actual cations per 4 O})}{3} \times 100\%$$

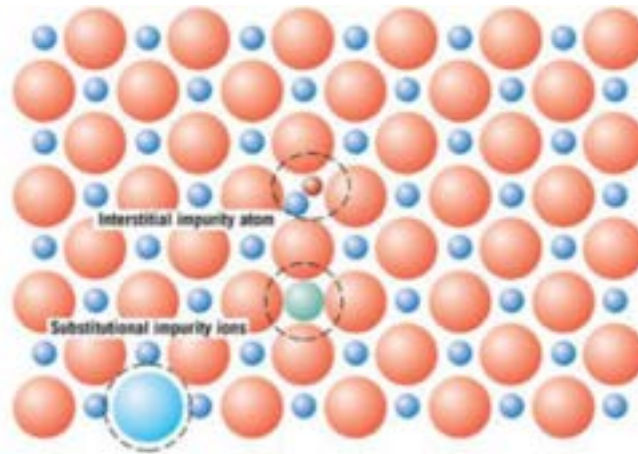
Charge balance check The total positive charge must equal total negative charge:

$$\sum (\text{cation valence} \times \text{number of cations}) = 8 \text{ (for 4 } \text{O}^{2-}\text{)}$$

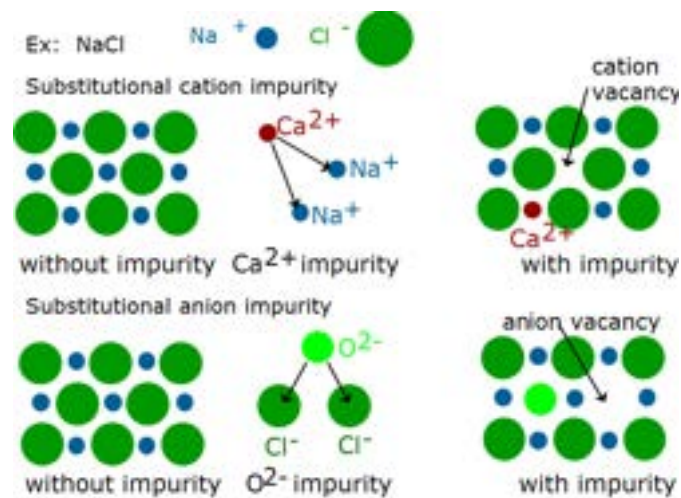
This confirms the defect type.

Impurities in Ceramics

- Impurities can form solid solutions
- Size and charge must be close to the host ion, else defects will form
- Substitutional and interstitial

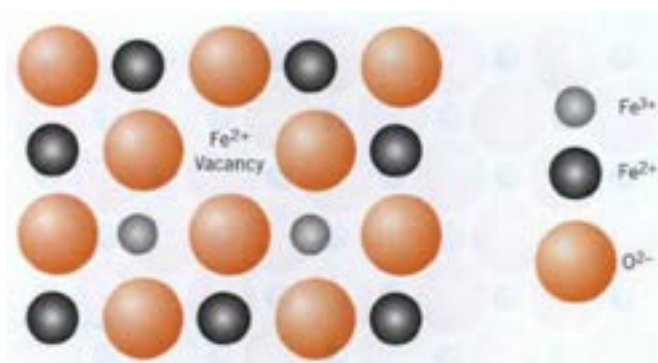


- Electronegativity (charge balance) must be maintained when impurities are present.



Stoichiometry

- Exact ratio of cations to anions as predicted by chemical formula for a given compound
- Any deviation from this leads to non-stoichiometric compound



- Fe^{2+} vacancy in FeO from Fe^{3+}
- FeO becomes Fe_{1-x}O

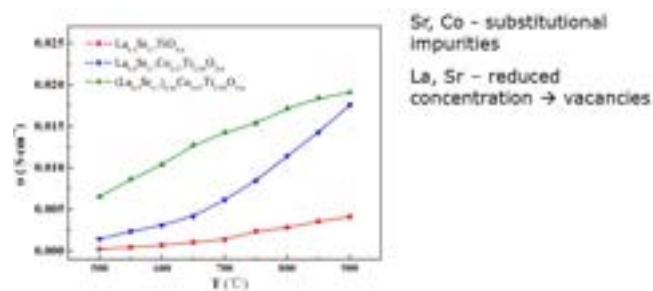
Imperfections in Ceramics

- Pores

- Unwanted in dense ceramics
- Wanted in porous ceramics
- Can be controlled
- Cracks
 - Usually unwanted ceramic components, negative impact on mechanical properties
 - Should be avoided

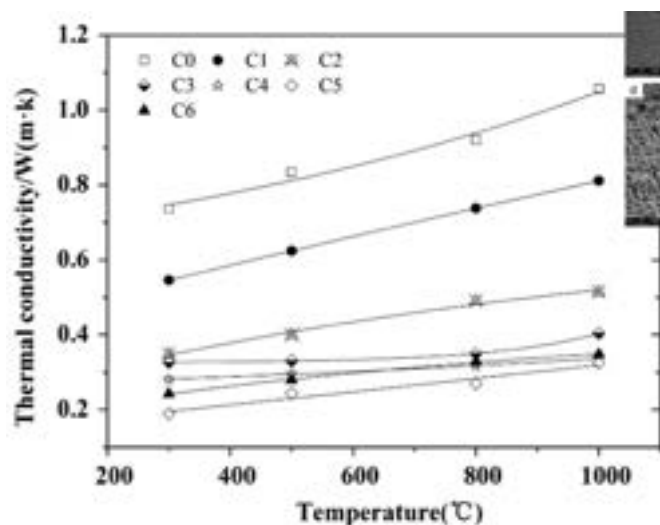
Introduction of point defects

- (La, Sr)TiO₃ typo compound
- Introduction of substitutional impurities lead to introduction of vacancies
- Increased conductivities



Introduction of pores

- Al₂O₃ ceramics, impact of pores on thermal conductivity
- C0, with a low porosity 30%
- C6, with a high porosity 65%
- High porosity leads to low thermal conductivity which entails better thermal insulation



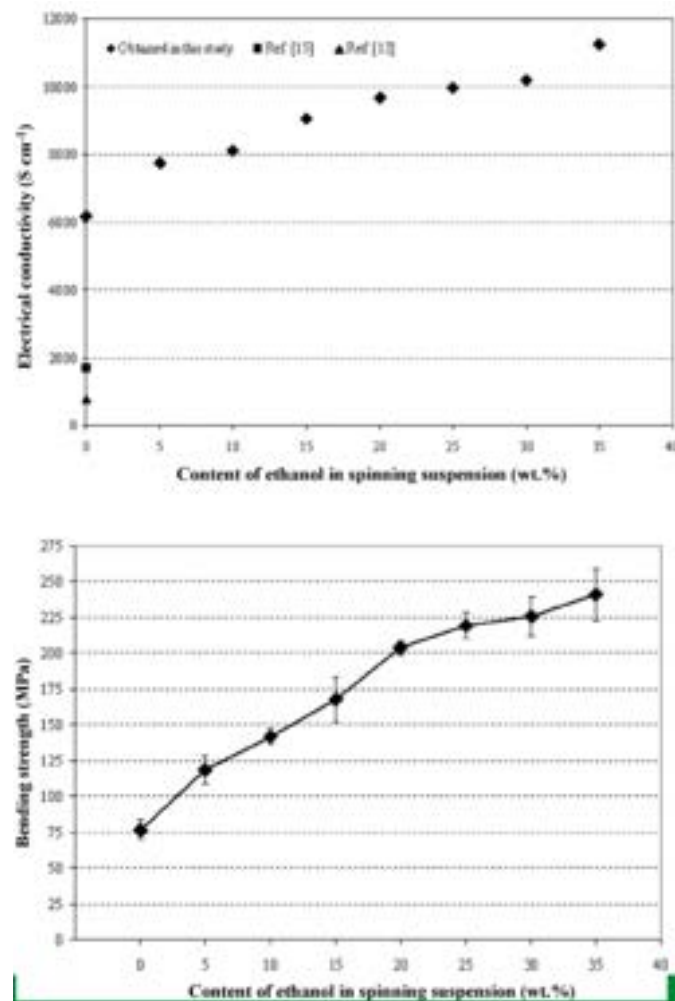


Figure 26: Electrical properties, based on the microstructure

Density of ceramics

- Crystallographic (the perfect lattice of one phase)
 - XRD
- Bulk (combined density of all phases, porosity, etc.)
 - Geometry
 - Archimedes
- True (if no porosity)
 - Pycnometry
- Relative (your material's bulk density vs. theoretical)
- Tap (how big of a breaker do I need for this powder)

Density computations for ceramics

$$\rho = \frac{n'(\sum A_c + \sum A_A)}{V_C N_A}$$

Where, n' is the number of formula units/unit cell's,

$\sum A_C$ is the sum of atomic weights of all cation in the formula unit,

$\sum A_A$ is the sum of atomic weights of all anion in the formula unit.

V_C is the volume of the unit cell and,

N_A is Avogadro's number

Density of Material classes

In general $\rho_{\text{metals}} > \rho_{\text{ceramics}} > \rho_{\text{polymers}}$, this happens because,

- Metals have close packing (metallic bonding), and often large atomic masses
- Ceramics have less dense packing, and often lighter elements
- Polymers have low packing density (often amorphous), and lighter elements (C, H, O)
- Composites have intermediate values

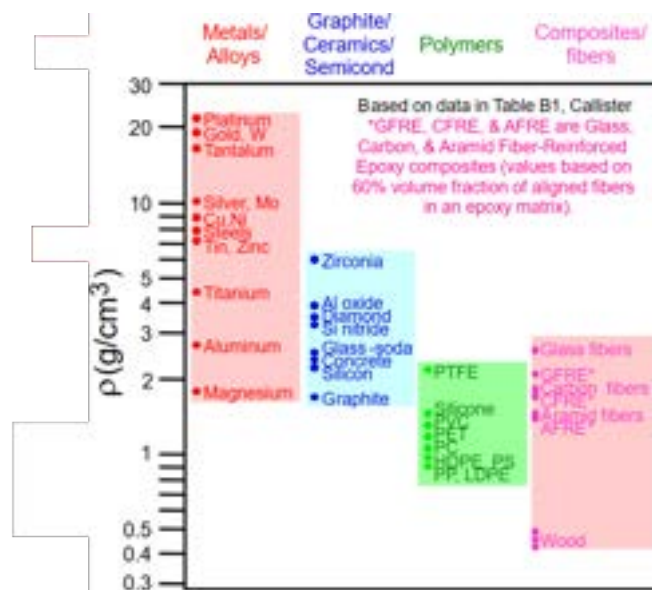


Figure 27: Densities of Material Classes

1.9 Processing of ceramics



Figure 28: Typical ceramic component fabrication (glass ceramics have a different fabrication process)

Powder synthesis

Need to produce the required compound from starting materials

- Finalize composition
- Decide the physical characteristics
- Decide the physical characteristics
- High sintering activity
- Economic and environmentally friendly process

Different synthesis methods exist, classified based on reaction schemes

- Solid phase reactions
- Liquid phase reactions
- Gas phase reactions

Outputs → Powders, films

Synthesis of ceramics

Solid-Solid reactions

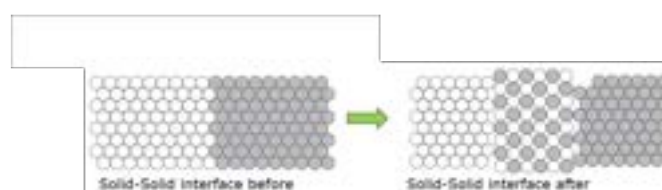


Figure 29: Solid-Solid interface before and after Solid-Solid reactions

$D(T) = D(0)e^{-E_a/RT}$ and $\langle \Delta x^2 \rangle = 2Dt$ form the basis for Solid-Solid reactions

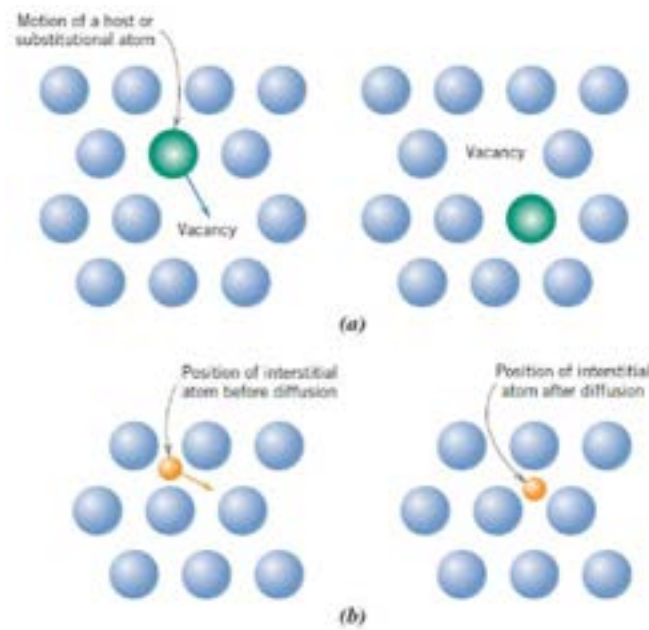
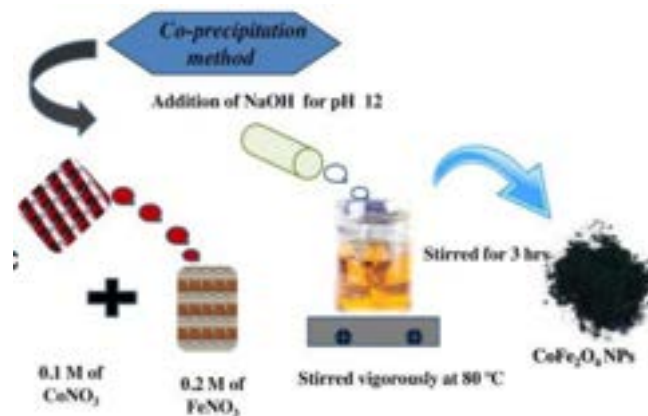


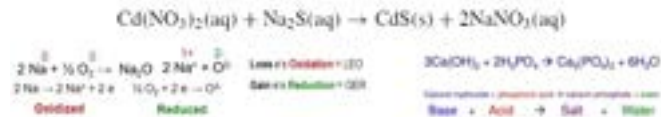
Figure 30: (a) Vacancy diffusion. (b) Interstitial diffusion.

Powder size, mixing (homogenization), temperature and time are critical parameters.

Solid-Liquid: Precipitation

- When the product of any of these solution-phase chemical reactions is an insoluble product, it is called a precipitation reaction.
- Versatile method due to control of kinetic with temperature and solvent type / concentration.
- Typical reactions involved are:
 - Double exchange reactions
 - Redox reactions
 - Acid-base reactions





Solid-Vapour Reactions

- Vaporization of the precursors are condensation on substrates.
- Mostly used to form the thin films and coatings.
- Temperature driven process.
- Film growth stages: (1) Nucleation, (2) Crystal growth, (3) Grain growth.

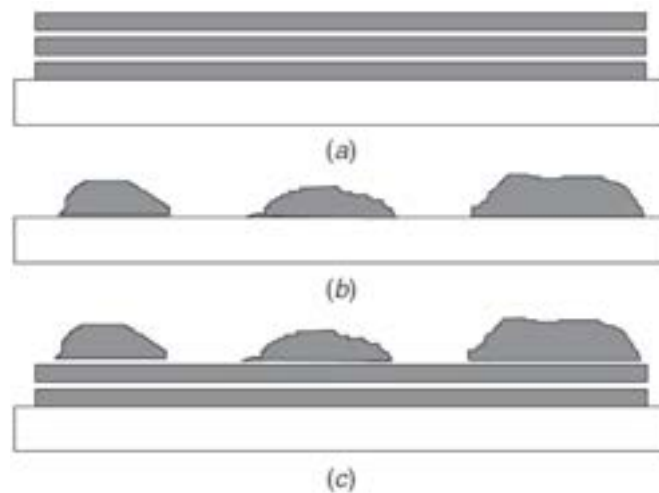
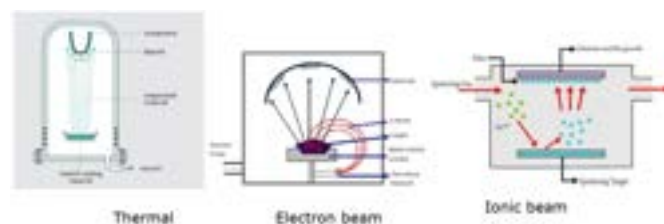


Figure 31: Different types of Growth:(a) Layered growth. (b) Island growth. (c) Combined growth.

Solid-Vapour: Physical Vapour Deposition

- The process involves evaporation of material and condensation as thin film on a substrate.
- Different sources of energy are used to evaporate the material: thermal, electron beam, ionized gas etc.
- Requirements: Source material must have a reasonable vapor pressure. The composition of the vapor produced must be the desired chemical formula and the vapor must stick to the substrate



Solid-Vapour: Chemical Vapour Deposition

- Chemical vapor deposition: $\text{A}(\text{g}) \rightarrow \text{A}(\text{surface}) \rightarrow \text{B}(\text{surface})$
- Surface is referred to as substrate and vapour phase as precursor.

- Preparation of precursor in vapour phase as precursor
- Preparation of precursor in vapour form is key and quite demanding
- Important issues: structural coincidence, chemical reactivity, thermal compatibility, and substrate morphology.
- Epitaxial films, film structure is the same as substrate.
- Homoepitaxial, film material is the same as the substrate.
- Heteroepitaxial films, film material is not the same as the substrate

Powder processing

Particle size and size distribution are the most important parameters.

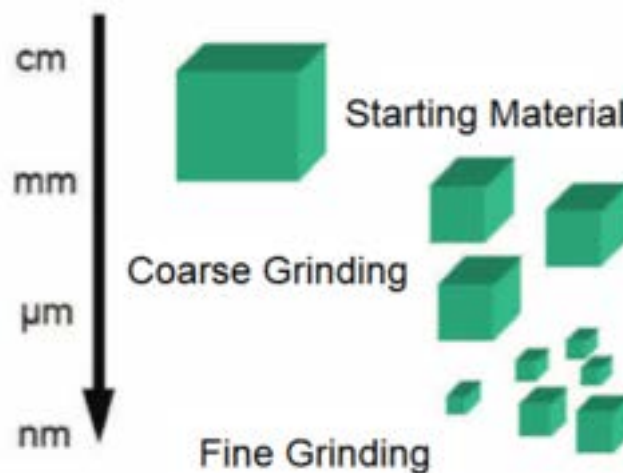
Important to have suitable physical characteristics from powders

- Modification of physical properties
 - Grinding, milling → results in required powder size and distribution.
- Heat treatment
 - provide heat energy → increase in size
- Powder conditioning
 - Required additives → possible to shape them as required

Grinding

In the context of comminution, grinding refers to the reduction of small pieces after crushing to fine powder.

Accomplished by abrasion, impact, and/or compaction by hard media such as balls or rolls.



Examples of Grinding:

Roller Mill:

Stock is compressed against flat horizontal table by rollers riding on the table surface.

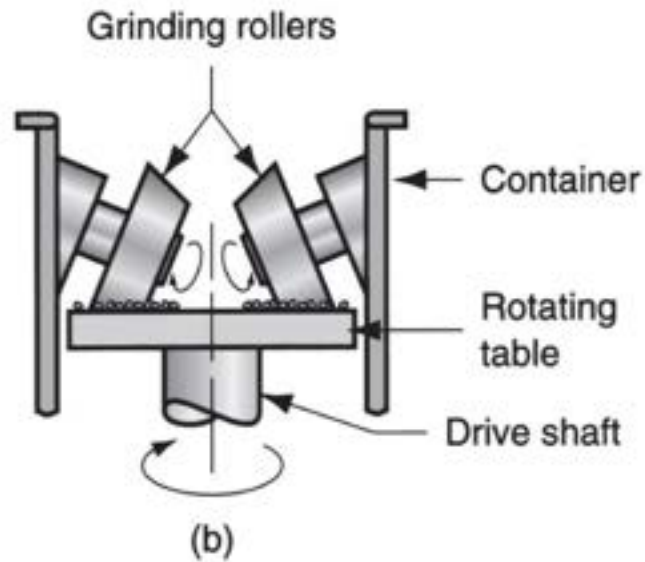


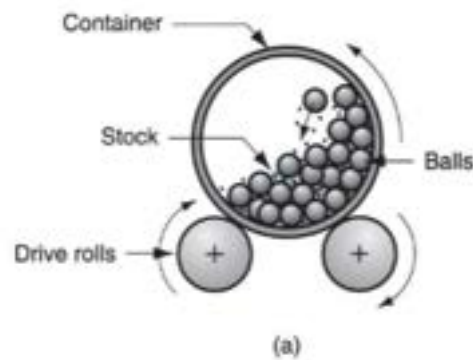
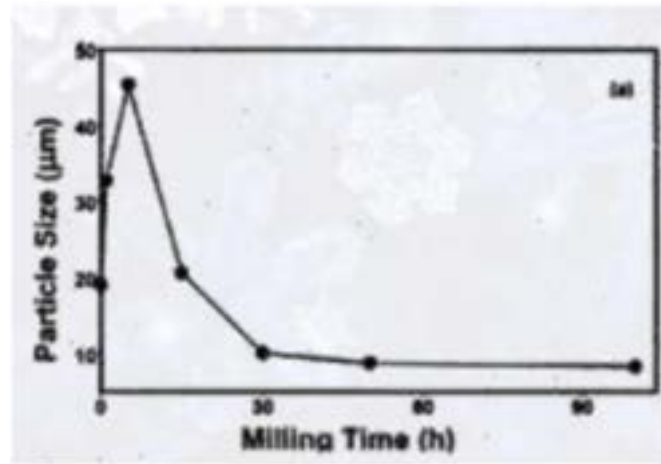
Figure 32: Roller Mill

Ball Mill:

Hard spheres mixed with stock are rotated inside large cylindrical container.

Mixture is carried upwards in container as it rotates, then dropped by gravity to accomplish grinding action.

Challenge: contamination with milling media.



Particle size modification, Heat treatment

- It may be necessary to increase the particle size
- Can be done by heating the powder-heat treatment/calcination.
- Temperature, time and atmosphere are critical parameters
- Influences the crystallinity
- Influences the sintering activity

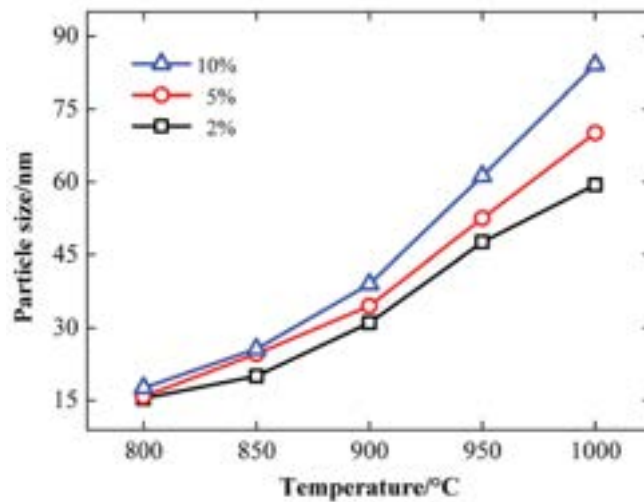


Figure 33: CeO₂ synthesis, effect of heat treatment temperature

Powder conditioning and shaping

Shaping and Forming

- To be used → the powders need to be shaped
- Dry / wet shaping process
- Simple geometries to very complex shapes
- Different thicknesses
- *Different shaping processes:*
 - Compaction
 - Slip casting
 - Tape casting
 - Screen printing
 - Extrusion
- Drying/additives removal, need to remove the additives, quite complex process.

To form/ shape component, the ceramic powder need to be transformed into slurry / paste / ink, introduce plastic characteristic.

Requires addition of other organic materials to ceramic to achieve this.

Additive	Function	Examples
Dispersant	Keep the particles dispersed and stable in solvents	Polyvinylpyrrolidone (PVP), Polyacrylates and Citrates (< 1 wt.%)
Binder	Provides the binding action between the ceramic particles	Polyvinyl alcohol (PVA), Polyvinyl butyral (PVB), Ethyl Cellulose and clay particles (< 5 wt.%)
Plasticizer	Increase the plasticity the green component	Polyethylene glycol (PEG) and glycerol (0.1 – 1 wt%)

Figure 34: Types of additives

Stabilization of ceramic powders

- Important for ceramic particles to be in suspended state for forming them into defect free components
- Faster → non-uniform slurries / inks → defects in components
- Requires stability for days / weeks / months.
- Without stability → cannot be used
- Different additives can be used to stabilize the particles through different mechanisms Inherent surface characteristics and interaction with solvents are very important.



Figure 35: Stabilization of ceramic powders, Mechanisms

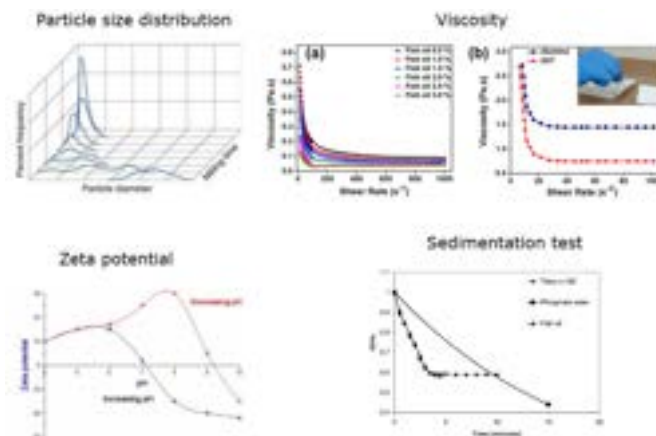


Figure 36: Slurry characterization

Forming and shaping

- Powders/slurries/pastes/inks will be formed / shaped into the required or close to be required component shape and dimensions by using one of the following processes.

Compaction,

- Powder compaction process
- Uniaxial or isostatic pressure application
- Density distribution can be different depending on pressure mode
- Required to maintain aspect ration in uniaxial mode.

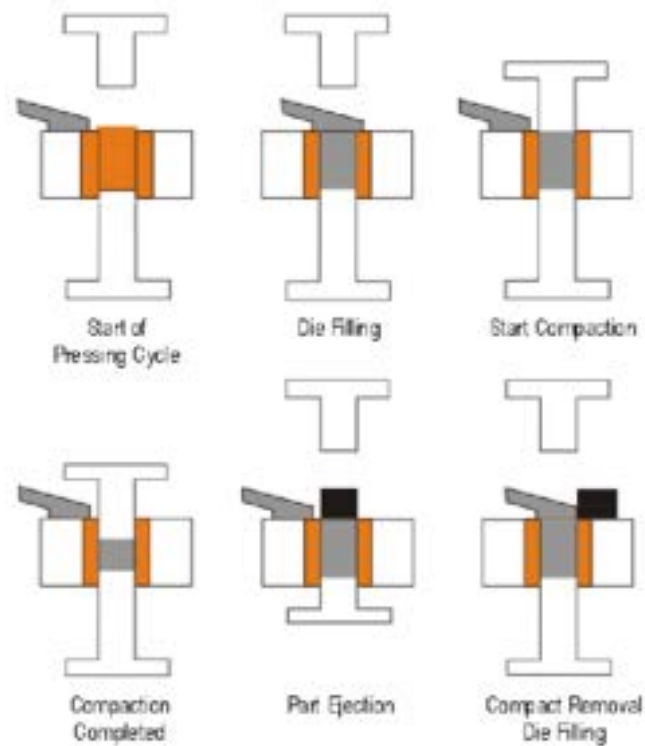


Figure 37: Compaction

Slip casting,

- Mostly used in traditional ceramics fabrication
- Suitable for producing complex shaped
- Accurate thickness control is difficult

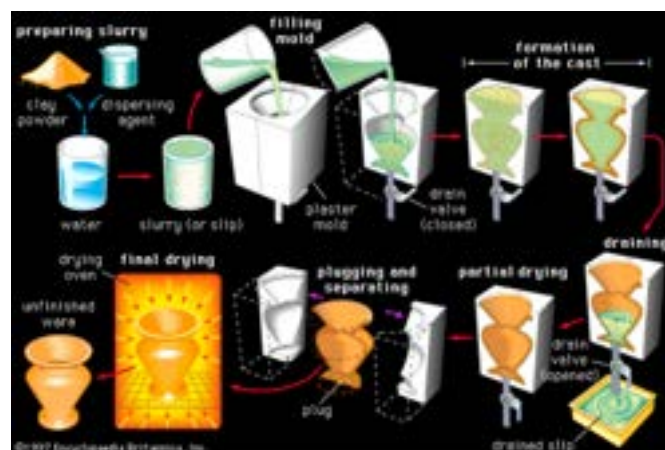


Figure 38: Slip casting

Tape casting,

- Also known as doctor blade process
- Common process to produce ceramic thick layers (0.005 mm to 1 mm)
- Suitable for mass production

- Used for the fabrication of capacitors, solar cells, solid oxide cells, batteries

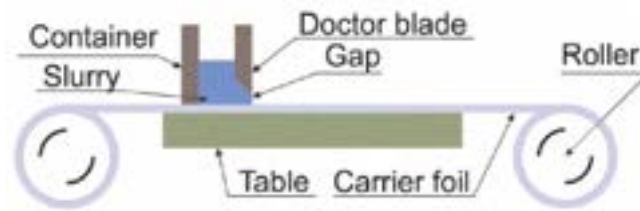


Figure 39: Tape casting

Screen printing,

- Used for more than 1000 years
- Traditionally used in printing of images in cloth
- Suitable for mass production
- Thickness from 0.005 mm to 0.1 mm possible
- Used for fabrication of solid oxide cells, semiconductors, electronic circuit boards, capacitors, solar cells etc.



Figure 40: Screen printing

Extrusion,

- Versatile method used right from food industry to metals to ceramics
- Uniform cross sectional area
- Can be continuous process
- Product is called extrudate
- The extrusion material is thermoplastic

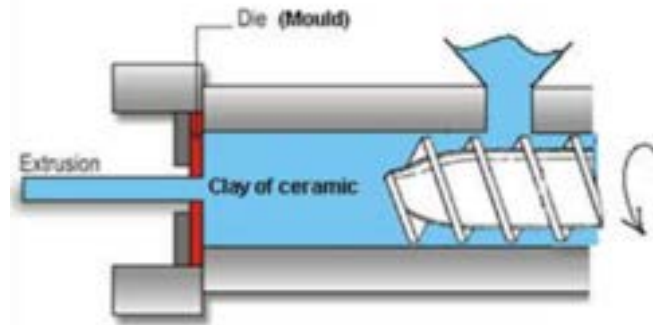


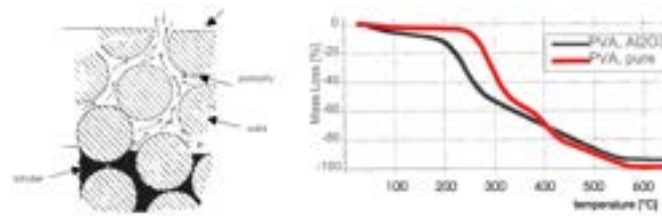
Figure 41: Extrusion

Drying

- *Drying*, elimination of free and adsorbed water (included capillary water)
- The processes results in shrinkage of the component (reduction in the dimensions)
- *Stages in drying*:
 - Initial increase in drying rate
 - Constant drying rate
 - Final decrease in drying rate
- *Types of drying*:
 - Conventional drying (room temperature)
 - Oven drying
 - Microwave drying
- *Problems*:
 - Liquid evaporation in the constant rate period should not be too fast, otherwise the liquid flow from the interior could not compensate the liquid loss and the surface would dry and shrink preferentially → stresses → cracking or deformation (warping)
 - Similar problems can arise when in the body there is a gradient in moisture or particle

Organics removal (Debinding)

- *Organics removal*, the elimination of organic binders and additives, in the pre-firing stage
- *Methods of removal*
 - Melting of organics and absorbing them in powder bed
 - Removal of organic materials by dissolving in solvents followed by thermal decomposition
 - Thermal decomposition of organic materials.
- Thermal decomposition: Breaking of bonds in the organics and remove them in gaseous form. This method is most commonly used.



- Temperature: room temperature to 1000 °C
- Important to control the rate of removal
- Can result in cracks, same as in drying.

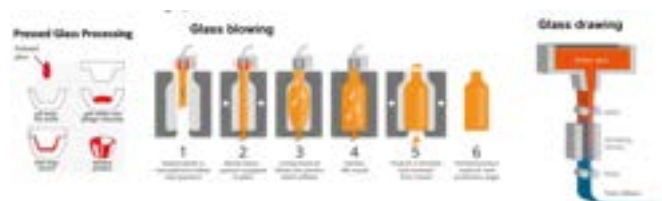


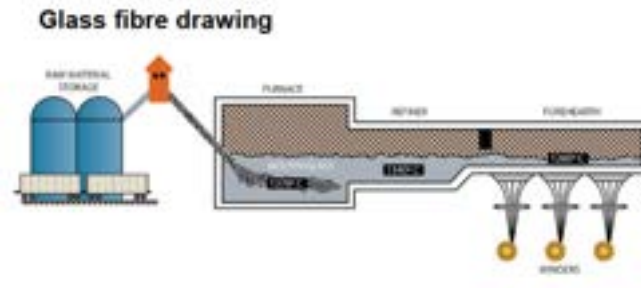
Figure 42: Batch ovens, suitable for limited volume of samples



Figure 43: Continuous ovens, suitable for large volume (mass production) of samples.

Glass forming processes





Glass tempering

- To increase the strength of the glass
- Intentionally induce compressive surface stresses
- Cooling from liquid to solid → Stresses → Annealing

Table 1: Comparison of ceramic shaping methods

Method	Thickness range	Shape complexity	Production volume	Common applications
Tape casting	0.005–1 mm	Flat, large area	High	Capacitors, substrates, solid oxide cells, multilayer ceramics
Screen printing	0.005–0.1 mm	Flat, patterned	High	Circuits, sensors, solar cells, thick films
Slip casting	1–10 mm	Complex (hollow)	Low–medium	Tableware, sanitaryware, art ceramics
Extrusion	>1 mm	Constant cross-section	High	Tubes, bricks, honeycomb catalysts, rods
Dry pressing	>0.5 mm	Simple to moderate	High	Small components, tiles, spark plugs
Injection molding	>0.1 mm	Very complex	High	Turbocharger rotors, precision parts
Gel casting	>0.5 mm	Complex	Low–medium	Near-net shape complex parts

- **Thickness control:** Tape casting and screen printing offer the best control for thin layers ($< 100 \mu\text{m}$)
- **Area:** Tape casting is ideal for large-area flat components
- **Complexity:** Injection molding and slip casting excel for 3D complex shapes
- **Cost:** Dry pressing is economical for high-volume simple shapes

How to choose the right shaping method

When selecting a ceramic shaping equipment, consider these factors in order:

1. Desired thickness:

- $< 10 \mu\text{m}$: Screen printing, spin coating, or PVD/CVD
- $10 - 100 \mu\text{m}$: Screen printing or tape casting

- $100\text{ }\mu\text{m} - 1\text{ mm}$: Tape casting
- $> 1\text{ mm}$: Dry pressing, slip casting, extrusion, or injection molding

2. Component geometry:

- Flat plates/disks: Tape casting, screen printing, dry pressing
- Tubes/rods: Extrusion
- Complex 3D shapes: Injection molding, slip casting, gel casting
- Hollow shapes: Slip casting

3. Production volume:

- Laboratory/small scale (< 100 parts): Slip casting, gel casting
- Medium scale ($100 - 10,000$ parts): Dry pressing, extrusion
- Mass production ($> 10,000$ parts): Tape casting, screen printing, injection molding

4. Tolerance requirements:

- Tight tolerances ($\pm 1\%$): Dry pressing, injection molding
- Moderate tolerances: Tape casting, screen printing
- Loose tolerances: Slip casting

5. Material considerations:

- Some methods require specific rheological properties (slip for casting, paste for screen printing)
- Particle size and distribution affect packing and final properties

For the exercise requiring a $225\text{ }\mu\text{m}$ TiO_2 layer ($100 \times 150\text{ mm}$) and a $10\text{ }\mu\text{m}$ Al_2O_3 layer on top:

- TiO_2 : Tape casting is ideal because:
 - Thickness $225\text{ }\mu\text{m}$ is within tape casting range ($5\text{ }\mu\text{m} - 1\text{ mm}$)
 - Large area ($100 \times 150\text{ mm}$) is efficiently produced
 - Excellent thickness uniformity across large area
- Al_2O_3 : Screen printing is suitable because:
 - $10\text{ }\mu\text{m}$ is typical for screen printing
 - Can deposit precisely on top of existing layer
 - Pattern can be controlled by screen design
 - Cost-effective compared to PVD/CVD

Sintering and interplay between materials

Sintering

- Sintering, a process of consolidating the ceramic powder by the application of heat or pressure or both without melting the material
- Necessary to achieve the required mechanical, thermal, electrical and other functional properties in ceramic
- Critical parameters
 - Powder characteristics

- Phase composition
- Powder composition
- Temperature, time, heating rate, pressure and atmosphere

Sintering types

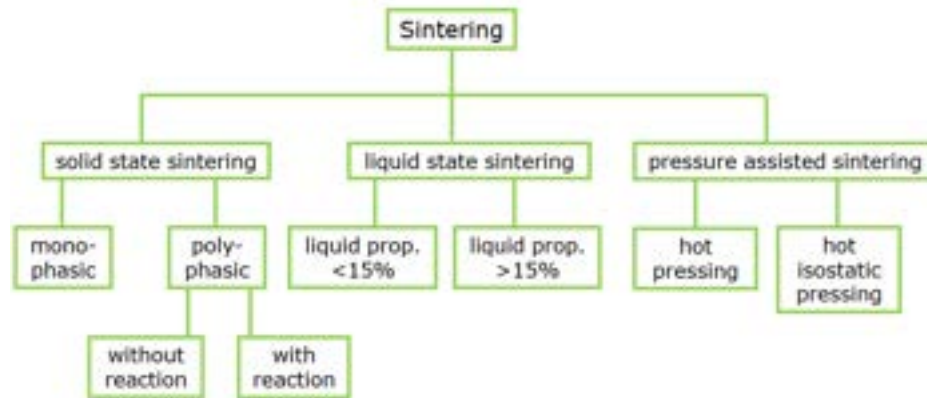


Figure 44: Sintering classification

Solid state sintering.

- Mass transport is through solid state diffusion predominantly
- Little ($< 2\text{vol.}\%$) or no liquid phase present
- Most non-silicate ceramics are sintered by them method

Liquid state sintering

- The predominant densification mechanism is through rearrangement of the solid particles "submerged" in and wetted by the low viscosity liquid
- Can be through dissolution and re-precipitation of the solid
- Most refractory carbide systems are sintered by this method (e.g., WC/Co)

Pressure assisted Sintering

- increases the contact pressure of particles and thus the driving force for sintering, compared to pressure-less sintering
- The magnitude of pressure can be 20-40 MPa for hot pressing and 200-300 MPa for hot isostatic pressing (HIP)

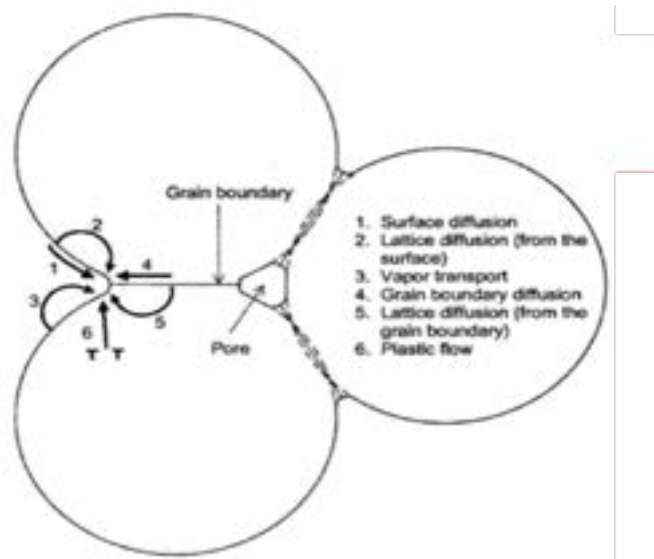
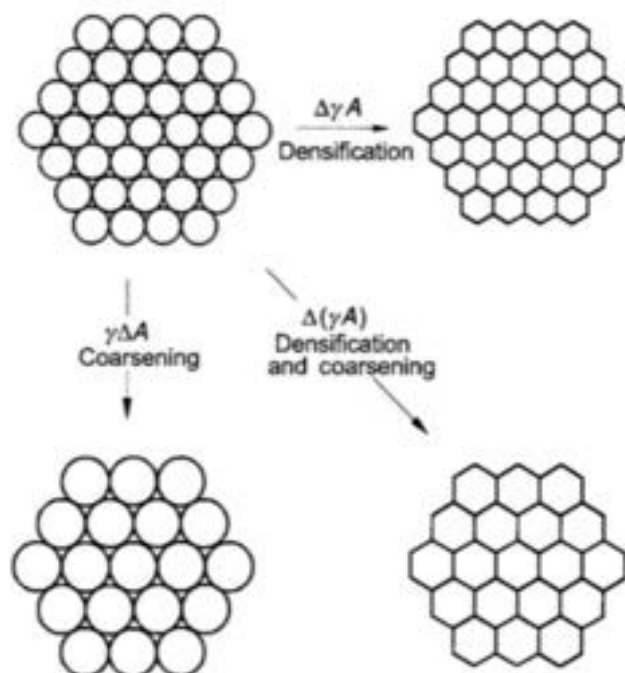


Figure 45: Sintering mechanism

Six distinct mechanisms can contribute to the sintering of consolidated mass of crystalline particles:

1. Surface diffusion
2. Lattice diffusion from the surface
3. Grain boundary diffusion
4. Lattice diffusion from the grain boundary
5. Plastic flow

The fundamental driving force for sintering is the reduction of internal interface/surface energy

Figure 46: Basic phenomena occurring during sintering under the driving force for sintering $\Delta(\gamma A)$

$$\Delta(\gamma A) = (\Delta\gamma)A + \gamma(\Delta A)$$

Where γ is the interface energy density and A is the interface area.

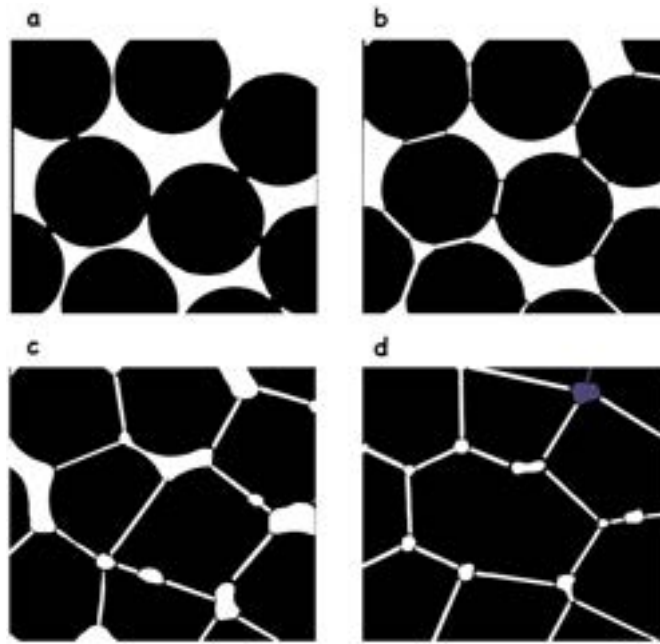
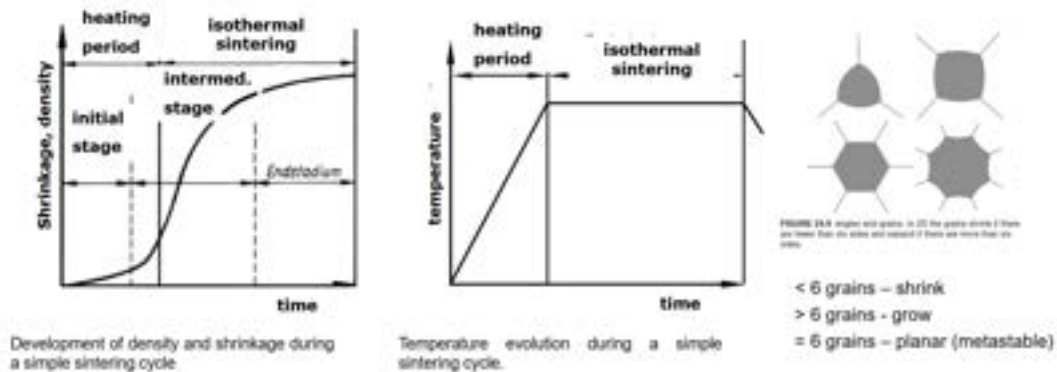
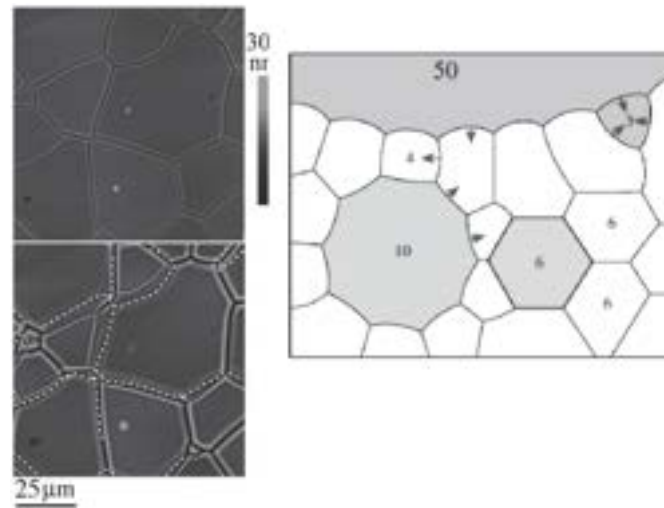


Figure 47: (a) Green body, loose powder. (b) Initial stage: increase of the interparticle contact area from 0 to 0.2 grain diameter, increase of the density from 60 to 65%. (c) Intermediate stage: further increase of the contact area, stage characterized by continuous pore channels along three grain edges, increase of the density from 65 to 90%. (d) Elimination of the pore channel along three grain edges, increase of the density to 95-99%

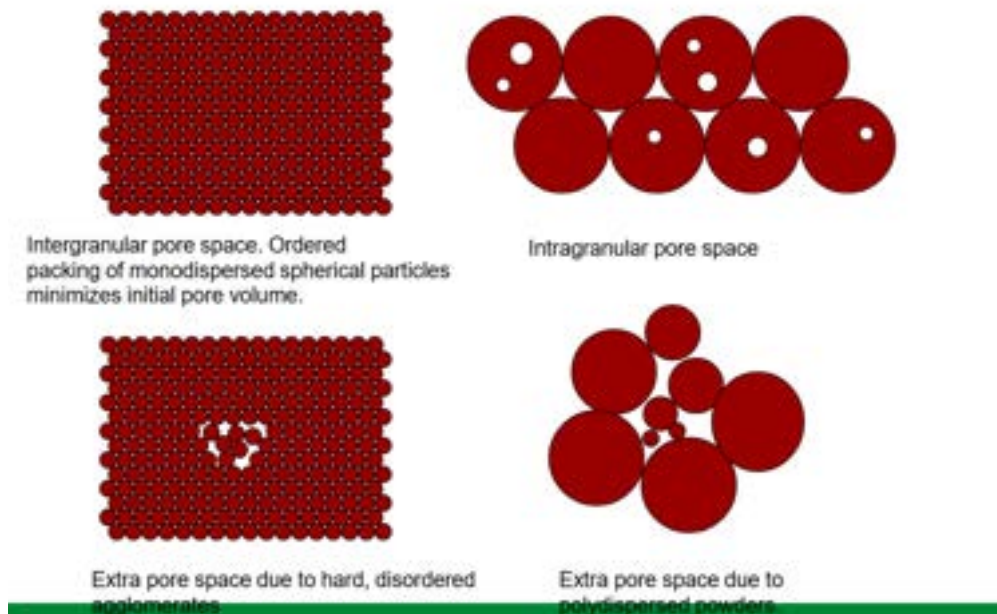


$$\frac{\Delta L}{L_0} = \text{Linear shrinkage}, \quad \frac{\Delta V}{V_0} = \text{Volumetric shrinkage}$$



Porosity of sintered bodies

Pores are important and unwanted elements of a ceramic microstructure. The final pore space in an sintered ceramic is mainly a function of the pore volume in the greenbody. Source of pores



Evolution of porosity

Pores can like grains grow or shrink. The two parameters affecting pore growth is the number of surrounding grains and dihedral angles. Generally pores with few neighboring grains tend to shrink.

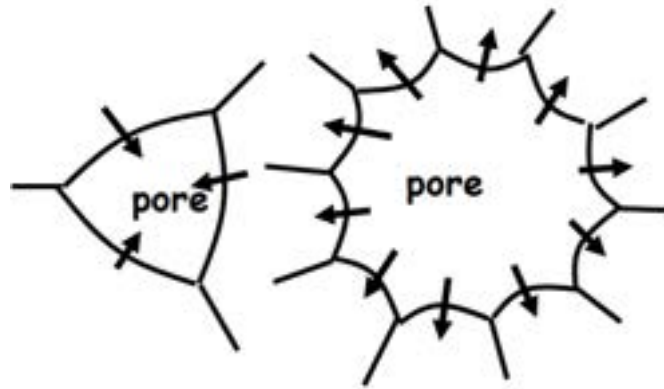


Figure 48: (Small) pores with few neighboring grains have concave surfaces and tend to shrink, whereas (large) pores with many neighboring grains have convex surfaces and tend to grow

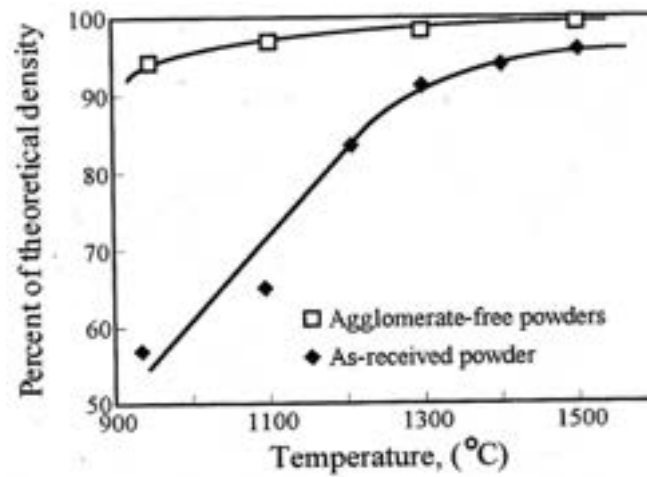


Figure 49: Sintered density of yttrium stabilized zirconia as function of powder processing. Eliminating agglomerates (through milling) enhances the sintering kinetics and allows to achieve higher densities.

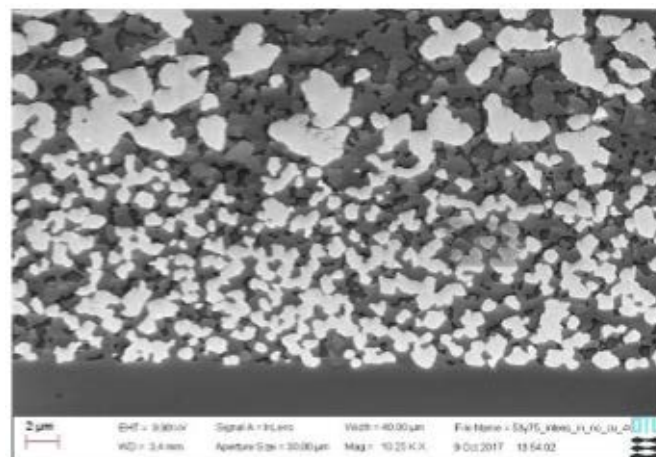
1.10 Ceramic Composites

Ceramic-Metal composites



Figure 50: WC–Co system

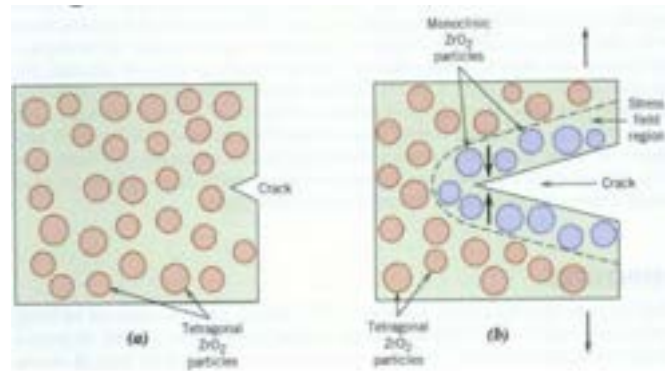
- Carbides such as WC are brittle and difficult to process
- Addition of metals such as Co in low amounts → ease of processing and improved toughness
- Extensively used in cutting tools

Figure 51: Y_2O_3 – ZrO_2 –Ni system

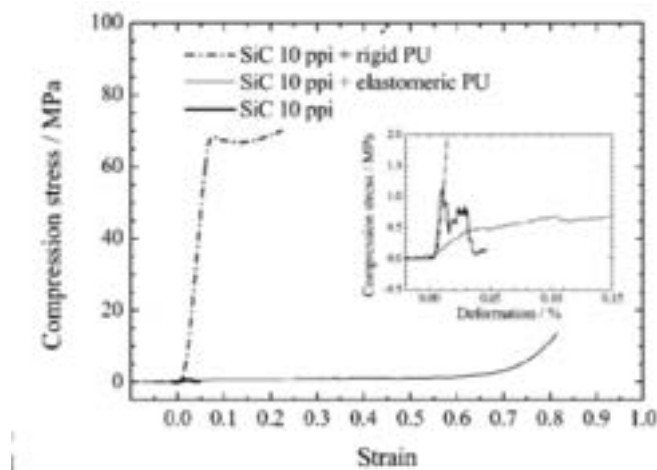
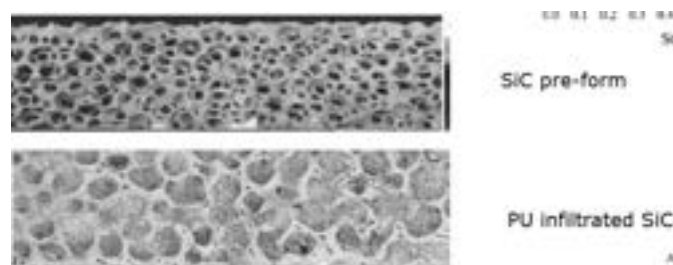
- Used as an anode in solid oxide fuel cell
- Pure Ni not enough strength and only electronic conductivity
- Addition of Y_2O_3 – ZrO_2 phase → improved strength and introduces ionic conductivity

Ceramic-Polymer composites

SiC, Polyurethane system

Figure 52: ZrO_2 system

- SiC pre-form infiltrated with PU
- significant increase in compressive strength
- Developed for ballistic applications



Ceramic-Ceramic composites

Transformation toughening.

Stress $\rightarrow$ phase transformation from tetragonal to monoclinic $\rightarrow$ volume expansion $\rightarrow$ compressive stress $\rightarrow$ shutting of crack

Whisker Content (wt%)	Fracture Strength (MPa)	Fracture Toughness (MPa $\sqrt{\text{m}}$)
0	—	4.5
10	435 $\pm$ 25	7.1
20	655 $\pm$ 135	7.5-8.0
40	850 $\pm$ 130	6.0

Figure 53: Al_2O_3 , SiC system

Cutting tool inserts, longer life time than cermet tools.

Nano composites

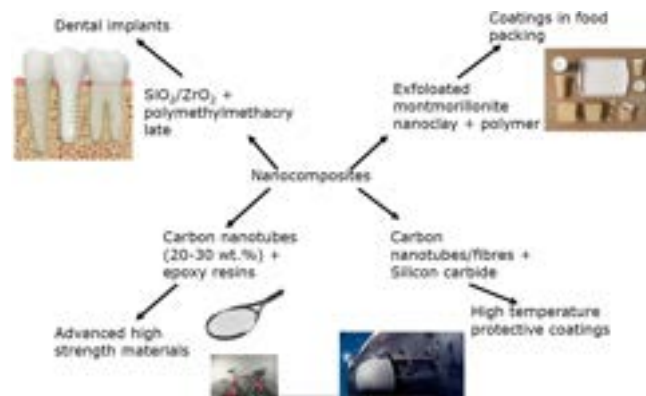
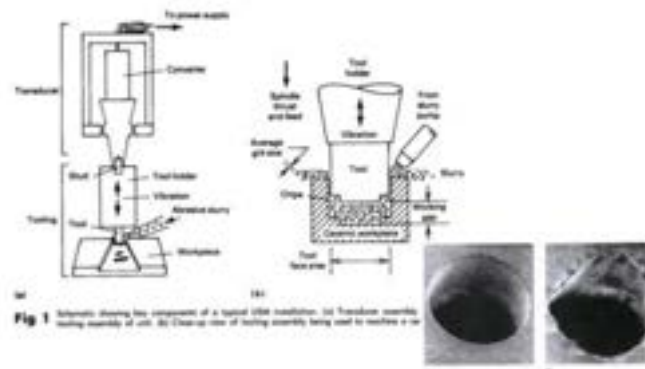


Figure 54: Nano composites

Machining and finishing

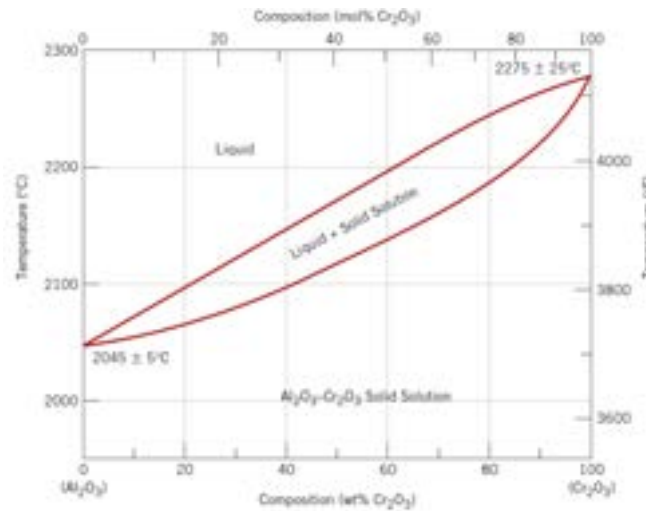
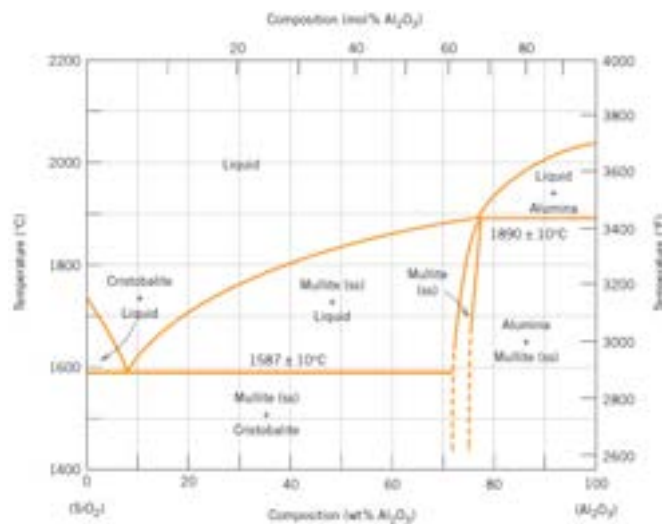
- Most ceramics are hard and brittle, so machining is difficult
- Machining before firing is called green machining
- Machining after firing are typically nonconventional machining processes
 - Grinding, lapping, polishing, drilling, cutting, ultrasonic, laser, electron beam, water-jet, and chemical



1.11 Properties and Degradation

Ceramic Phase Diagrams

- Configuration and interpretation similar to metal phase diagrams
- Mostly experimentally determined
- Tertiary diagrams are more complicated
- Common element between two end components

Figure 55: Example of Ceramic Phase Diagram $\text{Al}_2\text{O}_3 - \text{Cr}_2\text{O}_3$ Figure 56: Example of Ceramic Phase Diagram $\text{SiO}_2 - \text{Al}_2\text{O}_3$

1.11.1 Physical Properties

Important physical properties for manufacturing of ceramic components

Particle size, size of the primary ceramic particles



Figure 57: Visualization on the effects of different effect of average particle size on SiC powder

- Important to know the size, particularly for reaction chemistry, shaping and sintering of components.
- Numerous methods exist such as, Microscopy, light scattering, sieving, etc.

Surface area, external surface area of the ceramic particles

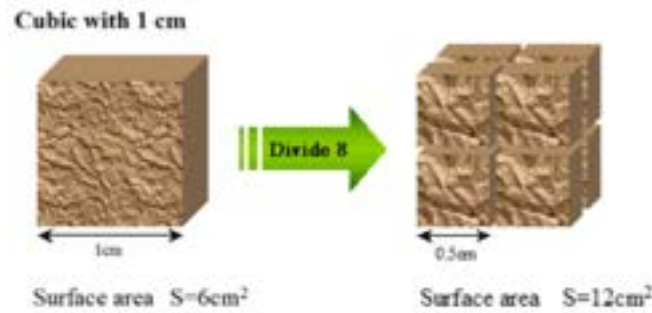
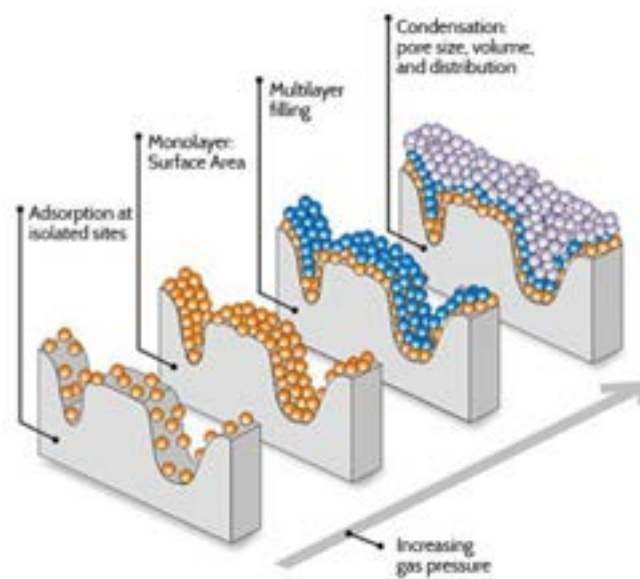


Figure 58: As can be seen above particles with the same mass $\rightarrow$ similar volume $\rightarrow$ can have significantly different surface area



- As can be seen above gas absorption methods can be used to measure the surface area of ceramic powders
- A monolayer of gas molecules $\rightarrow$ calculate the gas volume $\rightarrow$ surface area calculation
- Important to know the size, particularly for reaction chemistry, shaping and sintering of components
- Usual gasses used is Nitrogen, but it depends on whether the ceramic reacts or not with the gas, therefore an inert gas such as helium is preferred

Porosity, measure of the empty/void space in a material



Figure 59: Examples of different types of porosities found in ceramics

Hg-Porosimetry,

- Sample is exposed to Hg under viable pressures

- Measure intruded volume of Hg into the sample vs. pressure
- Pore diameter calculated from the pressure (Washburn equation):

$$p = \frac{4\gamma \cos \theta}{d}$$

Where p is the pressure, γ is the surface tension of Hg, θ is the contact angle, and d is the pore diameter

- Hg is used since it has a very small surface tension
- The only way to measure the porosity is done using Archimedes principle
- This test however is not able to analyze the closed pores
- The use of ceramics that let gas pass through it through tiny paths can be used to purify the air
- The use of ceramics that let liquid pass through it through tiny paths can be used to purify liquids

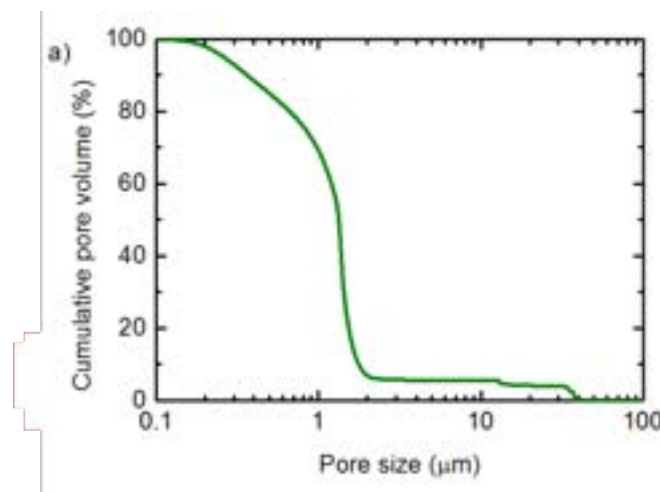


Figure 60: Example of Plot generated from Hg-Porosimetry

1.11.2 Mechanical Properties

Fracture behavior in ceramics

- Characterized commonly as brittle fracture
- Can be transgranular or intergranular
- Flaws acts as stress raisers

Brittle fracture of ceramics, formation and propagation of cracks perpendicular to applied load.

The measured fracture strength is lower than theoretical strength due to presence of microscopic flaws.

Fracture toughness

- Measure of a ceramic materials ability to resist fracture, in the presence of a crack/defect.
- Smaller than metals, since ceramics are known to be brittle and very sensitive to defects

$$K_{IC} = T\sigma\sqrt{\pi a}$$

Where K_{IC} is the fracture toughness, Y is the sample and crack geometrical parameter, σ is the applied strength and a is the crack length.

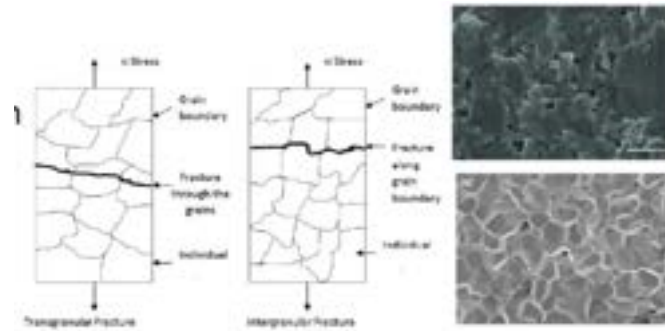
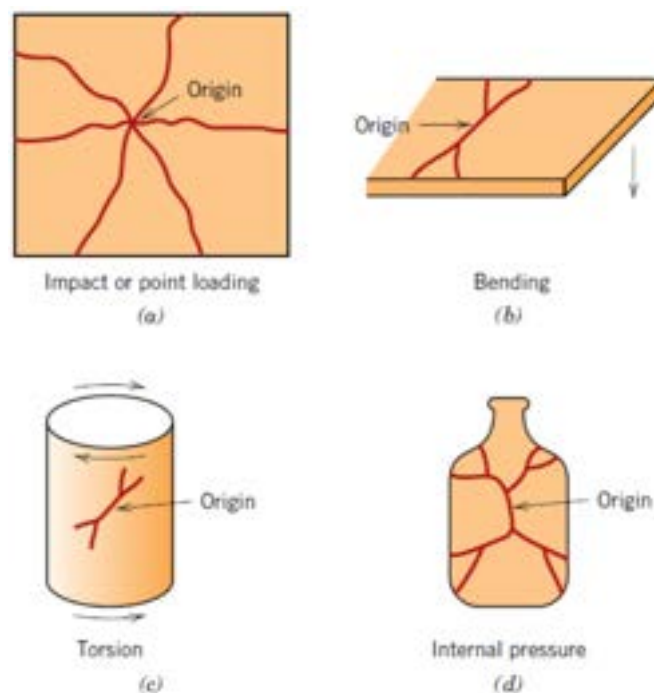


Figure 61: Crack propagation: transgranular or intergranular. Measured fracture strength is lower than predicted value.

- Slow propagation of the crack is possible under static loads → delayed fracture
- Fracture strength is dependent of fabrication method, size, shape, heat treatments, etc.
- Higher strengths (up to $10\times$) in compression compared to tension

Fractography

- Fractography is a part of failure analysis
- Investigated the formation and propagation behavior of flaws
- Focus on location, type of the defect/flaw
- Convergence of cracks → origin
- Increase in stress → increase in branches
- Interaction of crack growth with microstructure → distinctive features in microstructure
- Microstructure examination of the fractured surface → information about crack origin and propagation
- Useful in taking measures to reduce similar fractures



Stress-strain behavior

- Normal tensile testing for ceramics don't work well as they lack any plastic deformation, since there are no dislocation in ceramics opposed to metals where there are these dislocations and plastic deformation.
- The fact there are mainly ionic bonds in ceramics and their little room for deformation lead to no dislocation which leads to no plastic deformation
- Normal tensile test are not possible for ceramics, difficult to prepare samples and strains are quite low (very brittle)
- The results can often be unreliable and depend on porosity of the material as well as their location
- The use of 3 or 4-point bending is commonly used to calculate the strength at fracture → flexural strength
- There tests are called flexural tests and the stress at fracture is called the flexular strength / bend strength / modulus of rupture
- It is also know for ceramics that the flexural strength depends on sample size
- If managed, tensile strength is lower than flexural strength
- geometry and volume dependent

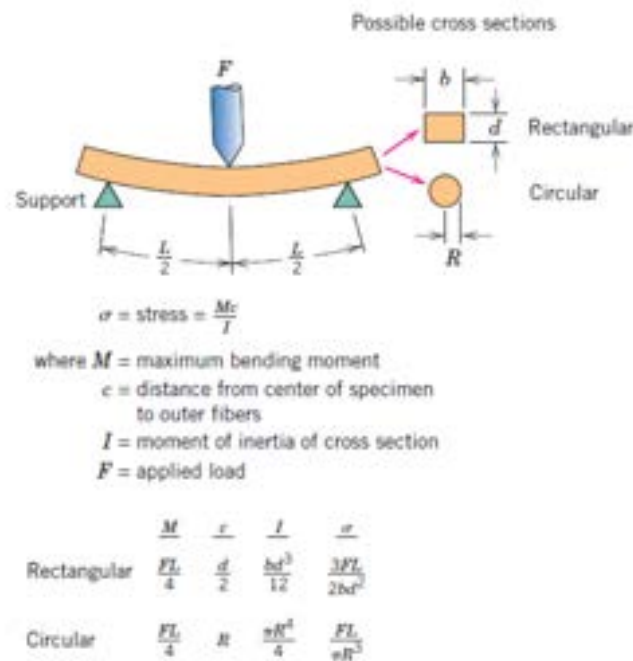


Figure 62: Example of 3 point bend test to find the flexural strength

Bending test formulas

Three-point bending Maximum stress at fracture (flexural strength):

$$\sigma_f = \frac{FL}{\pi R^3} \quad (\text{circular cross-section})$$

$$\sigma_f = \frac{3FL}{2bd^2} \quad (\text{rectangular cross-section})$$

Where:

- F = load at fracture (N)

- L = support span (m)
- R = radius (m)
- b = width (m), d = thickness (m)

Four-point bending

$$\sigma_f = \frac{F(L_i - L_o)}{\pi R^3} \quad (\text{circular})$$

Where L_i = inner span, L_o = outer span.

Statistical nature of ceramic strength Ceramic strength is not deterministic but follows Weibull distribution:

$$P_f = 1 - \exp \left[- \left(\frac{\sigma}{\sigma_0} \right)^m \right]$$

Where P_f = failure probability, m = Weibull modulus, σ_0 = scale parameter. Higher m means less variability in strength.

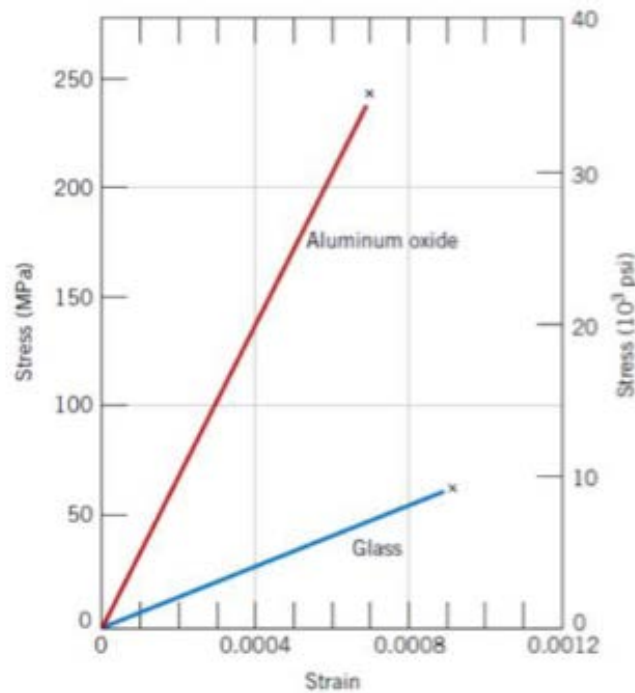


Figure 63: Example of the Stress-Strain curve of ceramics

Flexural strength

- The stress-strain behavior is similar to metals, in the initial stages
- No plastic deformation before the fracture at room temperature
- In crystalline ceramics, dislocation may exist but cannot move
- Non-crystalline ceramics will not have dislocations. Possible viscous flow at high temperatures.
- Slope in the elastic region is modulus of elasticity

Material	Flexural Strength		Modulus of Elasticity	
	MPa	ksi	GPa	10 ⁶ psi
Silicon nitride (Si ₃ N ₄)	250–1000	35–145	304	44
Zirconia ^a (ZrO ₂)	800–1500	115–215	205	30
Silicon carbide (SiC)	100–820	15–120	345	50
Aluminum oxide (Al ₂ O ₃)	275–700	40–100	393	57
Glass-ceramic (Pyroceram)	247	36	120	17
Mullite (3Al ₂ O ₃ ·2SiO ₂)	185	27	145	21
Spinel (MgAl ₂ O ₄)	110–245	16–35.5	260	38
Magnesium oxide (MgO)	105 ^b	15 ^b	225	33
Fused silica (SiO ₂)	110	16	73	11
Soda-lime glass	69	10	68	10

^aPartially stabilized with 3 mol% Y₂O₃.
^bSintered and containing approximately 5% porosity.

Figure 64: Table of Flexural strength and Modulus of elasticity for varioes ceramics

Porosity and flexural strength

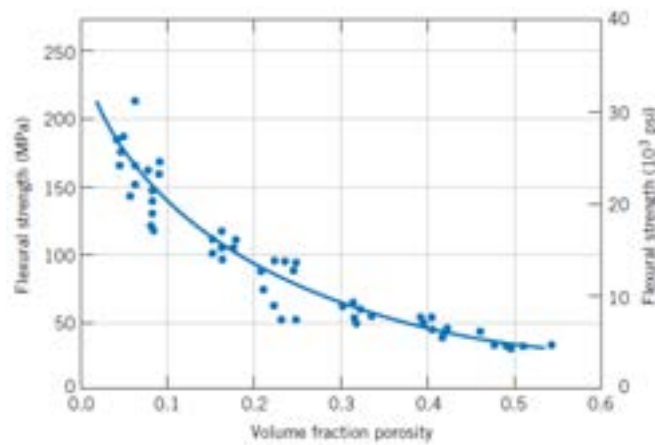


Figure 65: Example of flexural strength vs volume fraction of porosity

$$\sigma_{fs} = \sigma_0 \exp(-nP)$$

- σ_0 and n are experimental constants
- Flexural strength σ_{fs} decreases exponentially with volume fraction of porosity P
- Pore affect flexural strength badly
 - Reduce unit cross sectional area
 - Act as stress cooncentrators

$$E = E_0(1 - 1.9P + 0.9P^2)$$

- E_0 is the modulus of elasticity without pores
- Modulus o elasticity E decreases with increases pore volume fraction P

Deformation

- At room temperature, rarely measurable
- Similar mechanism as in metals
- In non-crystalline ceramics, deformation is through viscous flow

Impact of defects

- Influence of porosity, decreases the modulus of elasticity as well as flexural strength

Hardness

- Difficult to measure hardness using hardness indenters
- Vickers or Knoop techniques are used using pyramidal indenters
- Hardness decreases with increasing load, eventually flattening.

Material	Vickers Hardness (GPa)	Knoop Hardness (GPa)	Comments
Diamond (carbon)	130	160	Single crystal, (100) face
Boron carbide (B_4C)	44.2	—	Polycrystalline, sintered
Aluminum oxide (Al_2O_3)	36.5	—	Polycrystalline, sintered, 99.7% pure
Silicon carbide (SiC)	29.4	19.8	Polycrystalline, reaction bonded, sintered
Tungsten carbide (WC)	22.1	—	Fused
Silicon nitride (Si_3N_4)	18.0	17.2	Polycrystalline, hot pressed
Zirconia (ZrO_2) (partially stabilized)	11.7	—	Polycrystalline, 3 mol% Y_2O_3
Soda-lime glass	6.1	—	

Figure 66: Table of Hardness using Vickers and Knoop techniques for various ceramics

Porosity-property relationships

Table 2: Empirical equations for property dependence on porosity

Property	Equation	Notes
Elastic modulus	$E = E_0(1 - 1.9P + 0.9P^2)$ $E = E_0 \exp(-bP)$	Most common form Exponential approximation
Flexural strength	$\sigma = \sigma_0 \exp(-nP)$ $\sigma = \sigma_0(1 - P)^m$	n typically 4–7 Alternative form
Electrical conductivity	$\sigma_{el} = \sigma_0 \frac{1-P}{1+aP^2}$ $\sigma_{el} = \sigma_0(1 - P)^t$	$a \approx 11$ for spheres Percolation model
Thermal conductivity	$k = k_0(1 - P)$ $k = k_0 \frac{1-P}{1+\beta P}$	Simple linear More accurate

Where P = volume fraction porosity, subscript 0 = dense material property.

1.11.3 Thermal Properties

Thermal property is the response of the material to the application of heat. Where hat capacity, thermal conductivity and thermal expansion are the three key properties

Heat capacity

- Ability to absorb heat from surroundings
- Energy required to produce unit temperature rise
- Usually between metals and polymers
-

$$C = \frac{dQ}{dT}$$

Where dQ is the energy required to produce the temperature change of dT

- Heat capacity is measured at constant volume of the material C_V or constant pressure of the surroundings C_P . The difference between values is minimum for solids at room temperature.

Thermal expansion

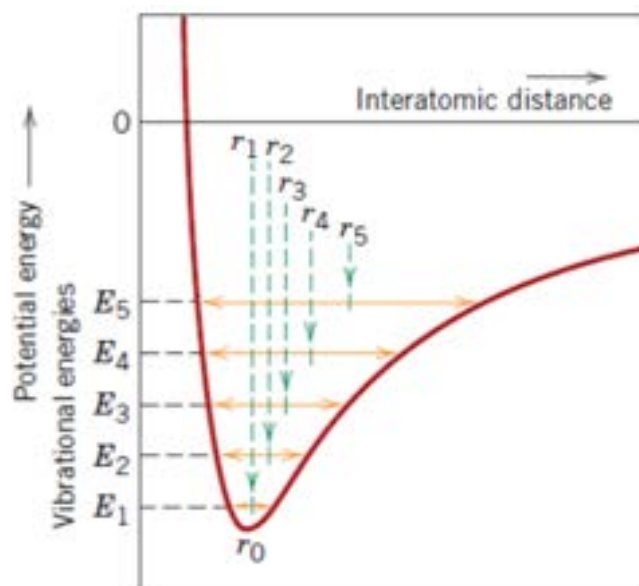
- Materials expand up on heating usually
- Expressed through the co-efficient of thermal expansion (CTE)
- Greater the bonding energy, lower the CTE
- Greater the bonding energy, lower the CTE
- Thermal shock with high CTEs
- Thermal expansion: Change in dimensions when heated.
- The linear thermal expansion, α_l is the linear co-efficient of thermal expansion used in the formulas as,

$$\frac{\Delta l}{l_0} = \alpha_l \Delta T$$

- The Volumetric thermal expansion, α_V is the volumetric co-efficient of thermal expansion used in the formula as,

$$\frac{\Delta V}{V_0} = \alpha_V \Delta T$$

- Important to note that α_V is 3 times larger than α_l for isotropic materials.



- Thermal expansion is an increase in the average distance between the atoms
- Greater the atomic bonding energy, lower the thermal expansion
- Magnitude of thermal expansion increases with increasing temperature.
- Low thermal expansion co-efficient for thermal shock resistance is required,

Thermal conductivity

- Ability of the material to transfer the heat

- Ceramics are usually thermal insulators
- Increase in $T \rightarrow$ increase conductivity, Increase in porosity $\rightarrow$ decrease in conductivity

-

$$k = \frac{QL}{A\Delta T}$$

- k is the thermal conductivity in W/(mK), Q is the amount of heat transfer through the material in J/s or W, L is the length of the body in m, A is the area of the body in m² and ΔT is the difference in temperature in K
- Heat conduction k happens through lattice vibration k_l and free electrons k_e ,

$$k = k_l + k_e$$

- Ceramics, lack of free electrons and defects like pores $\rightarrow$ low/no thermal conductivity
- Glass and amorphous ceramics $\rightarrow$ lower thermal conductivity than crystalline ceramics

Thermal stress

- Stresses due to change in temperature
- Ability to resist failure from these stresses $\rightarrow$ thermal shock in resistance

Table 20.1 Thermal Properties for a Variety of Materials

Material	c_p (J/kg·K) ^a	α_L (°C) ⁻¹ × 10 ⁻⁶	k (W/m·K) ^b	$\bar{\alpha}$ (°C·m/K) ^c × 10 ⁻⁶
Metals				
Aluminum	900	23.6	247	2.26
Copper	386	17.0	398	2.25
Gold	128	14.2	315	2.56
Iron	448	11.8	80	2.75
Nickel	443	13.3	90	2.08
Silver	235	19.7	426	2.13
Tungsten	136	4.5	170	3.26
1025 Steel	490	12.0	51.9	—
106 Stainless steel	502	16.0	15.9	—
Brass (70Cu–30Zn)	375	20.0	120	—
Kovar (54Fe–29Ni–17Cu)	460	5.1	17	2.80
Invar (64Fe–36Ni)	500	1.6	8.0	2.75
Super Invar (65Fe–32Ni–3Cu)	500	0.72	10	2.68
Ceramics				
Alumina (Al ₂ O ₃)	775	7.6	39	—
Magnesia (MgO)	940	13.5 ^d	32.7	—
Spinel (MgAl ₂ O ₄)	790	7.6 ^d	15.0 ^d	—
Fused silica (SiO ₂)	740	0.4	1.4	—
Soda-lime glass	840	9.0	1.7	—
Borosilicate (Pyrex) glass	859	3.3	1.4	—
Polymers				
Polyethylene (high density)	1850	106–196	0.40–0.50	—
Polypropylene	1825	145–180	0.12	—
Polystyrene	1770	95–130	0.13	—
Polytetrafluoroethylene (Teflon)	1880	128–216	0.25	—
Phenol-formaldehyde, phenolic	1500–1760	122	0.15	—
Nylon 6,6	1670	144	0.24	—
Polyisoprene	—	220	0.14	—

^aTo convert to cal/g·K, multiply by 2.39 × 10⁻⁴.
^bTo convert to cal/cm·K, multiply by 2.39 × 10⁻³.
^cValue measured at 100°C.
^dMean value taken over the temperature range 0°C to 1000°C.

Figure 67: Table of thermal properties for a variety of materials

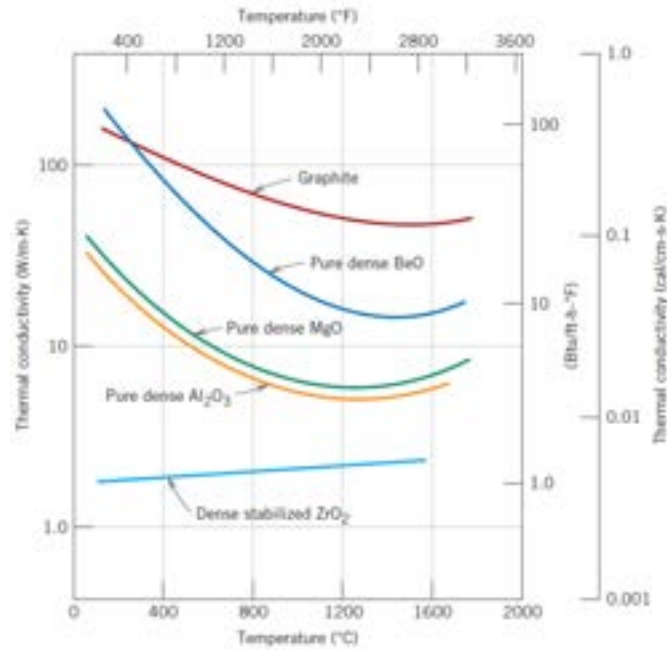


Figure 68: Visualization of thermal properties for a variety of ceramics over a range of temperature

1.11.4 Electrical Properties

Electrical conductivity

- Ease with which the material can conduct electric current
- Two components, electronic and ionic
- Can be tailored with impurities/defects
- Resistivity of materials,

$$\rho = \frac{RA}{l}$$

Where ρ is the resistivity, R is the resistance, A is the cross-sectional area and l is the distance between voltage probes

- Conductivity of materials,

$$\sigma = \frac{1}{\rho}$$

Where σ is the conductivity and ρ is the resistivity

- The conduction in ionic materials,

$$\sigma_{\text{total}} = \sigma_{\text{electric}} + \sigma_{\text{ionic}}$$

In ionic materials, due to the charge on anions and cations, the mobility under field results in conduction

- Mobility of ionic species,

$$\mu_1 = \frac{n_I e D_I}{kT}$$

Where n_I is the valence of the ion and D_I is the diffusion coefficient of the ion.

- We can also see that an increase in $T \rightarrow$ increase in ionic conductivity
- The lower the mobility $\rightarrow$ the higher the conductivity

Table 19.4 Typical Room-Temperature Electrical Conductivities for Various Nonmetallic Materials	<table> <tr> <th>Material</th><th>Electrical Conductivity [(Ω·m)⁻¹]</th></tr> <tr> <td>Graphite</td><td>$2 \times 10^2 - 2 \times 10^3$</td></tr> <tr> <td colspan="2">Ceramics</td></tr> <tr> <td>Concrete (dry)</td><td>10^{-10}</td></tr> <tr> <td>Soda-lime glass</td><td>$10^{-10} - 10^{-11}$</td></tr> <tr> <td>Pyrex glass</td><td>$10^{-10} - 10^{-11}$</td></tr> <tr> <td>Borosilicate glass</td><td>$< 10^{-10}$</td></tr> <tr> <td>Aluminum oxide</td><td>$< 10^{-10}$</td></tr> <tr> <td>Fused silica</td><td>$< 10^{-10}$</td></tr> <tr> <td colspan="2">Polymers</td></tr> <tr> <td>Phenol-formaldehyde</td><td>$10^{-10} - 10^{-11}$</td></tr> <tr> <td>Poly(methyl methacrylate)</td><td>$< 10^{-10}$</td></tr> <tr> <td>Nylon 6,6</td><td>$10^{-11} - 10^{-12}$</td></tr> <tr> <td>Polyethylene</td><td>$< 10^{-10}$</td></tr> <tr> <td>Polyethylene</td><td>$10^{-10} - 10^{-11}$</td></tr> <tr> <td>Polyvinylfluoride</td><td>$< 10^{-10}$</td></tr> </table>	Material	Electrical Conductivity [(Ω·m) ⁻¹]	Graphite	$2 \times 10^2 - 2 \times 10^3$	Ceramics		Concrete (dry)	10^{-10}	Soda-lime glass	$10^{-10} - 10^{-11}$	Pyrex glass	$10^{-10} - 10^{-11}$	Borosilicate glass	$< 10^{-10}$	Aluminum oxide	$< 10^{-10}$	Fused silica	$< 10^{-10}$	Polymers		Phenol-formaldehyde	$10^{-10} - 10^{-11}$	Poly(methyl methacrylate)	$< 10^{-10}$	Nylon 6,6	$10^{-11} - 10^{-12}$	Polyethylene	$< 10^{-10}$	Polyethylene	$10^{-10} - 10^{-11}$	Polyvinylfluoride	$< 10^{-10}$
Material	Electrical Conductivity [(Ω·m) ⁻¹]																																
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Concrete (dry)	10^{-10}																																
Soda-lime glass	$10^{-10} - 10^{-11}$																																
Pyrex glass	$10^{-10} - 10^{-11}$																																
Borosilicate glass	$< 10^{-10}$																																
Aluminum oxide	$< 10^{-10}$																																
Fused silica	$< 10^{-10}$																																
Polymers																																	
Phenol-formaldehyde	$10^{-10} - 10^{-11}$																																
Poly(methyl methacrylate)	$< 10^{-10}$																																
Nylon 6,6	$10^{-11} - 10^{-12}$																																
Polyethylene	$< 10^{-10}$																																
Polyethylene	$10^{-10} - 10^{-11}$																																
Polyvinylfluoride	$< 10^{-10}$																																

Table 19.1 Room-Temperature Electrical Conductivities for Nine Common Metals and Alloys	<table> <tr> <th>Metal</th><th>Electrical Conductivity [(Ω·m)⁻¹]</th></tr> <tr> <td>Silver</td><td>6.8×10^7</td></tr> <tr> <td>Copper</td><td>6.0×10^7</td></tr> <tr> <td>Gold</td><td>4.3×10^7</td></tr> <tr> <td>Aluminum</td><td>3.8×10^7</td></tr> <tr> <td>Brass (70 Cu-30 Zn)</td><td>1.6×10^7</td></tr> <tr> <td>Iron</td><td>1.0×10^7</td></tr> <tr> <td>Platinum</td><td>0.94×10^7</td></tr> <tr> <td>Plain carbon steel</td><td>0.6×10^7</td></tr> <tr> <td>Stainless steel</td><td>0.2×10^7</td></tr> </table>	Metal	Electrical Conductivity [(Ω·m) ⁻¹]	Silver	6.8×10^7	Copper	6.0×10^7	Gold	4.3×10^7	Aluminum	3.8×10^7	Brass (70 Cu-30 Zn)	1.6×10^7	Iron	1.0×10^7	Platinum	0.94×10^7	Plain carbon steel	0.6×10^7	Stainless steel	0.2×10^7
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Stainless steel	0.2×10^7																				

Figure 69: Electrical conductivity of materials

Electrical conductivity of porous ceramics

- Various models are proposed to estimate the electrical conductivity of porous ceramics. The models developed for composite materials is modified to make it application for porous ceramics
- Important to consider the shape, size and size distribution of the material and pores in these models. Consideration of these is the origin of many models.
- Among various models proposed, following is the simplified model to predict the electrical conductivity of the porous materials.

$$\sigma = \sigma_0 \frac{(1 - P)}{(1 + aP^2)}$$

- σ is the electrical conductivity of the porous body. σ_0 is the material electrical conductivity, P is the pore fraction and a is the pore shape and distribution constant, which is around 11.

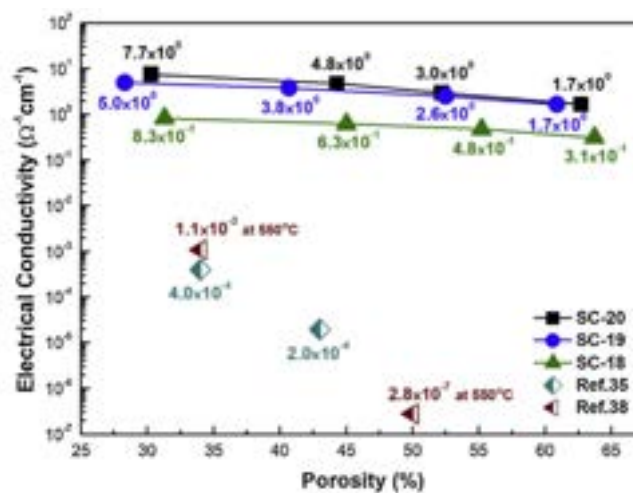


Figure 70: Example of the effect on porosity on the electrical conductivity

Dielectricity

- Electrically insulating and exhibits dipole structure

- Concept is used in capacitors, permittivity (dielectric constant) of the material is important
- A Dielectric material is an insulator with positive and negative electrically charged entities at molecular or atomic level
- Because of the dipole structure existence and their interaction with electric field, these materials are used in capacitors,
 - Capacitance,

$$C = \frac{Q}{V}$$

Where C is the capacitance, Q is the charge stored and V is the voltage applied

- In Vacuum,

$$C = \epsilon_0 \frac{A}{l}$$

Where ϵ_0 is the permittivity of vacuum, A is the area of the plates and l is the distance between plates

- With dielectric material

$$C = \epsilon \frac{A}{l}$$

Where ϵ is the permittivity of dielectric material

- The Dielectric constant ϵ_r ,

$$\epsilon_r = \frac{\epsilon}{\epsilon_0}$$

Material	Dielectric Constant		Dielectric Strength (V/mil)
	60 Hz	1 MHz	
	Ceramics		
Titanate ceramics	—	15–16,000	50–200
Mica	—	5.4–6.7	1000–2000
Silicates (MgO-SiO ₂)	—	5.5–7.5	200–250
Soda-lime glass	6.9	6.9	250
Porcelain	6.0	6.0	40–400
Fused silica	4.0	3.8	250
Polymers			
Phenol-formaldehyde	5.3	4.8	300–400
Nylon 6.6	4.0	3.8	400
Polystyrene	2.6	2.6	500–700
Polyethylene	2.3	2.3	400–500
Polytetrafluoroethylene	2.1	2.1	400–500

Figure 71: Dielectric constants of materials

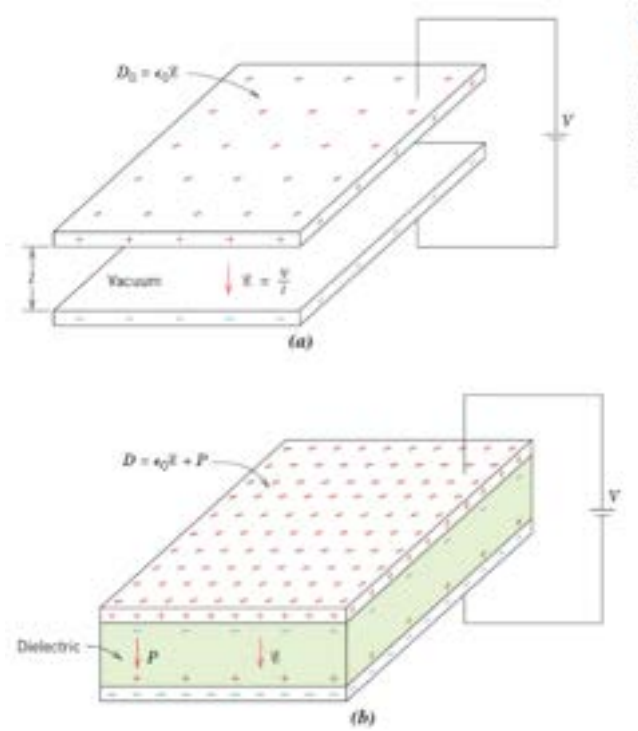


Figure 72: Visualization of Dielectric material

Ferroelectricity

- Spontaneous polarization, permanent electric dipoles
- Very high dielectric constants
- Types of polarization

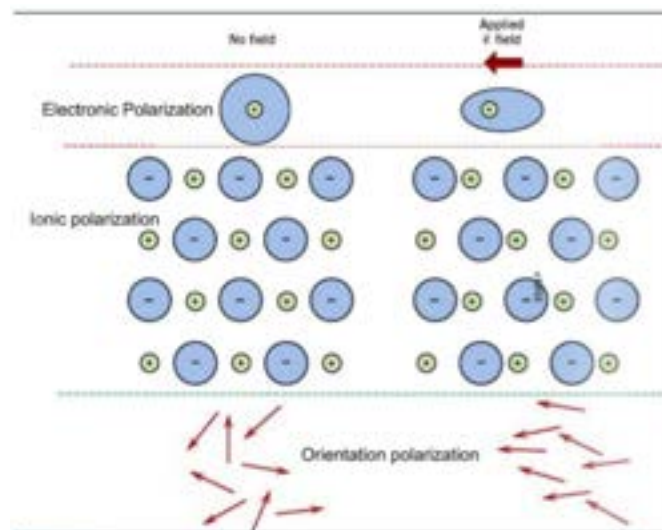
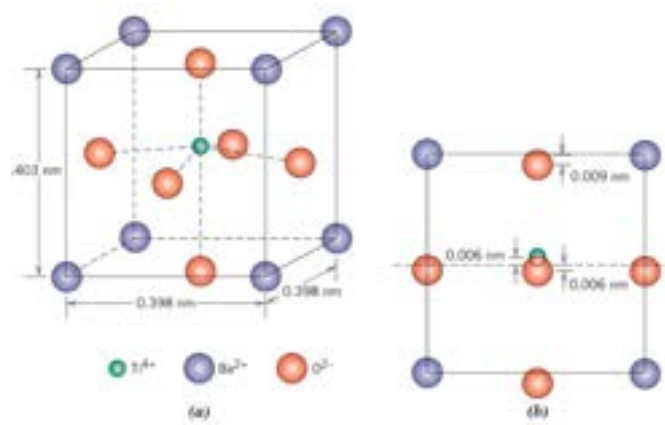


Figure 73: Visualization of the types of polarization

- Electronic polarization, can be induced in all atoms → displacement of negatively charged electron cloud relative to nucleus by an electric field
- Ionic polarization, needs to be ionic → dipole cations and anion in opposite directions → net dipole movement. Dipole movement = relative displacement $\times$ charge of the ion

- Orientation polarization, rotation of the permanent dipole moments in the direction of the applied field. Tendency is reduced with increased temperature
- The dielectric materials that exhibit spontaneous polarization (absence of electric field) are called ferroelectrics
- Permanent ionic dipole moment exists in these materials
- Exhibits extremely high dielectric constants at relatively low frequencies
- Thus capacitors made from these materials are much smaller compared to other dielectric materials
- Examples are: BaTiO_3 , KNbO_3 , $\text{Pb}(\text{Zr}, \text{Ti})\text{O}_3$, KH_2PO_4 , $\text{NaKC}_4\text{H}_4\text{O}_{64}\text{H}_2\text{O}$



Calculating polarization in ferroelectrics

For a ferroelectric material like BaTiO_3 , spontaneous polarization arises from ion displacement.

Step 1: Determine unit cell parameters

- BaTiO_3 at room temperature: tetragonal, $a \approx 4.00 \text{ \AA}$, $c \approx 4.03 \text{ \AA}$
- Ti^{4+} displaced from center by $\delta \approx 0.006 \text{ nm} = 6 \times 10^{-12} \text{ m}$

Step 2: Calculate dipole moment per unit cell

$$p = q \times \delta$$

Where q = charge of displaced ion = $4e = 4 \times 1.602 \times 10^{-19} \text{ C} = 6.408 \times 10^{-19} \text{ C}$.

$$p = (6.408 \times 10^{-19}) \times (6 \times 10^{-12}) = 3.84 \times 10^{-30} \text{ C} \cdot \text{m}$$

Step 3: Calculate unit cell volume

$$V_{\text{cell}} = a^2 \times c \approx (4.00 \times 10^{-10})^2 \times (4.03 \times 10^{-10}) = 6.45 \times 10^{-29} \text{ m}^3$$

Step 4: Calculate polarization

$$P = \frac{p}{V_{\text{cell}}} = \frac{3.84 \times 10^{-30}}{6.45 \times 10^{-29}} = 0.0595 \text{ C/m}^2$$

Note: This simplified model considers only Ti displacement. Actual polarization includes contributions from all ions and is typically $0.25 - 0.3 \text{ C/m}^2$ for BaTiO_3 .

Piezoelectricity

- Application of force induces electric field and vice versa
- Class of materials that exhibit mechanical strain when electric field is applied and electric field when mechanical strain is applied are called piezoelectric materials

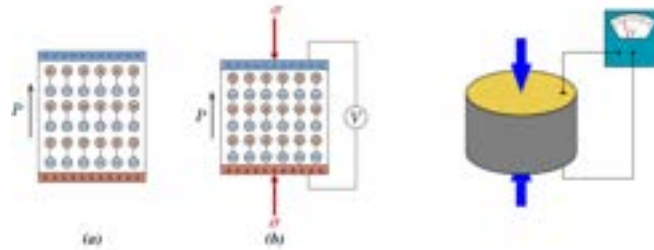


Figure 74: Visualization of Piezoelectricity

-
- These widely used as sonic transducers, position actuators, sensors for automotive vehicles, transformers, printers, drug delivery detectors
- Common materials, $\text{Pb}(\text{Zr}, \text{Ti})\text{O}_3$, KNbO_3 , etc.

Electrical and thermal conductivity

The Wiedemann-Franz law states that the ratio of thermal conductivity to the electrical conductivity of a metal (not usually applicable to ceramics) is proportional to its temperature,

$$LT = \frac{\kappa}{\sigma}$$

Where L is the proportionality constant (also known as the Lorenz number)

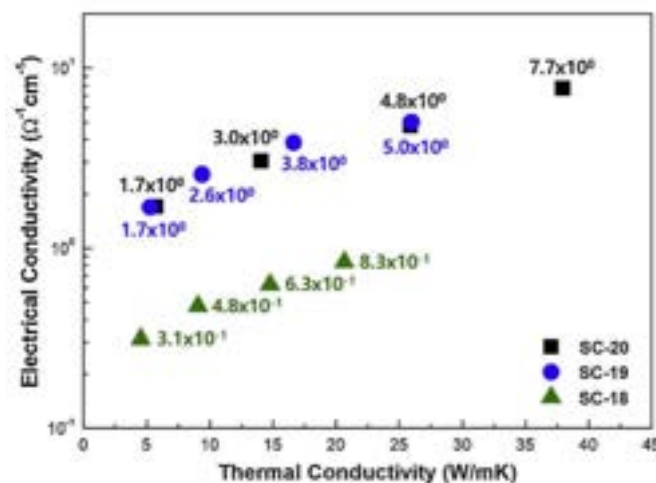
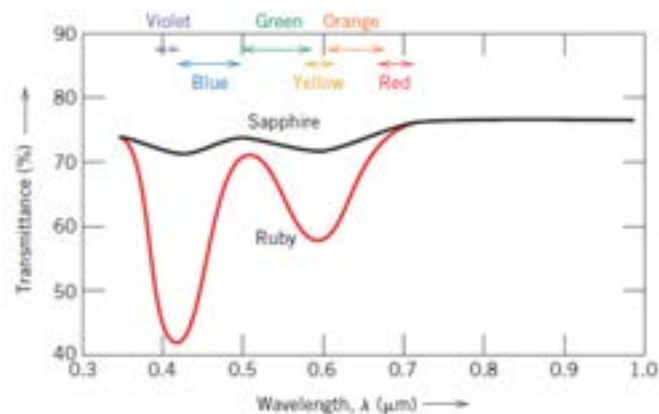


Figure 75: Visualization of Wiedemann-Franz law visualization

1.11.5 Visual Properties

Colours

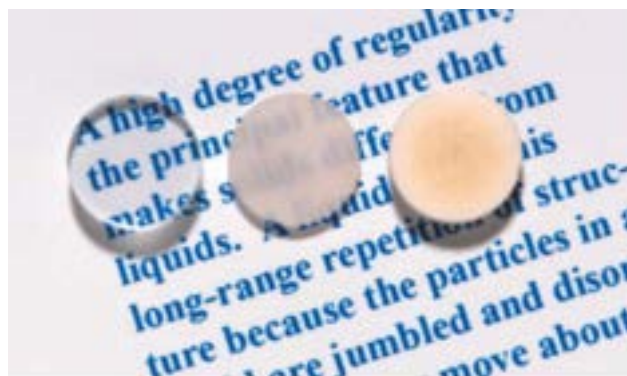
- Al_2O_3 with Cr_2O_3
- Formation impurity levels in band gap
- Emission of photons in different wavelengths
- Color in glasses:
 - Cu^{2+} - blue/green
 - Co^{2+} - blue/violet
 - Cr^{3+} - green
 - Mn^{2+} - yellow
 - Mn^{3+} - purple



•

Opacity and translucency

- Opacity results when scattering of light is extensive
- Scattering sources, polycrystalline, second phases, anisotropy of optical properties, porosity



The light transmittance of three aluminum oxide specimens. From left to right; single-crystal material (sapphire), which is transparent; a polycrystalline and fully dense (nonporous) material, which is translucent; and a polycrystalline material that contains approximately 5% porosity, which is opaque.

1.11.6 Degradation

Types of Degradation

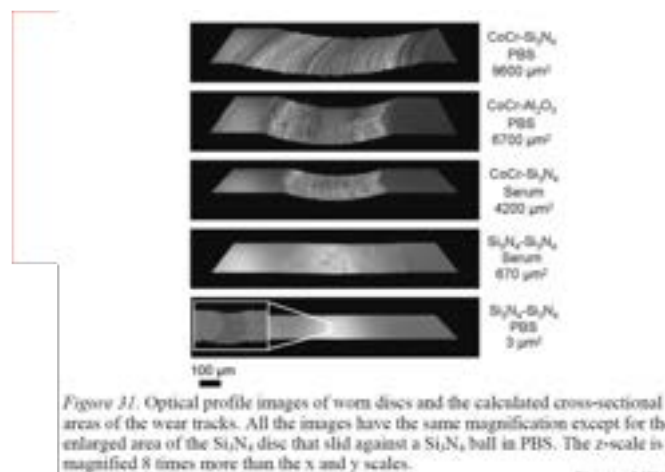
Physical degradation, Due to their fragility, damage to ceramics typically comes from mishandling and packing. However, other factors such as vandalism, frost, mold, and other similar occurrences can also inflict harm

Chemical degradation, Chemical degradation of objects occurs not in the physical structure of the object but rather in at the chemical or compound level. The degradation of the chemical component of an object will hinder or weaken the stability of the object when exposed to environmental factors such as water, air, pollution, heat humidity and the like.

Soluble salts, A common degradation issue in ceramics involves soluble salts can either enter the clay body from the environment, for example from being buried underground for decades, or they are already naturally occurring due to the components of the material or clay used.

Wear of Si_3N_4 ceramics, Hip joint

- Si_3N_4 evaluated in different body fluids
- With different combinations
- Si_3N_4 – Si_3N_4 was proven to be the best



Corrosion of refractories

Refractory ceramics:

- Silica r. (dinas) (refractoriness 1710-1750 °C), > 93% Silica quartz, tridymite, cristobalite, glass
- Fireclay r. (grog, chamotte) (1600-1750 °C), 15-46% Al_2O_3 - SiO_2 mullite $(\text{Al}_2\text{O}_3)_3(\text{SiO}_2)_2$
- Corundum (1850-2000°C), α - Al_2O_3
- Magnesite r. (> 2000 °C), MgO periclase
- Chrome-magnesite r. (1920-2000 °C), MgOCr_2O_3
- Fusion-cast corundum/baddeleyite r. (no open pores!), α - Al_2O_3 , ZrO_2 , SiO_2 glass phase

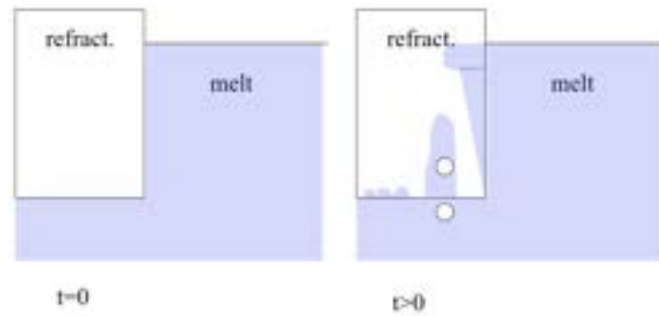
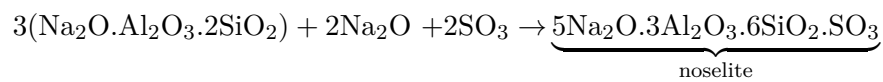
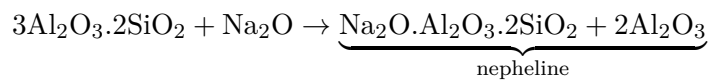


Figure 76: Influence of density or surface density gradient

- Corrosion of refractories, corrosion leading to mechanical degradation
- Regenerators in gas heated glass melting furnaces, preheating of gas by products of combustion,

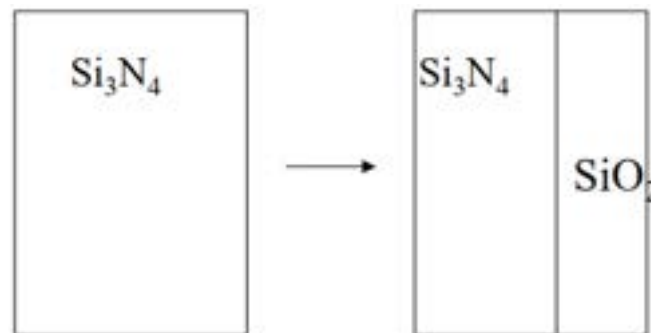


- Mullit $\rightarrow$ nepheline $\rightarrow$ noselite = volume expansion $\rightarrow$ mechanical stress
- Similar effect, changing oxidation/ reduction atmosphere $\text{MgO} \cdot \text{Fe}_2\text{O}_3$ and $\text{MgO} \cdot \text{FeO}$ or oxidation of $\text{FeO} \cdot \text{Cr}_2\text{O}_3$ and $\text{FeO} \cdot \text{Fe}_2\text{O}_3$

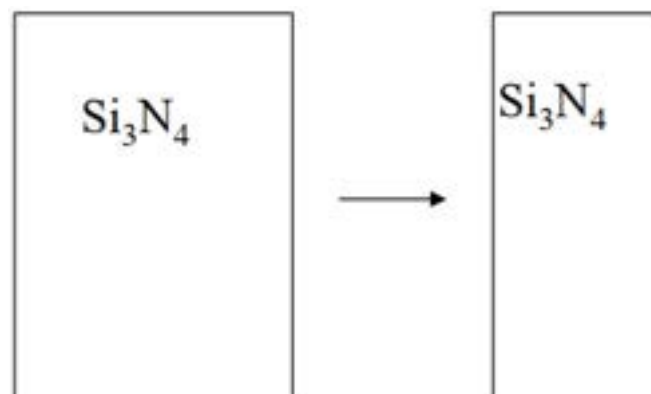
Corrosion of non-oxide

Active or passive mechanism:

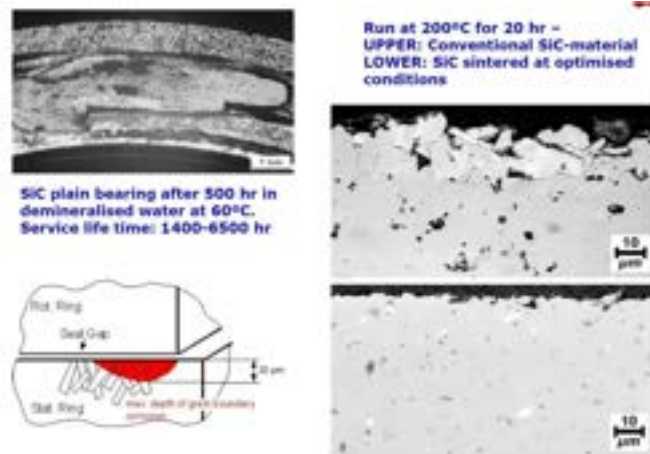
- Passive, high p_{O_2} , lower T $\text{Si}_3\text{N}_4(\text{s}) + 3\text{O}_2 \rightarrow 3\text{SiO}_2(\text{s}) + 2\text{N}_2$ controlled by diffusion



- Active, low p_{O_2} , higher T $\text{Si}_3\text{N}_4(\text{s}) + 3\text{O}_2 \rightarrow 6\text{SiO}(\text{g}) + 4\text{N}_2$ controlled by chemical reaction



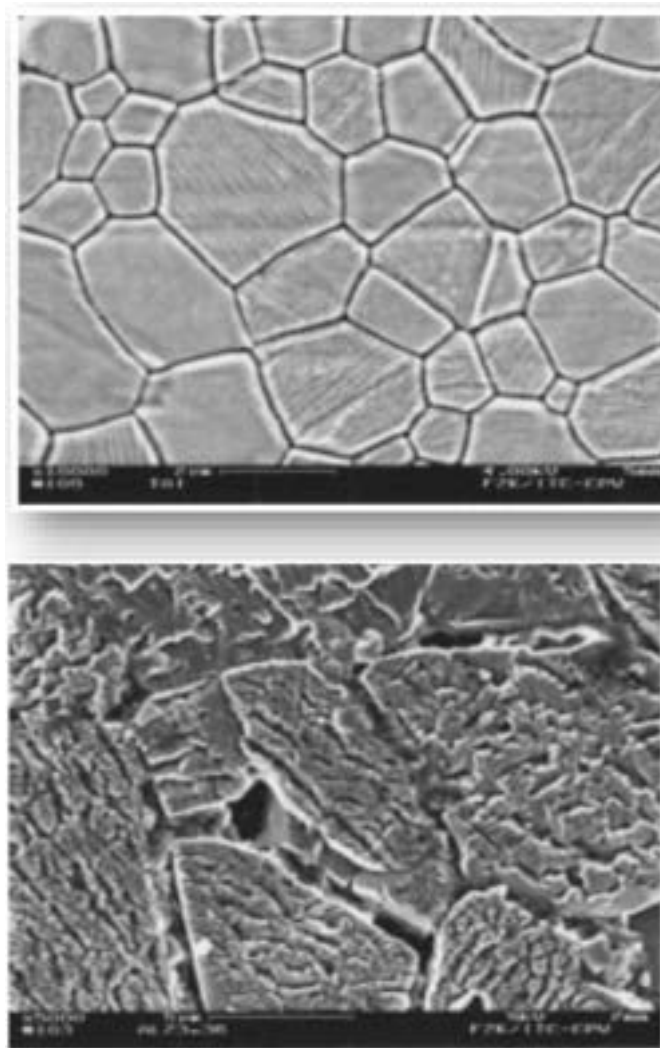
Corrosion of SiC in water



Corrosion of Alumina Ceramics

Often used as a reactor material in supercritical water oxidation at temperatures of 500 °C and pressures up to 30 MPa.

Impurities in the Alumina ceramic can leach into the solution and create acids.



Ceramics as protective coatings

- Ceramics reduce corrosion and are used to protect machines in extreme conditions

- Turbine blades
- Heat exchanger tubes

Case Study: Effect of porosity on multiple properties

Given a material's dense properties, we can predict how porosity affects:

- Elastic modulus: $E = E_0(1 - 1.9P + 0.9P^2)$
- Flexural strength: $\sigma = \sigma_0 \exp(-nP)$
- Electrical conductivity: $\sigma_{el} = \sigma_0 \frac{1-P}{1+aP^2}$

Example: SrTiO_3 with $E_0 = 270$ GPa, $\sigma_0 = 120$ MPa, $n = 5$, $\sigma_{el,0} = 211$ S/cm.

For different porosities:

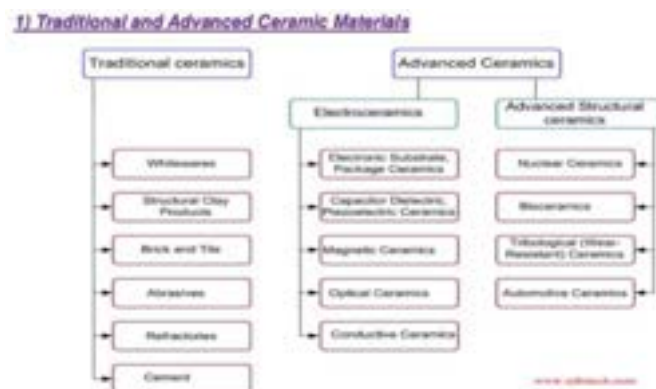
Porosity P	E (GPa)	σ_{fs} (MPa)	σ_{el} (S/cm)
0.624	44.5	5.30	15.0
0.674	34.6	4.14	11.5
0.707	28.8	3.49	9.50

Impact: Higher porosity significantly degrades all properties:

- E decreases by $\approx 85\%$ from dense to $P = 0.7$
- Strength decreases by $\approx 97\%$ (exponential dependence)
- Conductivity decreases by $\approx 95\%$

This demonstrates why dense ceramics are preferred for structural and electrical applications.

1.12 Applications of Ceramics



Whiteware

- Consists of clays, flint and feldspar. these are the silicates of Na, K, Ca, Fe and Al

Bricks and Tiles

- Extruded, moulded or dry-pressed

- Firing at high temperatures $< 1300\text{ }^{\circ}\text{C}$
- Usually consists of SiO_2 , Al_2O_3 , CaCO_2 , FeO , MgO , etc.

Abrasives

- Abrasive ceramics are used to cut or grind away the softer materials
- Hardness, toughness and wear resistance are important
- Usually bonded or coated on wheels or sheets
- Examples are Diamond, SiC , WC , Al_2O_3 and SiO_2 , etc.

Refractories

- Chemically and physically stable in aggressive environments Commonly used refractories are oxide based, Al_2O_3 , SiO_2 , ZrO_2 , MgO , CaO and mixtures there of
- Non-oxide ceramics (SiC , graphite, BN etc) cannot be used air
- HfC ($> 3890\text{ }^{\circ}\text{C}$) TaHfC ($> 4215\text{ }^{\circ}\text{C}$) are most highly refractory materials.

Cement

- Binder used to set and adheres to the materials binding them together
- Portland cement is largely produced. Clay, lime and gypsum are other types.
- Calcination of clay and lime mixtures at $> 1400\text{ }^{\circ}\text{C}$. Finally gypsum is added
- Ca_3SiO_5 , Ca_2SiO_4 etc. are the base compounds

Electronic substrates and packaging

- Insulating in nature
- Al_2O_3 is most commonly used ceramic. Posses best suited properties
- High dielectric constant and low thermal conductivity $\rightarrow$ problems
- Glass-ceramic compositions are gaining popularity

Dielectric and Piezoelectrics

- Dielectric materials, capacitors and microwave frequency & RF generators
- Piezoelectric materials
- Soft piezoelectric materials, actuators and sensors
- Hard piezoelectric materials, transducers
- Actuators for micropositioning and nanopositioning, sensors, such as conventional vibration detectors, ultrasonic transmitters and receivers.
- $\text{Pb}(\text{Zr}, \text{Ti})\text{O}_3$, BaTiO_3 , BiNaTiO_3 , etc.

Magnetic Ceramics

- Oxide materials with ferrimagnetism
- Ferrites with formula $M(\text{Fe}_x\text{O}_y)$, spinels are most common oxides, M is usually divalent metals.
- Others are hexagonal ferrites $\text{MFe}_{12}\text{O}_{19}$ and garnets $\text{M}_3\text{Fe}_5\text{O}_{12}$
- Applications in permanent magnets, transformers, telecommunications and information recording media
- Stealth vehicles

Optical Ceramics

- Most common are the the glass ceramics, windows of readers, infrared detectors, lasers, known as passive materials
- Examples are Sapphire Al_2O_3 , NaCl , CaF_2 , SrF_2
- Na discharge lamps with $\text{Mg}-\text{Al}_2-\text{O}_3$ transparent covers
- Phosphors, Lasers and electro-optical components are technical applications. Known as active materials
- Examples of activated phosphors are Pb, Mn, Pb and Eu for CRTs for Tvs and monitors

Conductive Ceramics

- Heating elements: SiC and MoSi_2
- Thermistors: Change in resistance with temperature, e.g., spinels of Fe, Co, mn
- Batteries: solid electrolyte in Na-S storage battery e.g., $\text{Na}_2 \cdot \text{O}11\text{Al}_2\text{O}_3$
- Fuel cells: anodes (NiO^+Y) and cathodes (LaSrMnO_3)
- Gas sensors: SnO_2 , Y, Ca
- Superconductors: $\text{YBa}_2\text{Cu}_3\text{O}_7$

Nuclear Ceramics

- Used in the generation of nuclear power and disposal of nuclear waste
- UO_2 and PuO_2 are the most common fuel pellets used in reactors because of their high melting point, inertness and open crystal structure.
- Carbides and nitrides are also developed for high thermal conductivity
- Nuclear waste is managed by encapsulating or blending with borosilicate
- Recent methods are to form synroc (tilinates + nuclear waste).

Bioceramics

- predominantly used in dental, bone implants and hip prosthetics
- Hardness and abrasion resistance, useful for bone and dental applications
- Inertness with body fluids (biocompatibility) useful for all other implants
- Commonly used materials, Al_2O_3 , ZrO_2 , Calcium phosphate, Aluminosilicates, carbon etc.

Automotive Ceramics

- Example components, spark plug insulators, catalyst supports and catalyst supports and catalysts for emission control, pressure and heat sensors etc.
- Al_2O_3 , $\text{Mg}_2\text{Al}_4\text{Si}_5\text{O}_{18}$ are common supports for catalyst convertors
- Oxides of Cr, Ni, Cu are the catalyst convertors
- Al_2O_3 and piezoelectric ceramic sensors are used for different sensors.
- Ceramic thermal barrier coatings for light weight, high temperature stability, hardness, wear resistance, corrosion resistance of metallic components in engine parts.

Ceramic Phase Diagrams

Phase diagrams show stability regions of phases as function of composition and temperature.

Key features in ceramic phase diagrams

- **Eutectic:** Liquid $\rightarrow$ two solids on cooling
- **Peritectic:** Liquid + solid $\rightarrow$ different solid
- **Compound formation:** Intermediate phases (e.g., spinel MgAl_2O_4)
- **Solid solution:** Single phase region with variable composition
- **Non-stoichiometry:** Deviation from ideal formula within a phase field

Using phase diagrams to find compositions

1. Identify the phase of interest at given T and composition
2. Read composition from tie lines
3. Convert between oxide mol% and formula units
4. Calculate defect concentrations from non-stoichiometry

Quick Reference: Common Ceramics and Their Applications

Material	Crystal structure	Key properties	Typical applications
Alumina (Al_2O_3)	Hexagonal	High hardness, electrical insulation, high melting point	Substrates, spark plugs, abrasives, hip implants, cutting tools
Zirconia (ZrO_2)	Monoclinic/tetragonal	High fracture toughness, oxygen ion conduction	Oxygen sensors, thermal barrier coatings, dental crowns, fuel cells
Titania (TiO_2)	Tetragonal	High dielectric constant, photocatalytic	Capacitors, photocatalysts, white pigment, solar cells
Barium titanate (BaTiO_3)	Perovskite	Ferroelectric, high dielectric constant	Multilayer capacitors, thermistors, actuators
PZT ($\text{Pb}(\text{Zr},\text{Ti})\text{O}_3$)	Perovskite	Piezoelectric	Sensors, actuators, ultrasonic transducers, inkjet printers
Silicon carbide (SiC)	Hexagonal/cubic	Extreme hardness, high thermal conductivity	Abrasives, high-temperature electronics, heating elements, armor
Silicon nitride (Si_3N_4)	Hexagonal	High strength at high temperature, thermal shock resistance	Bearings, cutting tools, engine parts, turbocharger rotors
Mullite ($3\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$)	Orthorhombic	Low thermal expansion, creep resistance	Refractories, kiln furniture, furnace linings
Cordierite ($\text{Mg}_2\text{Al}_4\text{Si}_5\text{O}_{18}$)	Orthorhombic	Very low thermal expansion	Catalyst supports, diesel particulate filters, cookware
Fused silica (SiO_2)	Amorphous	Very low thermal expansion, optical transparency	High-temperature lamp envelopes, telescope mirrors, optical fibers
Soda-lime glass	Amorphous	Easy to shape, transparent	Windows, bottles, containers
Borosilicate glass	Amorphous	Low expansion, chemical resistance	Lab glassware (Pyrex), cookware, piping
Ferrites (MFe_2O_4)	Spinel	Magnetic	Permanent magnets, transformers, recording media

Glossary of Key Ceramics Terms

- Amorphous** A non-crystalline solid with no long-range atomic order; glasses are amorphous.
- Archimedes method** A technique to measure density by weighing in air and in a liquid (usually water or ethanol).
- Binder** Organic additive that provides strength to the green body before sintering.
- Brittle fracture** Fracture that occurs with little or no plastic deformation; characteristic of ceramics at room temperature.
- Calcination** Heat treatment of powder to induce decomposition, phase transformation, or removal of volatile components.
- Ceramic loading** Volume fraction of ceramic particles in a slurry or suspension.
- Coordination number (CN)** Number of nearest neighbor ions surrounding an ion in a crystal structure.
- Debinding** Removal of organic additives (binders, plasticizers) before sintering.
- Density** Mass per unit volume. Types: theoretical (perfect crystal), bulk (including pores), green (before sintering), sintered (after sintering).
- Dielectric constant** Ratio of permittivity of a material to permittivity of vacuum; measure of ability to store electrical energy.
- Doctor blade** Another name for tape casting; a blade controls the thickness of ceramic slurry on a carrier film.
- Electroneutrality** The requirement that the total positive charge equals total negative charge in a ceramic compound.
- Ferroelectric** Material that exhibits spontaneous electric polarization that can be reversed by an electric field.
- Flexural strength** Stress at fracture in a bending test; also called modulus of rupture (MOR).
- Frenkel defect** A paired vacancy-interstitial defect where an ion moves from a lattice site to an interstitial site.
- Glass** An inorganic product of fusion that has cooled to a rigid condition without crystallizing.
- Glass transition temperature (T_g)** Temperature at which an amorphous material transitions from hard and brittle to soft and rubbery.
- Green body** Formed ceramic component before sintering (contains binders and porosity).
- Isotropic shrinkage** Shrinkage that is equal in all directions during sintering.
- Network former** Oxide that forms the glass network (e.g., SiO_2 , B_2O_3).
- Network modifier** Oxide that breaks up the glass network and lowers viscosity (e.g., Na_2O , CaO).
- Perovskite** Crystal structure with formula ABO_3 ; many functional ceramics (BaTiO_3 , PZT) have this structure.
- Piezoelectric** Material that generates electric charge when mechanically stressed, and vice versa.
- Plasticizer** Additive that increases flexibility of the green body by lowering the glass transition temperature of the binder.
- Porosity** Volume fraction of pores in a material; can be open (connected to surface) or closed (isolated).

Refractory Ceramic material that withstands high temperatures without melting or decomposing.

Schottky defect A paired vacancy defect where both a cation and an anion are missing from the lattice.

Screen printing A method where ceramic paste is forced through a mesh screen to form a pattern on a substrate.

Shrinkage Reduction in dimensions during drying and sintering; expressed as linear shrinkage ($\Delta L/L_0$) or volumetric shrinkage ($\Delta V/V_0$).

Sintering Consolidation of ceramic powder by heat (and sometimes pressure) without melting; driven by reduction of surface energy.

Slip casting Forming method where ceramic slurry (slip) is poured into a porous mold that absorbs liquid.

Solid loading Volume fraction of ceramic particles in a suspension; affects green density and final properties.

Tape casting Method to produce thin ceramic sheets by spreading slurry on a carrier film with a doctor blade.

Theoretical density Density of a material with no pores or defects; calculated from crystal structure.

Thermal shock Fracture caused by rapid temperature change; materials with low thermal expansion and high thermal conductivity resist thermal shock.

1.13 Surface Modification of Ceramics

1.13.1 Purpose of Surface Modification

Surface modification is used to change or improve only the surface of a ceramic component, while keeping the bulk material mostly unchanged. This is useful because many important interactions happen at the surface, such as wear, biological bonding, photocatalysis, or electrochemical reactions.

The main reasons for surface modification are:

- Improving aesthetics
- Improving mechanical properties
- Keeping surfaces clean
- Improving the performance of bioceramics
- Introducing specific functionality

In ceramics, surface modification is often used less for basic protection and more for adding a specific function to the surface.

1.13.2 Aesthetic Surface Modification

Surface modification can be used to improve the visual appearance of ceramic components. This is common in decorative ceramics, tiles, plates, and artistic ceramic products.

The surface can be modified by using glazes, coloured layers, textures, or patterns. These modifications mainly affect appearance, but they may also influence durability and cleanability.

1.13.3 Improved Mechanical Properties

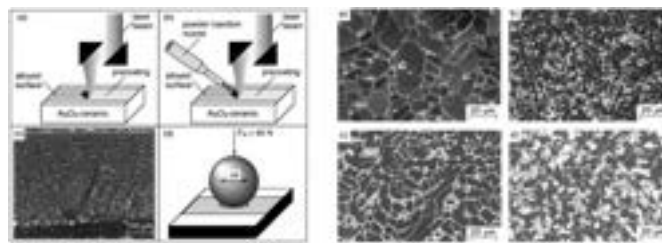
Ceramic surfaces can be modified to improve resistance against wear and surface damage. This is important because ceramics are often hard but brittle, and surface defects can strongly affect performance.

For domestic wear applications, ceramic glazes containing materials such as alumina and zirconia can be used to improve durability:

- Alumina: Al_2O_3
- Zirconia: ZrO_2

These glazes help protect the ceramic surface from scratching and wear during use.

1.13.4 Structural Wear of Alumina



Alumina ceramics are widely used because of their high hardness and good wear resistance. However, under severe structural wear conditions, the surface can still degrade.

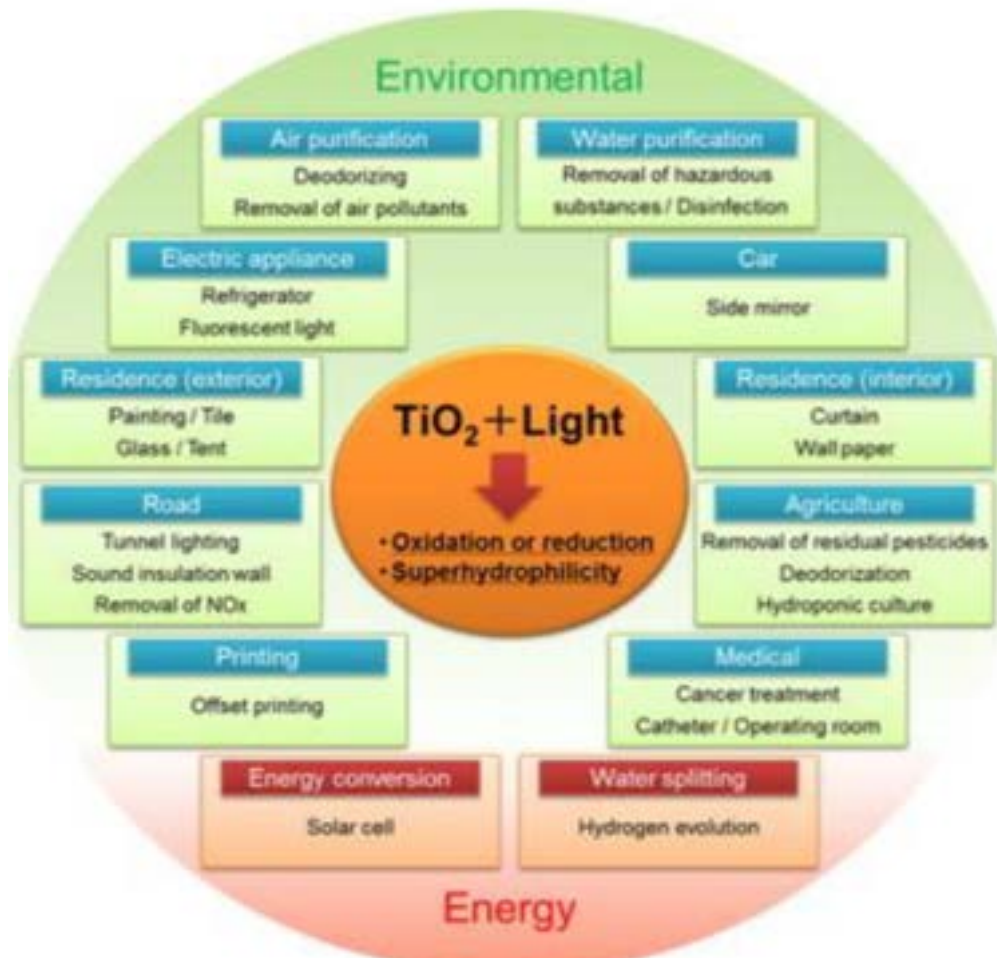
One possible solution is to modify the surface using laser alloying. In this method, a laser is used to locally melt or modify the surface together with added compounds. This changes the surface microstructure and can improve properties such as hardness and fracture toughness.

For alumina-based ceramics, laser alloying may include additions such as:

- Titanium nitride, TiN
- Hafnium oxide, HfO_2
- Tungsten and zirconia, $\text{W} + \text{ZrO}_2$

These additions can reduce grain size, introduce reinforcing phases, and improve fracture toughness. This shows how surface modification can be used to improve the mechanical performance of ceramic materials.

1.13.5 Clean Environment: Titanium Dioxide Coatings

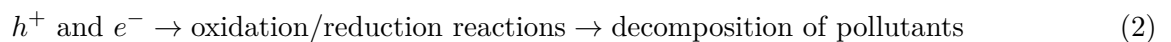


Titanium dioxide coatings are used for environmental and self-cleaning applications because of their photocatalytic behaviour.

When TiO_2 is exposed to light with sufficient energy, electron-hole pairs are generated:



The holes and electrons can then take part in oxidation and reduction reactions. These reactions can decompose organic pollutants on the surface.



Titanium dioxide coatings can be applied using methods such as:

- Physical vapour deposition, PVD
- Sol-gel processing
- Other thin-film coating methods

The coating thickness can range from a monolayer to a few nanometres.

1.13.6 Photocatalysis and Self-Cleaning Surfaces

Nano-TiO₂ coatings can be used on surfaces such as windows to create self-cleaning behaviour. Under UV light, the coating can decompose organic pollutants through photocatalysis.

Important effects include:

- UV light activates the nano-TiO₂ coating
- Organic pollutants are decomposed by oxidation reactions
- The surface becomes easier to clean
- Wetting behaviour can be changed using UV light

This is used in self-cleaning windows and surfaces exposed to outdoor environments.

1.13.7 Bioceramics: Calcium Phosphate Bone Materials

Calcium phosphate ceramics are important bioceramics because they have chemical similarity to bone. They are used for bone repair and implant applications.

Calcium phosphate materials are useful because they are:

- Suitable for bone repair
- Relatively inexpensive
- Chemically similar to bone mineral
- Biocompatible
- Bioactive
- Able to integrate with bone structure after implantation

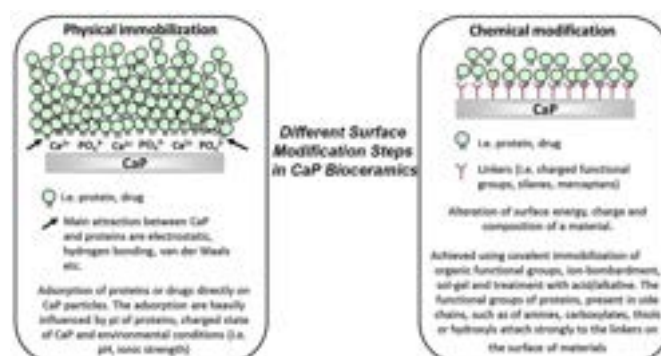
Surface properties are especially important for bioceramics because bonding with biological molecules takes place at the surface. Therefore, surface modification can strongly influence implant performance.

1.13.8 Calcium Phosphate Surface Modification

Calcium phosphate surfaces can be modified in different ways to improve interaction with proteins, drugs, or biological molecules.

Two main approaches are:

- Physical immobilization
- Chemical modification



Physical immobilization In physical immobilization, proteins or drugs are adsorbed directly onto the calcium phosphate surface. The interaction is mainly controlled by:

- Electrostatic attraction
- Hydrogen bonding
- Van der Waals forces

The adsorption depends on:

- Protein charge
- pH
- Ionic strength
- Surface charge of the calcium phosphate

Chemical modification In chemical modification, linker molecules are attached to the surface. These linkers can then bind proteins, drugs, or functional groups more strongly.

Examples of linkers include:

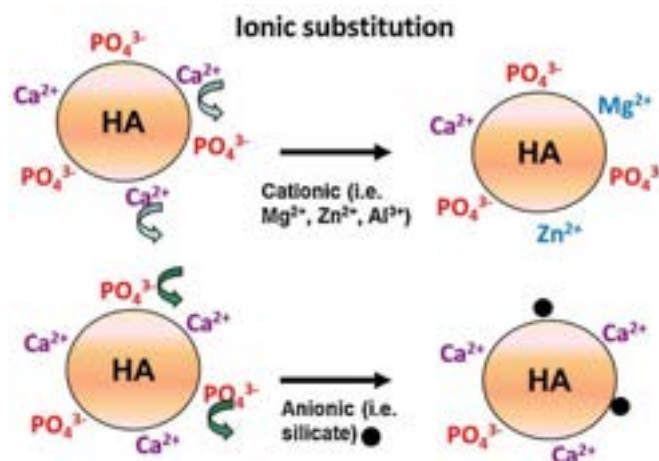
- Charged functional groups
- Silanes
- Mercaptans

Chemical modification changes the surface:

- Energy
- Charge
- Composition

It can be achieved using methods such as covalent immobilization, ion bombardment, sol-gel processing, or acid/alkaline treatment.

1.13.9 Surface Charge Alteration in Hydroxyapatite



Hydroxyapatite is a calcium phosphate material with the chemical formula:



Its surface charge can be altered by ionic substitution. This means that some ions in the material are replaced by other ions.

Cationic substitutions may include:

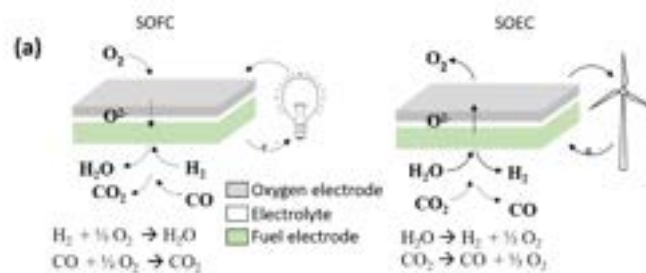
- Mg^{2+}
- Zn^{2+}
- Al^{3+}

Anionic substitutions may include:

- Silicate groups

Changing the surface charge affects how the material interacts with biological molecules, cells, and surrounding fluids. This is important for improving bone integration and bioactivity.

1.13.10 Specific Functionality: Solid Oxide Cells



Surface modification can also be used to introduce specific electrochemical functionality. One example is solid oxide cells.

Solid oxide cells are electrochemical energy conversion devices. They can operate in two modes:

- **Solid oxide fuel cell, SOFC:** converts fuels into electricity
- **Solid oxide electrolysis cell, SOEC:** converts electricity into fuels

Typical fuels include:

- Hydrogen, H_2
- Carbon monoxide, CO
- Methane, CH_4

For efficient electrochemical reactions, solid oxide cells require good electrocatalysts at the fuel and oxygen electrodes.

The performance depends strongly on:

- Shape of the electrocatalyst

- Size of the electrocatalyst
- Location of the electrocatalyst
- Distribution of catalyst particles on the ceramic surface

Examples include NiO coated or NiO exsolved doped SrTiO₃ ceramics. Exsolved nanoparticles can improve the catalytic activity because they form small, well-distributed active sites on the ceramic surface.

1.13.11 Summary

In ceramics, surface modification is often used to introduce new functionality rather than only to protect the material.

Common approaches include:

- Deposition of functional layers
- Formation of in-situ reactive layers
- Chemical surface modification
- Ionic substitution
- Nano- or microscale surface engineering

The modified region is often very thin, sometimes only at the nano- or microscale, but it can strongly affect the final performance of the ceramic component.

2 Polymers

Polymers are chains of repeating units, monomers.

Monomers, A small unit from which polymers are synthesized from.

Repeat unit, Are structural entities in long molecules, which are successively repeated along the chain.

Crosslink, Adjacent linear chains are connected to each other by these cross links (additives atoms or molecules) that are covalently bonded to the chains.

Branching, Polymers may be synthesized in which side-branch chains are connected to the main ones

Network, Multifunctional monomers forming three or more active covalent bonds make three-dimensional networks

With regard to behavior at elevated temperatures, polymers are classified as either **thermoplastic** or **thermosetting**.

Thermoplastic polymers have linear and branched structures; they soften when heated and harden when cooled.

In contrast, **thermosetting** polymers, once they have hardened, will not soften upon heating; their structures are crosslinked and network.

Degree of Polymerization

$$DP = \frac{\bar{M}_n}{m}$$

Where m is the molecular weight of the repeat unit and $\bar{M}_n$ is the number of average molecular weight.

2.1 Molecules and structure

Chemical formulae are representations of molecules:

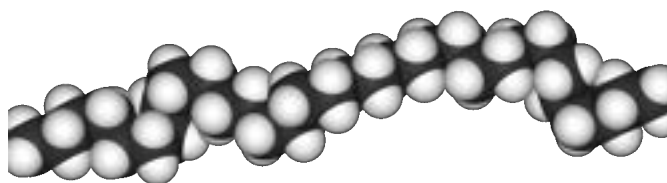


Figure 77: Space filling model emphasize shape

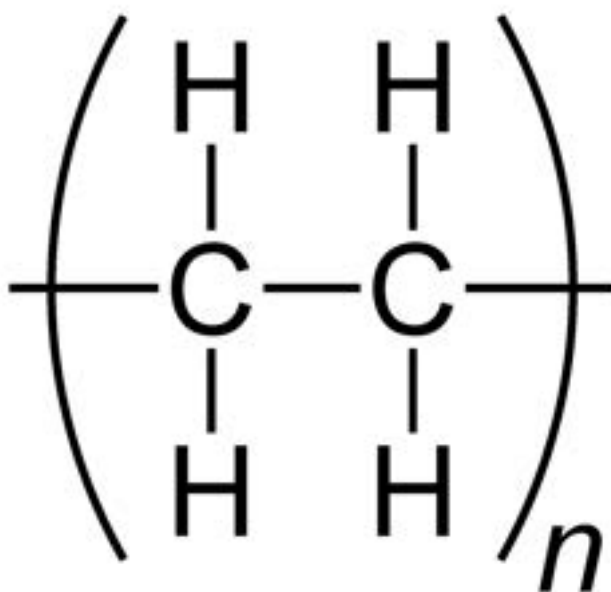


Figure 78: Various stick models emphasize connectivity (chemical bonds)

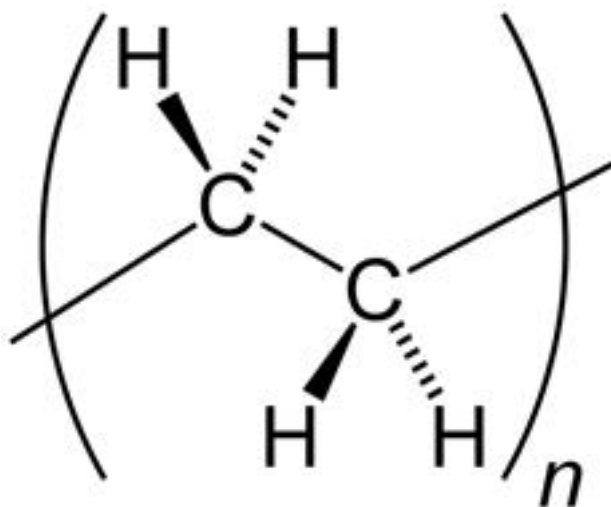


Figure 79: Or a compromise

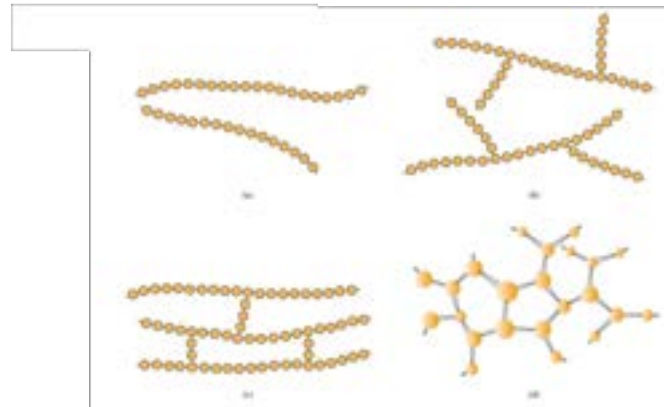


Figure 80: Polymer structure

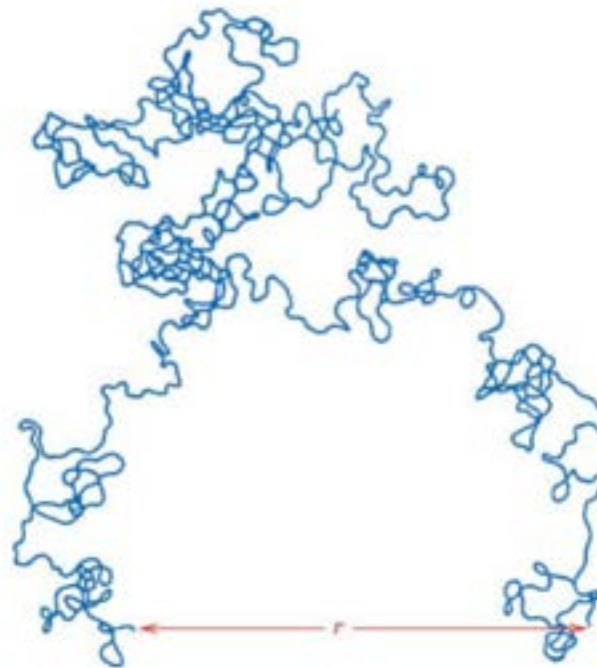
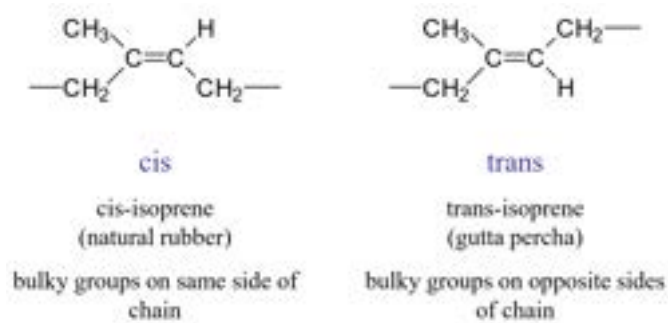


Figure 81: Random coil

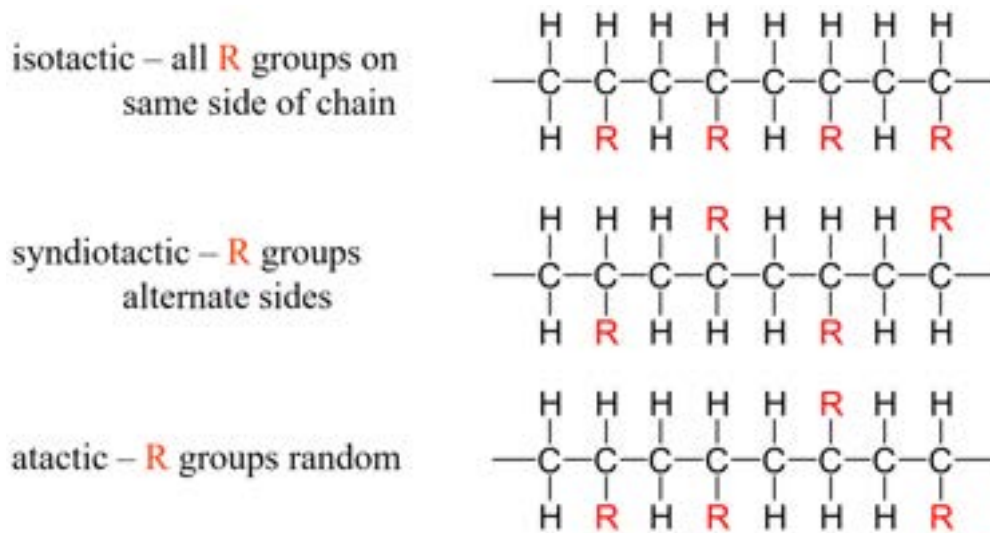
$$r = d\sqrt{N}$$

Where N is the number of units and d is length of a unit.

Cis/Trans isomerism



Tacticity, stereoregularity of chain



Copolymers, two or more monomers polymerized together

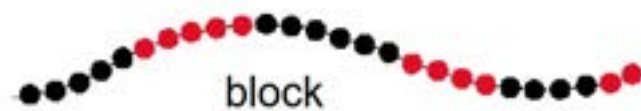
- Random, A and B randomly vary in chain



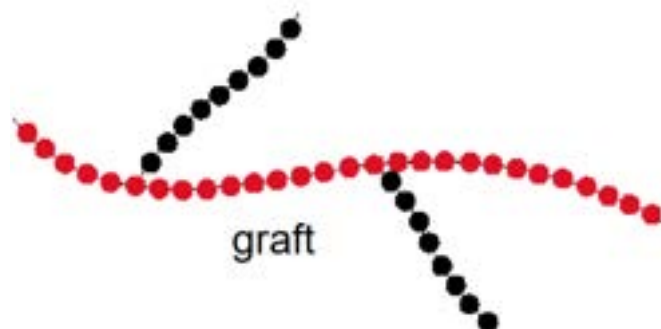
- Alternating, A and B alternate in polymer reaction



- Block, large blocks of A alternate with large blocks of B

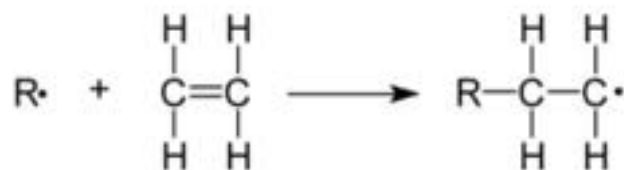


- Graft, chains of B grafted on to A backbone

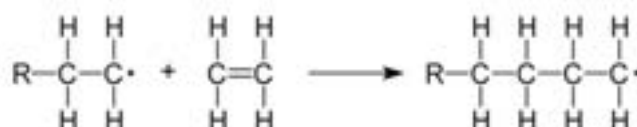


Addition (Chain) Polymerization

- Initiation



- Propagation



- Termination

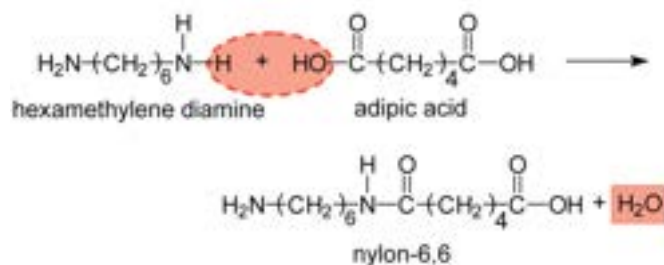
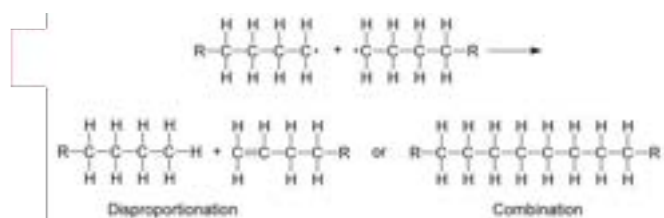


Figure 82: Condensation (Step) Polymerization

2.2 Polymer classification

Polymers are a very diverse materials class, where classification is based on:

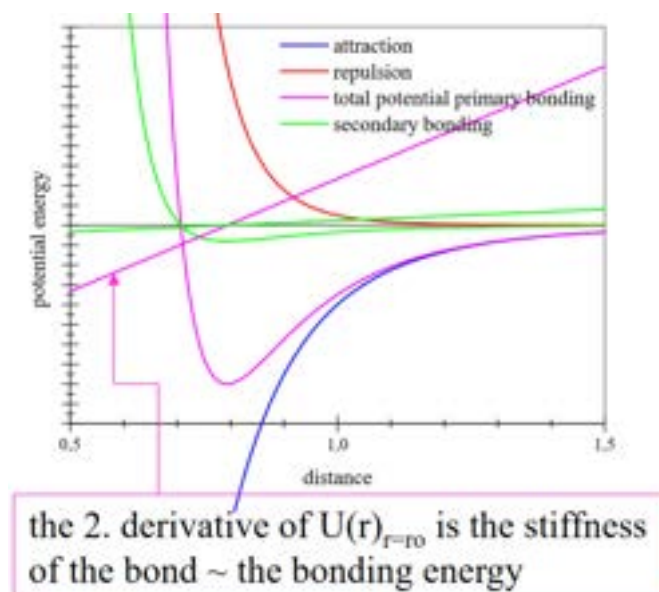
- Structure
 - Linear
 - branched
 - Cross-linked (network)
- Mode of polymerization
 - Addition
 - Condensation
- Structures that carry the stress in polymer solids

- Intramolecular (secondary bonding)
- Covalent (primary bonding) - tightly bonded networks and fibers
- Chain deformation (in rubbers)
- Source
 - Synthetic polymers
 - Naturally occurring polymers

Mode of polymerization

- Addition (or chain-growth)
 - Free radical polymerization
 - Typically very high molar mass - long chains
 - * Polyolefins
 - * Acrylics
 - * many others
- Condensation (or step-growth)
 - Release of small molecules in the polymerization reaction
 - Typically moderate molar mass - shorter chains
 - * Polyester
 - * Poly amides
 - * many others
- Cross-linking
 - During polymerization (eg epoxies and melamine)
 - After polymerization (most rubbers)

2.3 Bonding strength and moduli



Correlation between bonding energy and stiffness of materials.

High bonding energy $\Rightarrow$ High Young's modulus

Provided that one considers the bonding energy of the bonding that is deformed when the material is stretched.

Many bonds per unit volume $\Rightarrow$ High Young's modulus

That is, small atoms $\Rightarrow$ high Young's modulus

Bond type	Examples	Bond Stiffness S (N/m)	Young's Modulus E (GPa)
Covalent	Carbon-carbon bond	50 - 100	200 - 1000
Metallic	All metals	15 - 75	60 - 300
Ionic	Alumina, Al_2O_3	8 - 24	32 - 96
Van der Waals	Polyethylene	0 - 1	2 - 12
	Water	0.5 - 1	1 - 4

Figure 83: Note: Lowest predicted modulus ~ 1 GPa, Rubber ~ 1 -2 MPa

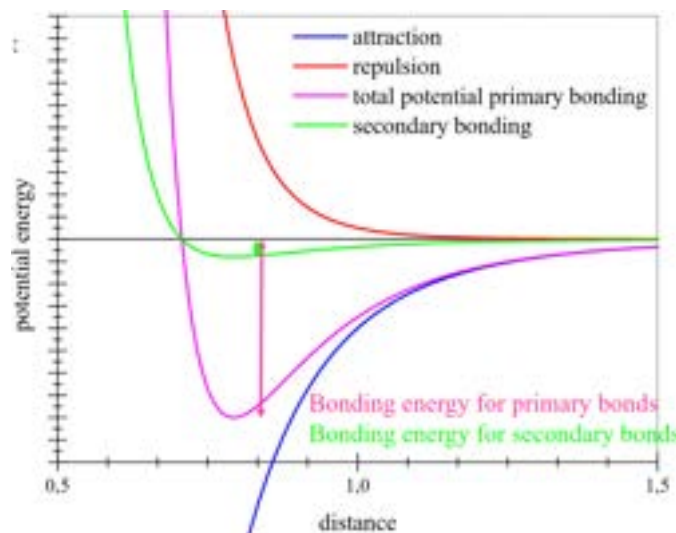


Figure 84: Bonding energy in primary and secondary bonds

Organic materials are kept together by secondary bonds,

- Secondary bonds are weak
- When secondary bond work cooperatively large stiffness is achievable
- The same mechanism gives approximately the same stiffness is achievable
- The same mechanism gives approximately the same stiffness ~ 1 GPa
- in polymers thermal expansion are primarily due to stretching of the secondary bonds

2.4 Microstructures in Polymer

A polymeric material consists of one or more type of domains (similar to grains in metals), one polymer chain can be a part of several domains:

- Crystalline: Solid, the chain cannot move due to binding in the crystal, the crystal is anisotropic mechanically:

$$E_{\parallel} \cong 240 \text{ GPa}$$

$$E_{\perp} \cong 3 - 9 \text{ GPa}$$

- Amorphous, glassy: Solid and transparent, the chain cannot move due to lack of free space the glass state is isotropic:

$$E \cong 1 - 4 \text{ GPa}$$

- Amorphous, rubbery/liquid: Liquid or rubbery depending on molar mass:

$$E \cong 0.1 - 10 \text{ MPa}$$

- Amorphous, cross-linked: The material is solid and in principle transparent, but often colored or reinforced, Thermosets. Similar to either "Amorphous, glassy" or "Amorphous, rubbery/liquid" depending on the degree of cross linking. Examples are epoxy and MF
- Semicrystalline, amorphous domains glassy, The material is solid and will likely look opaque or white, examples are PET and PVC
- Semicrystalline, amorphous domains rubbery, The material is solid and will likely look opaque or white, examples are PE and PP.
- Semicrystalline, amorphous domains rubbery and cross-linked, The material is solid and will likely look opaque or white, examples are PEX
- Polymer melt, temperature above both T_m and T_g . The material is a liquid (but very viscous), often transparent, examples are all thermoplasts at sufficiently high temperature, i.e. above the melt and glass transitions (T_m and T_g)

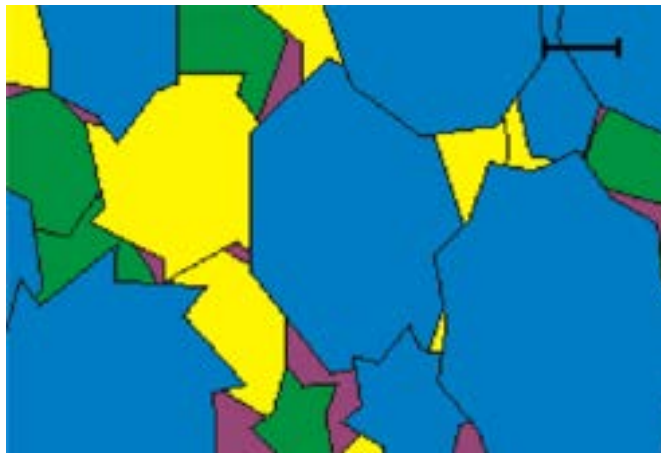


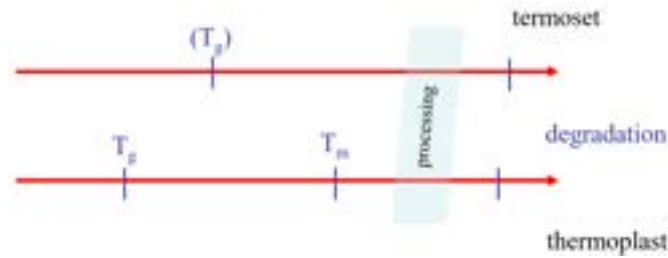
Figure 85: Schematic of materials microstructure

Most materials have a grain structure at some length scale (from $\sim 10\text{nm}$ to cm)

In metals and ceramics the grains are normally defined by different crystal orientations. However in polymers grains can be a variety of different structures.

Effect of temperature on molecular configuration and mobility

For crystallinity there exists the melting point, while for the glassy state there exists the glass transition. this happens because molecular mobility increases with temperature.



2.5 Secondary bonding

Dispersion forces:

Attraction between molecules because of interaction between the electrons (e.g. attraction between permanent dipoles).

Dispersion interaction is a surface phenomenon. Large surface gives larger dispersion interaction, large molecules have a higher boiling points than small

Hydrogen bonding:

A weak bonding between an electronegative atom (F, O, N) and hydrogen bonded to an electronegative atom.

2.6 World polymer consumption and source classification

2015 data extrapolated til 2026 with 2% yearly growth			
	Mton		kg/capita
LDPE, LLDPE	97	20.0%	11.7
HDPE	79	16.3%	9.5
PP	101	21.0%	12.3
PS	37	7.6%	4.4
PVC	57	11.8%	6.9
PET	49	10.2%	6.0
PUR	40	8.2%	4.8
Other	24	4.9%	2.9
total	483	100.0%	58.4
8271224750 https://www.worldometers.info/world-population/			
21.01.26 14:00			

Figure 86: World polymer consumption

The current for polymer production is the fossil resource oil $\sim 5\%$ of world oil usage goes into producing polymers.

Renewable carbon sources may be possible. Annual cellulose produced by all plants on earth ~ 180 Gton. 2021 bio based plastics production capacity is ~ 2.5 Mton/year.

2.7 Amorphous materials - the glassy state, glasses in polymers

Structure of glasses

- The structure of glass is *not* periodic
- One way to characterize this is the structure factor, $S(q)$
- $S(q)$ is basically the probability of finding an atom some distance from another atom

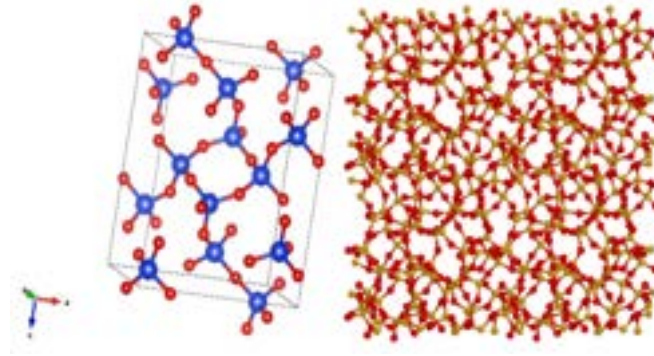


Figure 87: Illustration of crystalline and amorphous structures. Left: Cristobalite structure - one of several crystalline polymorphs of SiO₂. Right: Amorphous SiO₂.

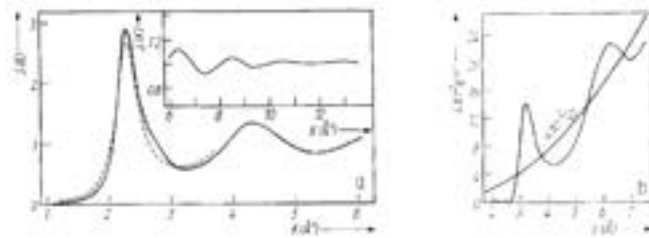


Figure 88: a) Structure factor $S(K)$ for a liquid In at 161 °C (full line) and at 190 °C (dotted line). The dashed line is $S(K)$ for liquid Cu (1090 °C) as shifted to fit the peak position of In. b) Radial distribution function for liquid In at 161 °C

Glass-forming liquid	Bonding
Silicates, borates	Primary: Covalent
KNO ₃ –Ca(NO ₃) ₂ mixtures	Primary: Ionic
Pd–Cu–Ni–P alloys, Cu–Zr alloys	Primary: metallic
Ortho-terphenyl	Secondary: Van der Waals
All glass forming polymers	Secondary; Van der Waals
Glycerol, glucose/water mixtures	Secondary: Hydrogen

Table 4: Examples of glass forming liquids

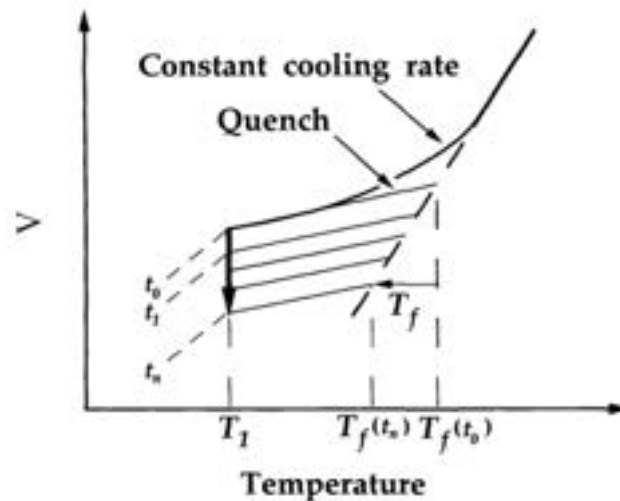


Figure 89: Illustration of the glass transition. The figure schematically shows the volume when a liquid is cooled from high temperature through the glass transition at different rates. At high temperature – the upper right corner – the sample is in equilibrium and the slope of the curve is the thermal expansivity of the liquid. At low temperature, the sample is **not** in equilibrium and the slope of the curve is the thermal expansivity of the glass. However, the exact location of the transition from liquid to glass depends on the rate at which the liquid is cooled. At a high rate, it will reach temperature T_1 at time t_0 . Whereas at increasingly lower rates it reaches T_1 at times $t_1, \dots, t_n$, with a different value of the volume – thus no longer in equilibrium since the state of the sample depends on the particular route by which the liquid arrived at this particular temperature. If the cooling at the high rate is stopped at T_1 the sample will start relaxing – an ageing process – where V decreases as a function of time as indicated by the vertical arrow at T_1 . If the linear slopes of both the solid (low temperature) and liquid (high temperature) V vs. T curves are extrapolated they meet at the so-called fictive temperature, T_f , which is the temperature at which the liquid is in equilibrium in that particular glassy state. Thus if a sample that has relaxed according to the arrow to the time t_n is heated to $T_f(t_n)$ it will be in equilibrium.

State functions and phase transitions

- State functions for a material is a function that depends only on the current state of the system; not on the way the system got there.
- The free energy, Gibbs function G , is a state function. The sign of the change in G , ΔG , between 2 states determines whether a process leading from one state to the other can occur spontaneously.
- Most phase transitions are 1st order transitions,
 - At 1st order transitions one observes a discontinuity in a 1st order derivative of G (entropy and volume).
 - At 2nd order transitions one observes a discontinuity in a 2nd order derivative of G (e.g. heat capacity and expansivity).

Differential Scanning Calorimetry (DSC)

- A sample and reference cell is heated (or cooled) at a prescribed rate.
- The difference in energy input required to do this reflects the properties of the sample

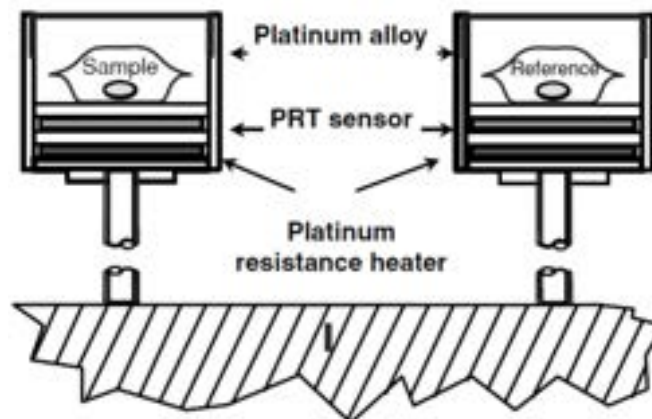


Figure 90: Diagram of a power compensation DSC. In this system, both the sample and reference furnaces are heated at a programmed heating/cooling rate. In order to maintain this rate when transitions occur in the sample, a power compensation circuit increases or reduces power to either furnace as required in order to maintain the heating rate. The power compensation circuit therefore reflects the energy changes occurring in the sample and is presented on the screen as a function of temperature or time. This technique measures energy changes directly.

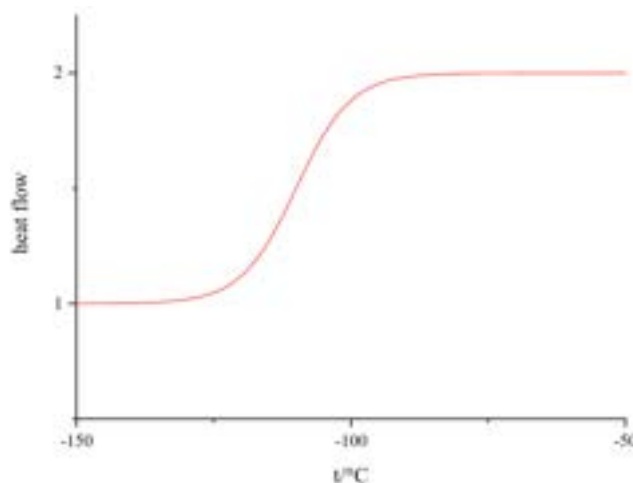


Figure 91: DSC glassy temperature T_g signature

A slowly cooled semi crystalline polymer sample (e.g. PE) heated from low temperature,

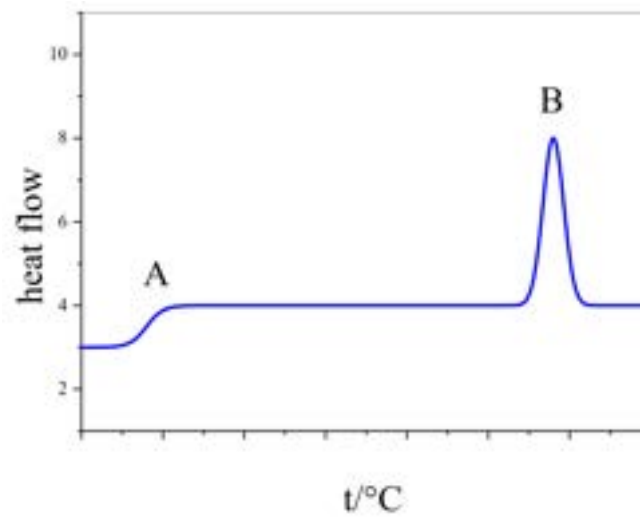
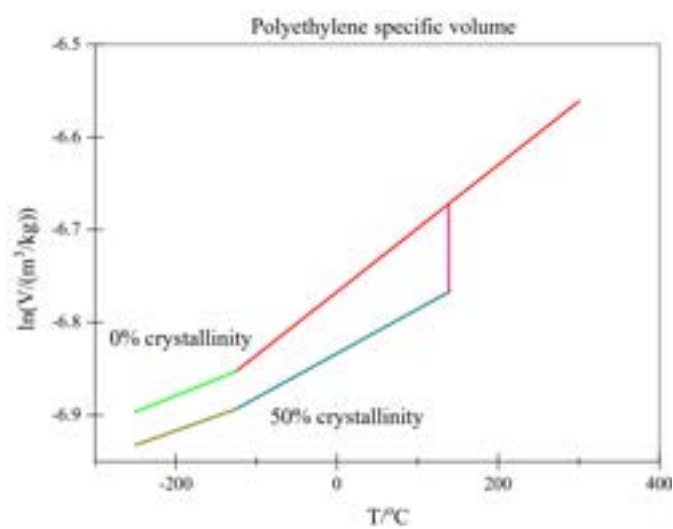
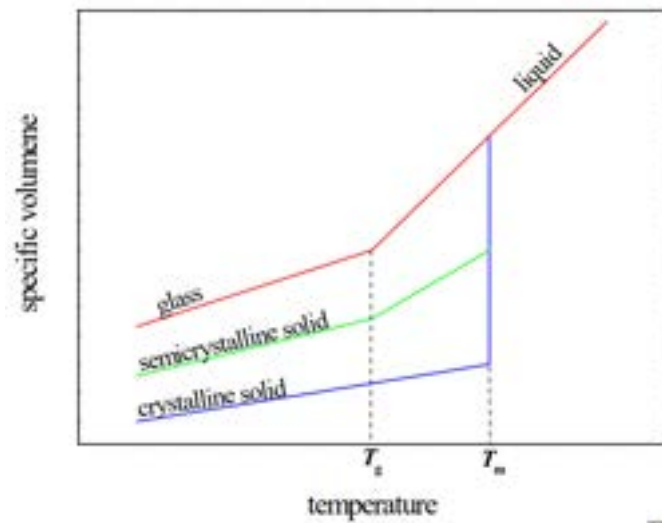


Figure 92: DSC glassy temperature T_g and melting temperature T_m signature



A quenched semi crystalline polymer sample (e.g. PE) heated from a low temperature,

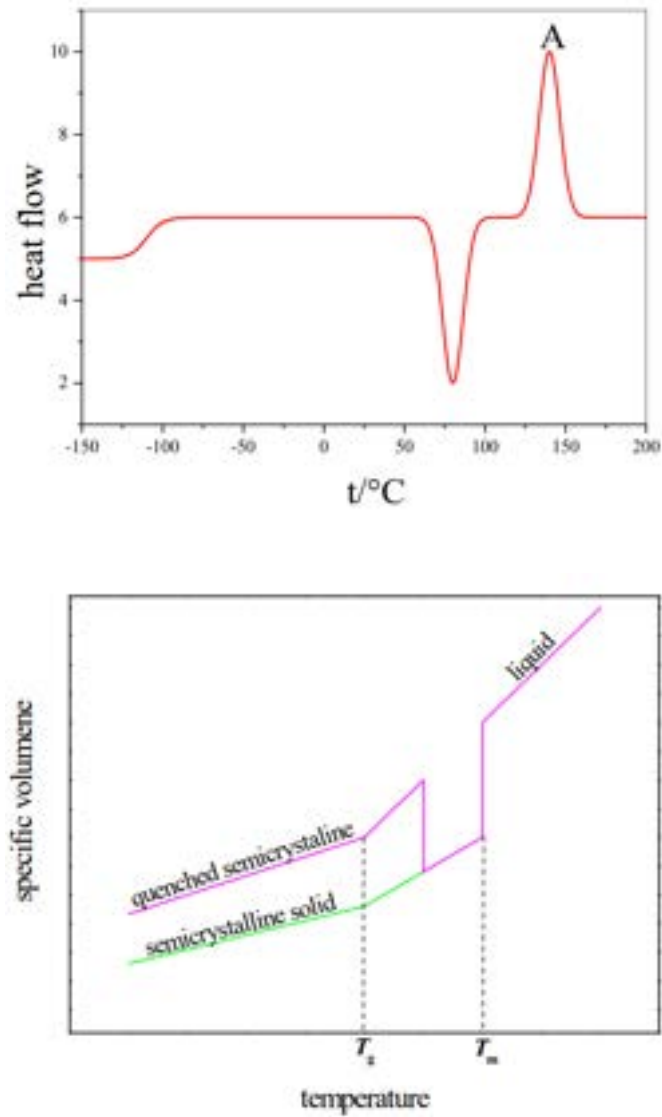


Figure 93: Quenched semicrystalline polymer volumetric response

DSC phase transition signatures

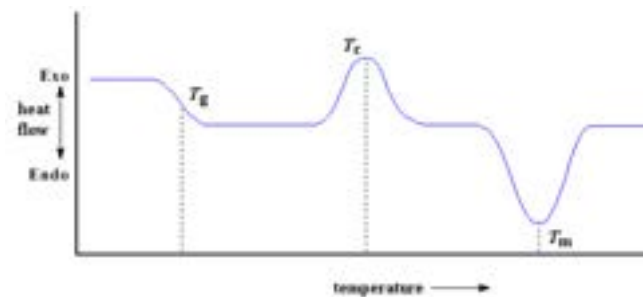


Figure 94: Caution: There is not a universal agreement on the way the heat flow is represented. You have to think!

Glasses in Polymers

Plasticizers for polymers

- A plasticizer is a T_g lowering additive (but added in substantial amounts). The plasticizer must be able to mix with the polymer to work.

- The glass transition of a mixture is described by the Flory-Fox equation

$$\frac{1}{T_{g,\text{mix}}} = \frac{\Phi_A}{T_{g,A}} + \frac{\Phi_B}{T_{g,B}}$$

- In medical grade PVC the disputed plasticizer DEHP is often used
- Global plasticizers production (2017): 7.5 MT (65% is phthalates)

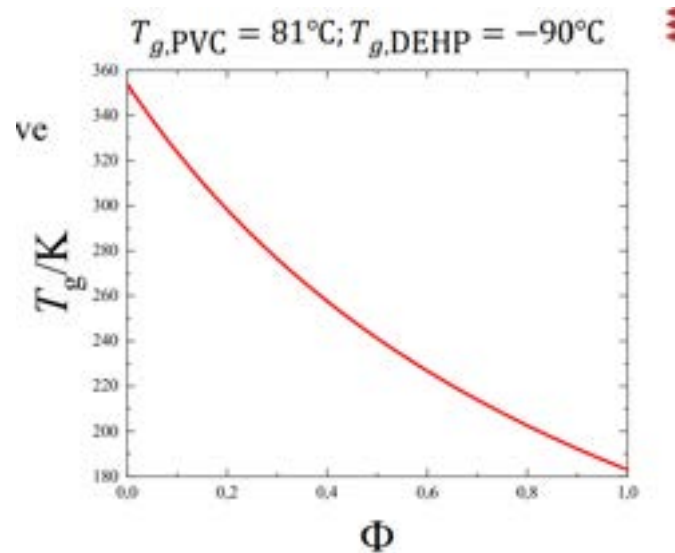


Figure 95: Plasticizers for Polymers

Polymer products examples:

- Polyvinylchloride - Domains (crystalline)
 - If plasticized (amorphous - rubbery/liquid)
 - If not plasticized (amorphous - glassy)
- Polystyrene - Domains (amorphous - glassy)
- Polycarbonate - Domains (amorphous - glassy)
- Polyethylene terephthalate - Domains (crystalline) (amorphous - glassy)

2.8 Polymer Additive Manufacturing

- Utilizing thermoplastics:
 - Crystallization
 - Glassification
- In-situ chemical reaction often termed "curing";
 - Photo polymerization
 - Photo cross-linking

Dimensionality

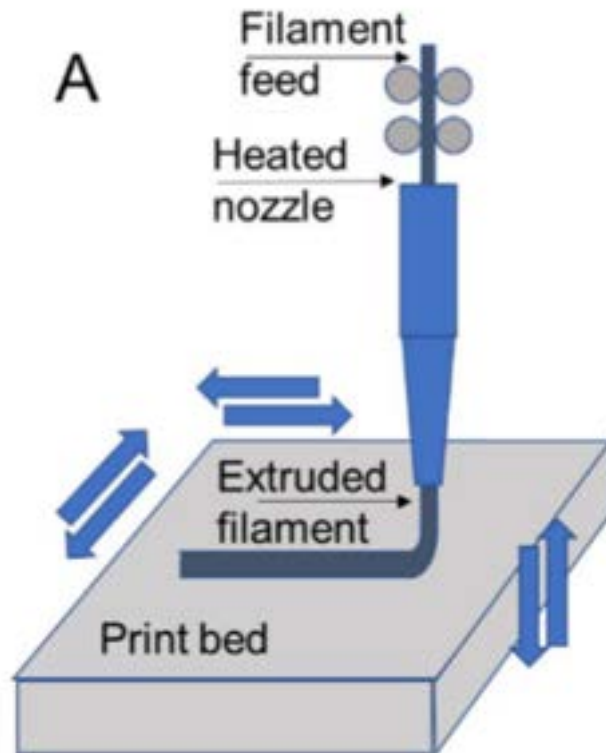
- 0D: Methods based on point to point deposition or writing
- 1D: Methods based on line scanning
- 2D: Methods based on layer by layer stereolithography
- The higher the dimensionality the speed

Fused Deposition Modelling (FDM)

The material is added drop by drop by passing a filament of the polymer through a heated nozzle.

- 0D or 1D method
- in principle all thermoplastics can be used: (ABS, PC, PLA, PEEK, PP, PA(s))

Drop by drop addition (thermoplastic) FDM (Fused Deposition Modelling) or FFF (Fused Filament Deposition)

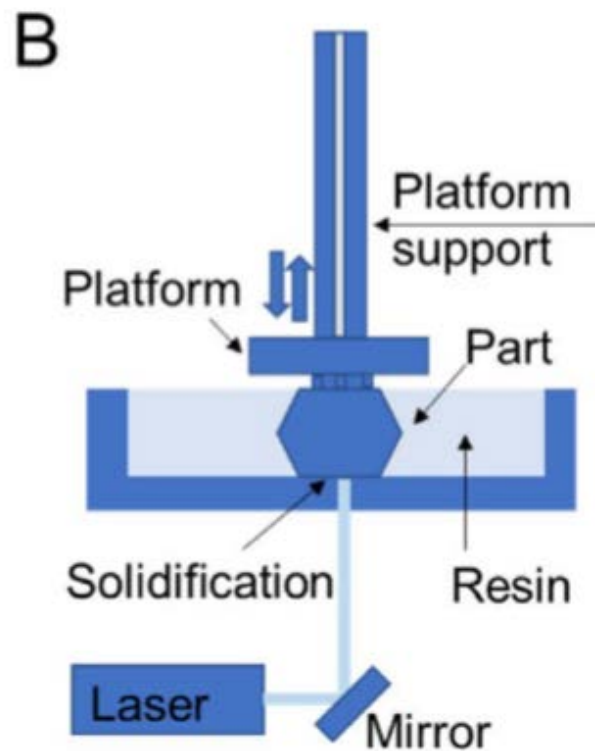


Stereolithography

The material is added layer by layer by curing, polymerizing, monomer in a plane

- 2D method
- Usually polymers with acrylate functionalities: resolution in xy plane and z direction in principle different.

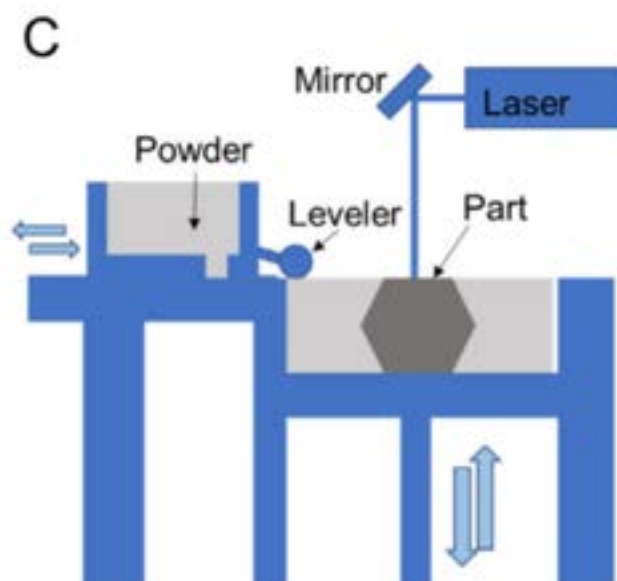
Stereolithography (thermoset photo cured)



Selective Laser sintering

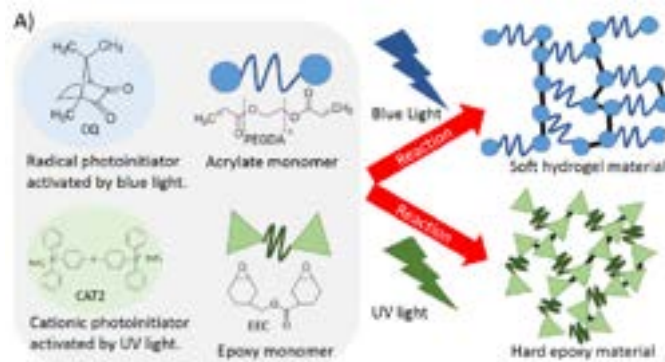
The material is added layer by layer as a powder and solidified either by curing, polymerizing or melting/sintering by position resolved illumination/heating

- 2D method
- Thermoset or thermoplastic materials
- Resolution in xy plane and z direction in principle different



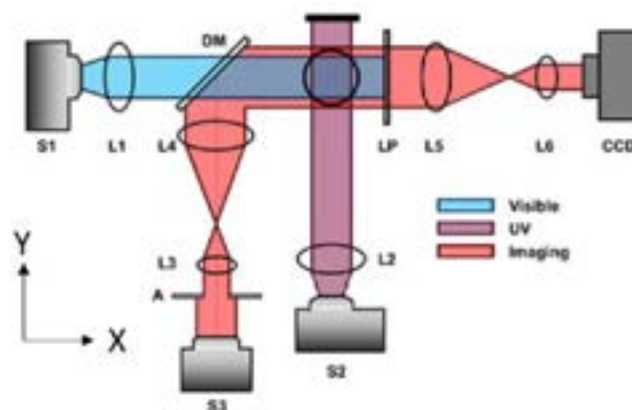
More than one material (multimaterial objects)

- You need chemistry that allows for 2 different polymerization processes.
- You need SLA setup with 2 different wavelengths of light controlled



Towards 3D methods

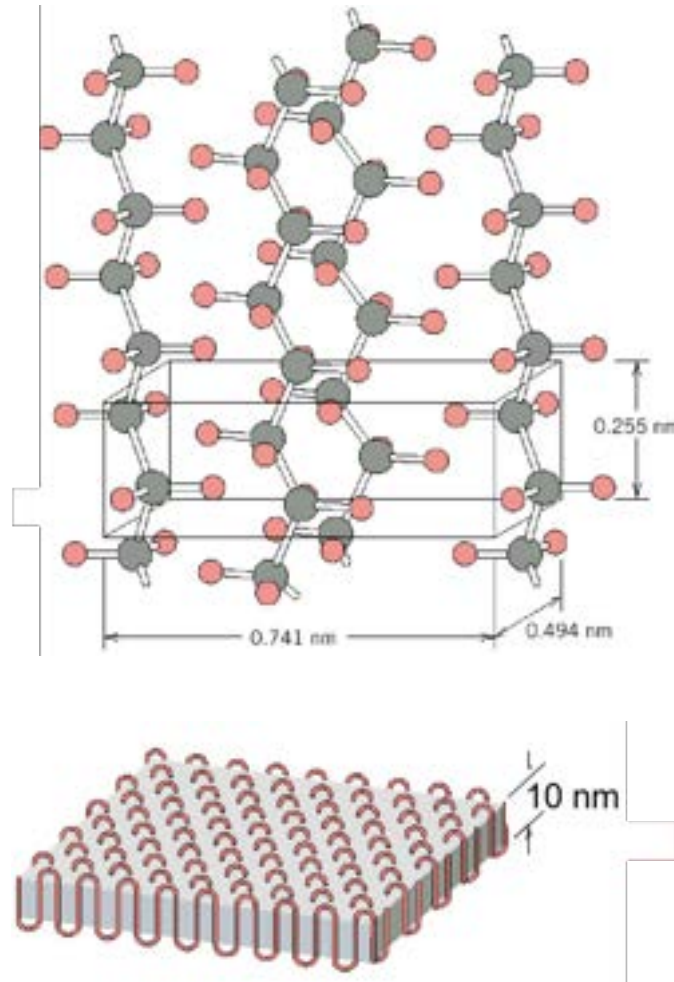
- Inverse computed tomography
- Tomographic Volumetric Printing (TVP)
- A 3D object is illuminated from several direction with a light pattern



2.9 Crystallinity



Example Polyethylene unit cell,



- Crystal must contain the polymer chains in some way
- Chain folded structure

Polymers rarely 100% crystalline

- Too difficult to get all those chains aligned
- % Crystallinity, % of material that is crystalline.
- TS (Tensile Strength) and E (Young's Modulus) often increase with crystallinity.
- Annealing causes crystalline regions to grow. Crystallinity increases.

Polymer Crystal Forms

Single crystals, only if slow careful growth,

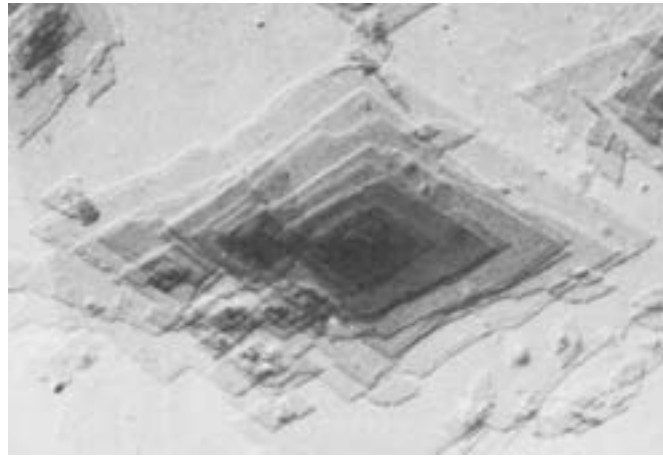


Figure 96: Example of Single Crystal growth in polymer

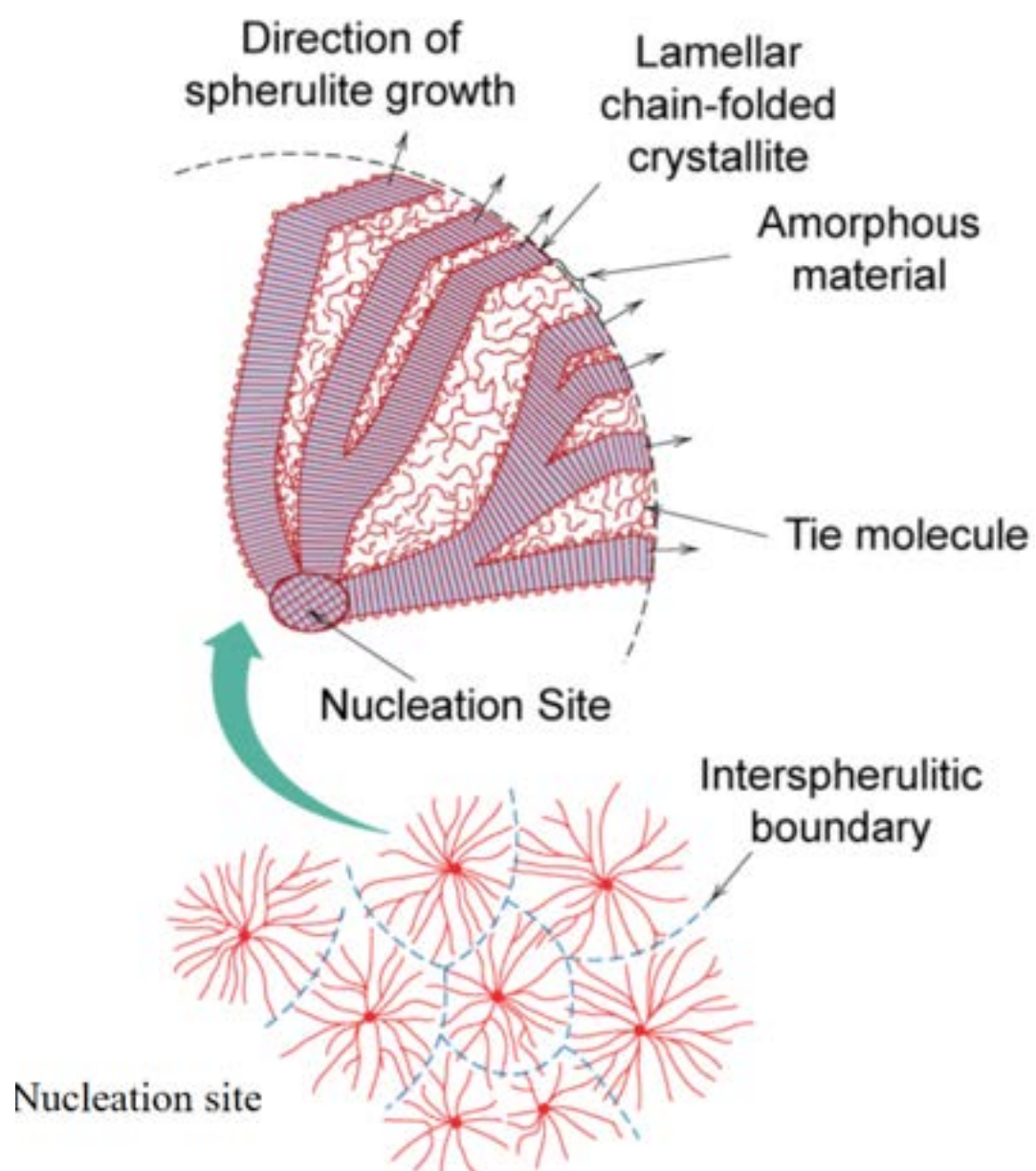


Figure 97: Spherulites, fast growth, forms lamellar (layered) structures

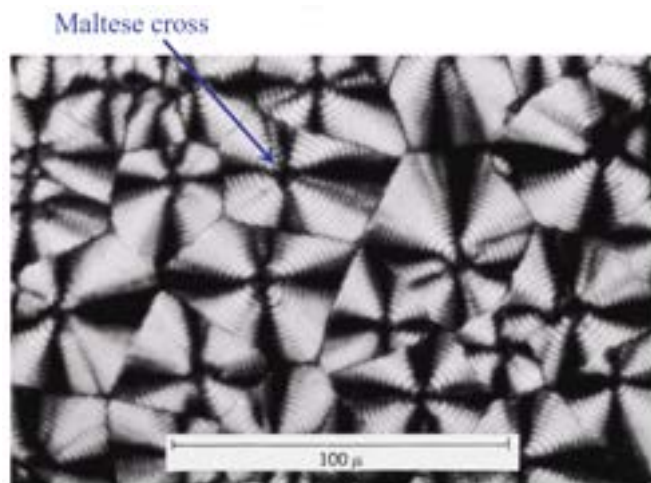


Figure 98: Spherulites, crossed polarizers

Molar Mass Average example

- two piles of stones equally big



100 g stones
200 kg pile



25 g stones
200 kg pile

- The average Size,

$$\bar{m} = \frac{100\text{g} \times 200\text{kg} + 35\text{g} \times 200\text{kg}}{200\text{kg} + 200\text{kg}} = 62.5\text{g}$$

$$\bar{m} = \frac{100\text{g} \times \frac{200\text{kg}}{100\text{g}} + 35\text{g} \times \frac{200\text{kg}}{25\text{g}}}{\frac{200\text{kg}}{100\text{g}} + \frac{200\text{kg}}{25\text{g}}} = 40\text{g}$$

Molar Mass Averages and Distribution Width

- The number of average molecular weight (Treats every polymer molecule equally),

$$M_n = \frac{\sum n_i M_i}{\sum n_i}$$

- The weight average molecular weight (Gives more weight to heavier molecules),

$$M_w = \frac{\sum w_i M_i}{\sum w_i} = \frac{\sum n_i M_i^2}{\sum n_i M_i}$$

- The width of the distribution is,

$$D = \frac{M_w}{M_n}$$

This ration is historically known a the polydispersity index (PDI) or for short polydispersity.

- Polydispersity, polymer sample contains many different chain lengths, typical for almost all synthetic polymers
- Monodispersity, polymer sample contains one molecular weight,
- In general $M_n < M_w$ (for almost all real polymers), only on special case $M_n = M_w$ (all chains have the same length)
- M_w is usually larger since it gives more weight to heavier molecules.

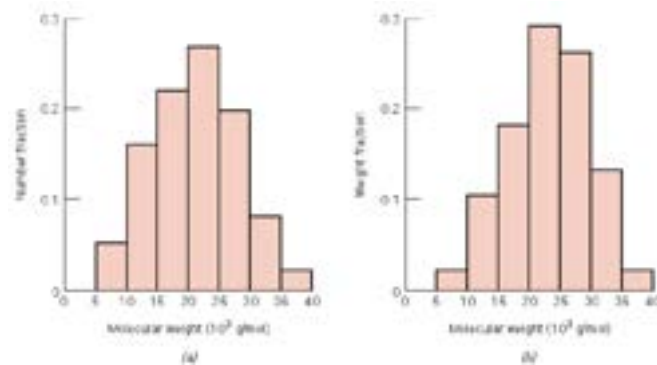
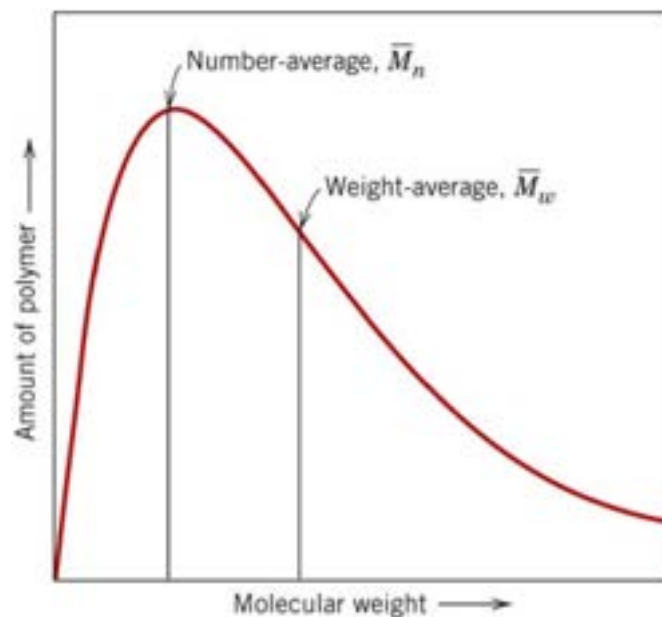


Figure 99: A hypothetical polymer molecule size distribution on the bases of (a) number and (b) weight fraction of molecules



2.10 Polymer Types

Elastomers - rubber

- Cross linked materials
- Natural rubber
- Synthetic rubber and thermoplastic elastomers
 - SBR, styrene-butadiene rubber (random copolymer)
 - SBS, styrene-butadiene thermoplastic elastomer TPE (block copolymer)

- Silicone rubber, PDMS, cross-linked

Fibers - length/diameter > 100

- Textiles are main use
 - Must have high tensile strength
 - Usually highly crystalline & highly polar
 - Formed by spinning
 - ex: Extrude polymer through a spinnerette
 - Pt plate with 1000's of holes for nylon
 - ex: rayon - Dissolved in solvent then pumped through die head to make fibers
 - The fibers are drawn
 - leads to highly aligned chains, fibrillar structure

Coatings, thin film on surface, i.e. paint, varnish

- To protect item
- Improve appearance
- Electrical insulation
- Adhesives, produce bond between two adherands
 - Usually bonded by Secondary bonds or Mechanical bonding
- Films, blown film extrusion
- Foams, gas bubbles in plastic

Advanced Polymers

- Ultrahigh molecular weight polyethylene (UHMWPE)
- Molecular weight of 4×10^6 g/mol
- Excellent properties for variety of applications: Bullet proof vest, golf ball covers, hip joints, etc.

2.11 Processing of plastics

Thermoplastic

- Can be reversibly cooled & reheated, i.e. recycled
- Heat till soft, shape as desired, then cool
- ex: polyethylene, polypropylene, polystyrene, etc.

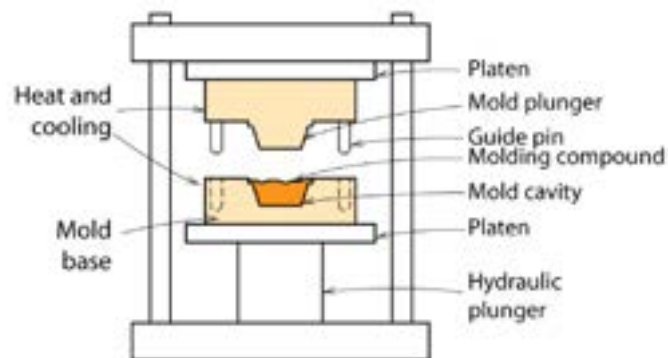
Thermoset

- When heated form a network
- Degrades (in contrast to melts) when heated
- Mold the prepolymer then allow further reaction
- ex: Urethane, epoxy

2.11.1 Molding

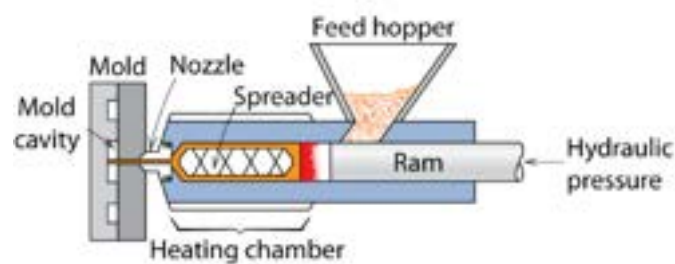
Compression and transfer molding

- thermoplastic or thermoset



Injection molding

- Thermoplastics & some thermosets



2.11.2 Extrusion

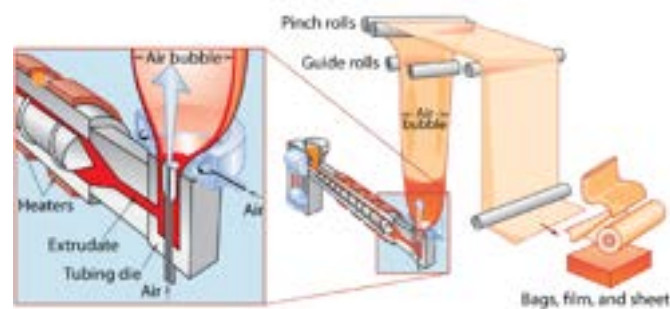
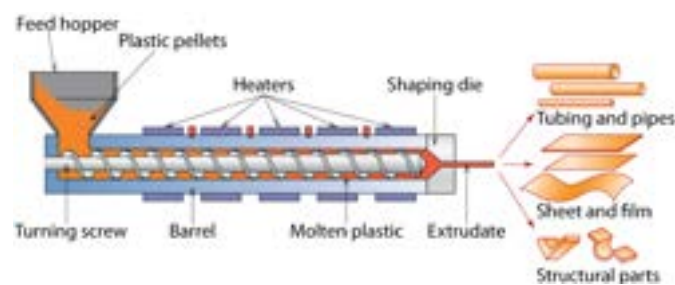
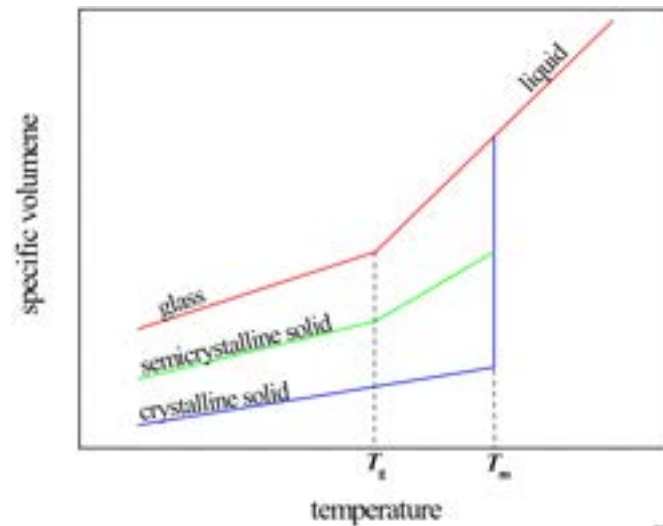


Figure 100: Blown film extrusion

2.11.3 Specific Volume vs Temperature for different types of polymers

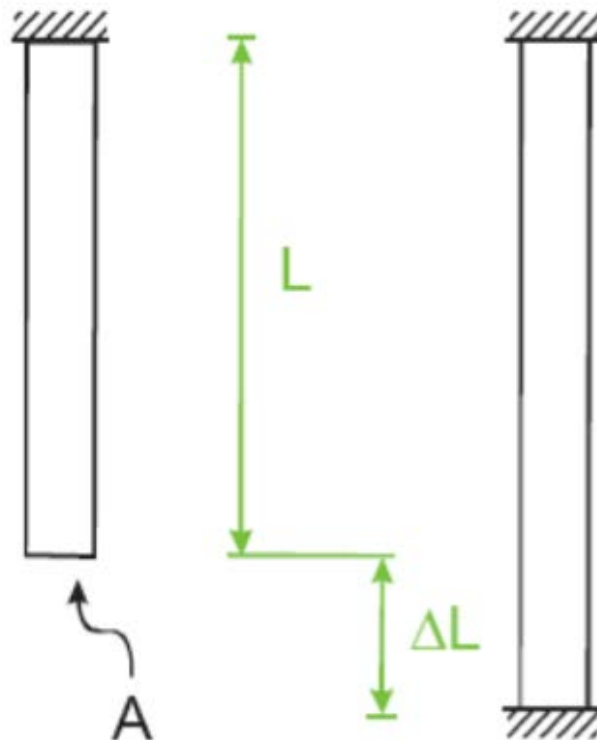


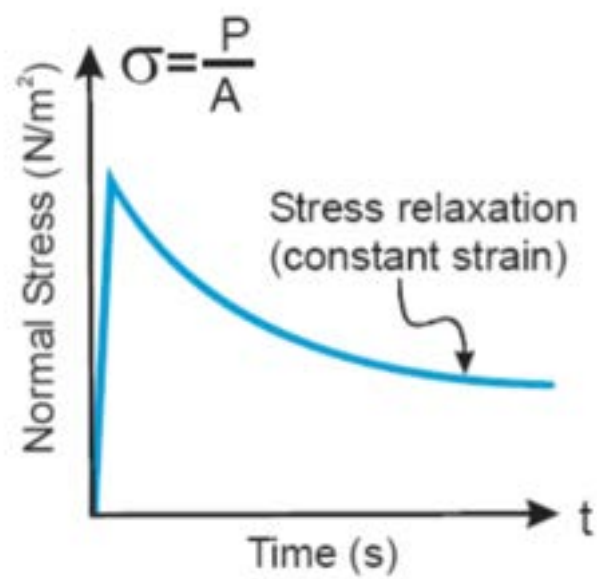
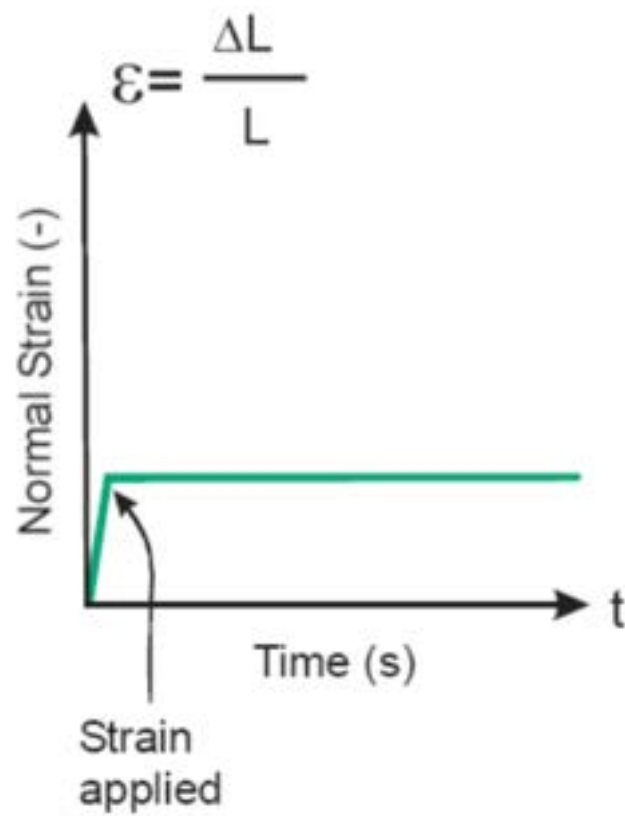
- Glass: Only amorphous glassy domains
- Crystalline: Only crystalline domains
- Semicrystalline: both crystalline and amorphous domains Glassy below T_g , rubbery above T_g

2.12 Polymer Melt flow, processing modification and properties

Stress, strain and stress relaxation

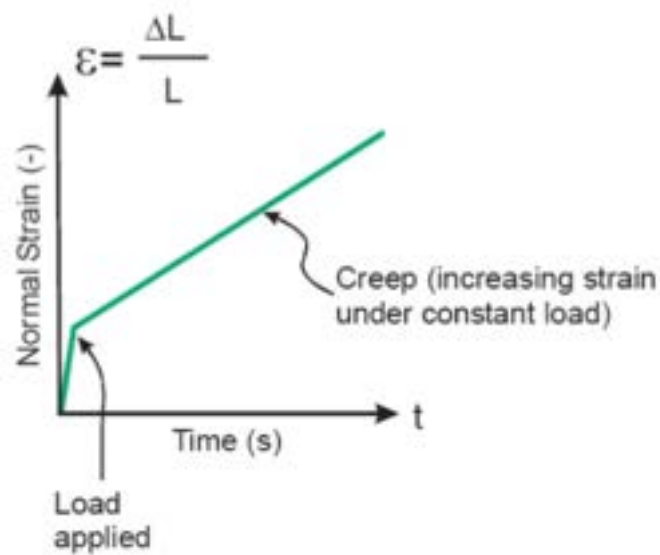
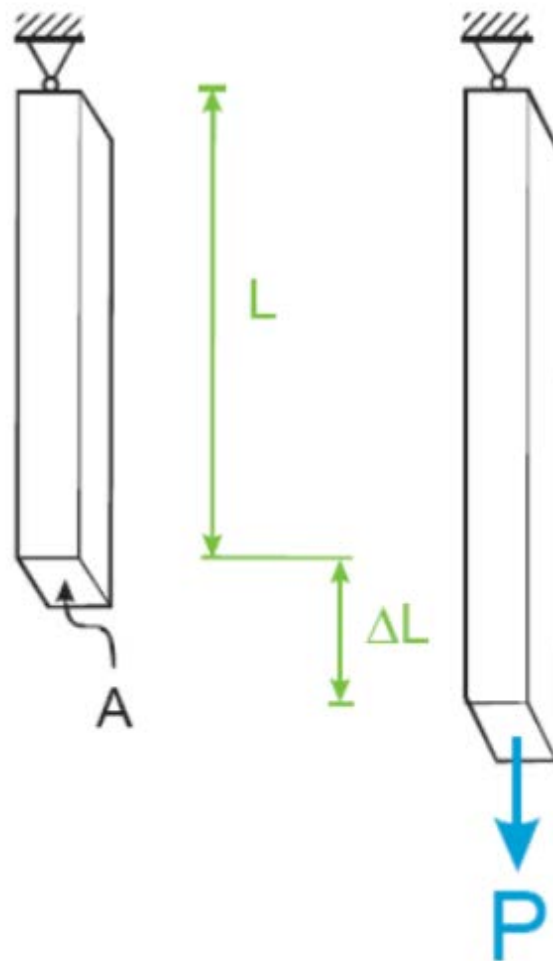
Stress relaxation, another viscoelastic effect





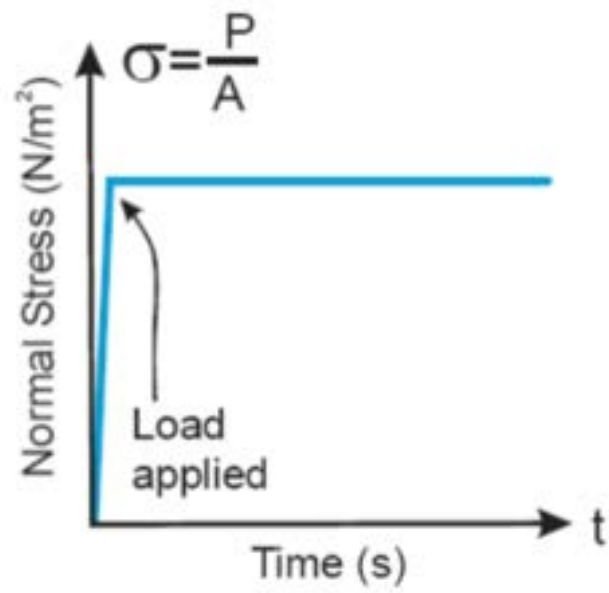
From this we see that stress decreases under constant strain.

Creep, a viscoelastic effects



The strain increases under constant stress.

The Maxwell Element



The spring is a perfect solid (elastic) element

$$\sigma = E\varepsilon \quad \text{or} \quad \frac{d\sigma}{dt} = E \frac{d\varepsilon}{dt}$$

The dashpot is a perfect viscous (liquid) element,

$$\sigma = \eta \frac{d\varepsilon}{dt} \quad \text{in elongation or,} \quad \tau = \eta \frac{d\gamma}{dt} \quad \text{in shear}$$

Note that the Maxwell elements is the simplest possible model for a viscoelastic liquid.

- In stress relaxation ($\varepsilon = \varepsilon_0$ for $t > 0$) the Maxwell element can be analuzed as,

$$\frac{d\varepsilon}{dt} = \frac{d\varepsilon_1}{dt} + \frac{d\varepsilon_2}{dt} = \frac{1}{E} \frac{d\sigma}{dt} + \frac{\sigma}{\eta}$$

since $d\varepsilon/dt = 0$ in stress relaxation

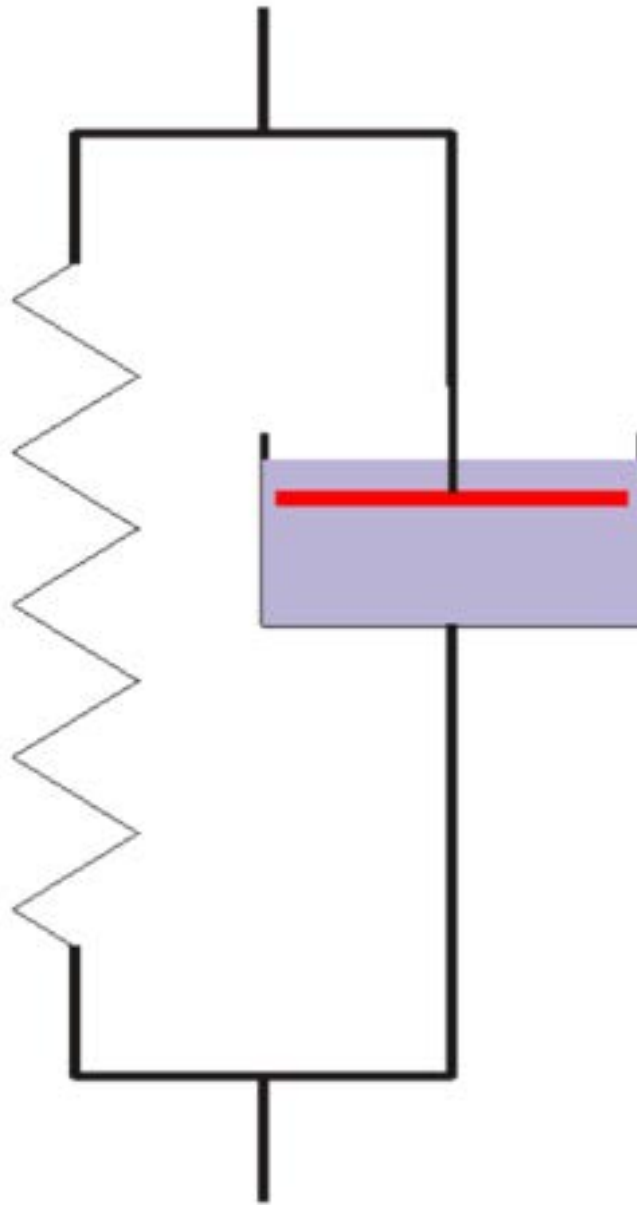
$$\frac{d\sigma}{\sigma} = -\frac{E}{\eta} dt \Rightarrow \sigma = \sigma_0 \exp\left(-\frac{t}{\eta/E}\right) = \sigma_0 \exp\left(-\frac{t}{\tau}\right) = \sigma_0 \exp\left(-\frac{t}{\tau_0}\right)$$

Where σ_0 is the initial stress and τ_0 is a relaxation time. Thus the Maxwell is a viscoelastic liquid with one relaxation time.

- In a creep experiment, the spring immediately deforms according to $\varepsilon = \sigma_0/E$. The dashpot follows Newton's law $\dot{\varepsilon} = (\sigma_0/\eta)t$. Note that $\tau_1 = (\eta/\sigma_0)$ is a characteristic time for the system.

$$\varepsilon = \varepsilon_1 + \varepsilon_2 = \frac{\sigma_0}{E} + \frac{t}{\tau_1}$$

The Voigt Element



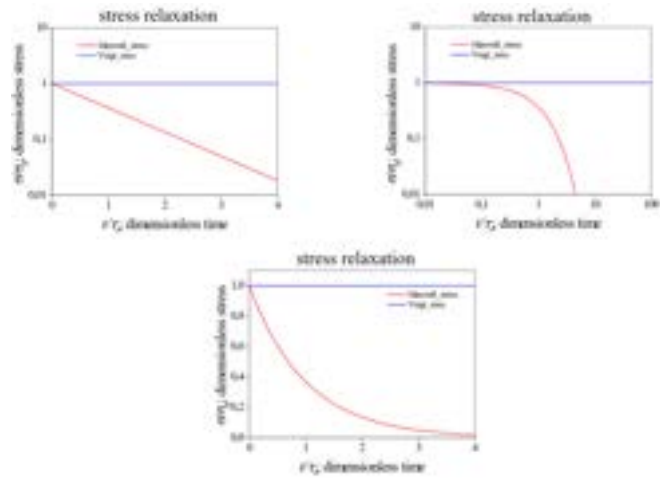
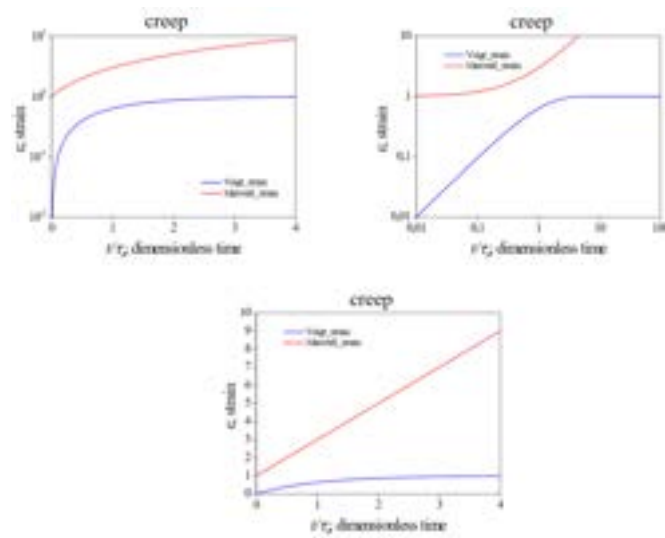
Spring and Dashpot in parallel. Note that the Voigt element is the simplest possible model for a viscoelastic solid

$$\sigma = \sigma_1 + \sigma_2 = E\varepsilon + \eta \frac{d\varepsilon}{dt} \Rightarrow \frac{d\varepsilon}{dt} = \frac{\sigma}{\eta} - \frac{1}{\eta/E} \varepsilon = \frac{\sigma}{\eta} - \frac{1}{\tau_1} \varepsilon$$

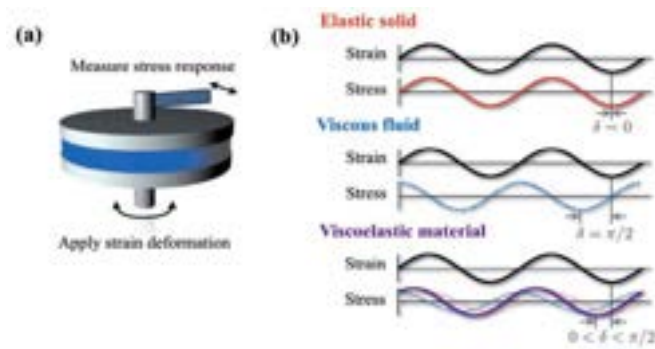
In creep $\sigma = \sigma_0$ for $t > 0$) the Voigt element can be analyzed.

$$\frac{d\varepsilon}{dt} = \frac{\sigma_0}{\eta} - \frac{1}{\eta/E} \varepsilon \Rightarrow \varepsilon = \frac{\sigma_0}{E} \left(1 - \exp \left(- \frac{t}{\tau_0} \right) \right)$$

Where σ_0 is the constant stress and τ_0 is a relaxation time. Thus the Voigt element is a viscoelastic solid with one relaxation time.

Figure 102: Stress relaxation over dimensionless time in linear and log x and y axis scalesFigure 103: Creep over dimensionless time in linear and log x and y axis scales

Oscillatory deformation



When testing materials this is often done in oscillatory mode.

The applied strain,

$$\gamma(t) = \gamma_0 \sin(\omega t)$$

Where ω is the angular frequency in the oscillatory motion.

The measured stress will (in the linear regime) also be oscillatory,

$$\sigma(t) = \sigma_0 \sin(\omega t + \delta)$$

Dynamic mechanical response functions

The stress-strain relation,

$$\sigma(t) = \gamma_0 [G'(\omega) \sin(\omega t) + G''(\omega) \cos(\omega t)]$$

$G'(\omega)$ storage or elastic modulus, $G''(\omega)$ loss modulus,

$$\tan \delta = \frac{G''}{G'}$$

It is possible to applied stress and measure a strain,

$$\begin{aligned}\sigma(t) &= \sigma_0 \sin(\omega t) \\ \gamma(t) &= \gamma_0 \sin(\omega t + \delta)\end{aligned}$$

The strain-stress relation,

$$\gamma(t) = \sigma_0 [J'(\omega) \sin(\omega t) - J''(\omega) \cos(\omega t)]$$

$J'(\omega)$, $J''(\omega)$, storage and loss compliance, Note that G is not the reciprocal of J ,

$$J' = \frac{G'}{G'^2 + G''^2}$$

Maxwell element under oscillatory motion

It is easier to analyze this in complex notation. The controlled strain,

$$\varepsilon^*(t) = \varepsilon_0^* \exp(i\omega t)$$

And the resulting stress,

$$\sigma^*(t) = \sigma_0^* \exp(i\omega t)$$

From,

$$\frac{d\varepsilon}{dt} = \frac{1}{E} \frac{d\sigma}{dt} + \frac{\sigma}{\eta} \Rightarrow E \frac{d\varepsilon}{dt} = \frac{d\sigma}{dt} + \frac{E}{\eta} \sigma = \frac{d\sigma}{dt} + \frac{1}{\tau} \sigma$$

$$E(i\omega) \varepsilon_0^* \exp(i\omega t) = (i\omega) \sigma_0^* \exp(i\omega t) + \frac{1}{\tau} \sigma_0^* \exp(i\omega t) = \sigma_0^* \left(i\omega \frac{1}{\tau} \right) \exp(i\omega t)$$

$$E^* = \frac{\sigma_0^*}{\varepsilon_0^*} = \frac{E(i\omega) \exp(i\omega t)}{(i\omega + \frac{1}{\tau}) \exp(i\omega t)} = \frac{E(i\omega)}{(i\omega + \frac{1}{\tau})} = \frac{E(i\omega)}{1 + i\omega\tau}$$

$$E^* = E' + iE'' = E \left(\frac{(\omega\tau)^2}{1 + (\omega\tau)^2} + i \frac{(\omega\tau)}{1 + (\omega\tau)^2} \right)$$

The complex modulus E^* can also be written,

$$E^* = \frac{\sigma^*}{\varepsilon^*} = \frac{\sigma_0 \exp(i(\omega t + \delta))}{\varepsilon_0 \exp(i\omega t)} = \frac{\sigma_0}{\varepsilon_0} \exp(i\delta) = \frac{\sigma_0}{\varepsilon_0} (\cos \delta + i \sin \delta) = \frac{\sigma_0}{\varepsilon_0} \cos \delta + i \frac{\sigma_0}{\varepsilon_0} \sin \delta = E' + iE''$$

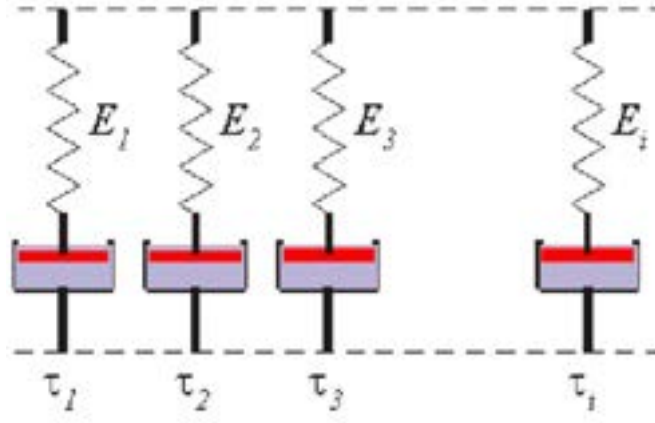
Thus,

$$E' = \frac{\sigma_0}{\varepsilon_0} \cos \delta \quad \text{and} \quad E'' = \frac{\sigma_0}{\varepsilon_0} \sin \delta$$

and,

$$\tan \delta = \frac{E''}{E'} = \frac{E\omega\tau}{E(\omega\tau)^2} = \frac{1}{\omega\tau}$$

Generalized models



If several Maxwell elements are put in parallel (the generalized Maxwell Model).

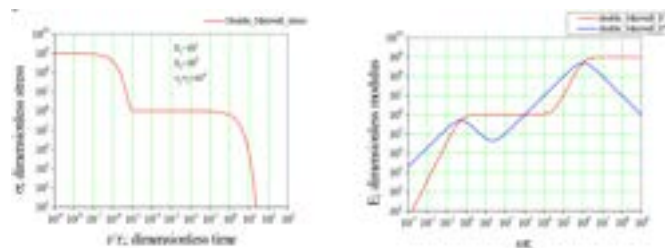
Stress relaxation,

$$\sigma(t) = \epsilon \sum_i E_i \exp\left(-\frac{t}{\tau_i}\right)$$

Dynamic response,

$$E' = \sum_i \frac{E_i(\omega\tau_i)^2}{1 + (\omega\tau_i)^2}, \quad E'' = \sum_i \frac{E_i(\omega\tau_i)}{1 + (\omega\tau_i)^2}$$

For just two Maxwell Elements,



Because of the relation $\log \omega \sim \log \frac{1}{\tau} \sim T^{-1}$

There is an approximate equivalence between increasing frequency and decreasing temperature.

These relations assume the validity of the Boltzmann superposition principle which states that incremental deformations can be added. Or: the response to a deformation is independent of the previous deformation history.

Polymer melts mechanical behavior, stress relaxation response

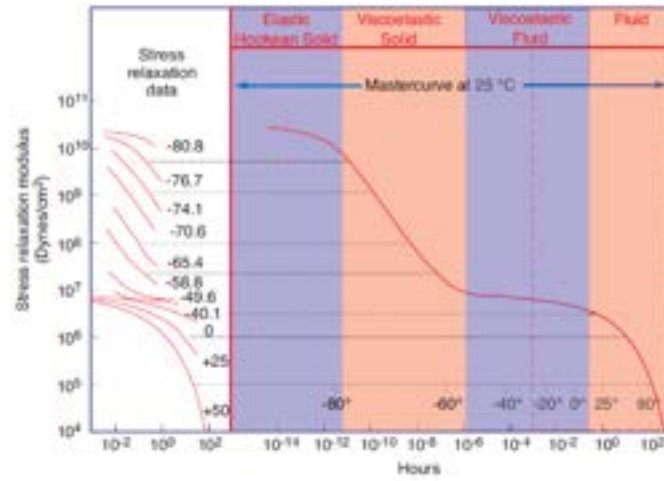


Figure 104: Relaxation modulus curves for polyisobutylene and corresponding master curve at 25 °C

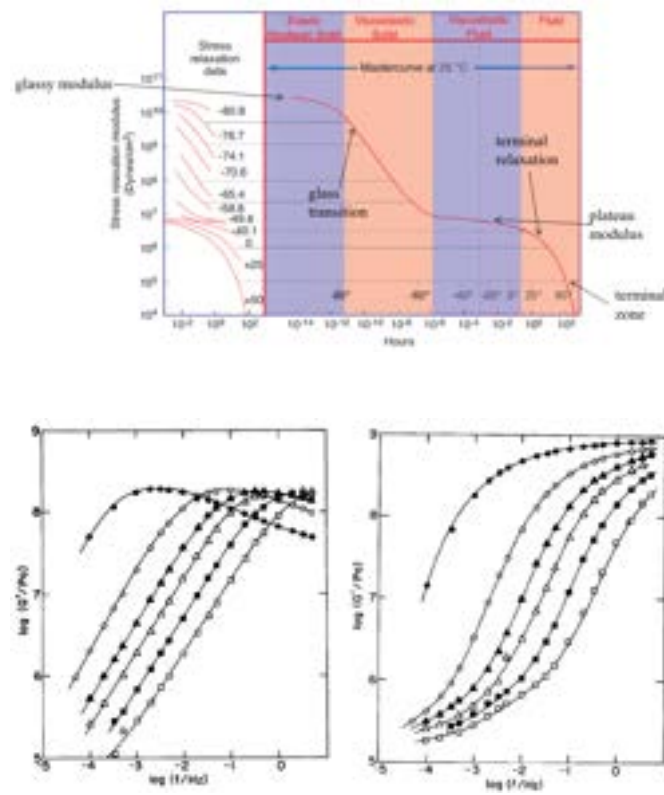


Figure 105: Dynamic mechanical data on polystyrene (molecular weight 98000) for several temperature. (a) The imaginary part of shear modulus, G'' . (b) the real part of shear modulus, G' . The data are $\bullet$ at 359.7 K, $\circ$ at 364.5 K, $\blacktriangle$ at 367.5 K, $\triangle$ at 369.0 K, $\blacksquare$ at 371.6 K, and $\square$ 373.9 K.

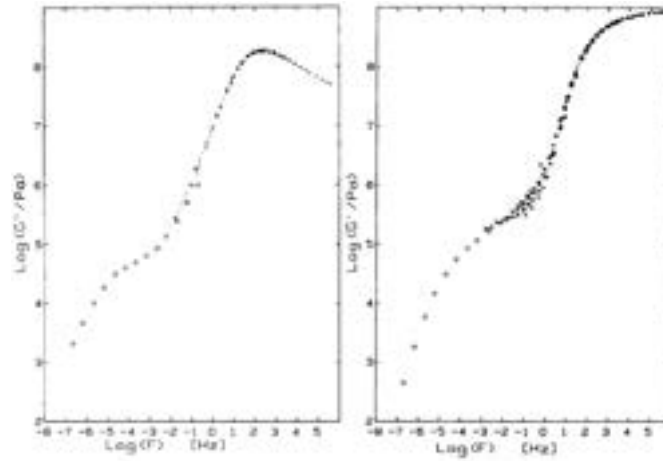


Figure 106: Atactic polystyrene

Time-temperature superposition

If the relaxation time in a material is due to an activated process then,

$$\tau = c \exp\left(\frac{A}{RT}\right), \quad \text{or} \quad \log \omega \sim \log \frac{1}{\tau} \sim T^{-1}$$

Where A is the activation energy (and c is a constant).

One would expect the influence of temperature to be describes through the standard Arrhenius equation,

$$\ln a_T = \frac{A}{R} \left(\frac{1}{T} - \frac{1}{T_0} \right)$$

It turns out that this equation for glass forming liquids (such as polymers) should be replaced by,

$$\ln a_T = \frac{-C_1(T - T_{ref})}{C_2 + (T - T_{ref})}$$

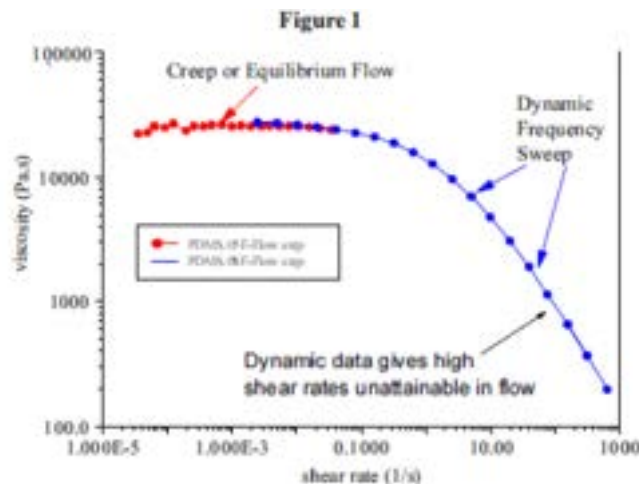
This is equivalent to replacing $\tau = c \exp\left(\frac{A}{RT}\right)$ with $\tau = c \exp\left(\frac{A}{R(T - T_{VFT})}\right)$

Which is offset the zero-point of the temperature scale.

The Cox-Merz rule

Measuring viscosities at high $\dot{\gamma}$ is difficult. Empirical rule (The Cox-Merz),

$$\eta^* = \frac{G^*}{i\omega}, \quad \text{thus} \quad |\eta^*| = \frac{|G^*|}{\omega}$$



2.13 Polymer Melts, Processing and Viscosity

Polymer melt as the starting point

A polymer melt is defined as,

- An ensemble of polymer chains (possibly branched but not cross linked)
- No solvent
- Above T_g
- Above T_m (if the polymer can be semi crystalline)

The material is a thick (high viscosity) liquid. It is in this state that thermoplastic polymers are processed.

The reptation (to move as a snake) model views the chains as moving in a tube created by the other chains. Between the chains one finds a free volume which allows the chain to move.



Figure 107: **Entanglement effects** Modelling the internal constraints on the chain motion imposed by the entanglement by a 'tube'. The average over the rapid wriggling motion within the tube defines the 'primitive path' (continuous dark line)

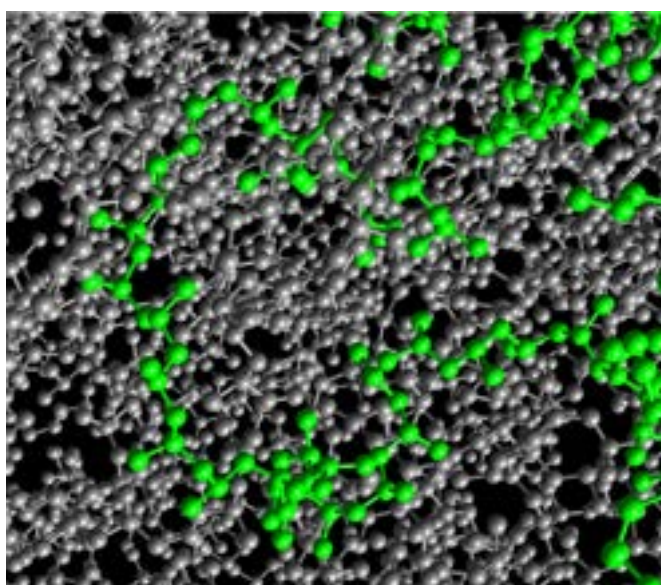


Figure 108: A ball and stick model of polypropylene

Introduction to Rheology

Rheology: The science of "the flow of matter" (rheo is greek for flow)

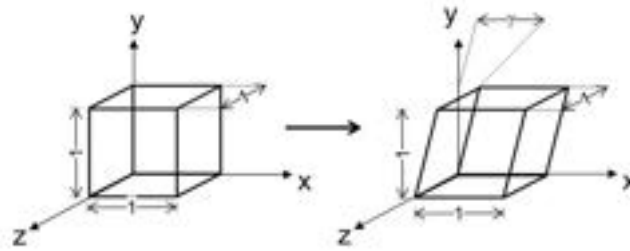
Modes of deformation:

Figure 109: Shear

Shear flow:

$$\begin{aligned} v_x &= \dot{\gamma}(t)y = \frac{dx}{dt} \\ v_y &= 0 = \frac{dy}{dt} \\ v_z &= 0 = \frac{dz}{dt} \end{aligned}$$

Shear rate: $\dot{\gamma}(t)$

$$\dot{\gamma}(t) = \frac{d\gamma}{dt}(t)$$

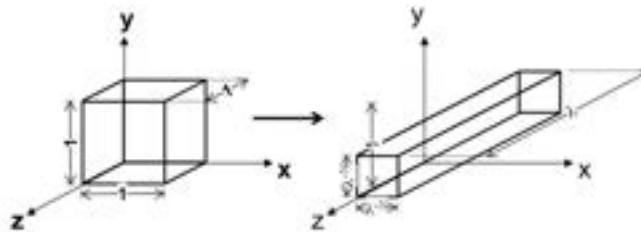


Figure 110: Elongation

Elongation flow:

$$\begin{aligned} v_x &= -\frac{1}{2}\dot{\epsilon}(t)x = \frac{dx}{dt} \\ v_y &= -\frac{1}{2}\dot{\epsilon}(t)y = \frac{dy}{dt} \\ v_z &= +\dot{\epsilon}(t)z = \frac{dz}{dt} \end{aligned}$$

A particle follows,

$$z = z_0 \exp(\dot{\epsilon}_0 t)$$

The elongation ratio and elongational rate is related to,

$$\lambda = \exp(\epsilon), \quad \epsilon = \dot{\epsilon}_0 t$$

Processing consideration

	Casting dipping	Laminating winding	Press molding	Injection molding	Extrusion	Blow-moldi
$\log\left(\frac{\dot{\gamma}}{S-1}\right)$	1-2	1-2	0-1	3-4	2-3	2-3

Table 5: Relevant shear rates

Flow curve showing Viscosity or Shear stress Vs. Shear Rate. Newtonian for $\dot{\gamma} \lesssim 0.1 \text{ s}^{-1}$ depending on the temperature

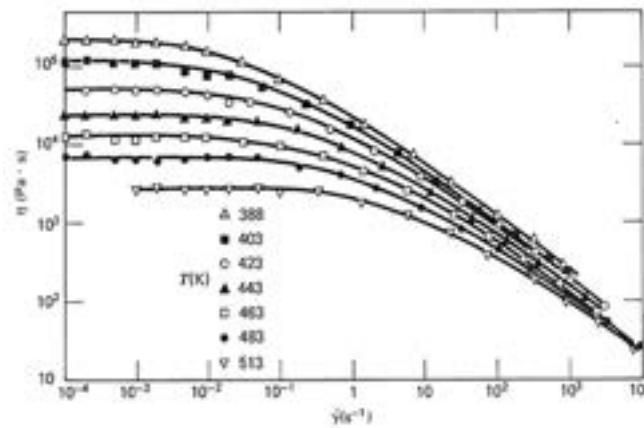


Figure 111: Non-Newtonian viscosity η of low density polyethylene melt (Melt I) at several different temperatures. The data taken at shear rates below $5 \times 10^{-2} \text{ s}^{-1}$ were obtained with a Weissenberg Rheogoniometer; the viscosity at higher shear rates was determined using a capillary viscometer.

Viscosity lower at higher shear rates. Energy saved by producing at higher shear rates.

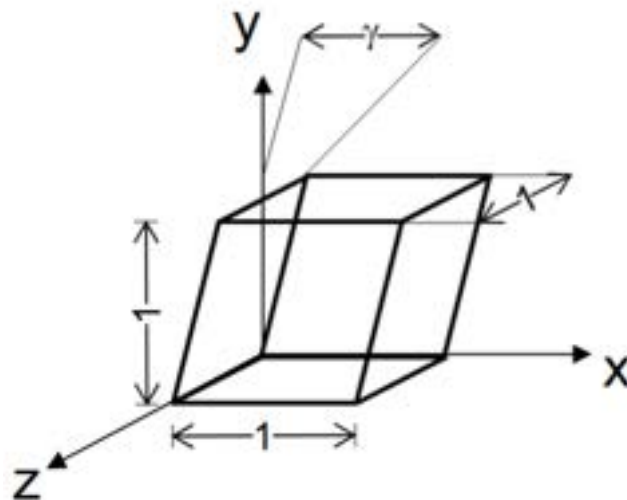
Viscosity and Newtonian liquids

Stress

$$\sigma = \frac{f}{A}, \quad \text{Where } f \text{ is the force and } A \text{ the area the force acts on}$$

Shear viscosity function

$$\sigma = \tau_{yx} = \eta(\dot{\gamma})\dot{\gamma}$$



In general shear stress is a non-linear function of shear rate.

The simplest case, $\eta(\dot{\gamma})$, constant, is followed (by definition) by Newtonian liquids

Shear thinning liquids, Careau model

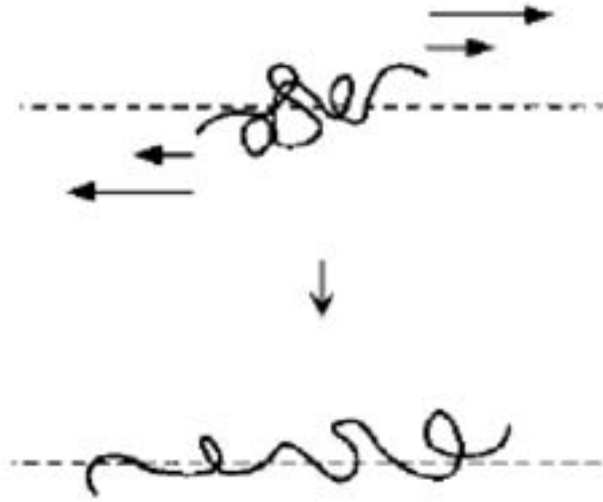


Figure 112: Single chain conformation under strong shear flow

- Slow shear, chain conformation is unaffected
- Strong shear, chain conformation is stretched
- Very strong shear, chain conformation is completely stretched
- Thus,

Slow shear	Newtonian liquid, η_0
Strong shear	Shear thinning
Very strong shear	Newtonian liquid, η_∞

$$\eta = \eta_\infty + (\eta_0 - \eta_\infty)[1 + (\lambda\dot{\gamma})]^{(n-1)/2}$$

- λ , n are system specific constants. As $\eta_\infty \rightarrow 0$

$$\eta = \eta_0[1 + (\lambda\dot{\gamma})^{(n-1)/2}]$$

- λ , n are system specific constants. Alternatively as $\eta_\infty \rightarrow 0$, $ab = 2$, $1/b = (n-1)/2$, $\lambda = 1/\dot{\gamma}_{crit}$,

$$\frac{\eta}{\eta_0} = \left[1 + \left(\frac{\dot{\gamma}}{\dot{\gamma}_{crit}} \right)^{ab} \right]^{1/b}$$

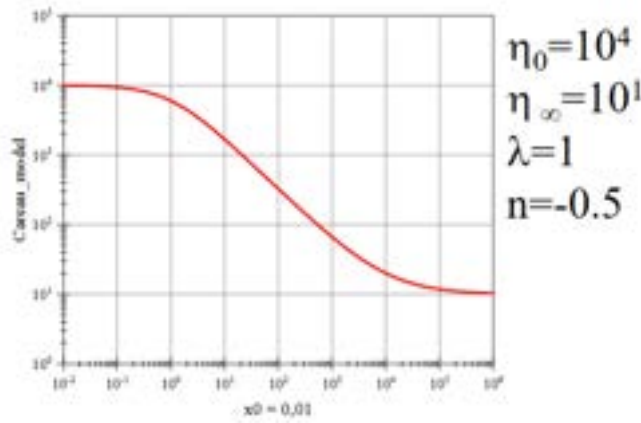


Figure 113: Visualization of Careau model

The deformed polymer melt can only relax by chain reptation. The process is strongly chain length dependent, from the tube model the longest relaxation time τ_m ,

$$\tau_m \propto M^3$$

For unentangled, short chain polymer melts,

$$\eta \propto M$$

For entangled polymer melts, experimentally,

$$\eta \propto M^{3.4}$$

Where M is the molecular weight of the polymer.

Temperature dependence of viscosity

From text on glasses,

$$\eta = \eta_0 \exp\left(\frac{DT_0}{T - T_0}\right) = \eta_0 \exp\left(\frac{E_a}{RT}\right)$$

Where $T_0 = 0$ is the Arrhenius expression.

Melt Flow Index MFI

The Melt Flow Index (MFI) is an easily measured quantity for a polymer melt.

Used as a simple characterization of polymer melts to describe processing properties.

The MFI measured how much polymer melt flows through the die in 10 minutes at a specified temperature (between 125 °C and 300 °C) given a specified load, (between 0.325 kg and 30.4 kg)

In the view of the Careau model the MFI is, a crude characterization which neglects the flowrate dependence of viscosity.

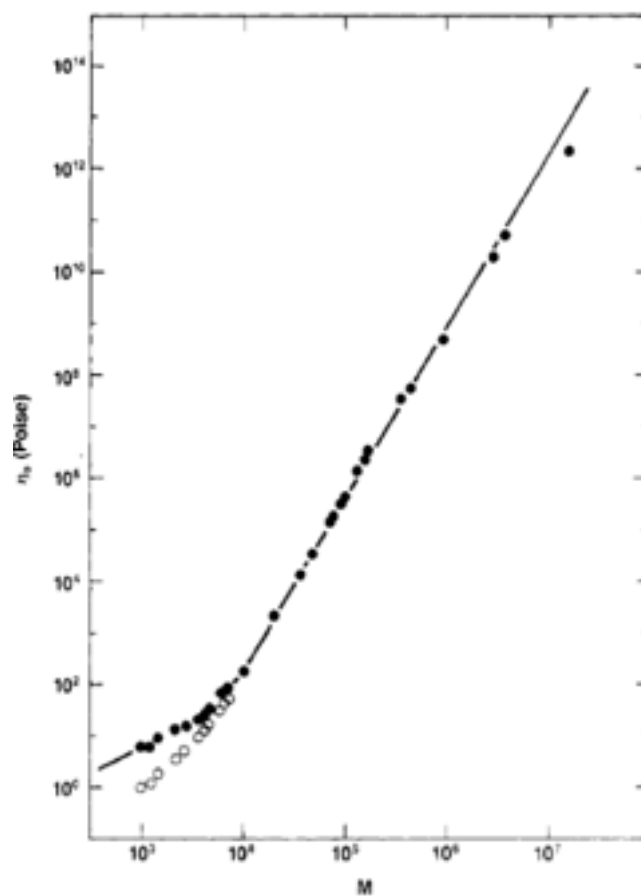


Figure 114: Viscosity vs. molecular weight for polybutadiene at 25 °C. The filled circles indicate the data at the iso-free-volume state. The open circles indicate the unadjusted results. The lines have been drawn with slopes of 1.0 and 3.4

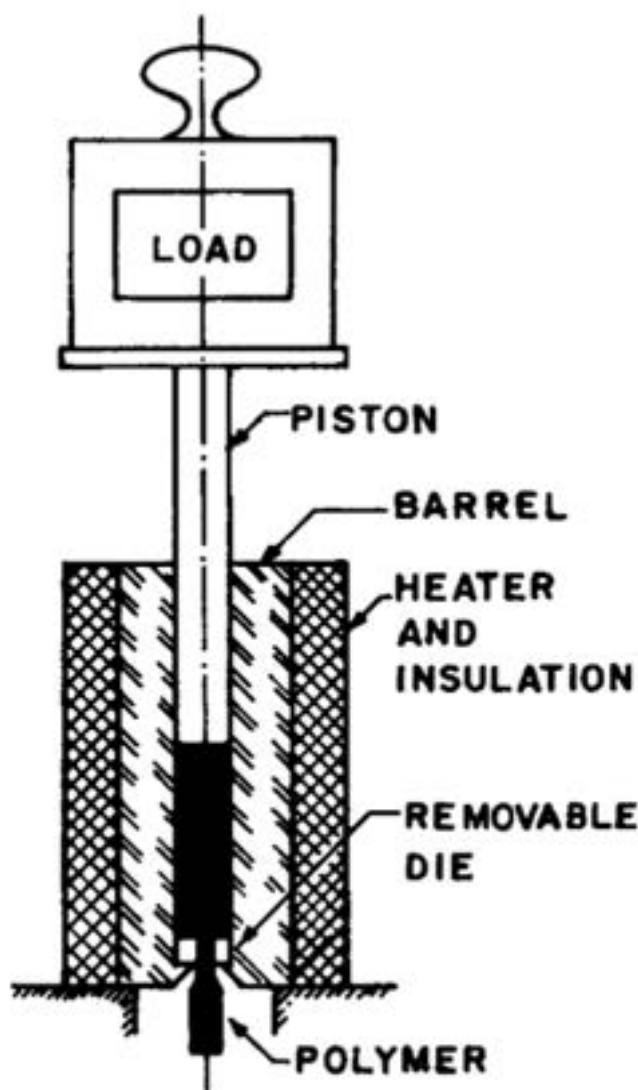


Figure 115: Schematic diagram of the melt flow index apparatus showing a cross-sectional view of the important parts.

Polymer solubility

Mixtures

- Polymers are basically never miscible,

$$\Delta_{mix}S = -nR \sum_i x_i \ln x_i \approx 0$$

$$\Delta_{mix}H \neq 0 \text{ and normally } \Delta_{mix}H > 0$$

For a molar mass of e.g. $10^6 \text{ g}\cdot\text{mol}^{-1}$ the number of moles, n , is really small

- Polymers are soluble in solvents. However, since the enthalpy in most cases is against dissolution the choice of solvent must ensure that the enthalpy contribution is small.
- For a partially crystalline polymer the enthalpy of melting will contribute to the positive enthalpy of mixing. Thus crystalline polymers are insoluble in general.

Solids

- Perfect elastic material deformed in shear (Hookes law),

$$\sigma = G\gamma$$

- G shear modulus
- In elongation

$$\sigma_t = E\varepsilon$$

- E , Young's modulus,

$$E = 2G(1 + \mu)$$

- μ Poisson's ratio
- For many soft materials (including all polymers including the solid states) $\mu = 0.5$ (for fundamental reason $\mu < 0.5$). Thus,

$$E = 3G$$

The stiffness of rubber, affine network model

- Rubber is similar to a polymer melt, but chains are chemically bonded.
- The Gaussian chain, end-to-end distance,

$$\langle R^2 \rangle = Nl^2$$

- Entropic spring,

$$F(\bar{r}) = \frac{3kT}{\langle R^2 \rangle} \bar{r}$$

- Affine deformation, all cross linking sites follow the macroscopic deformation



- The modulus is simply the sum of the contribution from the individual chains.
- Elastic modulus,

$$G = \frac{N}{V} kT$$

Where N is the number of elastically active chains in the volume V .

- The plateau modulus, G_N^0 , mean field approach, look at just one chain and assume the rest of the chain form a (temporary) network of cross-links (entanglements).

$$G_N^0 = a \frac{\rho RT}{M_e}$$

- ρ polymer melt density, R gas constant, M_e is the entanglement molar mass, that is mean chain size between the temporary cross-links in the melt, a is an order unity constant (general consensus is $a = 4/5$)

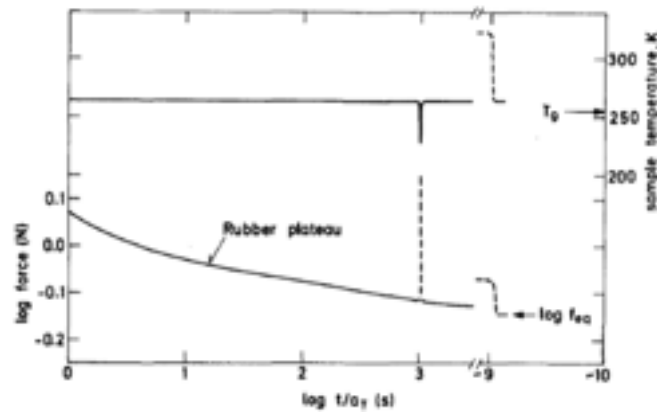


Figure 116: Poly-1,2-butadiene ($T_g = 255\text{K}$) stress-relaxation. Sample cross-linked at $\log(t/a_T) = 3$

Table 6: Comparison of Calculated and Measured Values of the Plateau Modulus at 413 K

Polymer	W_u^b	This work G_N^0 (MPa)	Measured	M_e	C_∞
PE	1.20	2.50	2.60	828	7.3
PEO	2.73	1.67	1.80	1624	5.5
1,4-PBd	2.15	1.01	1.25	1810	5.6
alt-PEP	1.20	0.76	0.85	2284	6.4
PEB-24.6	0.87	0.69	0.67	3276	7.1
alt-PEB	0.97	0.45	0.52	4226	6.6
a-PP	1.19	0.40	0.47	4623	6.0
1,4-PI	2.13	0.37	0.42	5429	5.0
PIB	0.75	0.30	0.32	7288	6.8
PMMA	0.31	0.29	0.31	10013	9.1
PEE	0.94	0.18	0.20	11084	5.9
PDMS	0.70	0.18	0.20	12293	6.3
PS	0.23	0.20	0.20	13309	9.6
PVCH	0.33	0.07	0.06	38966	7.6

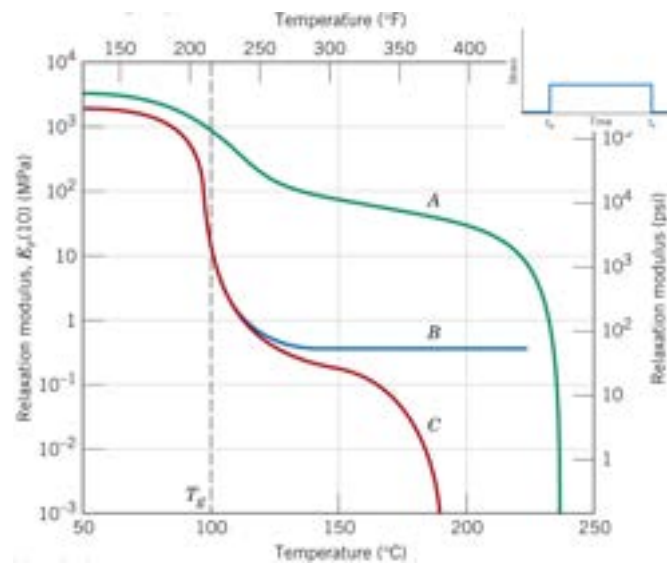


Figure 117: PS, isotactic, lightly c-linked, amorphous

2.14 Polymer mechanical failure

Failure modes

A material can fail (break) due to mechanical deformation,

- Stresses are induced exceeding the failure stress
- High stresses are concentrated in defects in the material (scratches, holes, ...)
- Failure is a measure of the most severe defect in a specimen

Failure can be,

- Brittle, no plastic deformation observed at the failure point
- Ductile, plastic deformation observed at the failure point

Fatigue, failure due to the cumulative effect of (many) smaller deformations.

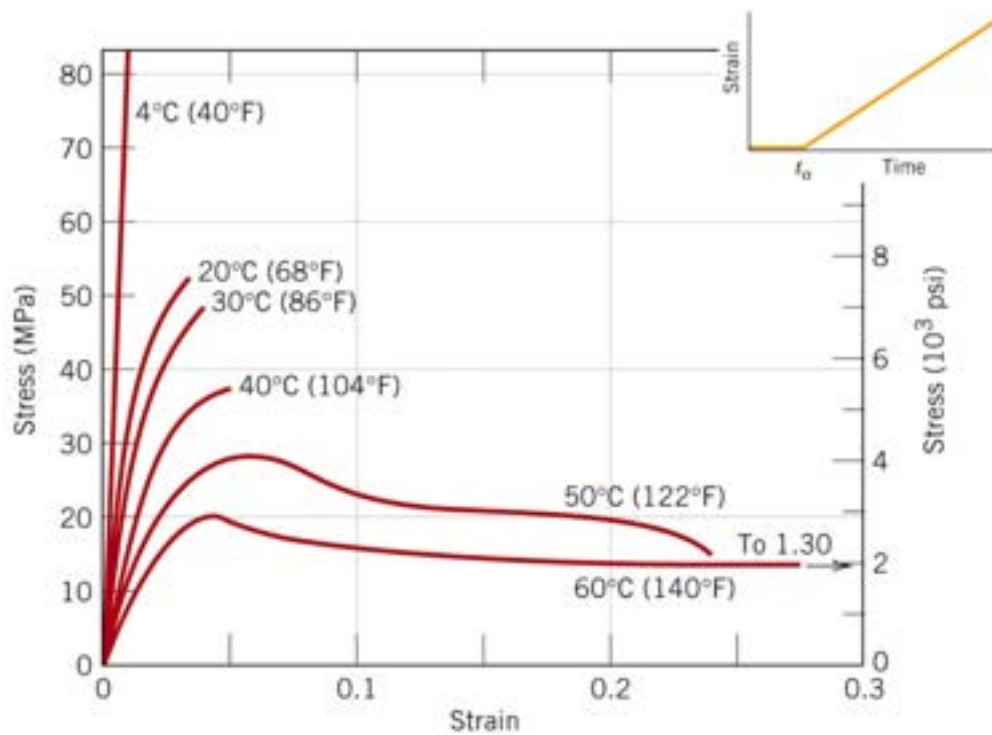


Figure 118: PMMA stress strain curves

Fatigue testing

- Cyclic deformation
- Mean stress,

$$\sigma_m = \left(\frac{\sigma_{\max} + \sigma_{\min}}{2} \right)$$

- Stress range,

$$\sigma_r = \sigma_{\max} - \sigma_{\min}$$

- Stress amplitude,

$$\sigma_a = \frac{\sigma_r}{2} = \frac{\sigma_{\max} - \sigma_{\min}}{2}$$

- The number of cycles, N , until failure is recorder for each stress level, S .

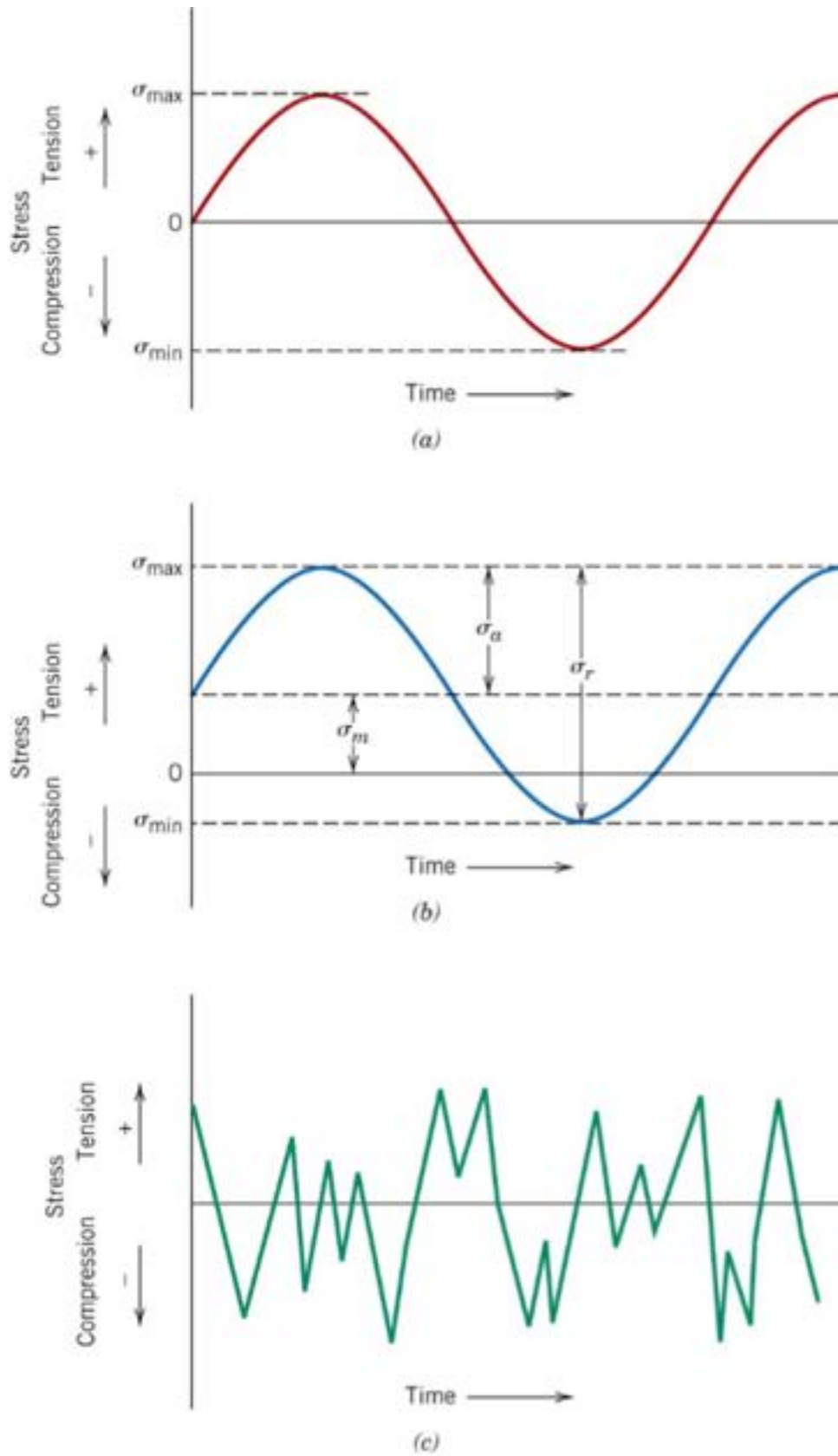


Figure 119: Fatigue testing

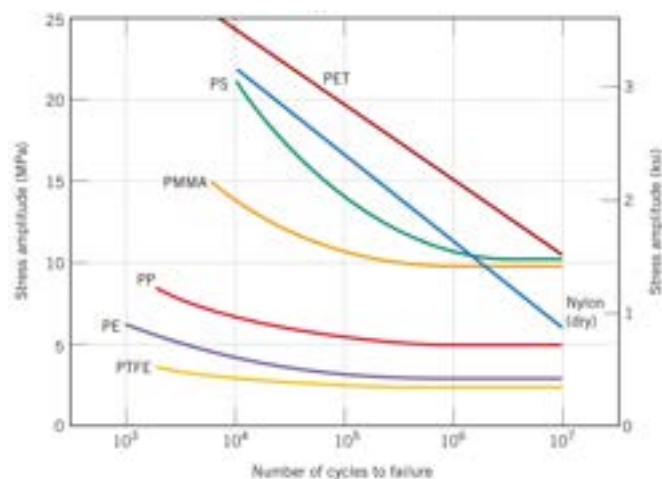


Figure 120: Stress amplitude Vs. number of cycles to failure

Necking, typical in semicrystalline polymers

- At yielding reorganization of the crystalline domains occurs
- Orientation of crystal is a potential strengthening mechanism
- The yield stress of the reorganized parts is substantially increased
- The force in any cross-section of the specimen is conserved
- The reorganized parts can sustain a substantially increased stress

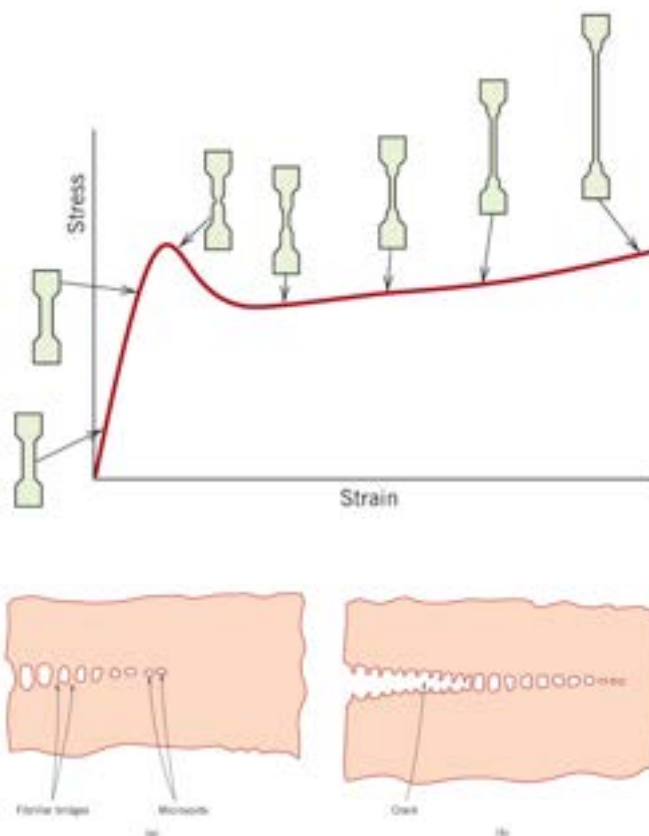


Figure 121: Visualization of Cracks and crazes

Material	Specific Gravity	Tensile Modulus [GPa (ksi)]	Tensile Strength [MPa (ksi)]	Yield Strength [MPa (ksi)]	Elongation at Break (%)
Polyethylene (low density)	0.917–0.932	0.17–0.28 (25–41)	8.3–31.4 (1.2–4.55)	9.0–14.5 (1.3–2.1)	100–650
Polyethylene (high density)	0.952–0.965	1.06–1.09 (155–158)	22.1–31.0 (3.2–4.5)	26.2–33.1 (3.8–4.8)	10–1200
Poly(vinyl chloride)	1.30–1.58	2.4–4.1 (350–600)	40.7–51.7 (5.9–7.5)	40.7–44.8 (5.9–6.5)	40–80
Polytetrafluoroethylene	2.14–2.20	0.40–0.55 (58–80)	20.7–34.5 (3.0–5.0)	—	200–400
Polypropylene	0.90–0.91	1.14–1.55 (165–225)	31–41.4 (4.5–6.0)	31.0–37.2 (4.5–5.4)	100–600
Polystyrene	1.04–1.05	2.28–3.28 (330–475)	35.9–51.7 (5.2–7.5)	—	1.2–2.5
Poly(methyl methacrylate)	1.17–1.20	2.24–3.24 (325–470)	48.3–72.4 (7.0–10.5)	53.8–73.1 (7.8–10.6)	2.0–5.5
Phenol-formaldehyde	1.24–1.32	2.76–4.83 (400–700)	34.5–62.1 (5.0–9.0)	—	1.5–2.0
Nylon 6,6	1.13–1.15	1.58–3.80 (230–550)	75.9–94.5 (11.0–13.7)	44.8–82.8 (6.5–12)	15–300
Polyester (PET)	1.29–1.40	2.8–4.1 (400–600)	48.3–72.4 (7.0–10.5)	59.3 (8.6)	30–300
Polycarbonate	1.20	2.38 (345)	62.8–72.4 (9.1–10.5)	62.1 (9.0)	110–150

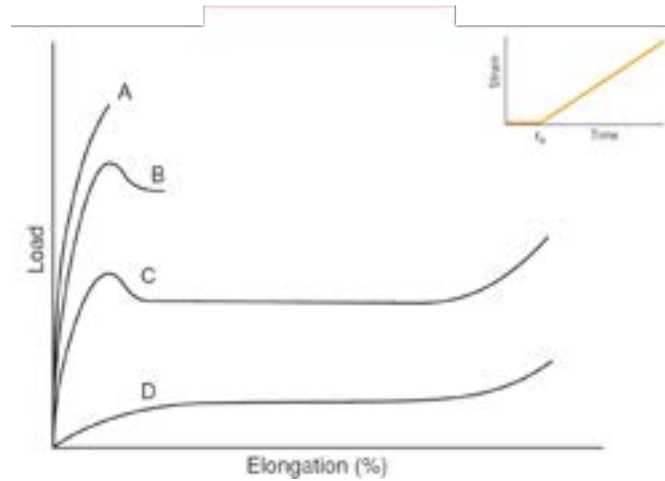


Figure 122: Polymer stress strain behavior

In the figure above we see four different behaviors of polymers that can be categorized,

- A: Brittle fracture
- B: Ductile failure
- C: Cold drawing (necking)
- D: Rubber-like behavior

Composite simple mechanics

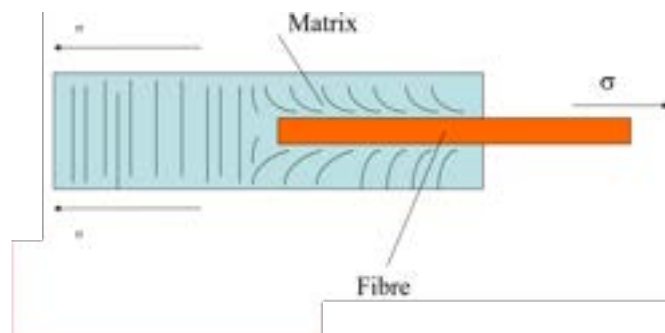


Figure 123: Critical fiber length, where the deformation pattern in the matrix around a fiber subjected to tensile load

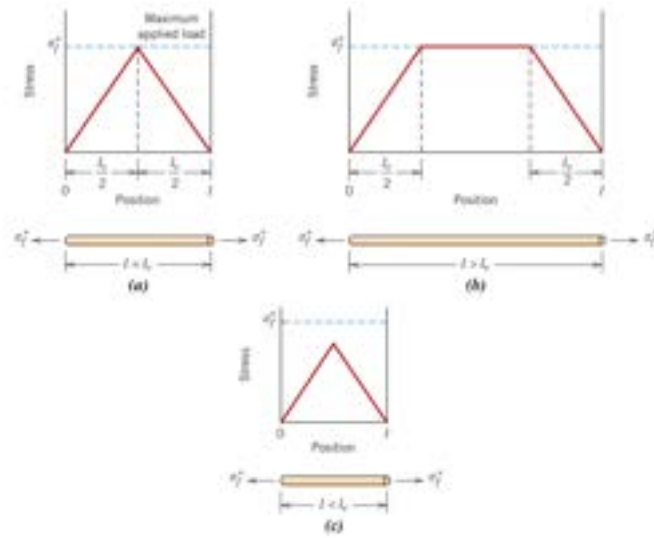
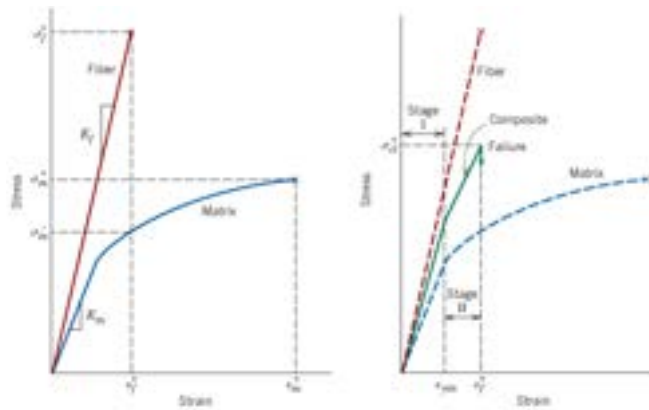


Figure 124: Stress-position profiles when the fiber length l (a) is equal to the critical length l_c , (b) is greater than the critical length, and (c) is less than the critical length for a fiber-reinforced composite that is subjected to a tensile stress equal to the fiber tensile strength σ_f^*

The critical fiber length can be calculated using,

$$l_c = \frac{\sigma_f^* d}{2\tau_c}$$

Where l_c is the critical fiber length, σ_f^* is the fiber tensile strength, d is the fiber diameter and τ_c is the fiber matrix bond strength.



2.15 Polymer additives and degradation

Additives

Plasticizers

Mainly used in PVC (80% of used additives) but also many others

- Poly-vinyl butyral), poly(vinyl acetate) (PVAc), acrylics cellulose molding compounds, poly(amide)s
- The polymer and plasticizers must be miscible for plastization to work

Types,

- Phthalates

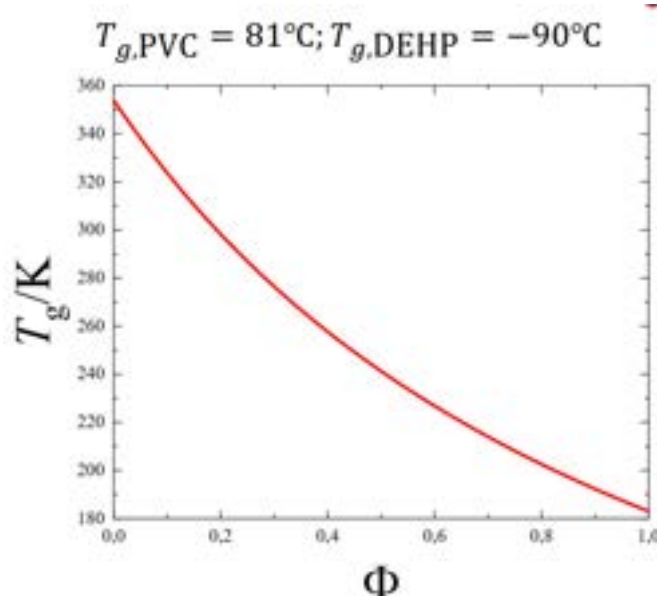
- Cyclohexanoic diesters
- Phosphates
- Adipates
- Azelates
- Sebacates
- Trimellitates
- Citrates
- Chlorinated hydrocarbons

Plasticizers for polymers

- A plasticizer is a T_g lowering additive (but added in substantial amounts). The plasticizer must be able to mix with the polymer to work
- The glass transition of a mixture is described by the Flory-Fox equation,

$$\frac{1}{T_{g,\text{mix}}} = \frac{\Phi_A}{T_{g,A}} + \frac{\Phi_B}{T_{g,B}}$$

- In medical grade PVC the disrupted plasticizer DEHP is often used



Stabilizers Used as,

- antioxidants
- UV Stabilizers
- Dehydrochlorination Stabilizers
- Metal deactivators
- Acid Scavengers
- Hydrolysis Stabilization

- Flame Retardants

Fillers

- Calcium carbonate
- Silicates
- Glass fibers
- Glass bulbs
- (Asbestos)
- Talc
- Kaolin
- Mica
- Barium sulfate
- Metal oxides and hydroxides
- Carbon black, graphite
- Wood flour
- Flours or fibers of other natural products
- Synthetic fibers

Others

- Lubricants
- Slip agents
- Nucleating agents
- Optical brighteners
- Colorants, dyes and pigments
- Antimicrobial additives
- Surface improvers
- Antifogging additives
- Antiblocking additives
- Blowing agents
- Compatibilizers

Degradation

Chemical Degradation: 3 (4) general processes, otherwise degradation chemistry depends on the chemistry of the individual polymer

- Photochemistry (e.g. sun light)
- Oxidation (e.g. O_2)
- Hydrolysis (major process for many biomaterials)
- Enzymatic degradation

Physical Degradation: At least 2 processes, same phenomena in all polymers, thus generalization is possible

- Fatigue
- Environmental stress cracking

Accelerated Aging consideration

Simple reaction The basic idea is that aging under "harsh" conditions to get "fast(er)" degradation and extrapolate to "use condition" using a theoretical relation between "harsh" and "use" condition. The most used "harsh" condition is elevated temperature.

The Arrhenius equation describes temperature dependence of the reaction rate,

$$k = A \exp\left(-\frac{E_A}{RT}\right)$$

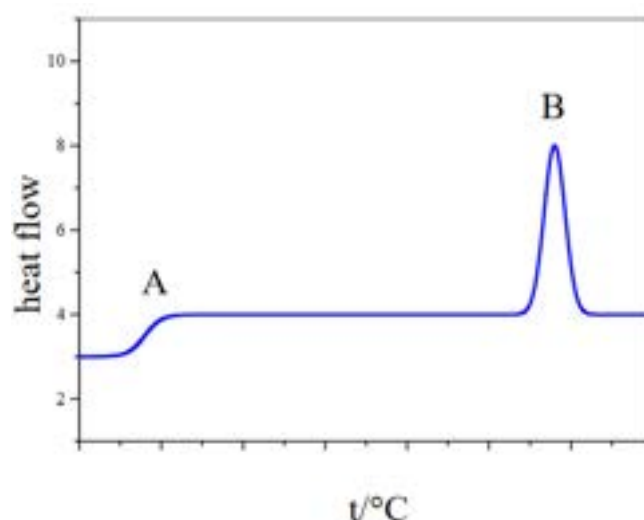
Complex reactions For a polymer with two simultaneous reactions each with their own rate constant the extrapolation on the degradation rate will be underestimated sometimes up to a factor of 3.

Extrapolation models Conversion of accelerated data from short time high temperature data to long time conditions of use prediction.

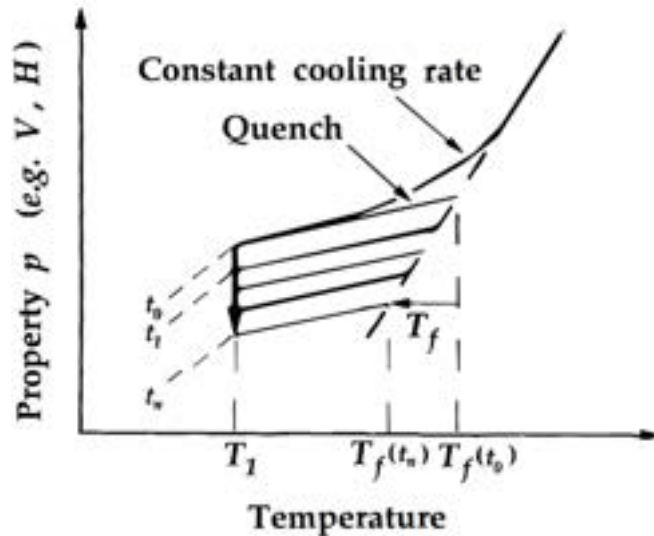
This is problematic when several degradation mechanisms with different temperature dependences or when a phase transition occurs between accelerated test temperature and use temperature.

Physical Aging

- Model DSC with a glass transition T_g and a melt transition T_m



- Below the glass transition the relaxation time of the material is long compared to the cooling rate used to get below T_g
- The system is out of equilibrium and will relax towards equilibrium over time
- Thus it may change properties
- The liquid falls out of equilibrium when the high temperature branch deviates from linearity



- The state of the liquid in the glass is determined by 3 temperatures: The actual temperature T the fictive temperature T_f and the temperature at which the liquid is in equilibrium.
- The main point of this is that it is required to have one more parameter on top of temperature to describe the system (classic time temperature superposition is an example)
- All glasses have the same general properties. There is nothing special with polymer glasses as compared to inorganic glasses or metallic glasses
- The time evolution of all linear, i.e. small deformation properties, such as thermal contraction, modulus, etc. can be predicted on this basis
- Changes in moduli are typically small ($\lesssim 10\%$)
- Can physical aging lead to significant changes in ultimate properties such as brittleness and elongation at break?
- Will a similar one parameter approach for ultimate properties work for predictions?

Physical degradation

- Swelling, the polymer absorbs solvent from the surrounding and changes properties
- Dissolution and washing out, the polymer is soluble in the surroundings, or Parts of the materials, e.g. additives are soluble in surroundings. The material changes properties.

Environmental stress cracking ESC

- Environmental stress cracking is a phenomenon where the combination of stress and an environmental influence leads to that a (polymeric) material fails.
- It is not a chemical reaction, but rather a physical interaction. It is thus possible to outline general criteria for when Environmental stress cracking is expected

- Stresses can be found as permanent residual stresses in polymeric materials due to the manufacturing method
- It turns out that compound that are borderline to be solvent for the polymer in question are especially prone to induce environmental stress cracking
- The potential for environmental stress cracking can be predicted for solubility parameters. Or it can be tested experimentally

2.16 Polymer Recycling

Types of recycling

- Primary recycling, products reused for their original purpose
 - e.g. beer bottles as beer bottles
 - * A deposit system is required
 - * Cleaning issues
- Secondary recycling, mechanical recycling
 - e.g. aluminum can are reprocessed to produce aluminum cans
 - Industrial polymer scrap reused at the same company, Municipal waste collected, fractionated, cleaned and reprocessed
 - * Cleaning issues
 - * Materials quality issues
- Tertiary recycling, chemical recycling
 - Polymers broken down to monomers through hydrolysis or pyrolysis
 - * e.g. PET hydrolyzed to phthalic acid and ethylene glycol
 - * PMMA pyrolyzed to MMA
 - Product purification not necessarily trivial
- Quarternary recycling, energy recovery incineration
 - high energy content in polymers
 - * Depending on the feed stock for polymer production problematic in global warming context

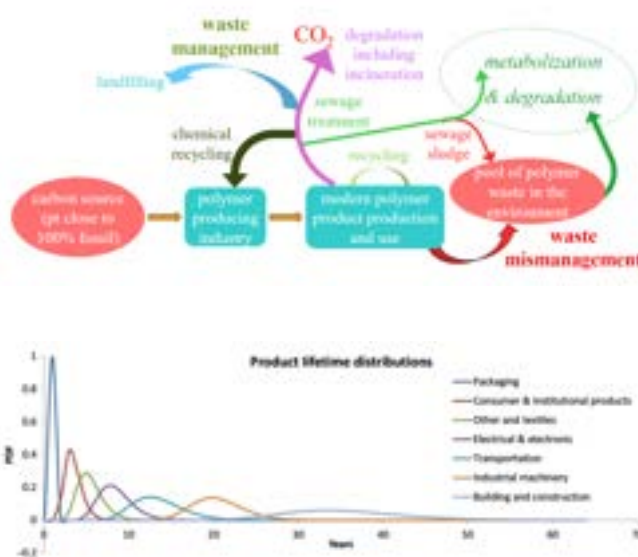


Figure 125: Product lifetime distributions for the eight industrial use sectors plotted as log-normal probability distribution.

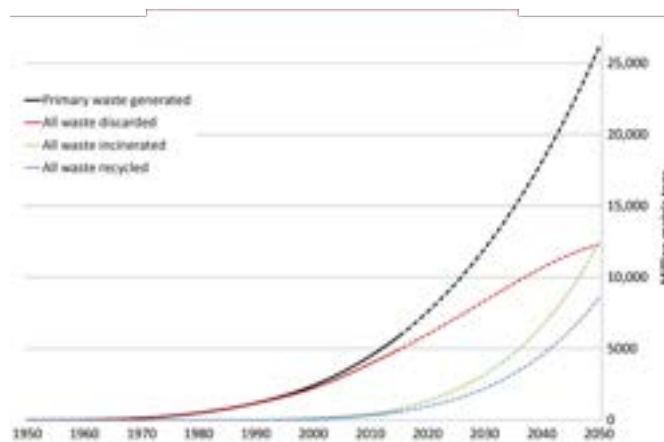


Figure 126: Cumulative plastic waste generation and disposal (in million metric tons). Cumulative plastic waste generation and disposal



Figure 127: Global production, use, and fate of polymer resins, synthetic fibers, and additives (1950 to 2015, in million metric tons)



Figure 128: Fate of polymers

Recycling challenge: Different polymers are in general not miscible

2.17 Polymer Surfaces

2.17.1 Polymer Surface Characteristics

Most polymers consist primarily of carbon and hydrogen, with smaller amounts of oxygen and nitrogen. As a result, many synthetic polymers are inherently hydrophobic.

Examples include:

- Polyethylene (PE)
- Polypropylene (PP)
- Polystyrene (PS)
- Polyethylene terephthalate (PET)
- Polyurethane (PU)

Native (untreated) polymer surfaces generally have the following characteristics:

- Hydrophobic
- Low stiffness
- Low yield strength

These properties arise because polymers are mainly held together by secondary bonds.

Therefore, common targets for surface modification are:

- Increasing hydrophilicity
- Increasing hardness

2.17.2 Surface Modification Techniques for Polymers

Surface modification of polymers can be divided into two main categories:

- Surface chemistry modification
- Hardness modification

Surface chemistry modification These methods aim to change the chemical composition of the surface.

- **Oxidative treatments:**
 - Flame treatment
 - Corona discharge (atmospheric plasma)
 - Plasma treatments
- **Irradiation methods:**
 - Photons
 - Electron beams
 - Ion beams
 - X-rays and gamma rays

Hardness modification These methods aim to improve the mechanical properties of the surface.

- Vapour deposition techniques (PVD, CVD)
- Solvent-based coatings

2.17.3 Plasma Treatment of Polymers

Plasma treatment is widely used to modify polymer surfaces.

Plasma is created by applying a high electric field to a gas, causing ionization.

Two main types are:

- **Atmospheric plasma (corona discharge):**
 - Low cost
 - Less control
- **Low-pressure plasma:**
 - More expensive
 - Better control of surface modification

Applications include:

- Surface preparation for printing
- Improvement of adhesion
- Use in cell culture (e.g., treated polystyrene)

2.17.4 Polystyrene in Cell Culture Applications

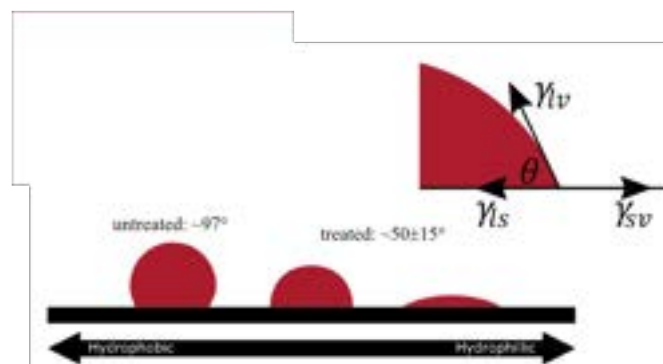
Polystyrene is widely used in cell culture applications, but its native surface is hydrophobic and not suitable for cell adhesion.

Surface modification techniques such as plasma treatment or chemical treatments are used to:

- Increase surface energy
- Improve wettability
- Enhance cell adhesion and growth

Surface chemistry and contact angle play a key role in determining how cells interact with the material.

2.17.5 Corona Treatment and Surface Wettability



Corona treatment introduces polar functional groups onto the polymer surface, making it more hydrophilic. This results in a change in contact angle:

- Untreated surface: approximately 97°
- Treated surface: approximately $50 \pm 15^\circ$

This reduction in contact angle indicates improved wettability and surface energy, which enhances adhesion and interaction with liquids or biological systems.

2.17.6 Surface Hardness Improvement

The surface hardness of polymers can be improved using coating techniques.

Vapour deposition methods

- **Physical Vapour Deposition (PVD):** typically used for metallic coatings
- **Chemical Vapour Deposition (CVD):** often used for ceramic coatings

Wet chemical methods

- Sol-gel coatings (nanoparticle-based)
- Particles dispersed in liquids
- Suspensions in curable liquids

These coatings can significantly improve:

- Surface hardness
- Wear resistance
- Functional properties (e.g., optical or electrical behaviour)

2.17.7 Functional Nanoparticle Coatings

Nanostructured coatings can introduce additional functionalities to polymer surfaces.

Typical materials include:

- SiO_2
- Al_2O_3
- ZrO_2
- TiO_2
- ZnO
- Indium tin oxide (ITO)
- Carbon nanotubes
- Graphene

These materials can provide:

- Increased hardness

- UV absorption
- Photocatalytic activity
- Electrical conductivity
- Antibacterial properties

3 Metals and alloys

3.1 Introduction

Metals are a class of elements that share certain characteristics. They are generally shiny, malleable, ductile, and good conductors of heat and electricity.

Metals can exist in various forms, from pure elements like gold or copper to complex alloys like steel or bronze.

Properties

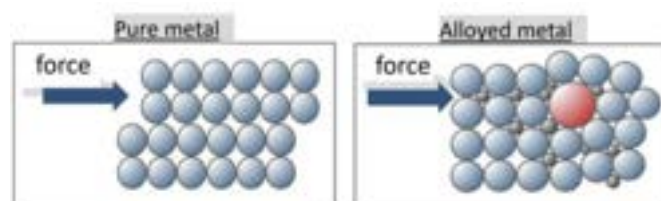
- **Strength and Ductility:** Metals are generally strong and ductile, allowing them to be shaped and formed without breaking.
- **Conductivity:** Metals are excellent conductors of heat and electricity, making them crucial for various applications.
- **Opacity:** Metals are opaque to light, reflecting most of the light that strikes their surface, giving them a shiny appearance.

Pure metals

- **Too ductile:** Metals can easily be drawn so will lack strength.
- **Corrosion:** Metals can rust and corrode over time.
- **Cost:** Metals can be expensive to produce and refine.
- **Flammability:** Some metals are flammable and can be dangerous.

Alloys

In pure metals the atoms can slip past each other, but in alloyed metals different atom sizes impede the atoms slipping.



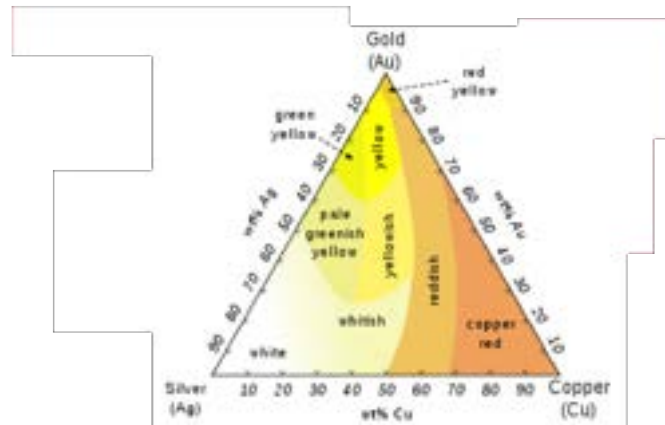


Figure 129: Metal colors and alloying. Note that Gold, Copper and Caesium are the only elemental metals which have a natural colour other than grey/white

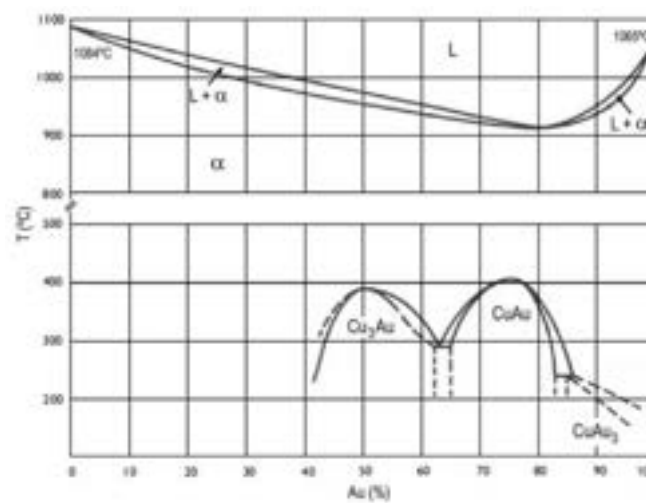


Figure 130: Kinetics and phase transformation

- Gold: Excellent corrosion resistance, very ductile and good electric conductivity.
- Silver: Good electric, thermal conductivity as well as optical reflectivity
- Copper: Good electric, thermal conductivity as well as bactericidal (substance that directly kills bacteria).
- Bronze (copper and tin): Harder than copper and low friction.
- Brass (copper and zinc): Easy to cast and easy to machine.
- Iron: Cheap, versatile but corrodes easily.
- Steel (iron with $< 2.1\%$ Carbon): in production since mid 1800, easier to produce in large scale.
- Stainless steel (iron with $> 12\%$ Chromium)
- Aluminium; Corrosion resistant, low density and high optical reflectivity.
- Aluminium alloys: stronger than aluminium and similarly light (table of strength of different aluminium alloys: [132](#))
- Magnesium (and its alloys): Very low density, flammable and biocompatible.
- Titanium (and its alloys): Strong, light as well as biocompatible.

Steel production today:

The process can be broadly divided into three main stages, Ironmaking, Steelmaking and Casting with a known yearly production in 2021 of 1950 million tons ≈ 62 tons/sec.

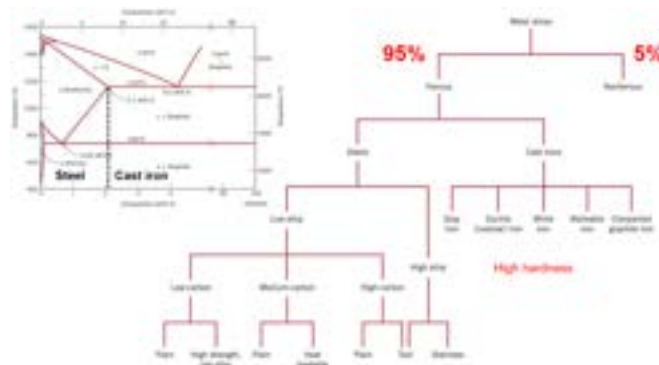


Figure 131: Heat treatment of steels

Purpose of heat treatments:

- Modify microstructure
- Change properties
- It however cannot remove impurities nor change the density

Strength, σ_y (MPa)	
Pure Al	10
0.5%Si	100
2.5%Mg	190
1%Mg 0.5%Si	280
4%Cu 1.5%Mg	340
5%Zn 2.5%Mg 2%Cu	500

Figure 132: Strength of aluminium alloys

High Entropy Alloys:

"Mix" of at least 5 different metals, HEAs typically give single phase materials (e.g. FCC or BCC)

3.2 Bulk Metallic Glasses

Definition of glass:

- A liquid which solidifies without nucleation and growth of the crystalline phases

- No long range atomic order
- Liquid becomes solid-like at *glass transition temperature* T_g demonstrating a dramatic increase in the viscosity at T_g
- Examples: SiO_2 glass (windows, glassware, ...), Plastics, Polymers



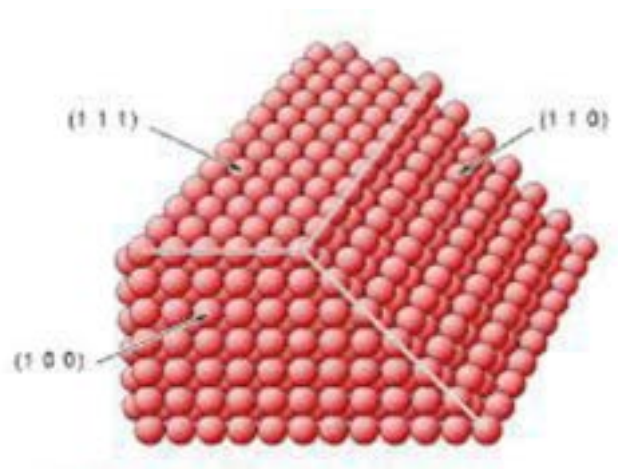
Figure 133: Crystalline and amorphous material visualization

Crystalline materials have a highly ordered arrangement of atoms

An **amorphous structure material** is a material with a disordered atomic-scale structure. Therefore amorphous materials are non-crystalline.

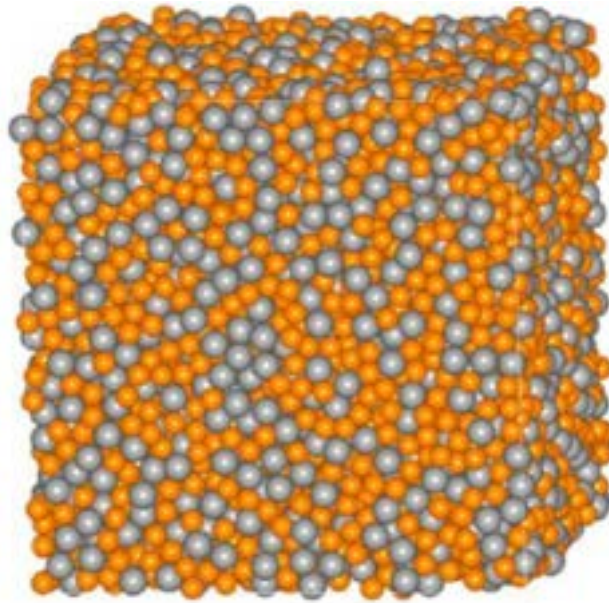
Metallic glass

Materials in which such a disordered structure is produced directly from the liquid state during cooling are called "glasses", and so *amorphous metals* are commonly referred to as "metallic glasses" or "glassy metals".



Crystalline metallic Cu

Figure 134: Crystalline metallic Cu



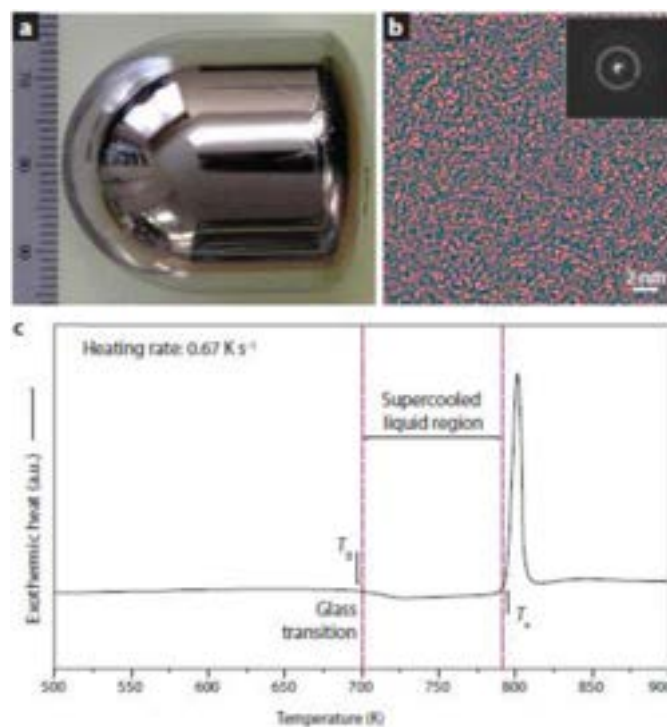
Amorphous $\text{Cu}_{46}\text{Zr}_{54}$

Figure 135: Amorphous $\text{Cu}_{46}\text{Zr}_{54}$

Transformation of the liquid into:

- Crystalline state achieved at low cooling rate
- Amorphous state achieved at high cooling rate.

At T_g the viscosity of the supercooled liquid becomes so high and atoms get "frozen-in".



Rapid solidification

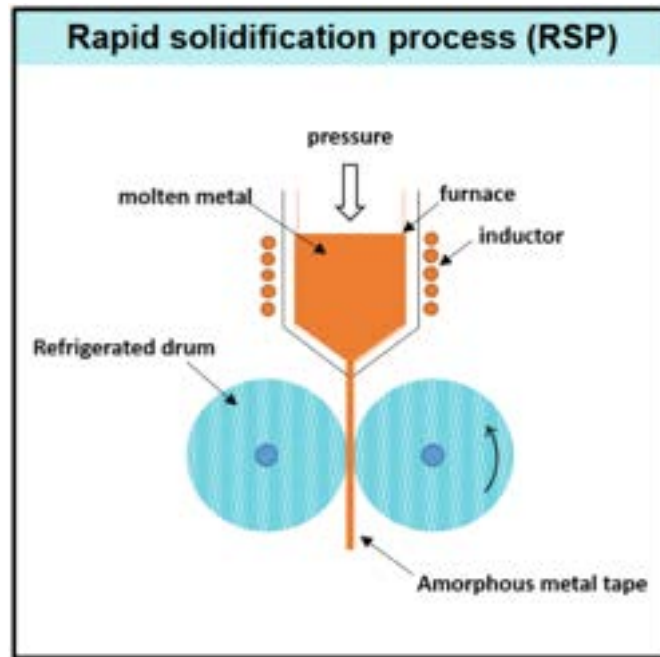


Figure 136: 1960, Au–Si (Eutectic with restricted solubility), rapid cooling rates (approximately 10^5 - 10^6 K/s)

Demerits of rapid solidification process (RSP)

- Low thermal stability
- Only thin ribbons can be produced (20-50 microns)
- Some alloys cannot be vitrified even at high cooling rates due to their low glass transition temperatures
- High cost of production



Amorphous ribbon

Developing "bulk" glassy metallic systems

- Lower cooling rates (10 - 10^3 K/s)

- Higher thermal stability and glass forming ability
- Production of parts thicker than 1 mm

Inoue's Criteria (Confusion principle)

- Existence of at least 3 alloying elements in the alloying system
- Mixing enthalpy should be negative
- Atomic radius difference of the main element should be higher than 12%
- Chemical composition adjacent to the eutectic composition

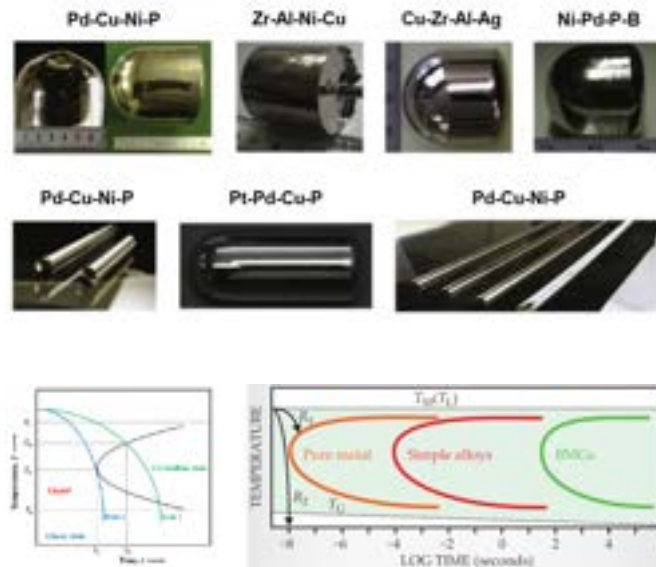


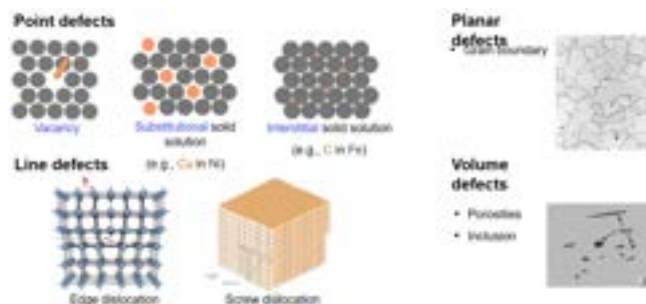
Figure 137: Bulk Metallic Glass (BMG)

Bulk metallic glass was first discovered by Atakan Peker. It was complex alloys of 5 metallic elements (Zr–Cu–Ni–Ti–Be) "Vitrelloy 1", with a cooling rate of 1 K/s

Amorphous metallic parts (*over 1 millimeter in thickness or diameter*) are known as bulk metallic glasses (BMG)

Mechanical properties

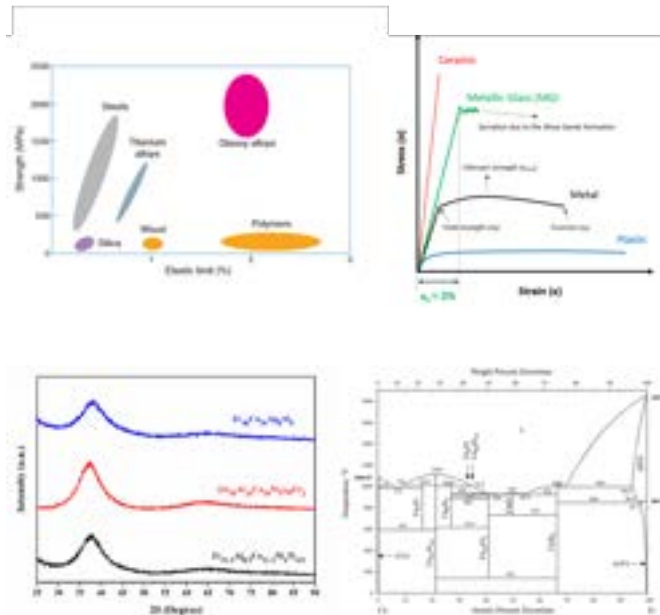
Crystal lattice imperfections in crystalline metals



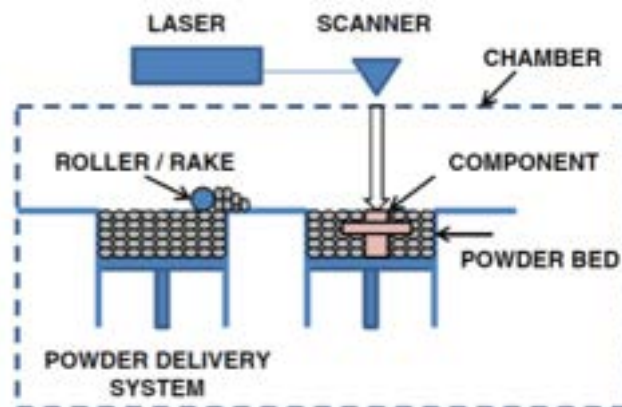
Mechanical properties of BMG's Superior mechanical properties such as high specific strength and hardness, high elastic limit and good corrosion and wear resistance

Metallic glasses exhibit low plasticity and fracture toughness.

- High strength and elastic limit
- Scratch and wear resistance
- Corrosion resistance
- Surface finish



BMG by additive manufacturing



3.3 Diffusion in Metals

Diffusion is driven by a gradient in Gibbs free energy or chemical potential. *Diffusion*

- **Definition:** The word diffusion derives from the Latin word, diffundere, which means "to spread out". Diffusion is the net movement of anything (for example, atoms, ions, molecules, energy) generally from a region of higher concentration to a region of lower concentration.
- **Importance:** It is critical in many metallurgical processes. These processes include heat treatment and alloying.
- **Mechanism:** Atoms move through the crystal lattice. This occurs by exchanging positions with vacancies.

Types of Diffusion

Interdiffusion: In an alloy, atoms tend to migrate from regions of high concentration to regions of low concentration.

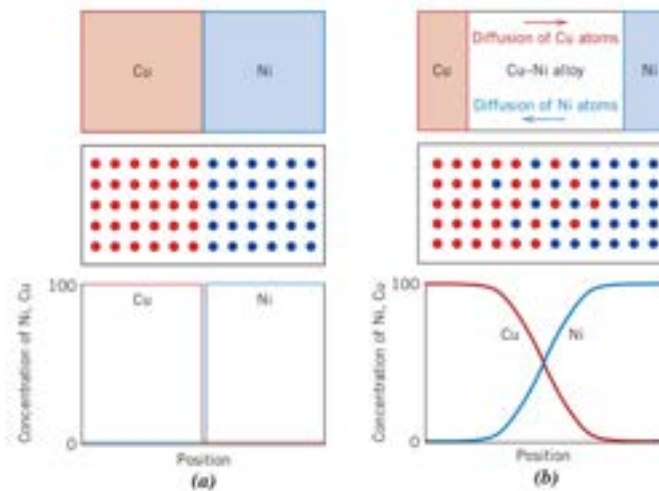


Figure 138: Interdiffusion

Self-diffusion: in an elemental solid, atoms also migrate.

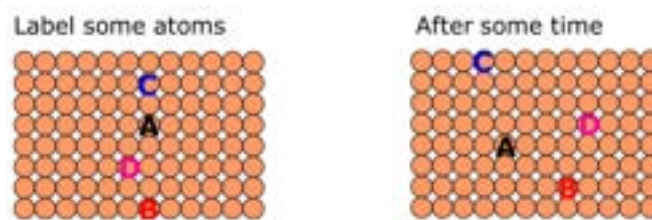


Figure 139: Self-diffusion

Mechanism for diffusion

Diffusion for Substitutional atoms:



Figure 140: Mechanism for diffusion of substitutional atoms.

Ernest Kirkendall, **Kirkendall Effect** (1947).

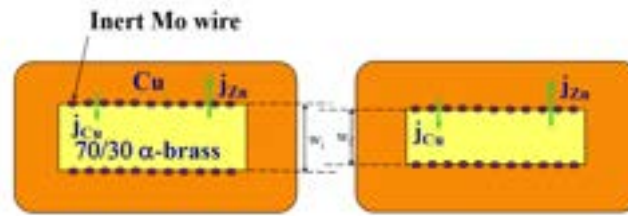


Figure 141: Kirkendall Effect

$$J_{\text{Cu}} = \frac{M_{\text{Cu}}}{At}, \quad J_{\text{Zn}} = \frac{M_{\text{Zn}}}{At}$$

Where J is denoted as the flux or rate of diffusion.

In Direct mechanism,

$$J_{\text{Cu}} = J_{\text{Zn}}$$

and in Vacancy mechanism,

$$J_{\text{Zn}} = J_{\text{Cu}} + J_{\text{Va}}$$

Diffusion for Interstitial atoms

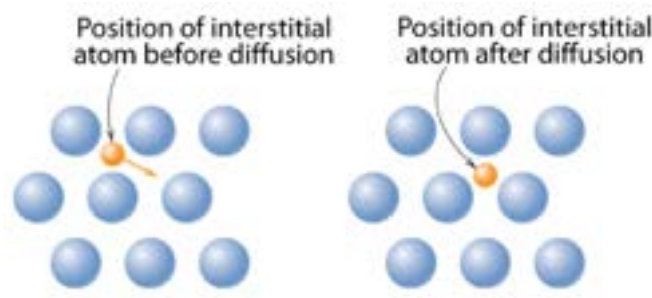


Figure 142: Diffusion for Interstitial atoms

Diffusion Flux

The amount or rate of diffusion J

- Diffusion flux refers to the quantity of a substance that passes through a unit area per unit time due to the process of diffusion.
- It is a measure of how much of a substance (such as particles, molecules, or ions) moves from an area of higher concentration to an area of lower concentration.
- Measured empirically:
 - Make thin film (membrane) of known surface area
 - Impose concentration gradient
 - Measure how fast atoms or molecules diffuse through the membrane

3.3.1 Phenomenology Fick's laws (1855)

Fick's 1st law: Flux is proportional with the concentration gradient

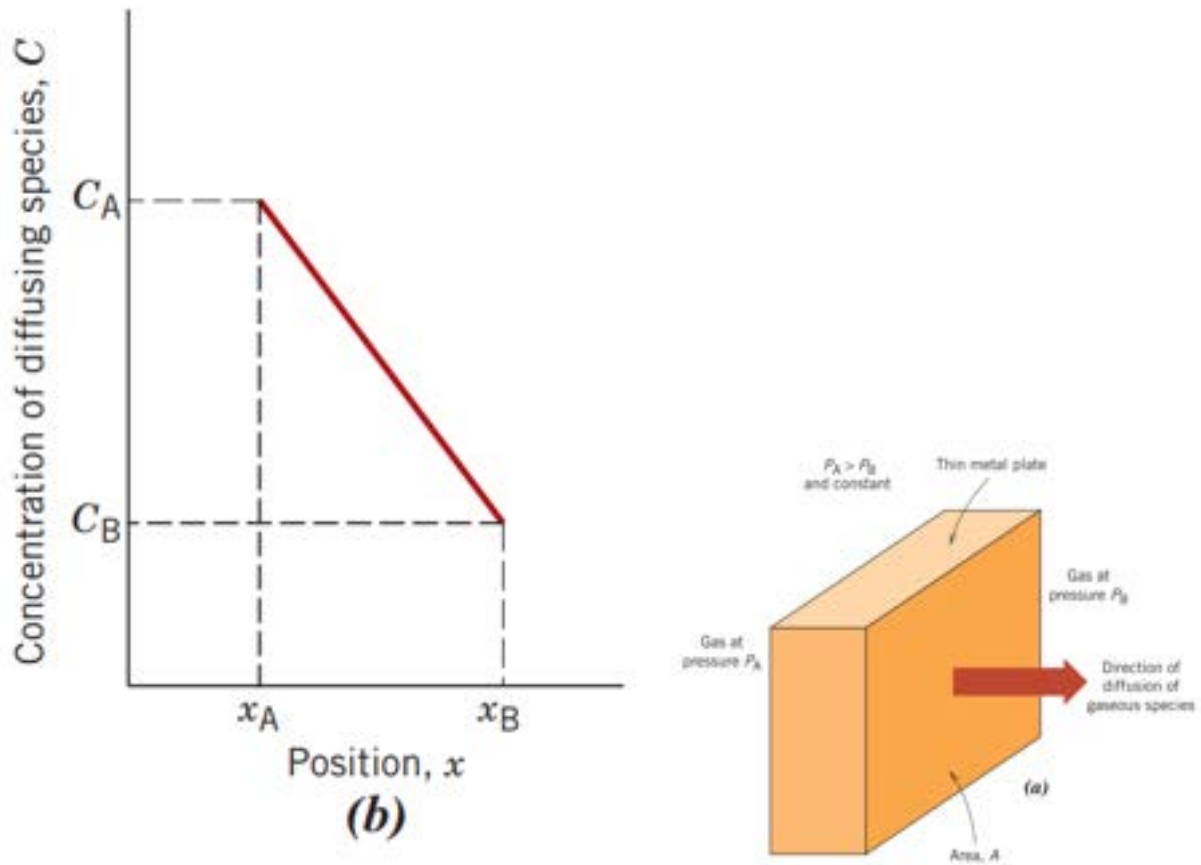


Figure 143: (a) Steady state diffusion across a thin plate. (b) A linear concentration profile for the diffusion situation in (a)

$$J = -D \frac{dC}{dx}$$

Fick's 2nd law: Non-stationary state

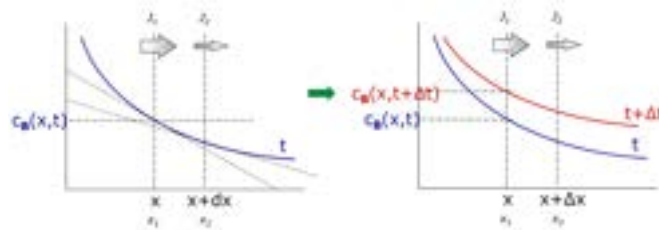


Figure 144: Fick's 2nd law

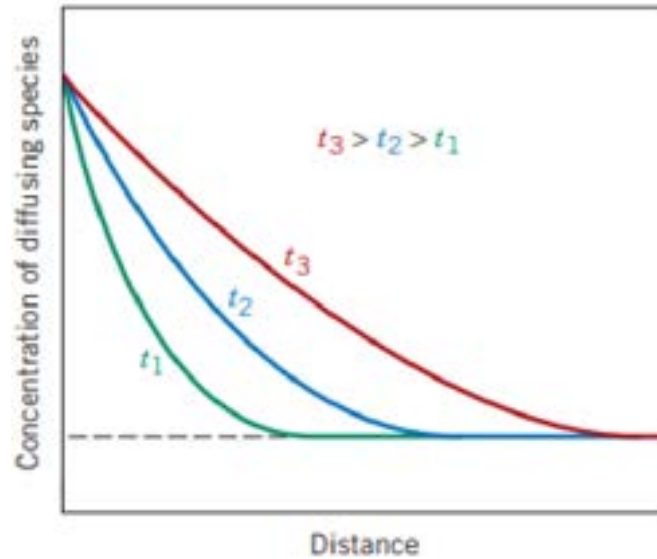
$$J_B(x, t) = -D_B \left. \frac{\partial c_B}{\partial x} \right|_{x, t}$$

$$\frac{\partial c_B}{\partial t} = -\frac{\partial J_B}{\partial x} = -\frac{\partial}{\partial x} \left(-D_B \frac{\partial c_B}{\partial x} \right) = \frac{\partial}{\partial x} \left(D_B \frac{\partial c_B}{\partial x} \right)$$

Fick's 1st law is combined with a mass balance.

Only for $D_B = \text{constant}$ this is reduced to,

$$\frac{\partial c_B}{\partial t} = D_B \frac{\partial^2 c_B}{\partial x^2}$$



In a non-steady state diffusion, the concentration of diffusing species is decreasing with respect to thickness.

Analytical solution to Fick's 2nd law for a semi-infinite system:

Assumptions:

- Source of the diffusing species is a gas phase, the partial pressure of which is maintained at a constant value.
- Before diffusion, any of the diffusing solute atoms in the solid are uniformly distributed with concentration of c_0 .
- The value of x at the surface is zero and increases with distance into the solid.
- The time is taken to be zero the instant before the diffusion process begins.

Initial conditions,

$$\text{For } t = 0, \quad c = c_0 \quad \text{at} \quad 0 \leq x \leq \infty$$

Boundary conditions,

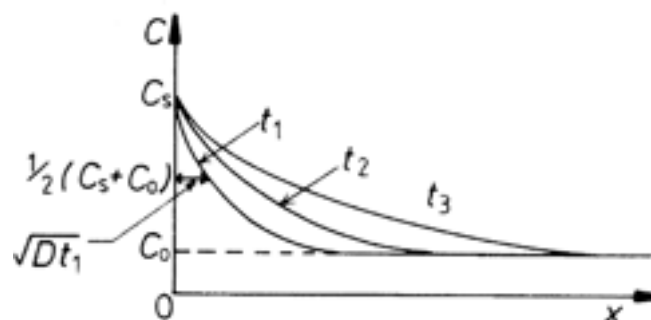
$$\text{For } t > 0, \quad c = c_s (\text{the constant surface concentration}) \quad \text{at} \quad x = 0$$

$$\text{For } t > 0, \quad c = c_0 \quad \text{at} \quad x = \infty$$

$$\frac{c(x, t) - c_0}{c_s - c_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

Where,

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$



z	$\text{erf}(z)$	z	$\text{erf}(z)$	z	$\text{erf}(z)$
0	0	0.55	0.5633	1.3	0.9340
0.025	0.0282	0.60	0.6039	1.4	0.9523
0.05	0.0564	0.65	0.6420	1.5	0.9661
0.10	0.1125	0.70	0.6778	1.6	0.9763
0.15	0.1680	0.75	0.7112	1.7	0.9838
0.20	0.2227	0.80	0.7421	1.8	0.9891
0.25	0.2763	0.85	0.7707	1.9	0.9928
0.30	0.3286	0.90	0.7970	2.0	0.9953
0.35	0.3794	0.95	0.8209	2.2	0.9981
0.40	0.4284	1.00	0.8427	2.4	0.9993
0.45	0.4755	1.10	0.8802	2.6	0.9998
0.50	0.5205	1.20	0.9103	2.8	0.9999

Table 7: Error function values $\text{erf}(z)$ **Factors influencing diffusion**

Diffusing Species, the diffusing species and the host material influence the diffusion coefficient

Temperature, the temperature dependence of the diffusion coefficients is,

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$$

Diffusing Species	Host Metal	D_0 (m ² /s)	Q_d (J/mol)
Interstitial Diffusion			
C ^b	Fe (α or BCC) ^a	1.1×10^{-6}	87,400
C ^c	Fe (γ or FCC) ^a	2.3×10^{-5}	148,000
N ^b	Fe (α or BCC) ^a	5.0×10^{-7}	77,000
N ^c	Fe (γ or FCC) ^a	9.1×10^{-5}	168,000
Self-Diffusion			
Fe ^c	Fe (α or BCC) ^a	2.8×10^{-4}	251,000
Fe ^c	Fe (γ or FCC) ^a	5.0×10^{-5}	284,000
Cu ^d	Cu (FCC)	2.5×10^{-5}	200,000
Al ^c	Al (FCC)	2.3×10^{-4}	144,000
Mg ^c	Mg (HCP)	1.5×10^{-4}	136,000
Zn ^c	Zn (HCP)	1.5×10^{-5}	94,000
Mo ^d	Mo (BCC)	1.8×10^{-4}	461,000
Ni ^d	Ni (FCC)	1.9×10^{-4}	285,000
Interdiffusion (Vacancy)			
Zn ^c	Cu (FCC)	2.4×10^{-5}	189,000
Cu ^c	Zn (HCP)	2.1×10^{-4}	124,000
Cu ^c	Al (FCC)	6.5×10^{-5}	136,000
Mg ^c	Al (FCC)	1.2×10^{-4}	130,000
Cu ^c	Ni (FCC)	2.7×10^{-5}	256,000
Ni ^d	Cu (FCC)	1.9×10^{-4}	230,000

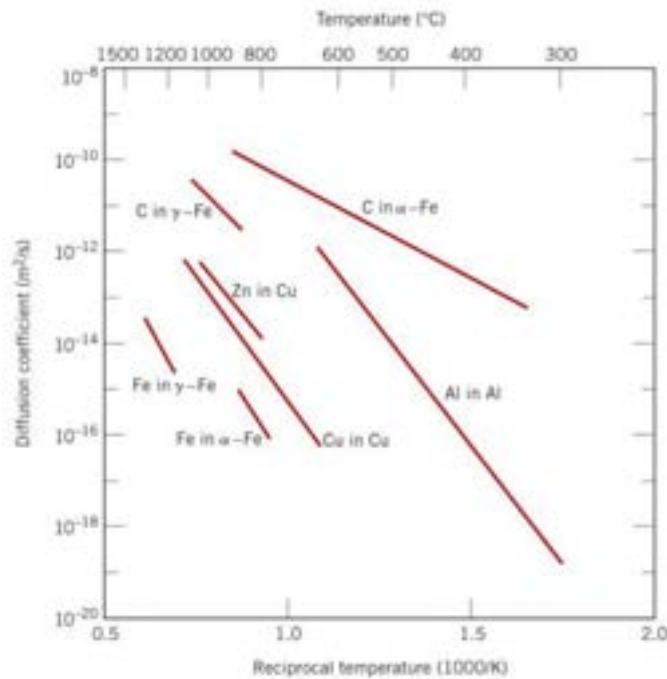
Table 8: Diffusion coefficients and activation energies for various systems

Diffusing Species	Host Metal	2^*D_0 (m^2/s)	Activation Energy Q_d		Calculated Values	
			kJ/mol	eV/atom	T(°C)	D (m^2/s)
Fe	α -Fe (BCC)	2.8×10^{-4}	251	2.60	500	3.0×10^{-21}
					900	1.8×10^{-15}
Fe	γ -Fe (FCC)	5.0×10^{-5}	284	2.94	900	1.1×10^{-17}
					1100	7.8×10^{-16}
C	α -Fe	6.2×10^{-7}	80	0.83	500	2.4×10^{-12}
					900	1.7×10^{-10}
C	γ -Fe	2.3×10^{-5}	148	1.53	900	5.9×10^{-12}
					1100	5.3×10^{-11}
Cu	Cu	7.8×10^{-5}	211	2.19	500	4.2×10^{-19}
Zn	Cu	2.4×10^{-5}	189	1.96	500	4.0×10^{-18}
Al	Al	2.3×10^{-4}	144	1.49	500	4.2×10^{-14}
Cu	Al	6.5×10^{-5}	136	1.41	500	4.1×10^{-14}
Mg	Al	1.2×10^{-4}	131	1.35	500	1.9×10^{-13}
Cu	Ni	2.7×10^{-5}	256	2.65	500	1.3×10^{-22}

Table 9: Diffusion parameters and calculated diffusion coefficients

Diffusion coefficient D in m^2/s

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right), \quad \ln D = -\frac{Q_d}{RT} \ln D_0$$



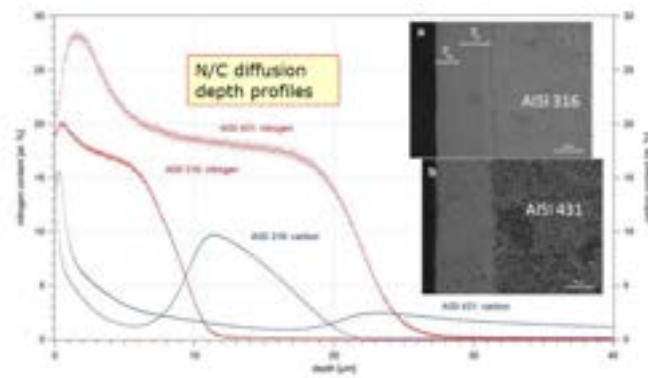


Figure 145: Low temperature surface hardening of stainless steel

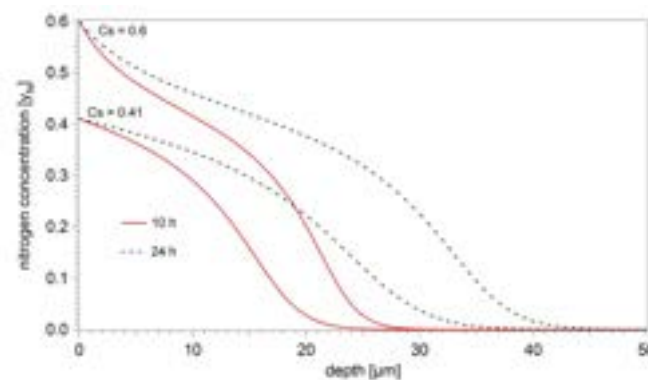


Figure 146: Diffusion profiles, concentration dependent diffusion coefficient

3.4 Strengthening Mechanisms in Metals

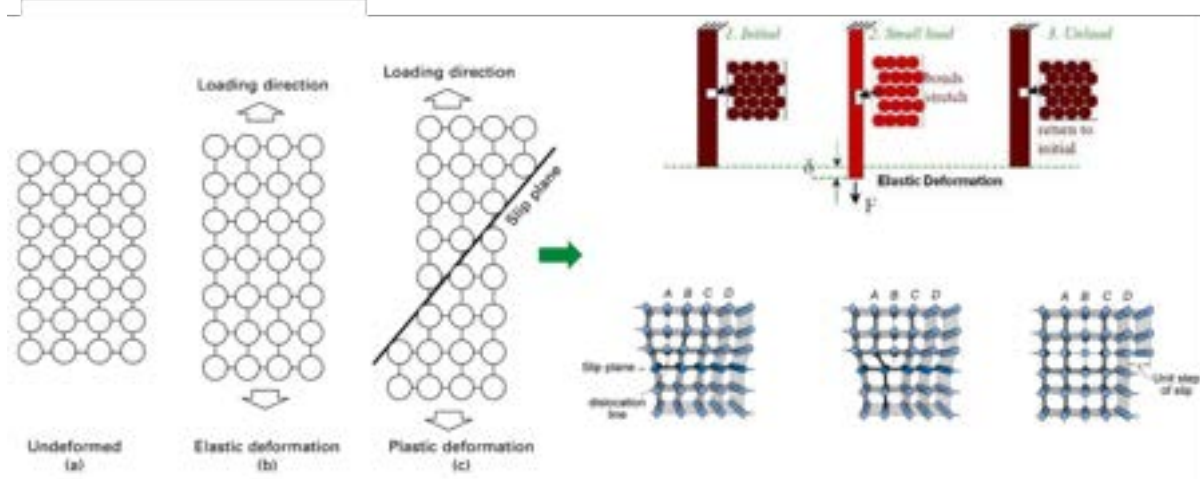


Figure 147: Deformation in Metals

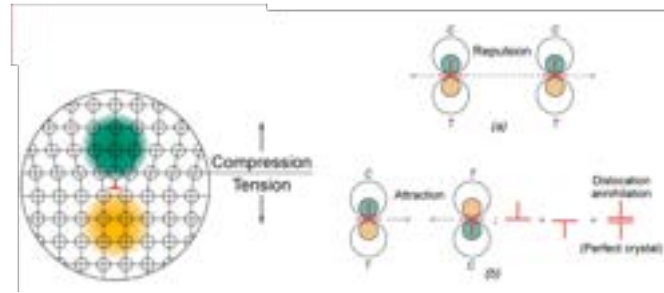


Figure 148: Characteristics of dislocations

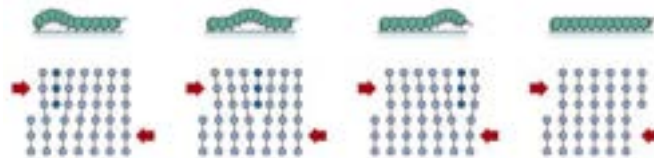


Figure 149: Dislocations, slip occurs by movement of dislocations

The material becomes stronger/harder if movement of dislocation is hindered, i.e. resistance against plastic deformation. This can be done by,

- Grain size hardening/refinement
- Work hardening
- Solid solution hardening
- Precipitation hardening

Principle behind strengthening of metals

- 0D: Point defect, solid solution strengthening
- 1D: Line-defects, work or strain hardening
- 2D: Planar-defects, grain boundary strengthening
- 3D: Volume-defects, precipitation or age hardening

3.4.1 Solid Solution Strengthening (0D)

Strengthen and harden metals is alloying with impurity atoms that go into either substitutional or interstitial solid solution.

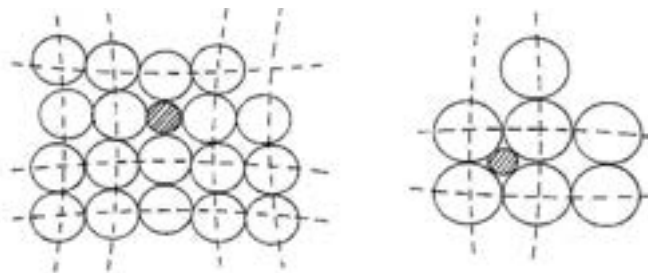


Figure 150: Left: Tensile lattice strains imposed on host atoms by a smaller substitutional impurity atom. Right: Compressive strains imposed on host atoms by a larger substitutional impurity atom

Elastic strains around the "solute" atom in the parent lattice. These strain fields interfere with dislocation movement leading to higher strength / hardness.

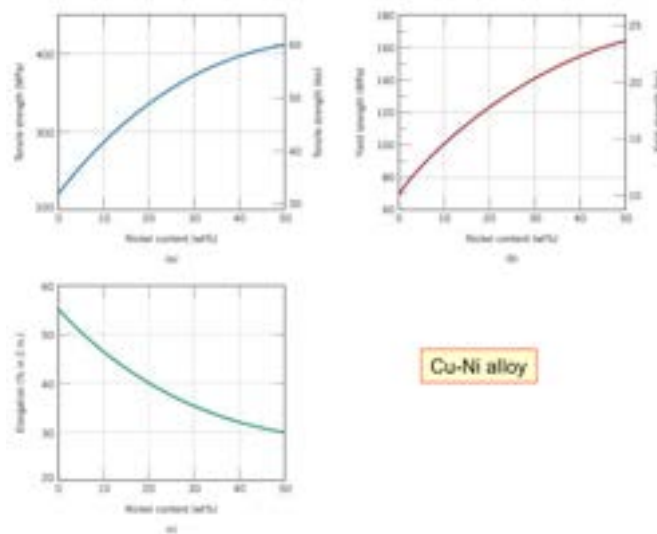


Figure 151: Example of solid solution strengthening in Cu-Ni alloy

3.4.2 Work hardening (1D)

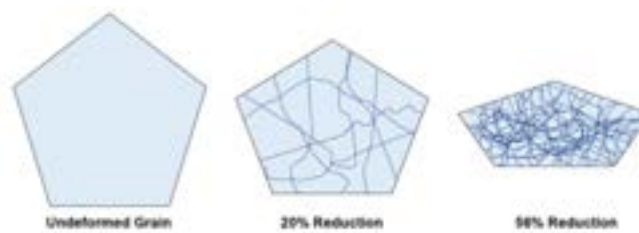
Strain hardening is the phenomenon by which a ductile metal becomes harder and stronger as it is plastically deformed.

- Plastic deformation leads to higher dislocation density
- Dislocations will block each other leading to higher strength

This is expressed as degree of cold work %CW

$$\%CW = \left(\frac{A_0 - A_d}{A_d} \right) \times 100$$

Where A_0 is the initial cross-sectional area and A_d is the final area.



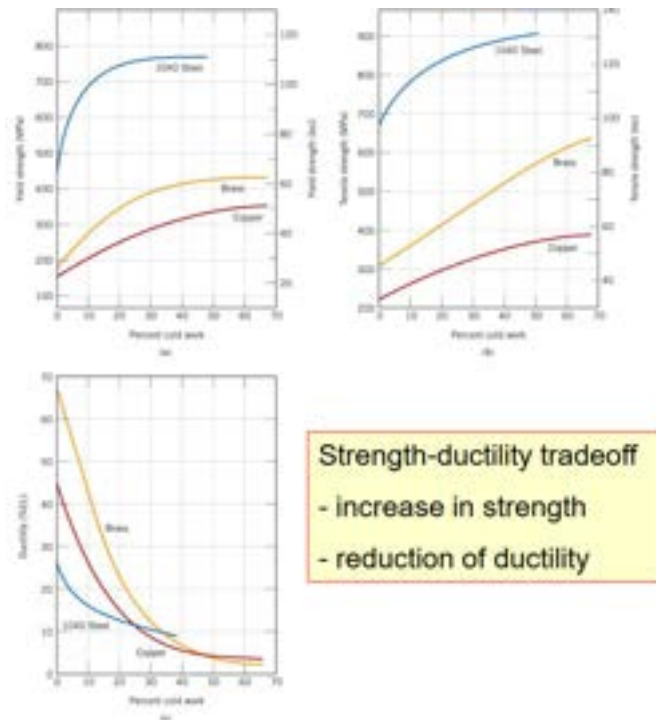


Figure 152: Cold Working Such as cold roll to "pancake" grains

There must be a strength-ductility trade-off since there is an increase in strength and reduction of ductility.

3.4.3 Grain boundary hardening (2D)

The grain boundary acts as a barrier to dislocation motion for two reasons:

- Because the two grains are of different orientations, a dislocation passing into grain B must change its direction of motion. This becomes more difficult as the crystallographic misorientation increases.
- The atomic disorder within a grain boundary region results in a discontinuity of slip planes from one grain into the other.

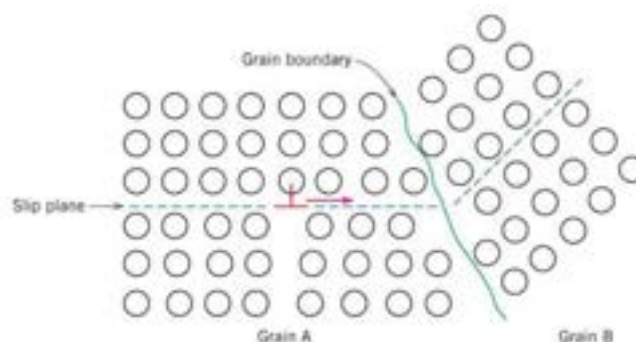
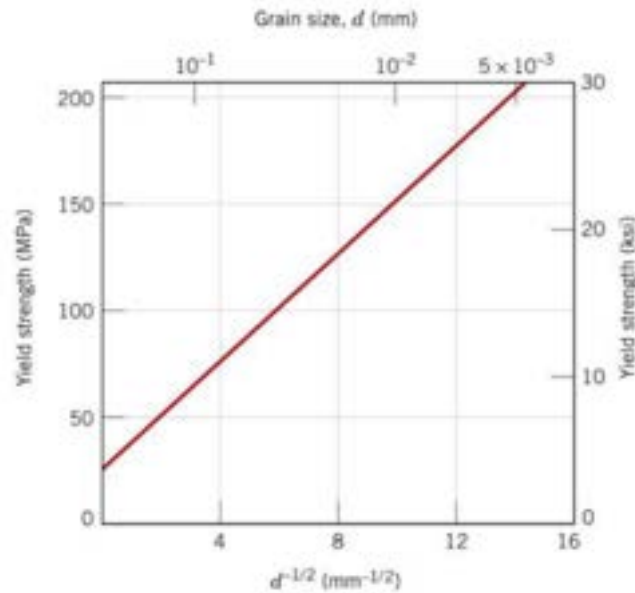


Figure 153: Dislocation movement through a grain boundary requires energy

Relationship between grain size and yield strength

$$\sigma_y = \sigma_0 + k \frac{1}{\sqrt{d}}$$

This relation is known as Hall-Petch, it is important to note it does not apply for nanometre sized grains and that it depends on the type of material through k



3.4.4 Precipitation (Dispersion) hardening (3D)

- Second phase provides a strengthening effect
- Dislocation movement is impeded by the precipitates = Strength
- Classical examples: Age hardening of aluminum alloys
- Fine precipitates are more effective in strengthening than coarse ones because, they have higher amount of surface area and require higher stress to cut

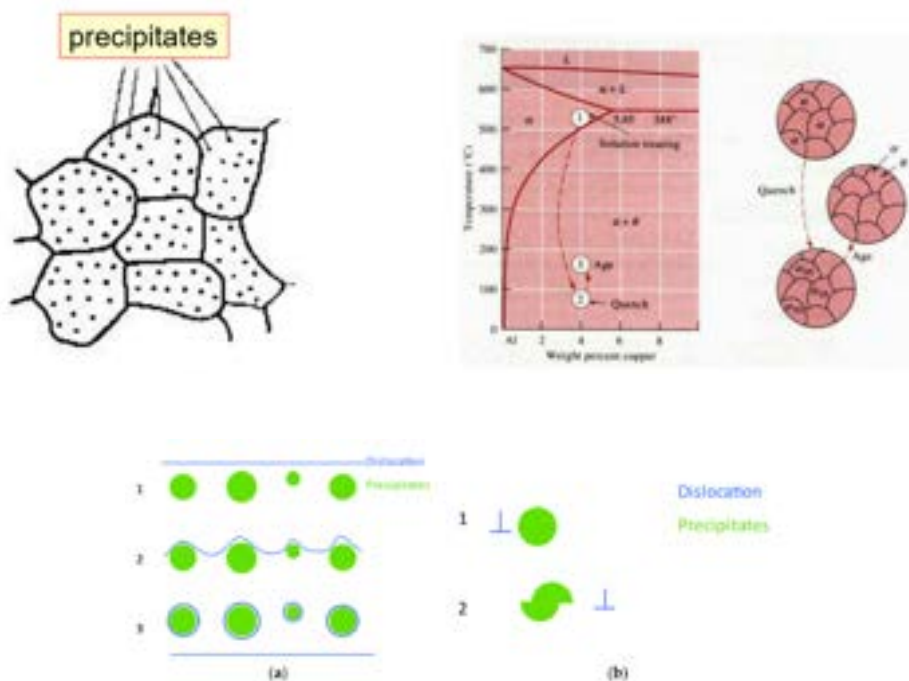


Figure 154: Mechanisms

3.4.5 Age hardening of Aluminum Alloys

Aluminum is very desired in aerospace due to its low density.

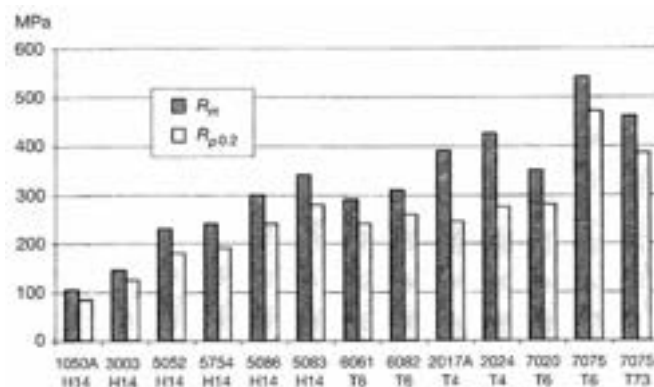


Figure 155: Strength of various aluminum alloys

Alloy designation	Main additions	Strengthening	Major applications
1xxx	Pure Al	No strengthening	Foil & electrical conductors
2xxx	Al-Cu	θ CuAl_2 type	Aircraft
3xxx	Al-Mn	Work-hardened	Cookware, building, packaging
4xxx	Al-Si	Si needles	Casting alloy
5xxx	Al-Mg	Work-hardened	Automotive, packaging, marine
6xxx	Al-Si-Mg	Mg_2Si	Automotive, building, marine
7xxx	Al-Zn-Mg	MgZn_2	High strength, aerospace, marine
8xxx	Other	—	—

Table 10: Classification of wrought aluminium alloys

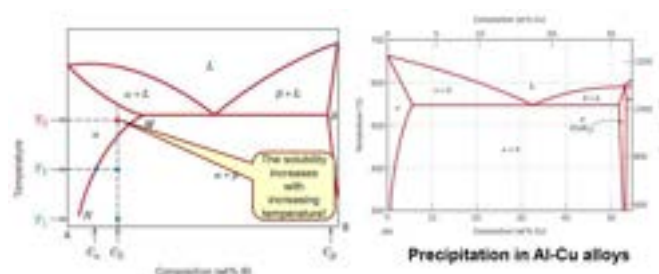
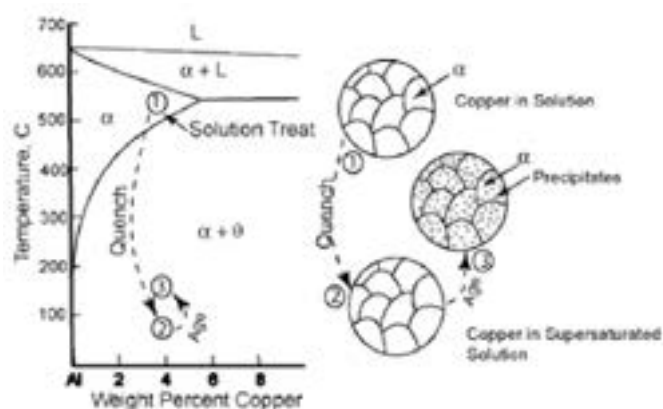


Figure 156: Precipitation hardening, aging

Precipitation hardening (example Al-Cu):



1. Solution treatment: Heating to 500-550°C to dissolve Cu atoms in solid solution
2. Quenching: Rapid cooling to "freeze" Cu atoms in solid solution
3. Aging: natural at room temperature or artificial at elevated temperatures (Usually 110-190°C) to form a large number of fine (nanosized) precipitates.

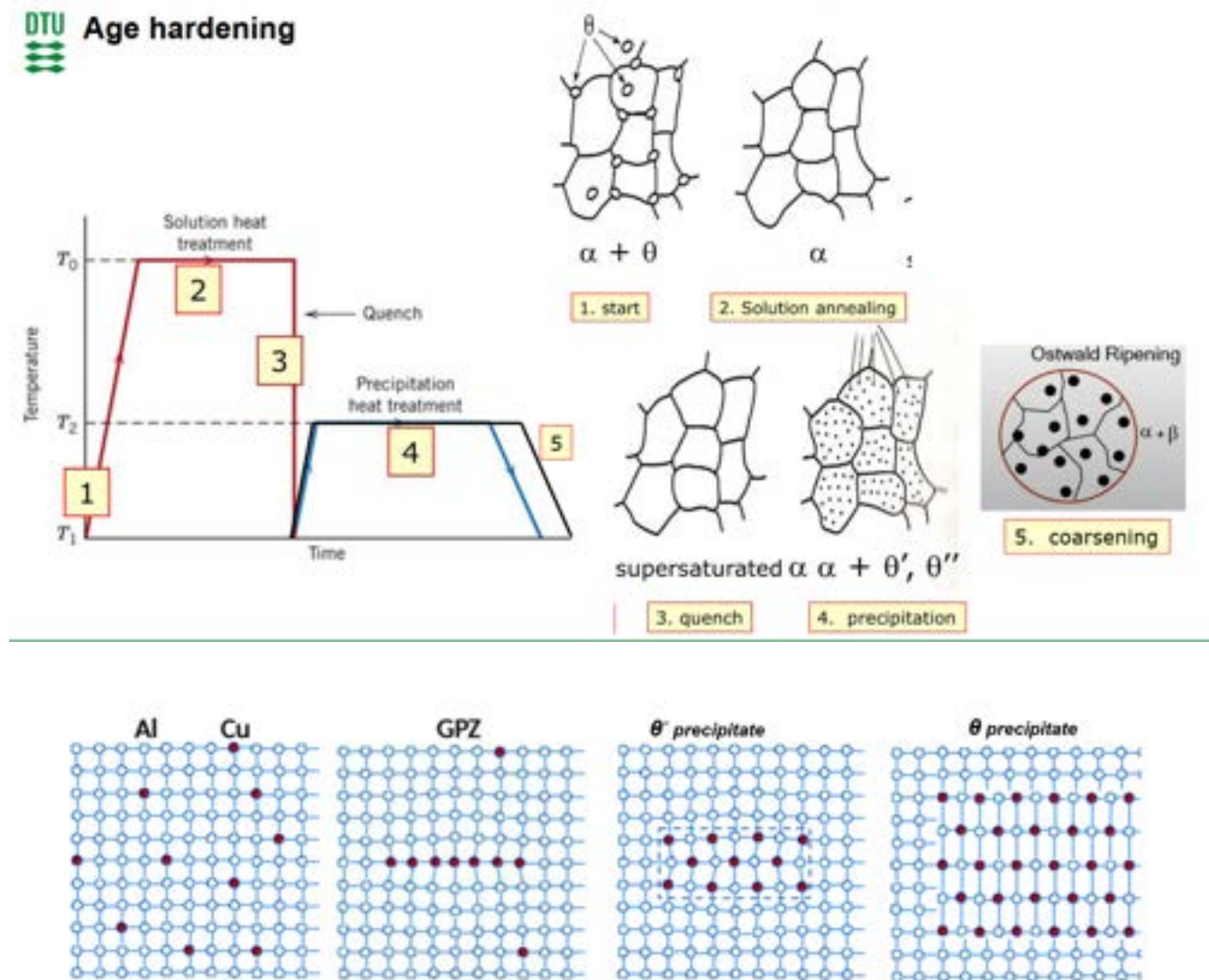


Figure 157: Supersaturated solid solution (SSSS) after quenching; Aging starts, Cu atoms cluster forming disks = Guinier-Preston (**GP**) zones; More aging, disks grow into semi-coherent θ' precipitates; More aging, coarser semi-coherent θ' and incoherent θ precipitates are formed

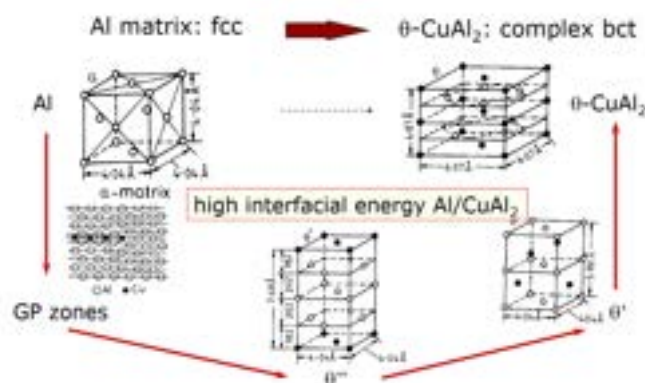
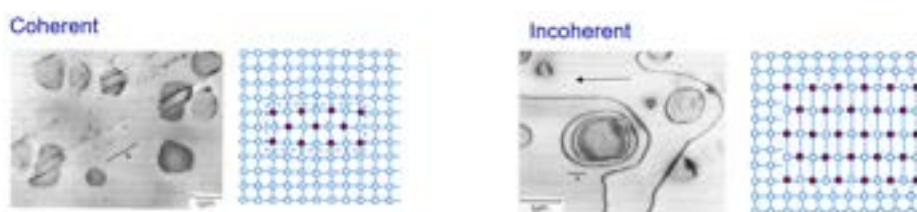


Figure 158: Precipitation in Al-Cu (Crystallography)

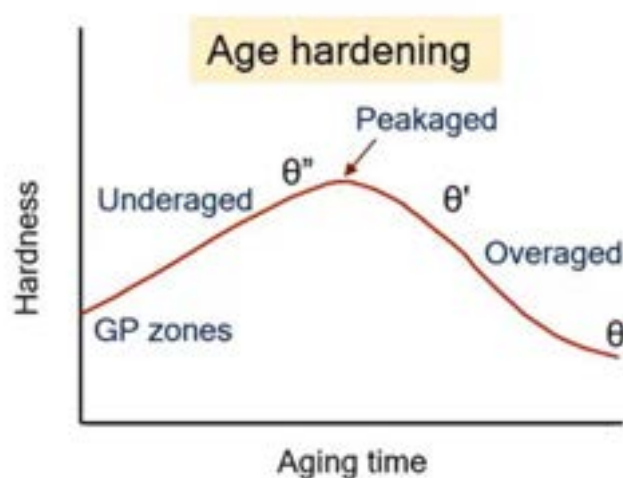


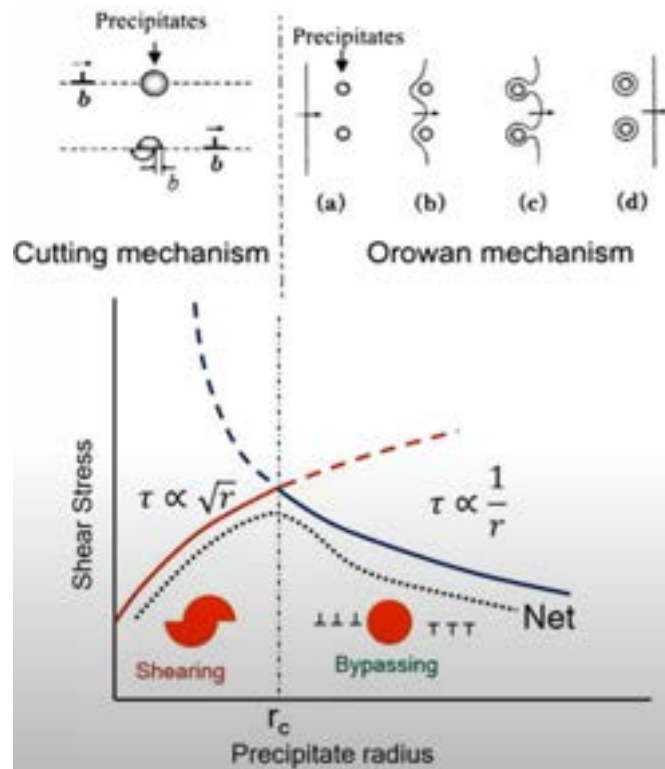
Coherent: As a dislocation approaches a coherent particle, dislocation can more through the particle and cut it. An increased stress is required to cut the particle leading to an increased hardness and strength.

Incoherent: Dislocation cannot cut through incoherent particles. Instead, they bypass such particles leaving dislocation loops.



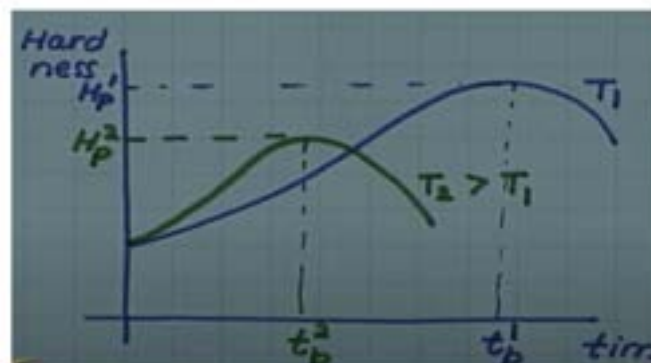
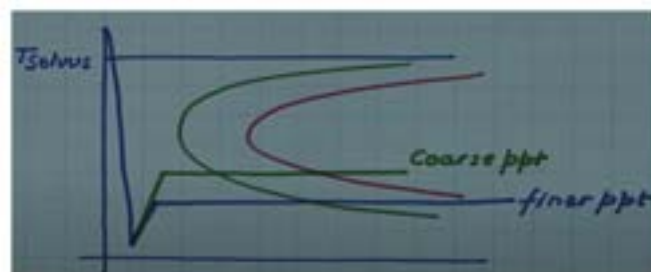
Evolution of Hardness





Effect of Temperature

- Lower Temperature = higher supersaturation + Finer precipitates
- Higher Temperature = Faster kinetics to finish precipitation



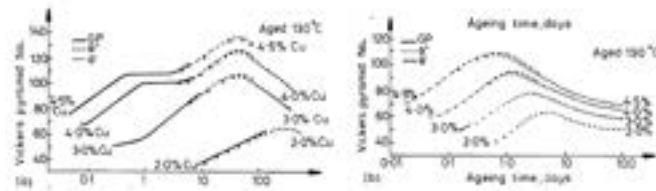


Figure 159: Impact of alloying element concentration

4 Kinetics of Phase Transformation

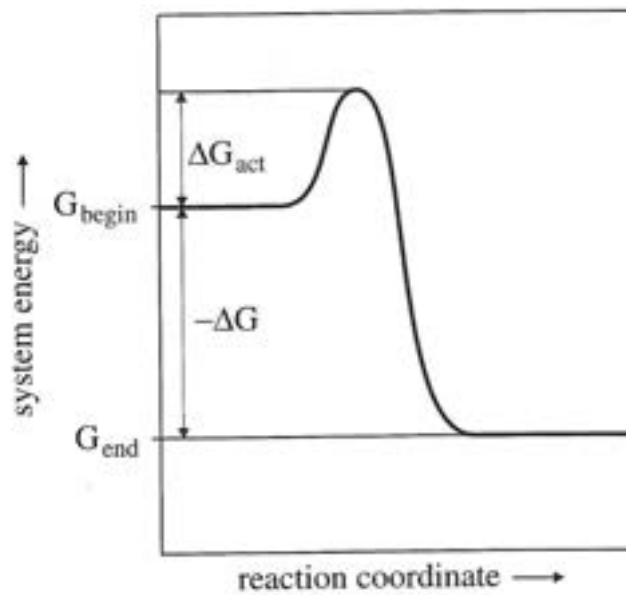
Origin of a phase transformation?

Thermodynamics of the phase transformation.

The energy difference of the system before the phase transformation and after the completed transformation: "driving force". The thermodynamic property that tells us in which direction a transformation proceeds spontaneously is the Gibbs energy (Enthalpy H minus Temperature T times Entropy S)

$$G = H - TS, \quad \Delta G = G_2 - G_1$$

- $\Delta G = 0$, Equilibrium, no net change.
- $\Delta G < 0$, Spontaneous change
- $\Delta G > 0$, Spontaneous reverse change

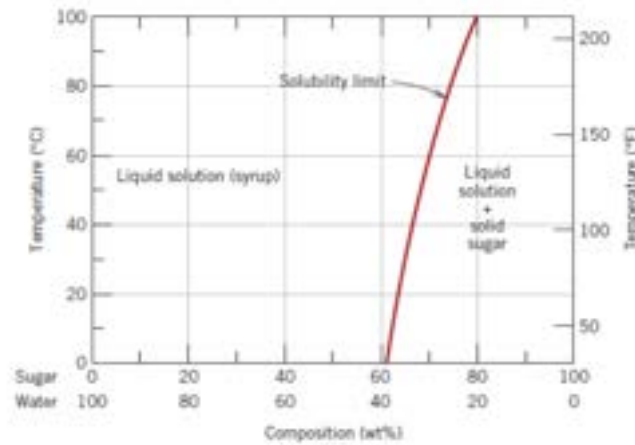


Kinetics of the phase transformation, The mechanism and rate of the phase transformation:

- Nucleation of the new phase
- Growth of the new phase

4.0.1 Thermodynamics of a phase transformation

Solubility limit The addition of solute in excess of this solubility limit results in the formation of another solid solution or compound that has a distinctly different composition.



Phases A phase may be defined as a homogeneous portion of a system that has uniform physical and chemical characteristics.

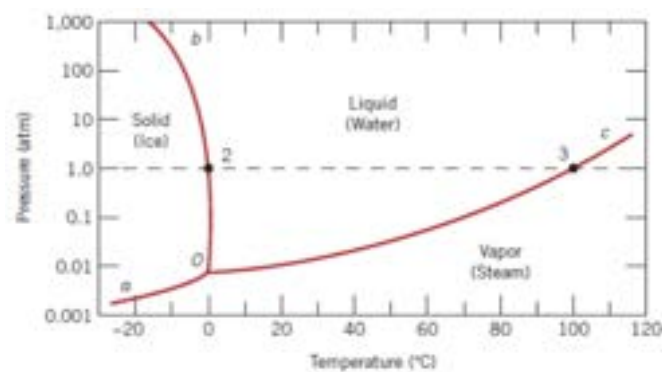


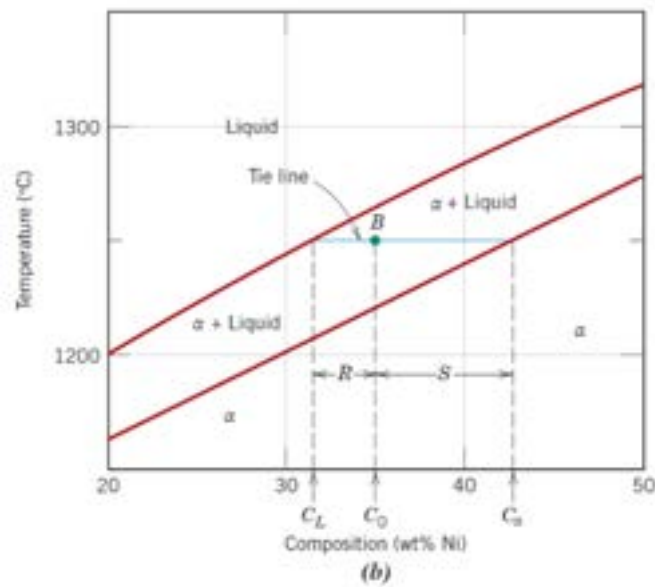
Figure 160: One-compound (Or unary) Phase Diagrams

Metastable phases Free energy considerations and diagrams provide information about the equilibrium characteristics, but they do not indicate the time period necessary for the attainment of a new equilibrium state.

4.0.2 Binary Isomorphous Systems

At temperatures below about 1080 °C,

- Copper and nickel are mutually soluble in each other in the solid state for all compositions
- This complete solubility is explained by the fact that both copper and Nickel have the same crystal structure (FCC), nearly identical atomic radii and electronegativities, and similar valences



$$W_L = \frac{C_\alpha - C_0}{C_\alpha - C_L}$$

The equilibrium concentrations of the two phases, the following procedure:

1. A tie line is constructed across the two-phase region.
2. The intersections of the tie line and the phase boundaries.
3. Perpendiculars from intersections determine composition of each phase.

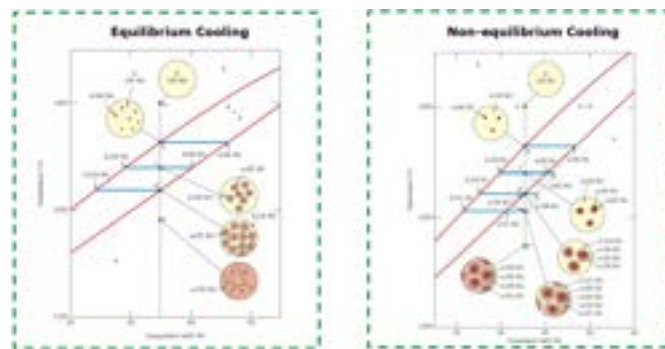
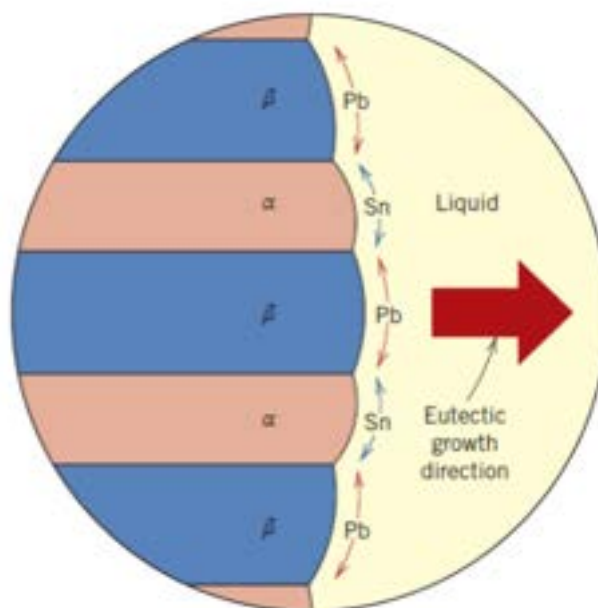
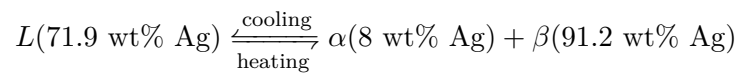
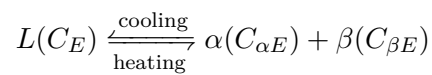
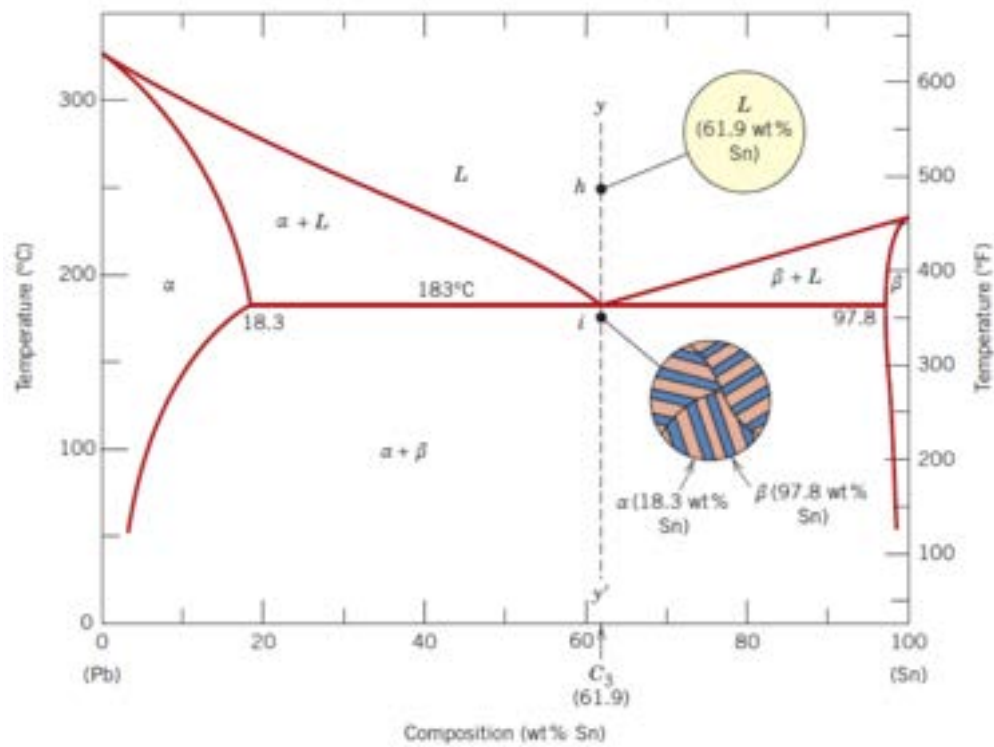


Figure 161: Development of Microstructure in Isomorphous Alloys

4.0.3 Binary Eutectic Systems



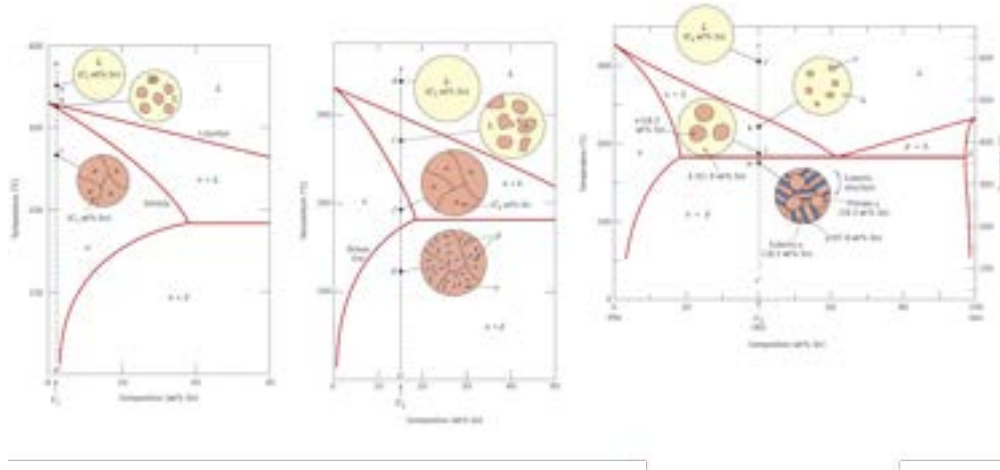


Figure 162: Development of Microstructure in Eutectic Alloys

$$\begin{aligned}
 W_e = W_L &= \frac{P}{P+Q} \\
 &= \frac{C'_4 - 18.3}{61.9 - 18.3} = \frac{C'_4 - 18.3}{43.6}
 \end{aligned}$$

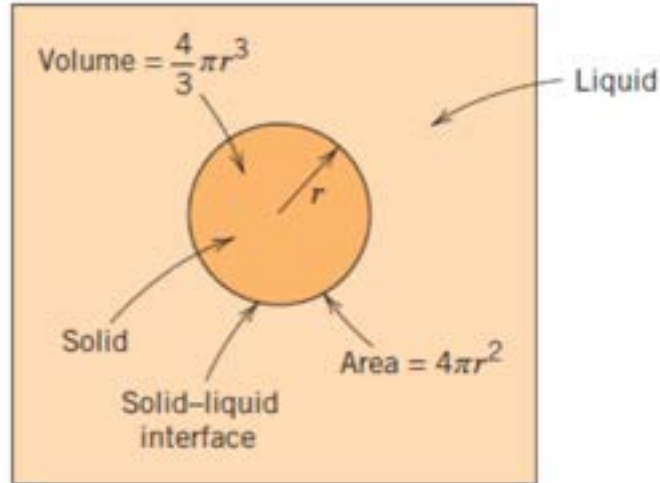
4.1 Rate of Phase Transformations

Kinetics measure approach to equilibrium vs. time Hold temperature constant and measure conversion vs. time.

This conversion is measured:

- Calorimetry: measurement of heat consumption/development
- Magnetometry: measurement of fraction of magnetic phase
- Dilatometry: measurement of volume changes
- Thermogravimetry: measurement of mass changes
- Resistometry: Measurement of resistivity changes
- X-ray diffractometry: direct measurement of phase fractions
- Microscopy: phase fractions and particle size

4.1.1 Energetics of nucleation Solidification of pure component: homogeneous nucleation



$$\Delta G = \frac{4}{3}\pi r^3 \Delta G_v + 4\pi r^2 \gamma$$

Where ΔG_v is the volume free energy change and γ is the surface free energy.

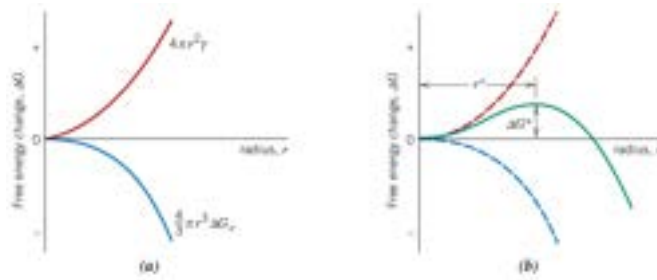


Figure 163: (a) Schematic curves for a volume free energy and surface free energy contributions to the total free energy attending the formation of a spherical embryo/nucleus during solidification. (b) Schematic plot of free energy versus embryo/nucleus radius, on which the critical free energy change ΔG^* and the critical nucleus radius r^*

$$\frac{d(\Delta G)}{dr} = \frac{4}{3}\pi \Delta G_v (3r^2) + 4\pi \gamma (2r) = 0$$

Which leads to the result,

$$r^* = -\frac{2\gamma}{\Delta G_v}$$

And substituting the expression of r^* into the change in Gibbs energy,

$$\Delta G^* = \frac{16\pi\gamma^3}{3(\Delta G_v)^2}$$

It can be shown that ΔG_v can be written as a function of temperature as,

$$\Delta G_v = \frac{\Delta H_f (T_m - T)}{T_m}$$

$$r^* = \left(-\frac{2\gamma T_m}{\Delta H_f} \right) \left(\frac{1}{T_m - T} \right), \quad \Delta G^* = \left(\frac{16\pi\gamma^3 T_m^2}{3\Delta H_f^2} \right) \frac{1}{(T_m - T)^2}$$

Temperature dependence

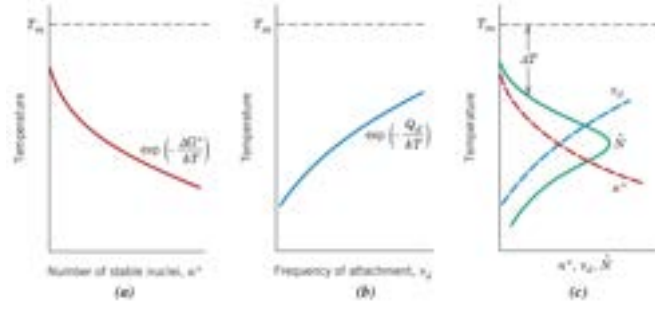


Figure 164: For solidification schematic plots of (a) number of stable nuclei versus temperature, (b) frequency of atomic attachment versus temperature, and (c) nucleation rate versus temperature (the dashed line curves are reproduced from a and b).

The number of stable nuclei n^* (having radii greater than r^*) is a function of temperature as,

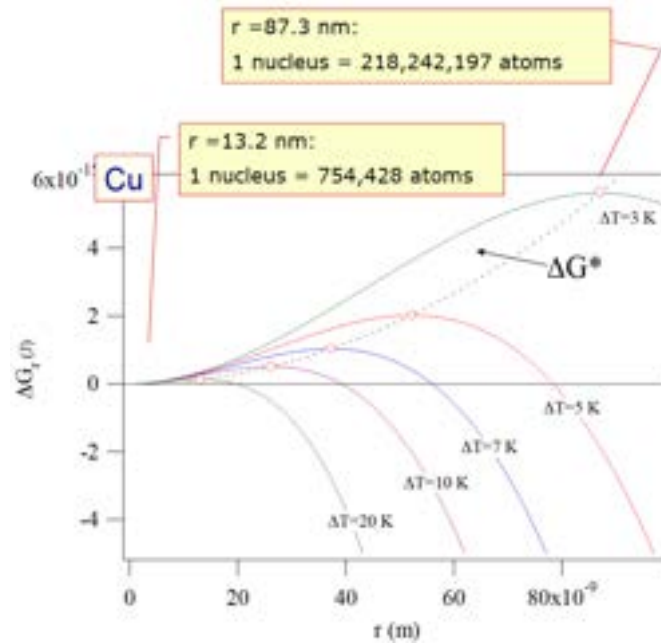
$$n^* = K_1 \exp\left(-\frac{\Delta G^*}{kT}\right)$$

Where K_1 is related to the number of nuclei of the solid phase.

The influence of temperature on the rate of diffusion D , this diffusion is related to the frequency at which atoms from the liquid attach themselves to the solid nucleus (frequency of attachment ν_d). The dependence of ν_d on temperature is the same for the diffusion coefficient,

$$\nu_d = K_2 \exp\left(-\frac{Q_d}{kT}\right)$$

Where Q_d is a temperature-independent parameter, the activation energy for diffusion and K_2 is a temperature independent constant.



4.1.2 Energetics of nucleation, heterogeneous nucleation

Radius of critical (critical radius) nucleus as can be seen below is not affected

$$r_{het}^* = r_{hom}^* = -\frac{2\sigma_{SL}}{\Delta G_v}$$

Taking a surface tension force balance in the plane of the flat surface leads to the following expression,

$$\gamma_{IL} = \gamma_{SI} + \gamma_{SL} \cos \theta$$

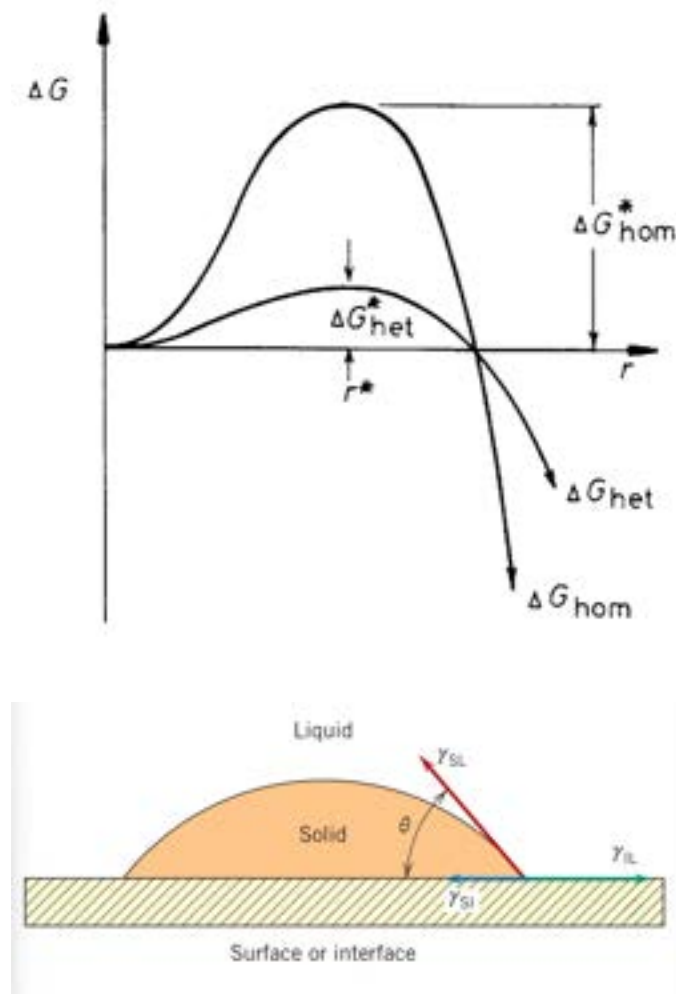


Figure 165: Heterogeneous nucleation of a solid from a liquid. The solid surface (γ_{SI}), solid-liquid (γ_{SL}), and liquid-surface (γ_{IL}), interfacial energies are represented by vectors. The wetting angle θ is also shown

The equation for r^* ,

$$r^* = -\frac{2\gamma_{SL}}{\Delta G_v}$$

As can be seen above there is lower activation energy barrier for formation of nucleus of critical size,

$$\Delta G_{het}^* = \frac{16}{3} \pi \frac{\sigma_{SL}^3}{\Delta G_v^2} \cdot S(\theta) = \Delta G_{hom}^* \cdot S(\theta)$$

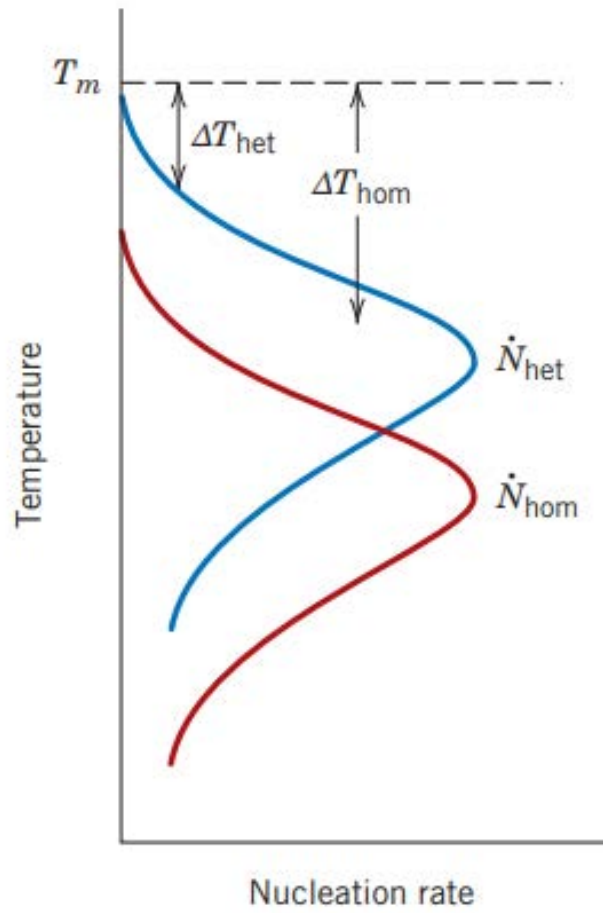
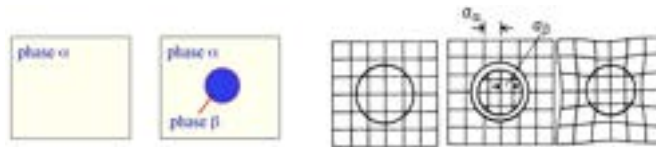


Figure 166: Nucleation rate versus temperature for both homogeneous and heterogeneous nucleation. Degree of supercooling (ΔT) for each is also shown.

Nucleation for solid-solid phase transformation



$$\Delta G = V_\beta \Delta G_v + A_{\alpha\beta} \sigma + V_\beta \Delta G_s$$

Total Gibbs energy, change on nucleation depends on size and composition of the stable embryo (nucleus).

4.1.3 Growth and overall transformation

Particle growth occurs by long-range atomic diffusion, which involves several steps: For example diffusion,

- through the parent phase
- Across a phase boundary
- Then into the nucleus

$$\dot{G} = C \exp \left(- \frac{Q}{kT} \right)$$

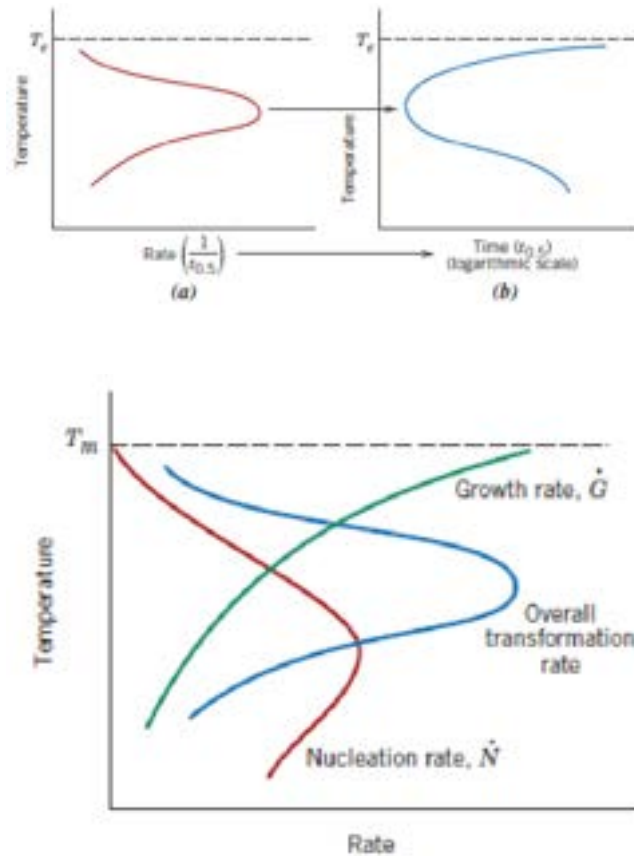
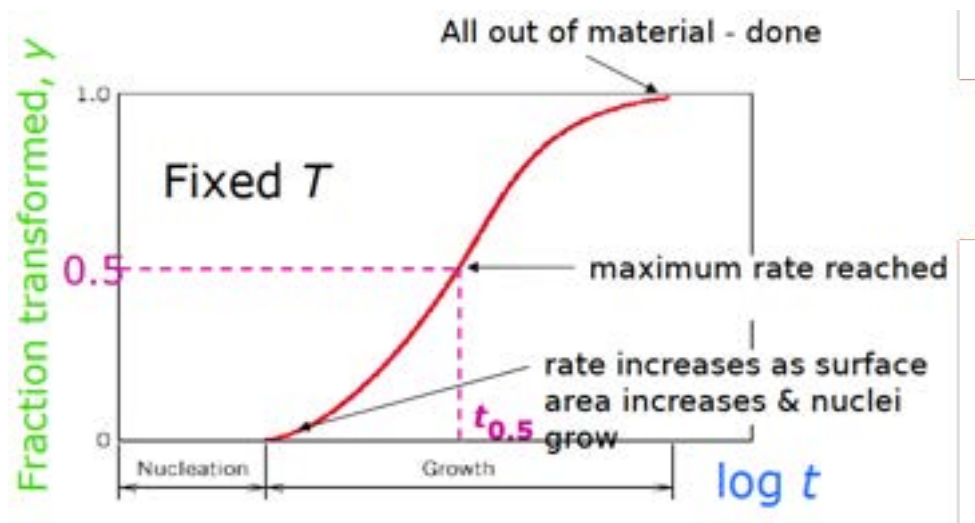


Figure 167: At a specific temperature, the overall transformation rate is equal to some product of $\dot{N}$ and $\dot{G}$

Rate of Phase Transformation



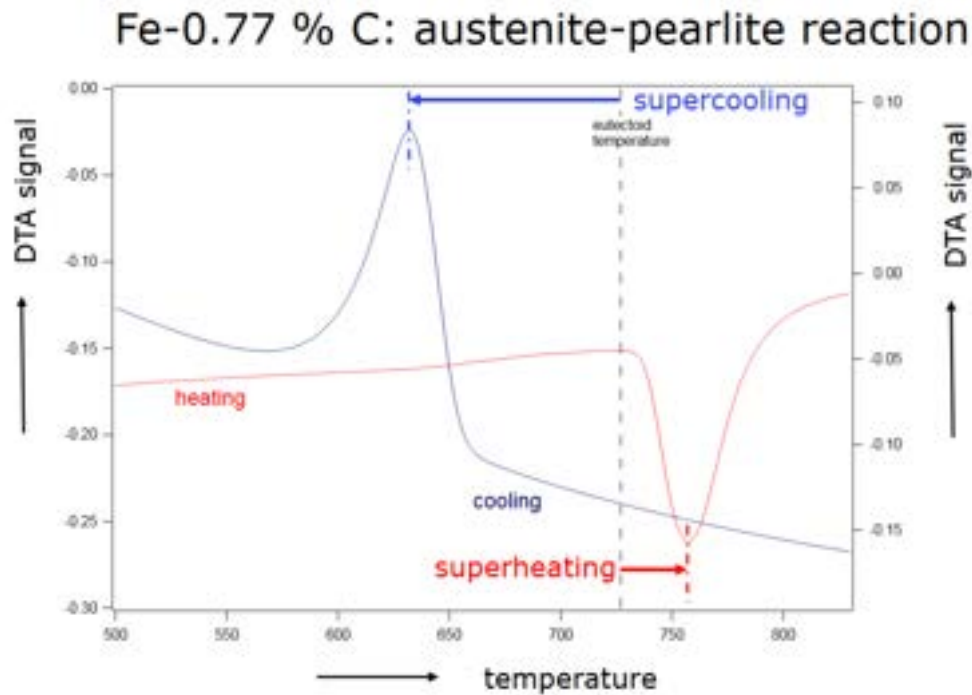
The fraction of transformation y is a function of time t using the Avrami rate equation,

$$y = 1 - \exp(-kt^n)$$

Where k and n are fit for a specific sample.

The rate of transformation is taken as the reciprocal of time required for the transformation to proceed halfway to completion, $t_{0.5}$. By convention the rate of transformation = $1/t_{0.5}$

Metastable versus Equilibrium States



The rate of approach to equilibrium for solid systems is so slow that true equilibrium structures are rarely achieved.

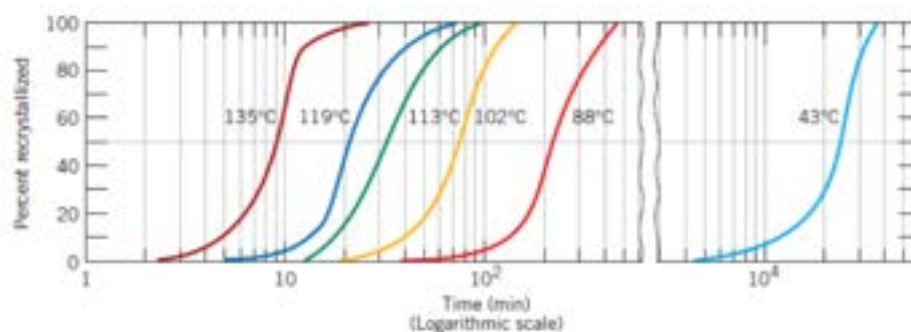
For non-equilibrium states,

- For cooling, transformations are shifted to lower temperatures than indicated by the phase diagram, supercooling.
- For heating the shift is to higher temperatures, superheating.

4.1.4 Overall transformation kinetics

Transformation Time Temperature diagrams. Transformation kinetics is determined by,

- Nucleation rate
- Growth rate
- Density/distribution of nucleation sites
- Overlap of diffusion fields
- The conditions that give the fastest transformation is a high ΔG and high diffusion



4.2 Heat treatment of steel

Processes used in Heat treatment of steel

- Full anneal
- Normalising
- Isothermal pearlite formation (patenting)
- Austempering
- Martempering

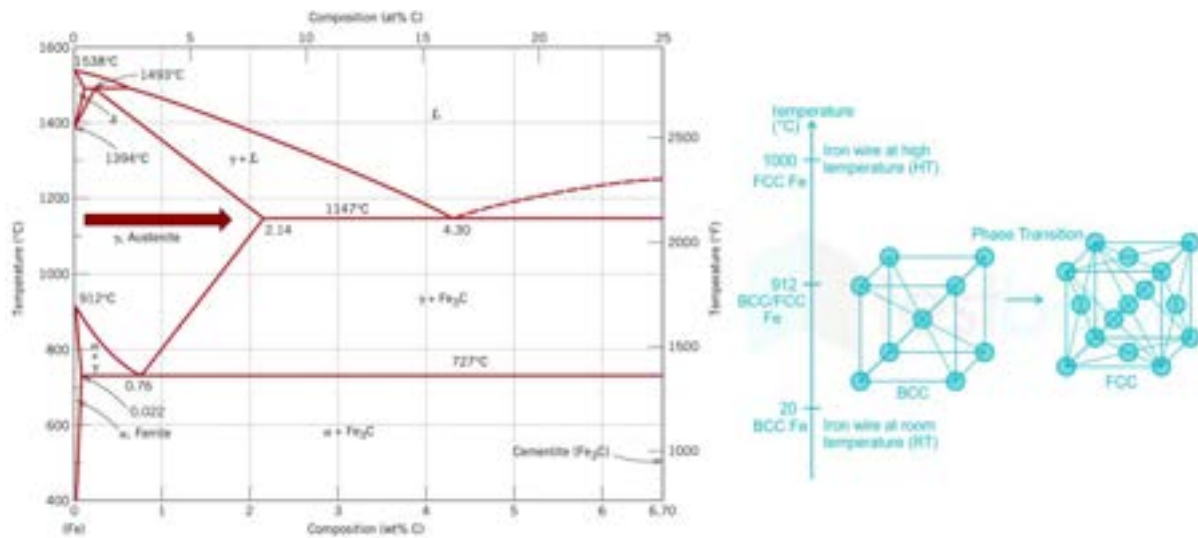
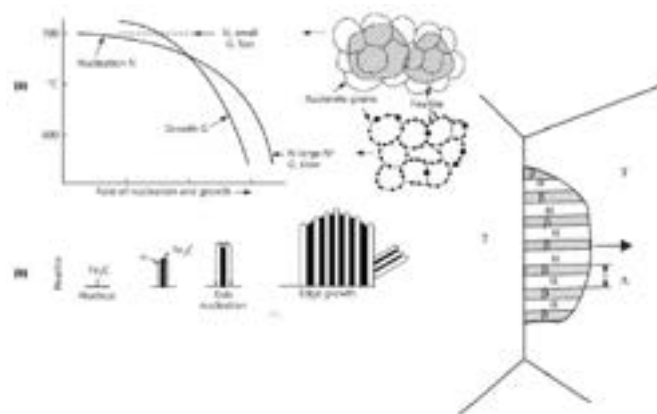
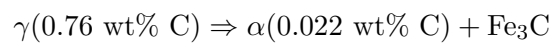
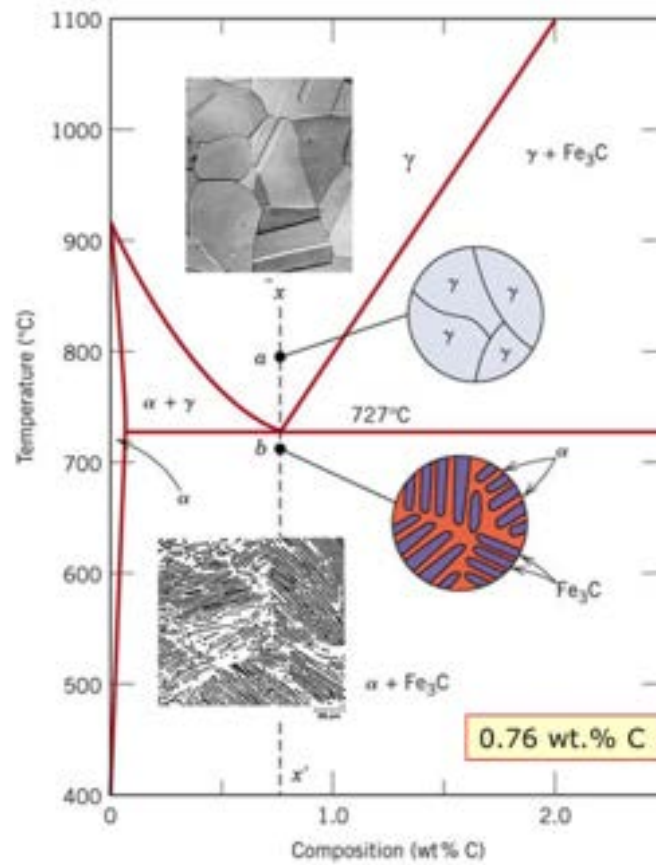


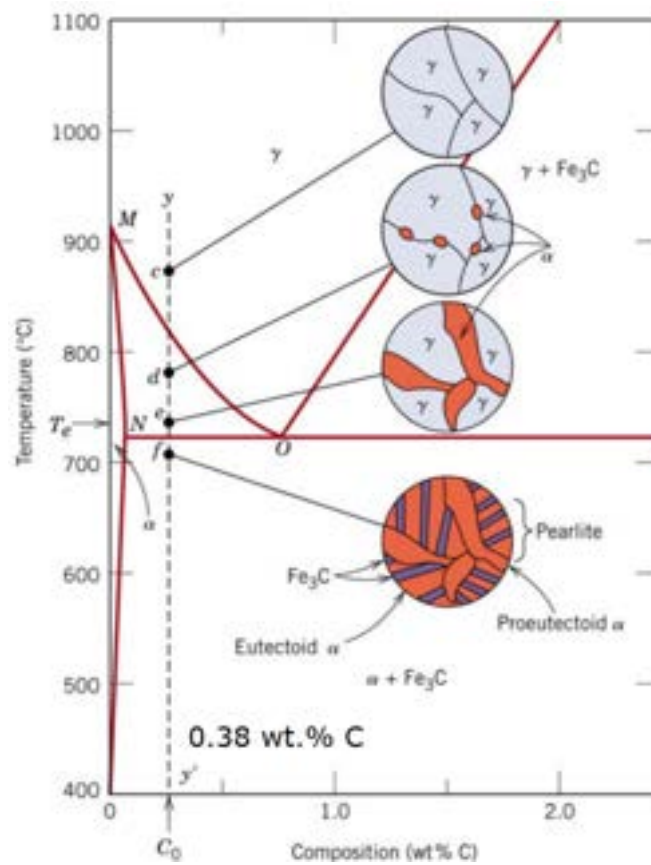
Figure 168: Fe-Fe₃C phase diagram

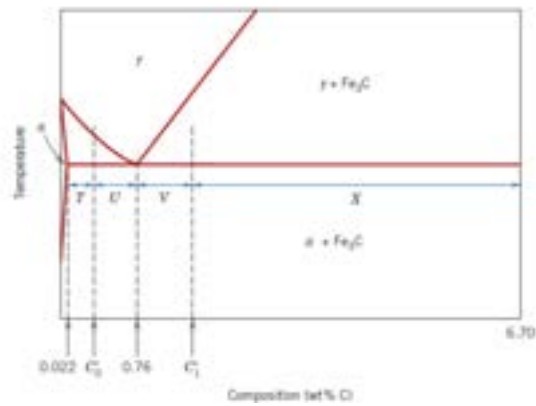
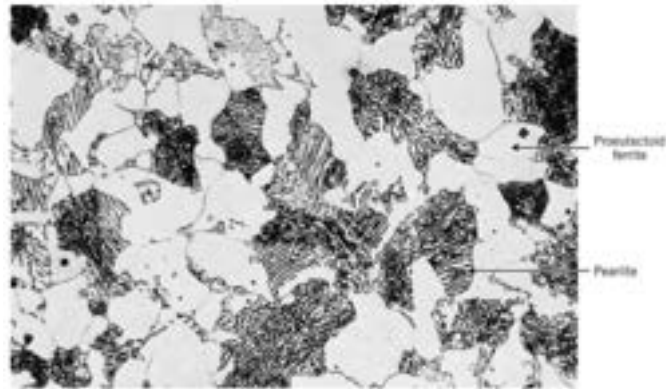
The eutectoid reaction



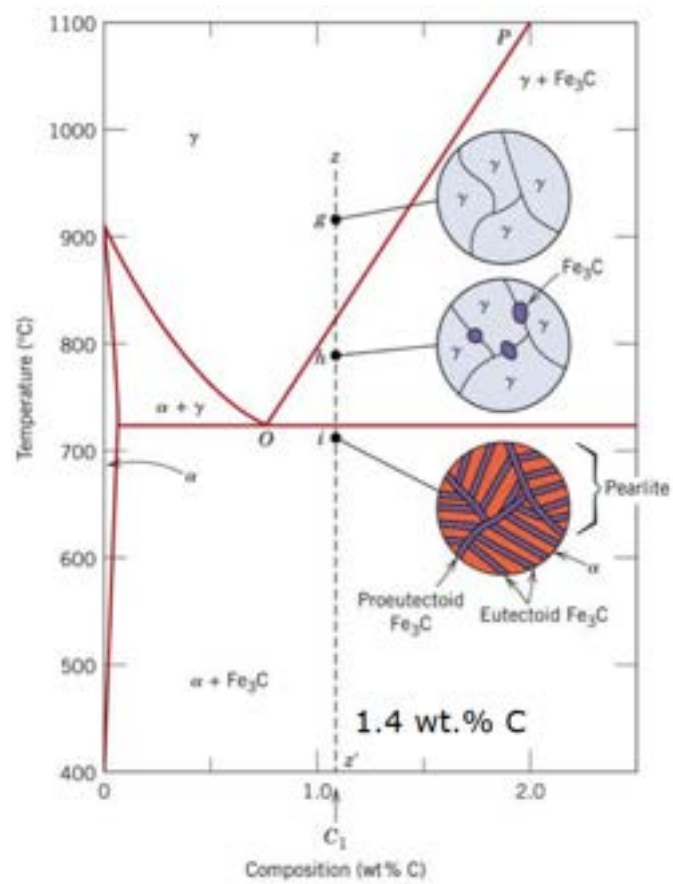


Hypoeutectoid alloy





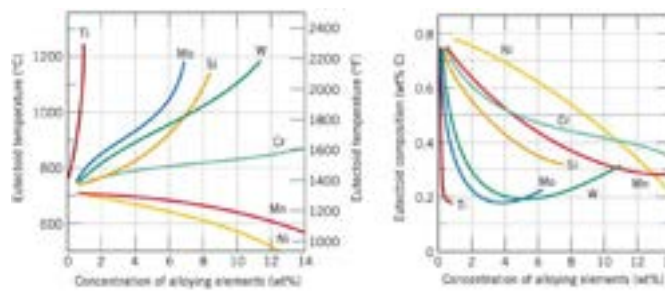
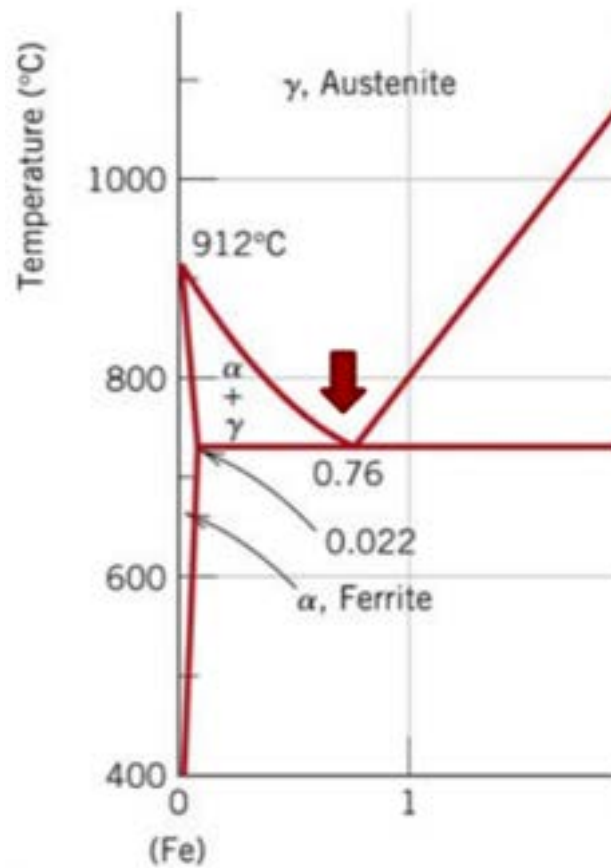
Hypereutectoid alloy



$$W_P = \frac{X}{V + X} = \frac{6.70 - C'_1}{6.70 - 0.76} = \frac{6.70 - C'_1}{5.94}$$

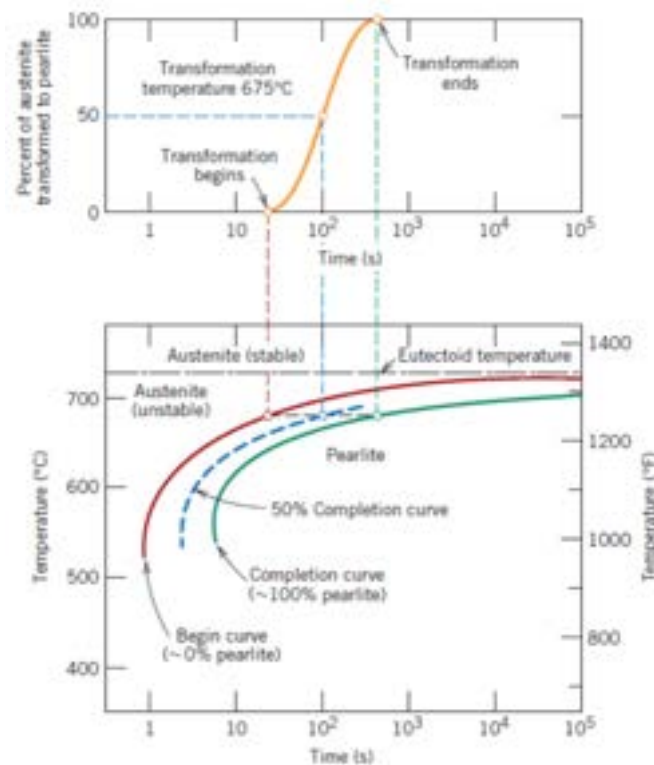
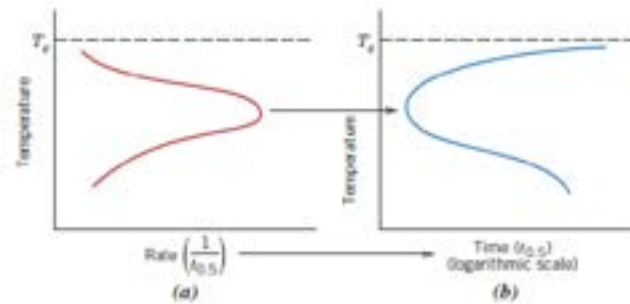
$$W_{\text{Fe}_3\text{C}'} = \frac{V}{V + X} = \frac{C'_1 - 0.76}{6.70 - 0.76} = \frac{C'_1 - 0.76}{5.94}$$

Eutectoid point



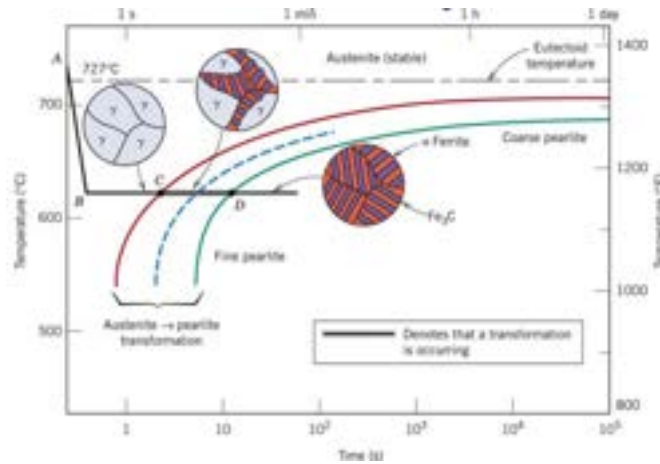
The extend of these alterations of the positions of the phase boundaries and the phase fields depends on the particular alloying element and its concentration

4.2.1 Overall transformation kinetics



- Nucleation rate
- Growth rate
- Density/distribution of nucleation sites
- Overlap of diffusion fields

Isothermal transformation diagram - Pearlite



Eutectoid composition,

- At temperatures just below the eutectoid, relatively thick layers of both α -ferrite and Fe_3C phases are produced, this microstructure is called coarse pearlite.
- With decreasing temperature, the carbon diffusion rate decreases, and the layers become progressively thinner. The thin-layered structure produced in the vicinity of 540 °C is termed fine pearlite

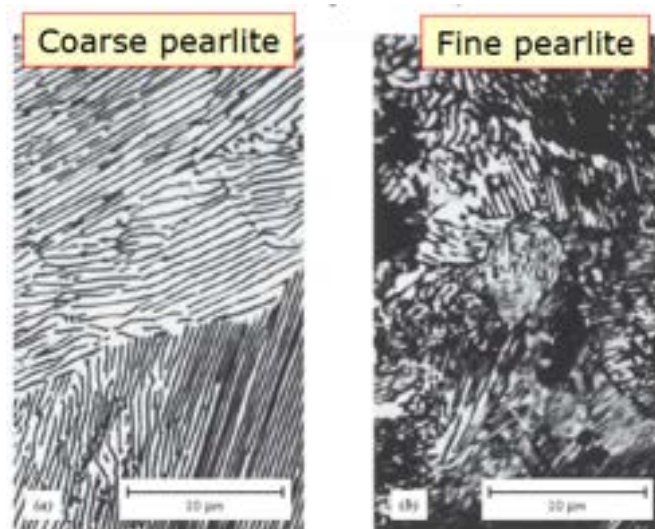


Figure 169: Visual representation of coarse and fine pearlite

Fine structures are formed from high nucleation rate and low growth rate. Fine pearlite forms at lower temperature because of higher nucleation rate lower growth and higher driving force ΔG_v

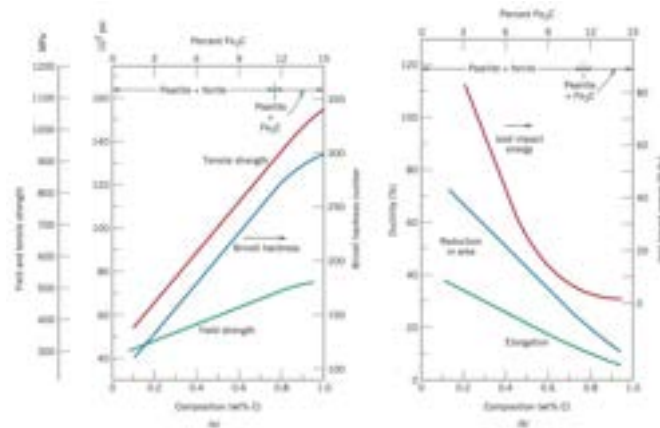


Figure 170: Mechanical properties of fine pearlite

Isothermal transformation diagram - Bainite

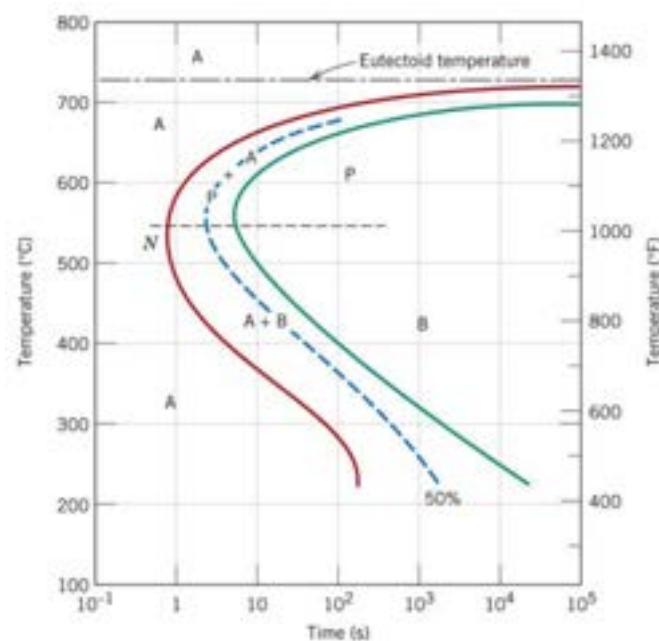


Figure 171: Where B is Bainite

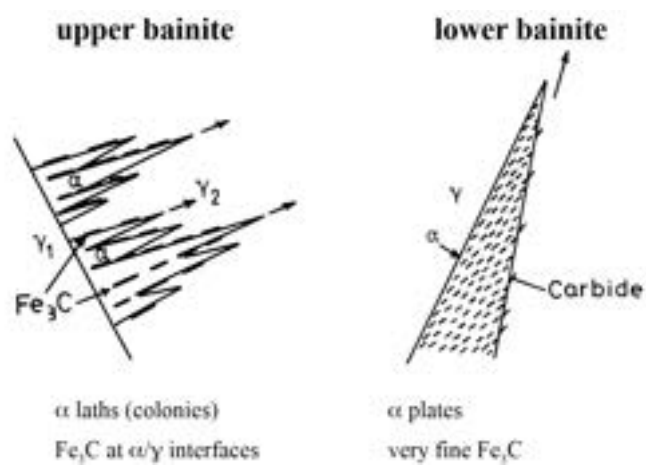


Figure 172: Visual representation of upper and lower bainite

Spheroidizing full anneal

If a steel alloy having either pearlitic or bainitic microstructures is heated to, and left at, a temperature below the eutectoid for a sufficiently long period of time, for example, at about 700 °C for between 18 to 24 hours, yet another microstructure will form called spheroidite. Long holding time close to the eutectoid temperature / slow furnace cooling.

- Instead of the alternating ferrite and cementite lamellae (pearlite) or the microstructure observed for bainite, the Fe_3C phase appears as sphere like particles embedded in a continuous α -phase matrix.
- This continuous occurs by additional carbon diffusion with no change in the compositions or relative amounts of ferrite and cementite phases,

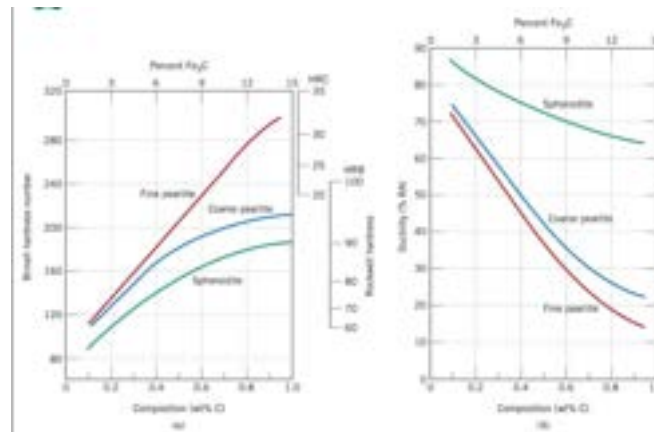


Figure 173: Comparison of mechanical properties for Fine pearlite, coarse pearlite and spheroidite

The following are in the decreasing order of hardness/strength,

- Lower bainite
- Upper bainite
- Fine pearlite
- Pearlite
- Coarse pearlite
- Spheroidite

Martensite

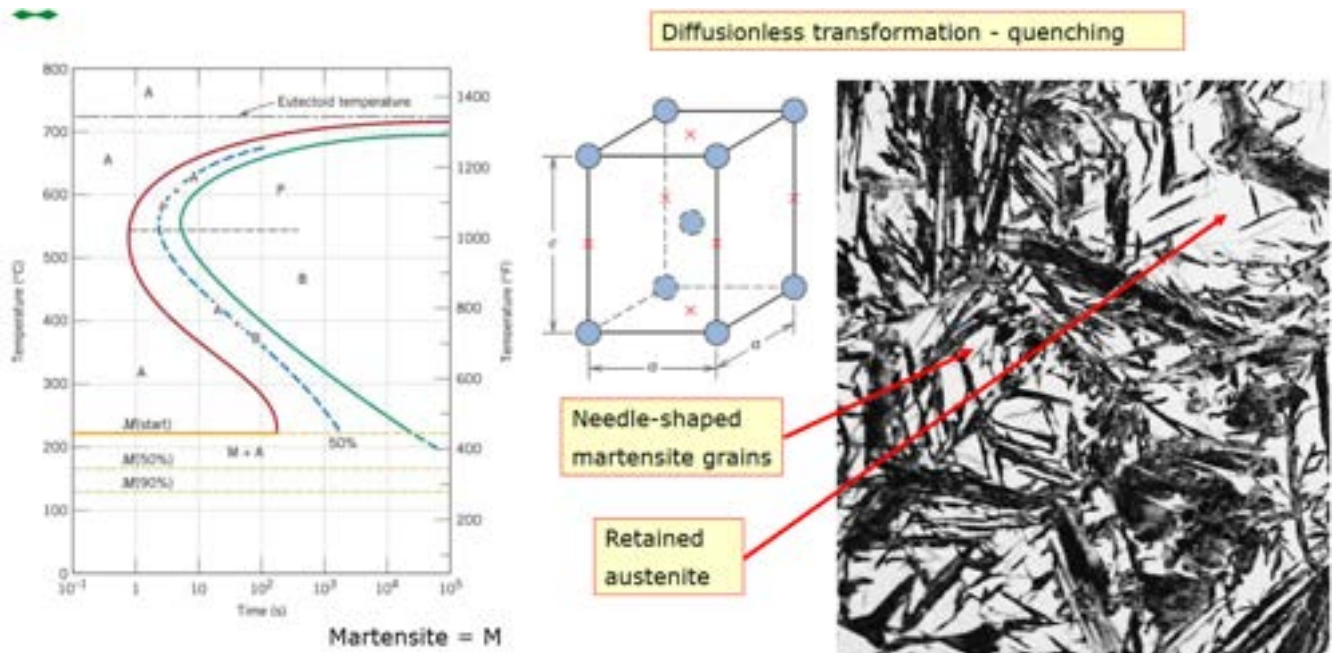


Figure 174: TTT diagram for an alloy steel type 4340

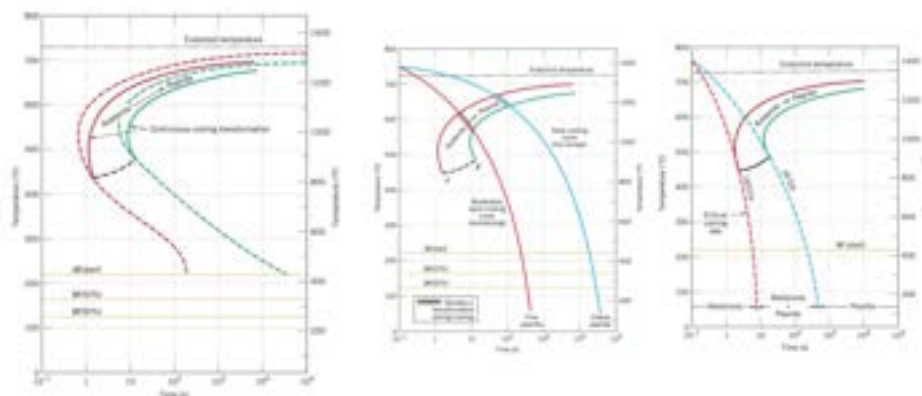


Figure 175: Continuous cooling transformation (CCT)

4.2.2 Tempering

Why tempering is needed:

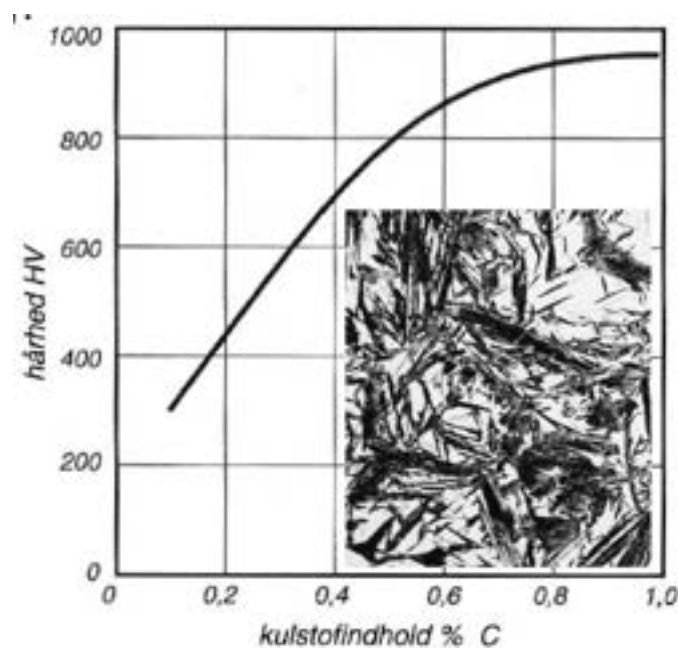
- Martensite is very hard but brittle ($C > 0.25 \text{ wt\%}$)
- Quenching introduces internal stresses
- These stresses weaken the material and can cause failure

Definition:

Tempering is a heat treatment process in which martensitic steel is heated to a temperature below the eutectoid temperature for a certain period of time.

Purpose of tempering:

- Reduce brittleness
- Relieve internal stresses
- Improve toughness
- Adjust hardness to a desired level



Temperature ranges and mechanisms for Martensite tempering:

- $\sim 100^\circ\text{C}$: Carbon segregation to defects (e.g. dislocations)
- $100\text{--}250^\circ\text{C}$: Formation of ϵ -carbide (structure still martensitic)
- $250\text{--}700^\circ\text{C}$: Decomposition into ferrite + cementite

Effect of tempering temperature:

- $180\text{--}250^\circ\text{C}$: High hardness, improved impact toughness
- $400\text{--}700^\circ\text{C}$: Increased ductility and toughness (lower hardness)

The microstructure of tempered martensite consists of extremely small and uniformly dispersed cementite particles embedded within a continuous ferrite matrix.



Martensite is a metastable phase $\Rightarrow$ equilibrium phases

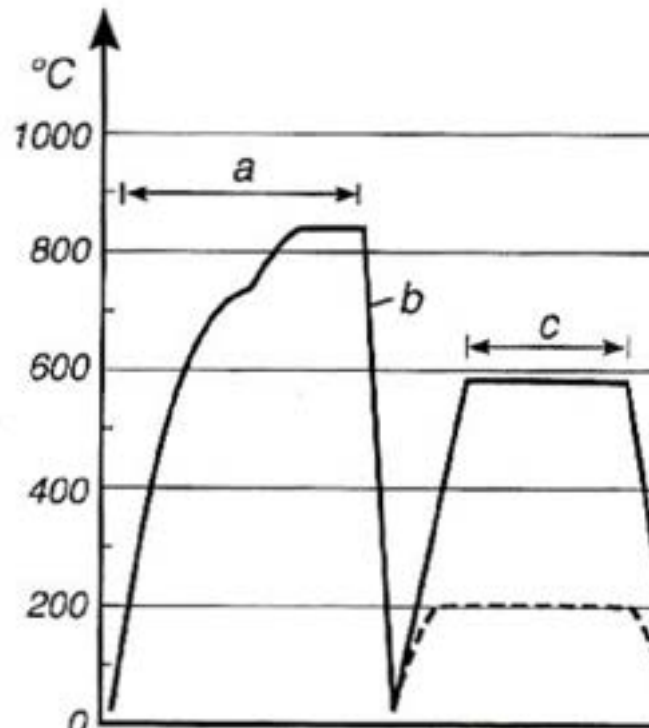


Figure 176: Typical heat treatment path for quenching and tempering of martensitic steel.

The tempering step (c) is performed after quenching to adjust the balance between hardness and toughness.

Influence of tempering temperature upon mechanical properties

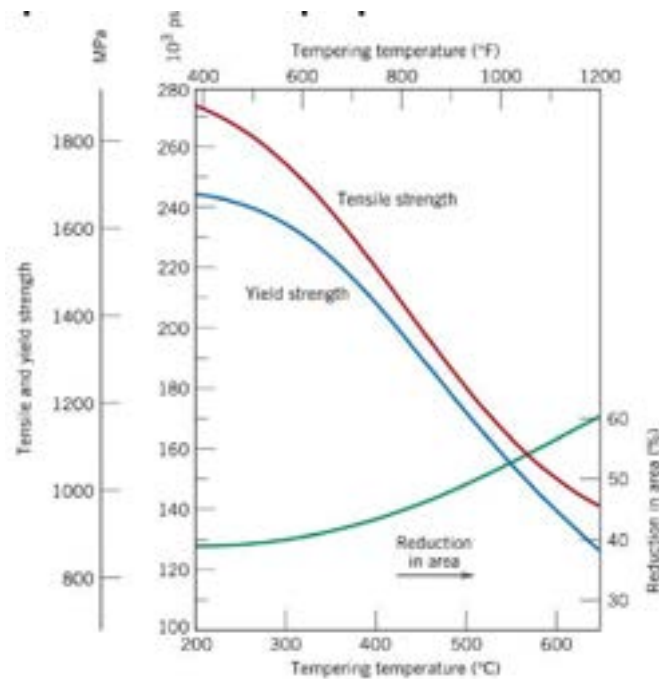


Figure 177: Influence of tempering temperature upon mechanical properties

- Tempered martensite may be nearly as hard and strong as martensite but with substantially enhanced ductility and toughness.
- The hardness and strength may be explained by the large ferrite-cementite phase boundary area per unit volume that exists for the very fine and numerous cementite particles
- The size of the cementite particles influences the mechanical behavior of tempered martensite; increasing the particle size decreases the ferrite-cementite phase boundary area and consequently, results in a softer and weaker material yet one that is tougher and more ductile.

Influence of tempering time upon mechanical properties

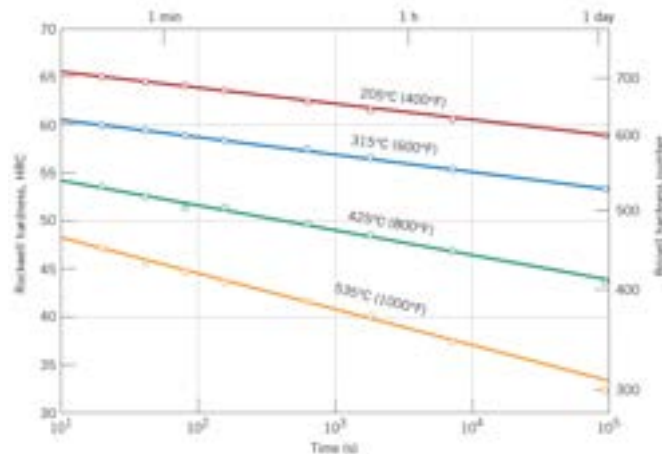
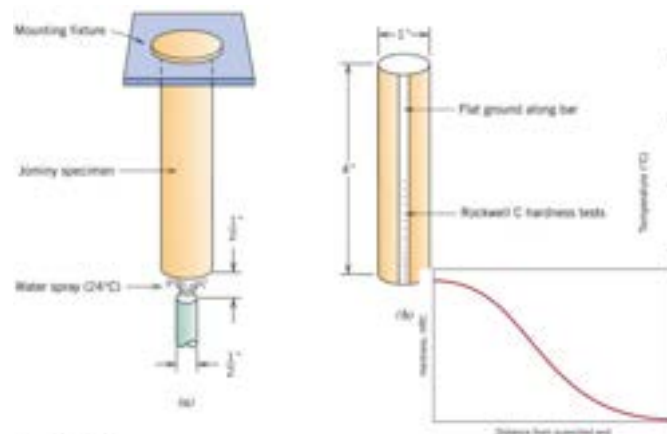


Figure 178: Influence of tempering time upon mechanical properties

With increasing time the hardness decreases, which corresponds to the growth and coalescence of the cementite particles

4.2.3 Hardenability - Jominy test



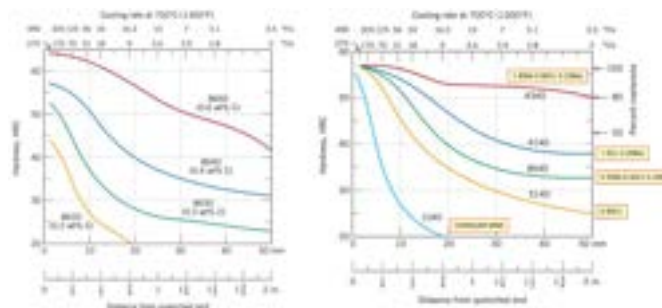
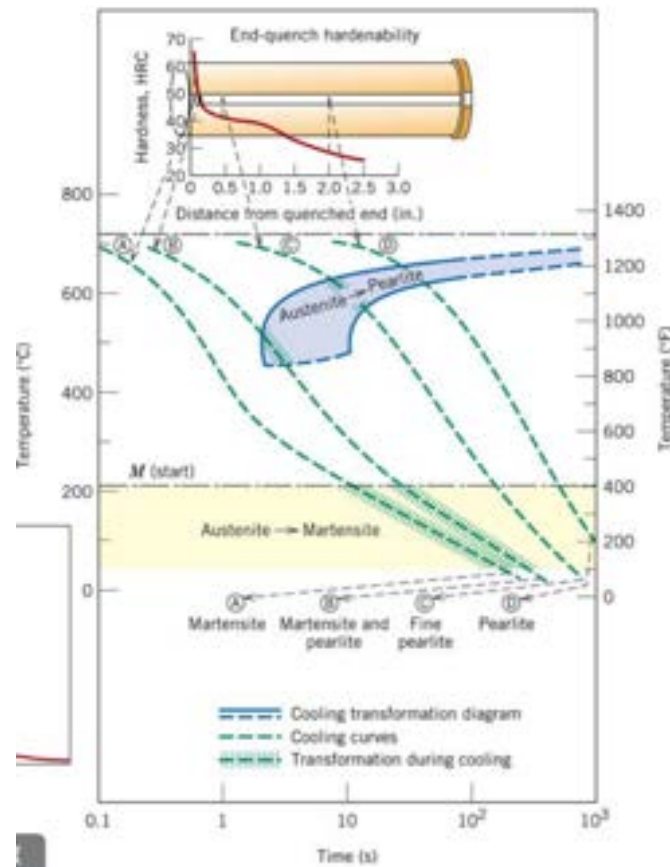


Figure 179: Effect of chemical composition on the hardness from Jominy test

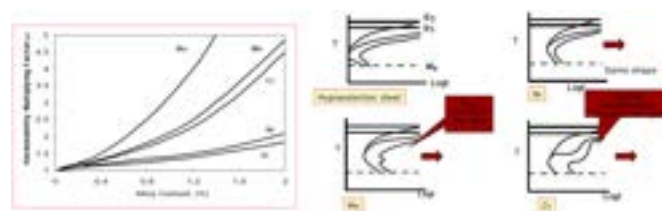
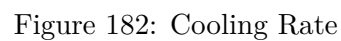
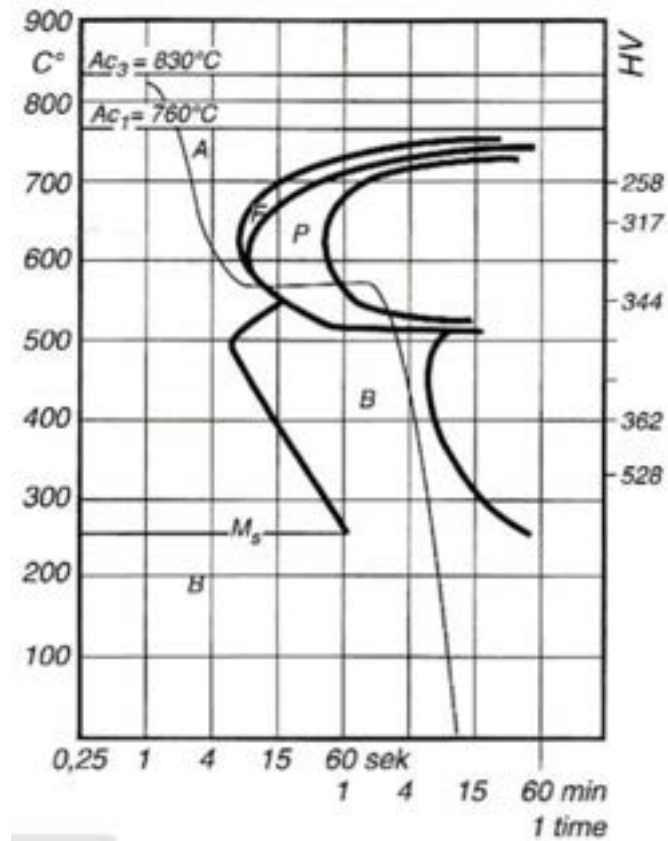


Figure 180: Influence of alloying elements from improved hardenability

Full anneal and normalizing

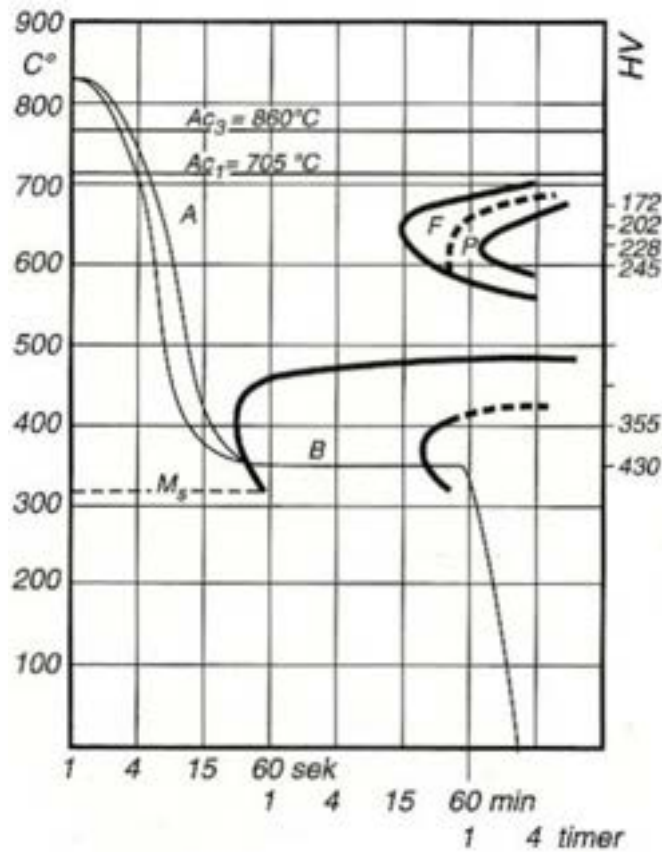


Isothermal transformation into fine pearlite (+ plastic deformation)



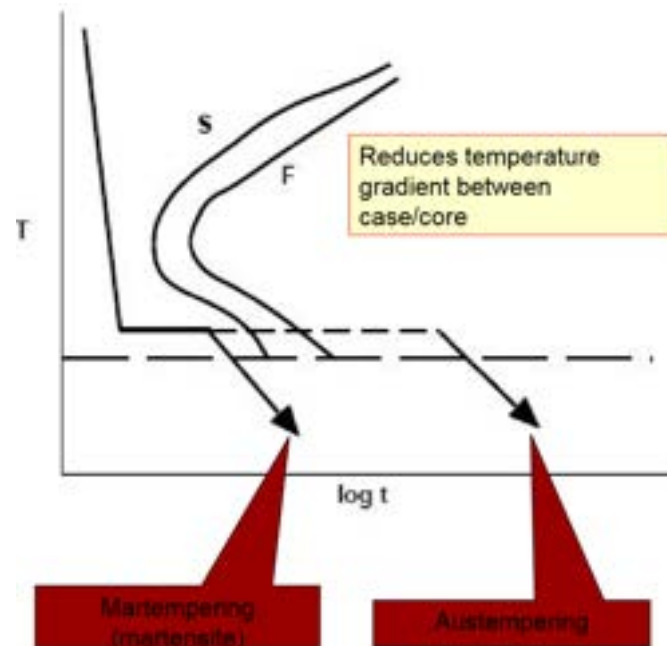
- Patenting consists of passing the wire through tubes in a furnace at about 970°C
- This high temperature treatment produces uniform austenite of rather long grain sizes.
- The subsequent cooling, in air or molten lead, is rapid since the sections treated are generally small (e.g. wire rods)
- So that the resulting structure consists of very fine pearlite preferably with no separation of primary ferrite

Austempering



- Austempering is a heat process for medium to high carbon ferrous metals which produce a metallurgical structure called bainite
- It is used to increase strength, toughness, and reduce distortion
- Parts are heated to the hardening temperature, then rapidly cooled rapidly enough to a temperature above the Martensite start (M_s) temperature and held for a time sufficient to produce the desired bainite microstructure.
- Higher ductility, toughness and strength for a given hardness
- Resistance to shock
- Reduced distortion, specifically with thin parts

Martempering



- In Martempering, steel is heated above the upper critical point (above the transformation range) and then quenched in a hot-oil, molten-salt, or molten-lead bath kept at a temperature of 150-300 °C
- Martempering is a method by which the stresses and strains generated during the quenching of a steel component can be controlled. The drawback of this process is that the large section cannot be heat treated by this process

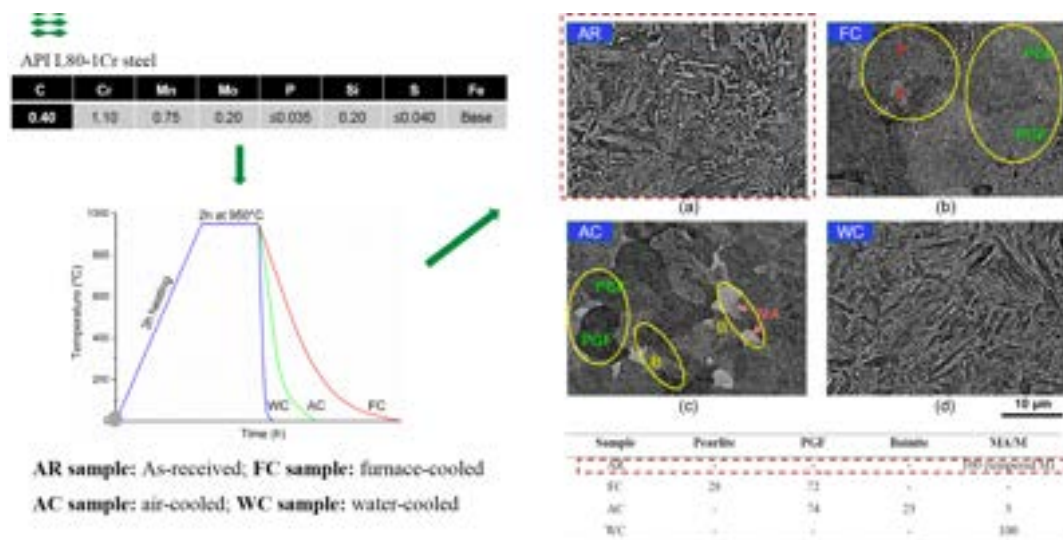


Figure 183: Effect of cooling rate induced microstructural changes on CO₂ corrosion behavior

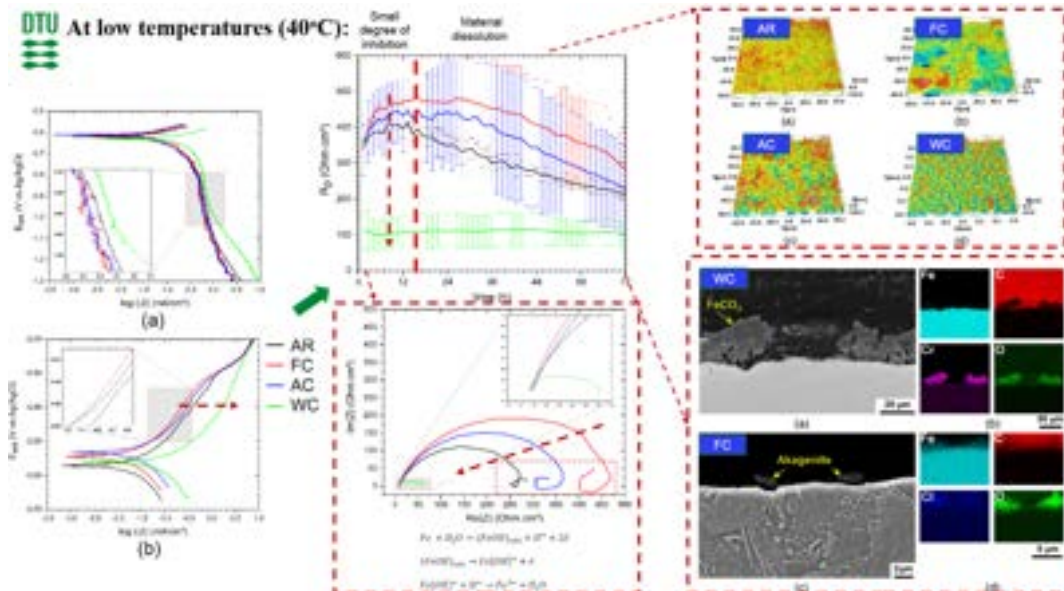
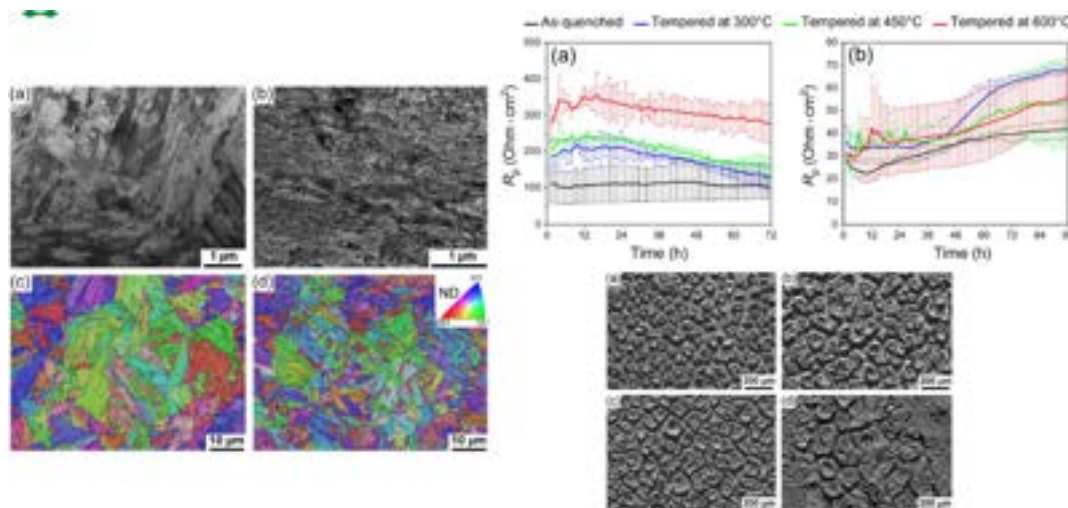


Figure 184: At low temperatures (40°)

Figure 185: Effect of tempering temperature on CO₂ corrosion behavior

4.3 Failure in Metallic materials: Fatigue, creep and wear

Fundamentals of fracture: Simple fracture is the separation of a body into two or more pieces in response to an imposed stress that is static (i.e., constant or slowly changing with time) and at temperatures that are low relative to the melting temperature of the material.

Classification in fracture: Classification is based on the ability of a material to experience plastic deformation.

- **Ductile metals**, typically exhibit substantial plastic deformation with high energy absorption before fracture
 - Ductile fractures is characterized by extensive plastic deformation in the vicinity of an advancing crack.
 - The process proceeds relatively slowly as the crack length is extended
 - Such a crack is often said to be stable that is, it resists any further extension unless there is an increase in the applied stress

- **Brittle metals:** there is normally little or no plastic deformation with low energy absorption.
 - Cracks may spread extremely rapidly, with very little accompanying plastic deformation
 - Such cracks may be said to be unstable, and crack propagation, once started, continues spontaneously without an increase in magnitude of the applied stress

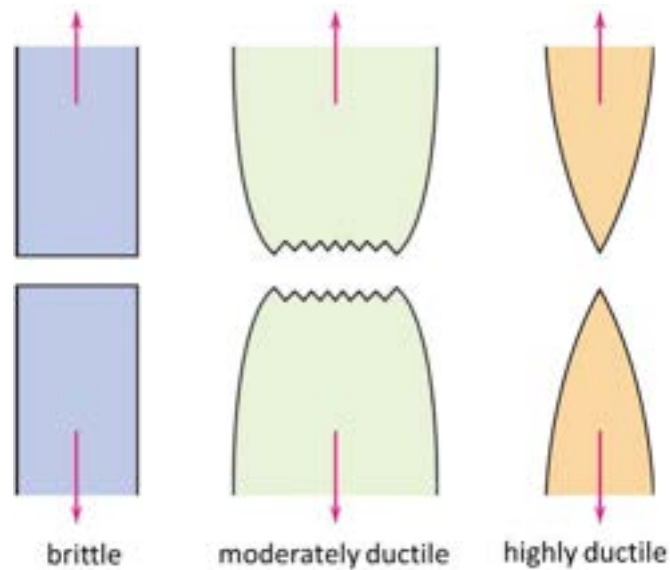


Figure 186: Examples of: Brittle, Moderately ductile and Highly ductile fractures

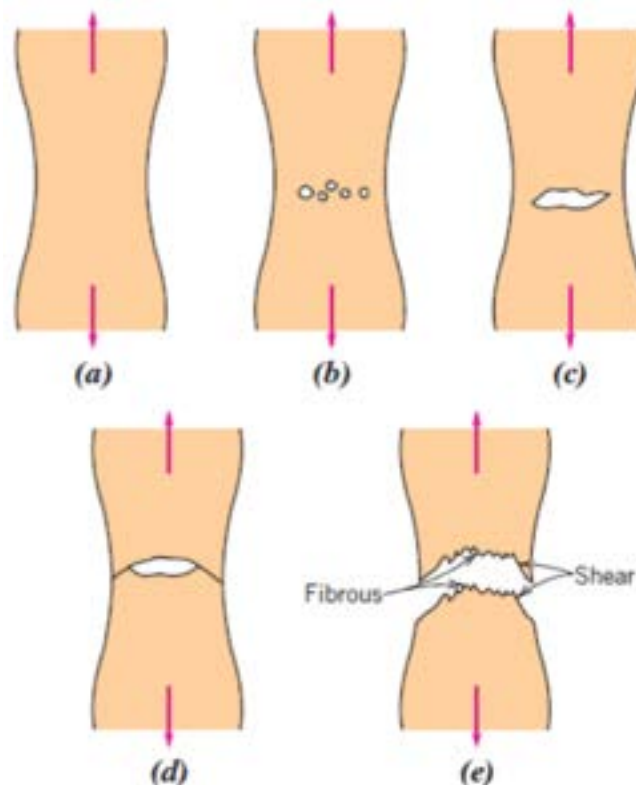


Figure 187: Stages in the cup-and-cone fracture. (a) Initial necking, (b) Small cavity formation, (c) Coalescence of cavities to form a crack, (d) Crack propagation, (e) Final shear fracture at a 45° angle relative to the tensile direction

Brittle fracture

- Fracture surfaces of materials that fail in a brittle manner have distinctive patterns, any signs of gross plastic deformation are absent
- Brittle fracture surfaces contain lines or ridges that radiate from the origin of the crack in a fanlike pattern
- For most brittle crystalline materials, crack propagation corresponds to the successive and repeated breaking of atomic bonds along specific crystallographic planes, such a process is termed cleavage. This type of fracture is said to be transgranular

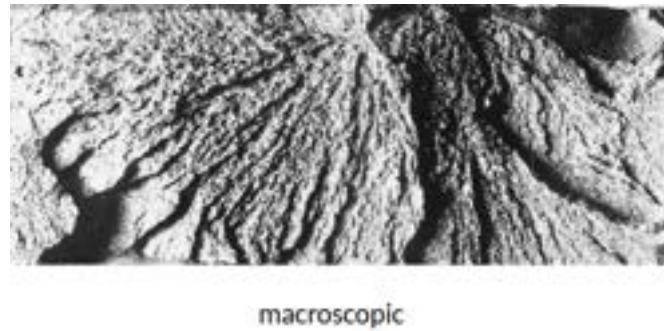


Figure 188: Macroscopic view of brittle fracture with the fanlike pattern

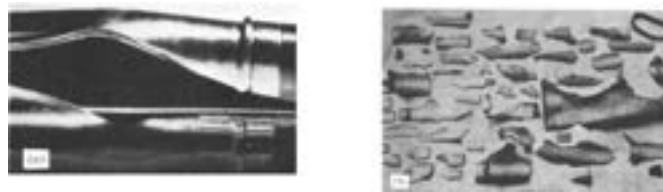
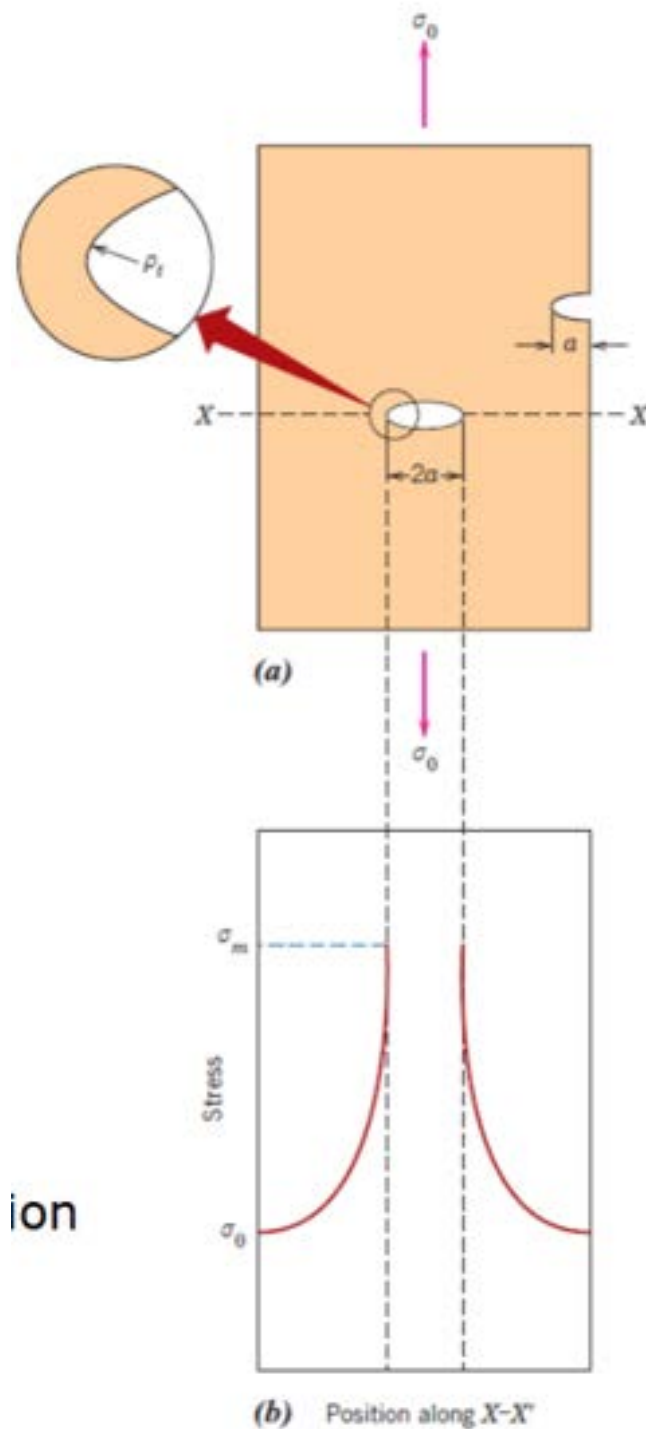


Figure 189: Failure of a Pipe: (a) Ductile failure, one piece and large deformation. (b) Brittle failure, many pieces and small deformation

4.3.1 Principles of fracture mechanics



Stress Concentration

- The measured fracture strengths for most materials are significantly lower than those predicted by theoretical calculations based on atomic bonding energies
- The discrepancy is explained by the presence of microscopic flaws or cracks that always exist under normal conditions at the surface and within the interior of a body of material
- These flaws are detrimental to the fracture strength because an applied stress may be amplified or concentrated at the tip

- The maximum stress, σ_m , occurs at the crack tip,

$$\sigma_m = 2\sigma_0 \left(\frac{a}{\rho_t} \right)^{1/2}$$

Where σ_0 is the magnitude of the applied tensile stress, and ρ_t is the radius of curvature of the crack tip

- The ratio σ_m/σ_0 is denoted the stress concentration factor K_t ,

$$K_t = \frac{\sigma_m}{\sigma_0} = 2 \left(\frac{a}{\rho_t} \right)^{1/2}$$

- The critical stress for crack propagation in a brittle material,

$$\sigma_x = \left(\frac{2E\gamma_s}{\pi a} \right)^{1/2}$$

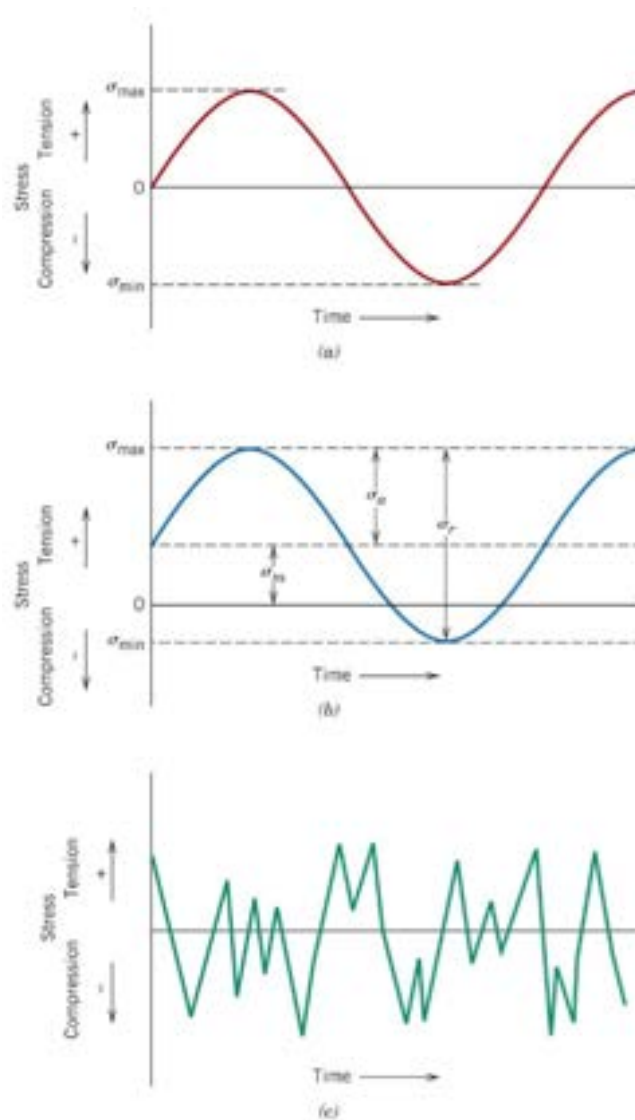


Figure 190: fatigue

4.3.2 Crack initiation and Propagation

The process of fatigue failure is characterized by three distinct steps,

1. **Crack initiation**, in which a small crack forms at some point of high stress concentration
2. **Crack propagation**, during which this cracks advances incrementally with each stress cycle, and
3. **Final failure**, which occurs very rapidly once the advancing crack has reached a critical size

Cracks initiation

- Cracks associated with fatigue failure almost always initiate (or nucleate) on the surface of a component at some point of stress concentration.
- Crack nucleation sites include surface scratches, sharp fillets, keyways, threads, dents
- In addition, cyclic loading can produce microscopic surface discontinuities resulting from dislocation slip steps that may also act as stress raisers and therefore as crack initiation site

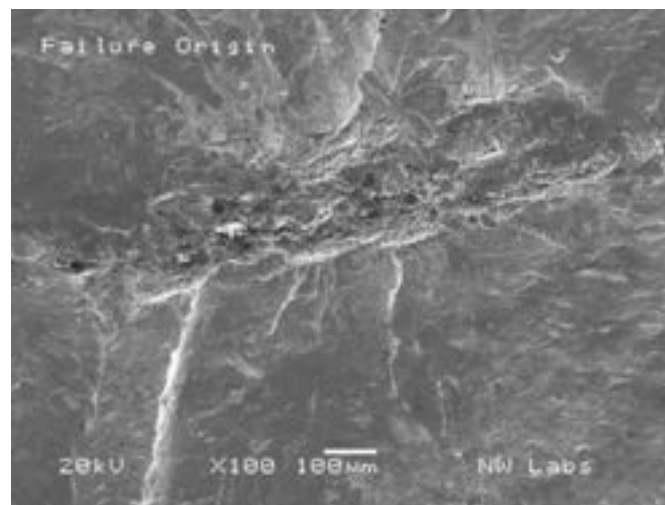
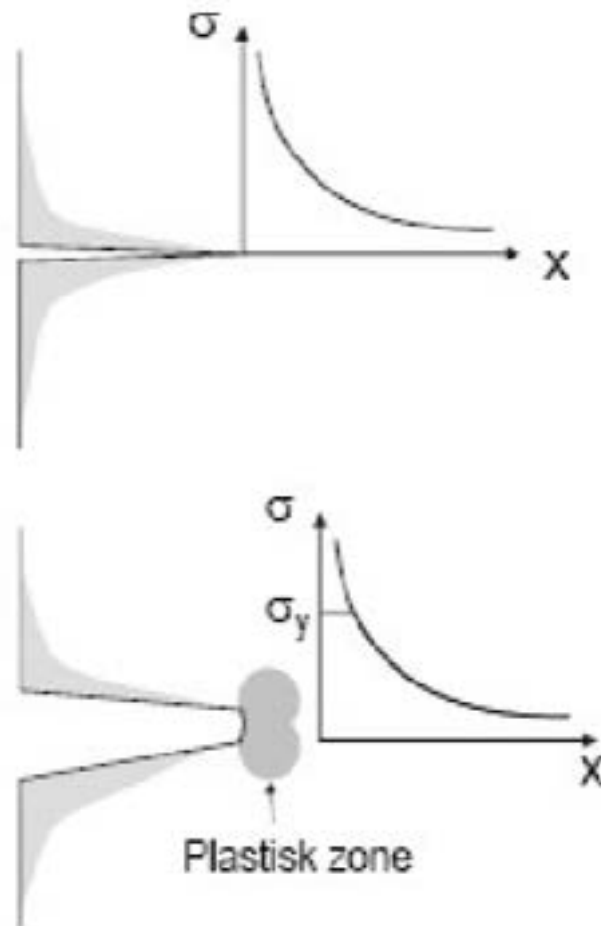


Figure 191: E.g. can initiate at a particle in the metal



Crack propagation

- The region of a fracture surface that formed during the crack propagation step may be characterized by two types of markings termed beachmark and striations
- Both features indicate the position of the crack tip at some point in time and appear as concentric ridges that expand away from the crack initiation site(s),



Figure 192: Visualization of benchmarks and striations

- Beachmarks
 - Beachmarks (sometimes also called clamshell marks) are of macroscopic dimensions
 - For components that experienced interruptions during the crack propagation stage
- Striations
 - Fatigue striations are microscopic in size and subject to observation with electron microscope
 - Each striation is thought to represent the advance distance of a crack front during a single load cycle

Final failure

- The presence of beachmarks and/or striations on a fracture surface confirms that the cause of failure was fatigue. Nevertheless, the absence of either or both does not exclude fatigue failure.
- Beachmarks and striations do not appear on the region over which the rapid failure occurs. Rather, the rapid failure may be either ductile or brittle, evidence of plastic deformation will be present for ductile failure and absent for brittle failure

4.3.3 Wohler (S/N curves) - different behavior

- The higher the magnitude of the stress, the smaller the number of cycles the material is capable of sustaining before failure.
- For some ferrous (iron-base) and titanium alloys, the S-N curve, becomes horizontal at higher N values, there is a limiting stress level, called the fatigue limit below which fatigue failure will not occur.
- Most nonferrous alloys (e.g. aluminum, copper) do not have a fatigue limit, in that the S-N curve continues its downward trend

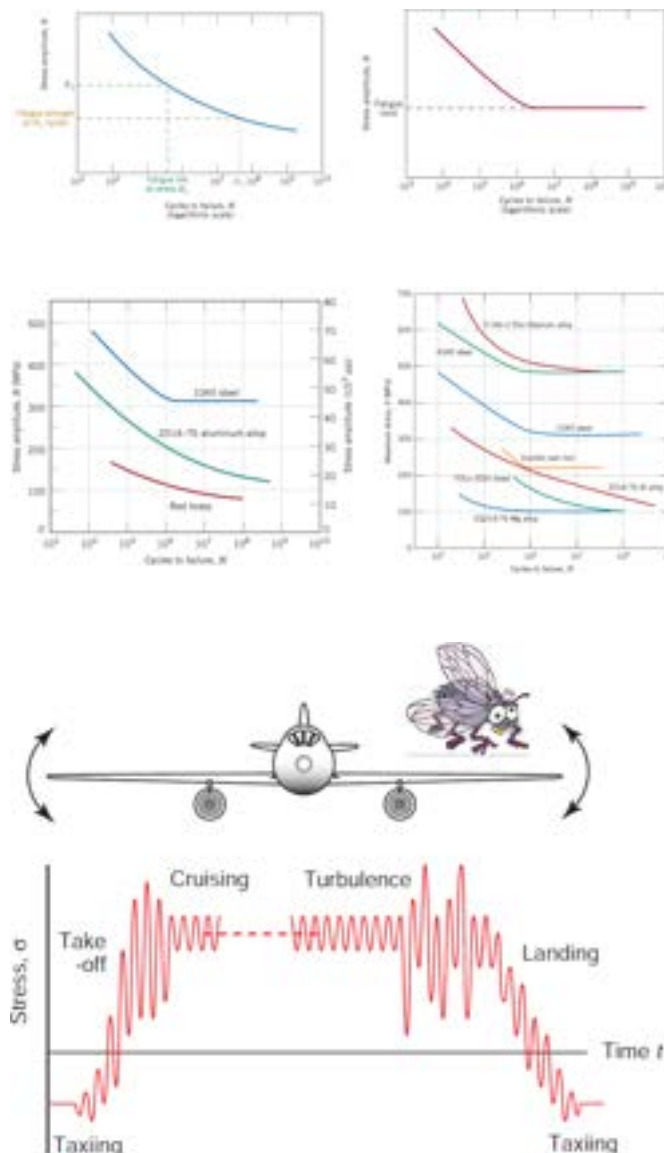


Figure 193: Al alloys, aerospace

How to prevent fatigue failure

- Avoid high stress concentrations: Design of parts and/or Surface finish (roughness, flaws, etc.)
- Minimize the stress amplitude
- Design
- Surface treatment → Introduce compressive residual stresses in the surface
 - Case hardening / diffusion processes: Carbon and nitrogen
 - Shot peening (work hardening)

4.3.4 Creep

- **Creep** is defined as the time independent and permanent deformation of materials when subjected to a constant load or stress at high temperatures
- Materials are often placed in service at elevated temperatures and exposed to static mechanical stresses
- (e.g., turbine rotors in jet engines and steam generators that experience centrifugal stresses, high pressure steam lines),
- Creep is normally undesirable phenomenon and is often the limiting factor in the lifetime of a part

Generalized Creep behavior, Creep is plastic deformation at a stress lower than yield strength of the material.

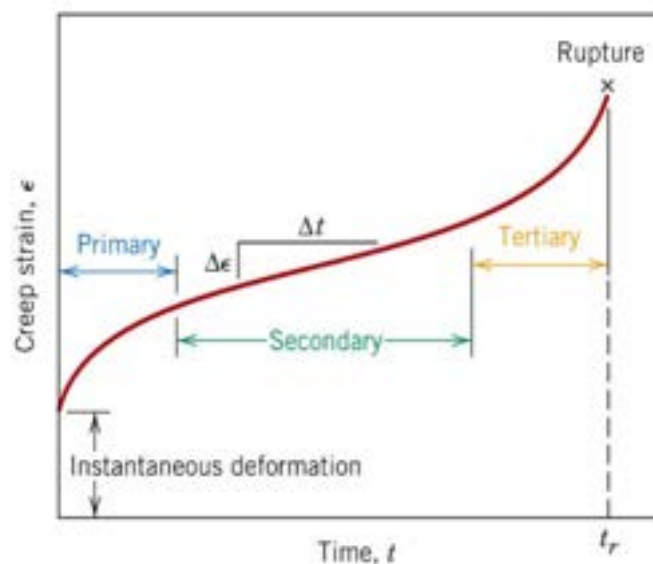


Figure 194: Creep curve for $T > 0.4T_m$

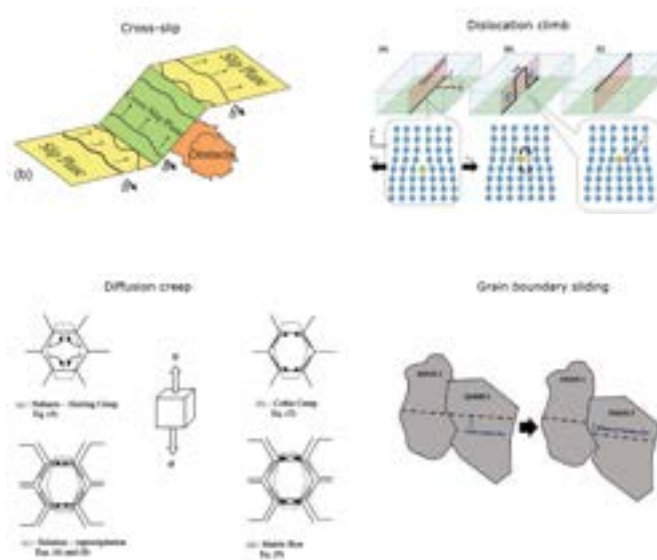
The creep curve consists of three regions

- Primary of transient creep, a continuously decreasing creep rate, that is, the slope of the curve decreases with time
- an increase in creep resistance or strain hardening, deformation becomes more difficult
- For secondary creep (steady-state creep): the rate is constant, the longest duration.

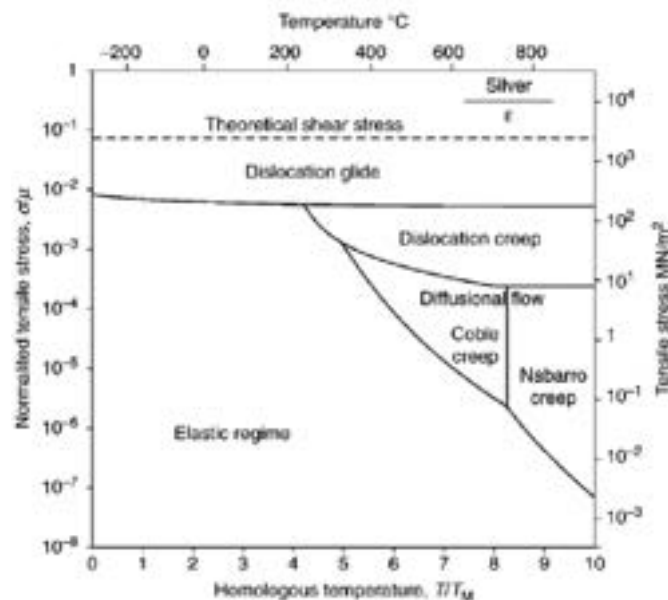
- The constancy of creep rate is explained on the basis of a balance between the competing processes of strain hardening and recovery
- Tertiary creep: an acceleration of the rate and ultimate failure
- This failure is frequently termed rupture and results from microstructural and/or metallurgical changes for example, grain boundary separation, and the formation of internal cracks, cavities and voids-

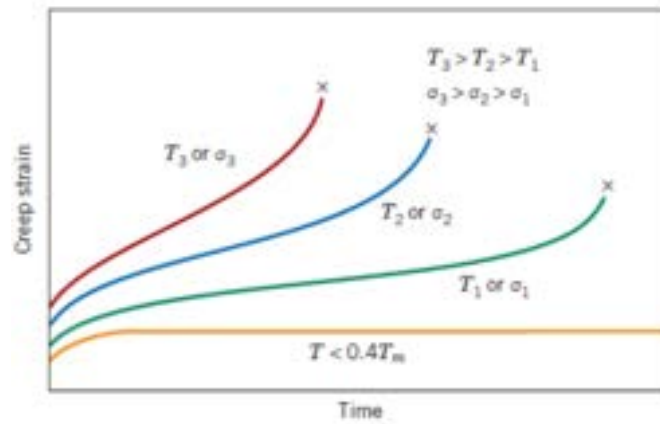
Mechanisms, creep can occur through 4 different mechanisms,

- Dislocation based mechanisms
 - Cross-slip
 - Dislocation climb
- Non-dislocation based mechanism
 - Diffusion
 - Grain boundary sliding

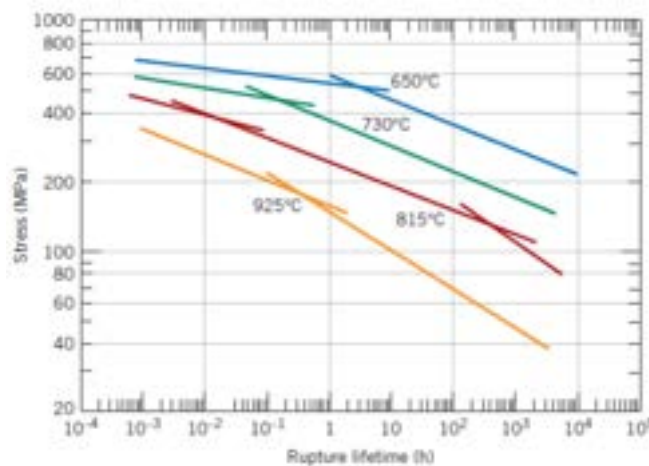


Impact of stress level and temperature





$$\dot{\epsilon}_s = K_1 \sigma^n$$



$$\dot{\epsilon}_s = K_2 \sigma^n \exp\left(-\frac{Q_c}{RT}\right)$$

How to prevent Creep:

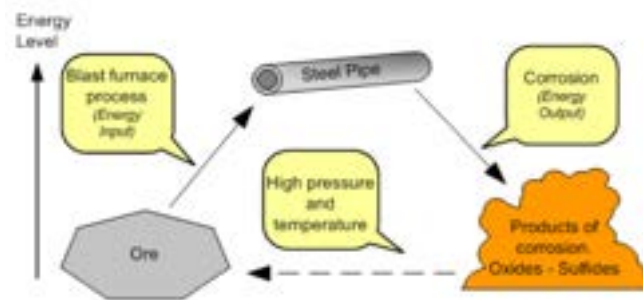
- Larger wall thickness $\rightarrow$ less effective stress
- Large grains in the material
 - Turbine blades: columnar crystals, single crystal
- Lower the temperature (if possible)
- Use special alloys
 - Age hardenable alloys, steels and nickel base alloys

4.4 Corrosion in Metallic Materials

Corrosion is the deterioration of a material, usually a metal, due to a chemical or electrochemical reaction with its environment. In many cases, corrosion can be understood as the metal returning to a more stable, lower energy state, similar to the state in which it is naturally found as an ore.

4.4.1 Why Metals Corrode

Most metals are not found naturally in their pure metallic form. Instead, they usually occur as ores, such as oxides or sulphides. Energy is supplied during extraction to convert these ores into metals. This means that the metallic state is often a higher energy and less stable state.



Corrosion is therefore the tendency of the metal to return to a more stable state. Noble metals, such as gold, are more stable in their metallic form and therefore show much better corrosion resistance.

Corrosion is strongly affected by the surrounding environment. Important factors include:

- Presence of water or humidity
- Presence of oxygen
- Acidic or alkaline conditions
- Dissolved ions in the electrolyte
- Sulphur-containing gases or other aggressive species

In most practical cases, corrosion occurs in the presence of a thin water layer. Therefore, the process is often electrochemical in nature.

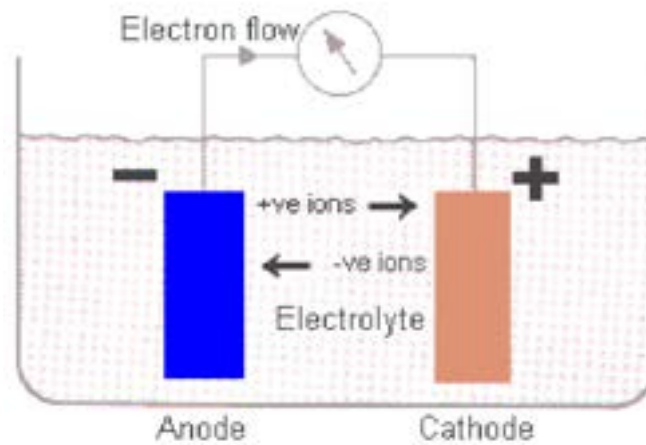
4.4.2 Types of Corrosion Environments

Corrosion can occur in different environments:

- **Aqueous corrosion:** corrosion in a liquid electrolyte, usually water-based. This is electrochemical corrosion.
- **Corrosion in other liquids:** for example corrosion in liquid metals.
- **Dry corrosion:** high-temperature corrosion in dry gases such as oxygen or nitrogen.

Aqueous corrosion is especially important because many metals are exposed to humidity, rainwater, seawater, or other electrolytes during service.

4.4.3 Electrochemical Nature of Aqueous Corrosion



Aqueous corrosion involves two simultaneous electrochemical reactions:

- **Anodic reaction:** oxidation of the metal.
- **Cathodic reaction:** reduction of species in the electrolyte.

At the anode, the metal dissolves into the solution:



For a divalent metal, this becomes:



The electrons released at the anode are consumed by a cathodic reaction. Common cathodic reactions include:



This reaction is common in oxygen-rich neutral or alkaline solutions.



This reaction occurs in oxygen-rich acidic solutions.



This reaction occurs in acidic media.



This reaction can occur in neutral or alkaline media.

4.4.4 Electrode Potential

The electrode potential describes the tendency of a metal electrode to lose or gain electrons when it is in contact with a solution containing its ions.

A metal with a strong tendency to lose electrons behaves as an anode and corrodes more easily. A more noble metal has a lower tendency to oxidize and is therefore more corrosion resistant.

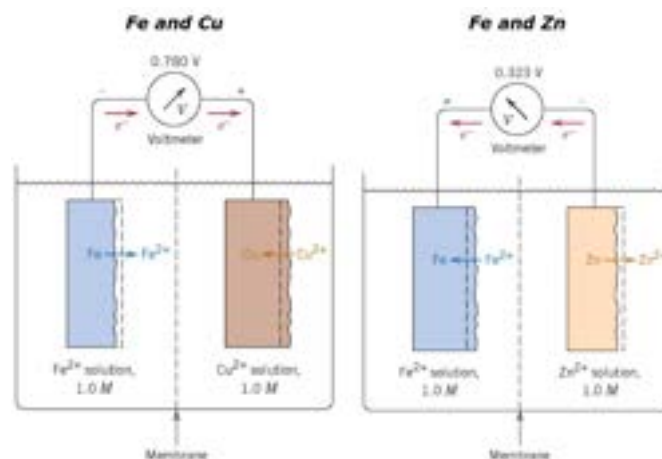
The standard electrode potential is measured under standard conditions:

- Ion concentration of 1 M
- Temperature of 25°C
- Pressure of 1 atm

The electrode potential depends on several factors:

- Metal type
- Ion concentration
- Electrode surface condition
- Temperature
- Pressure, for gases
- pH, for reactions involving H^+ or OH^-

4.4.5 Galvanic Corrosion



Galvanic corrosion occurs when two different metals are electrically connected and exposed to the same electrolyte. The less noble metal acts as the anode and corrodes faster, while the more noble metal acts as the cathode.

For example, when iron is coupled with copper, iron tends to corrode because copper is more noble. When iron is coupled with zinc, zinc corrodes preferentially and protects the iron.

The severity of galvanic corrosion depends on:

- Difference in electrode potential between the two metals
- Conductivity of the electrolyte

- Relative surface areas of the anode and cathode

A small anodic area connected to a large cathodic area is especially dangerous because the corrosion rate at the anode becomes very high.

Galvanic corrosion can be reduced by:

- Choosing metals that are close together in the galvanic series
- Avoiding a small anode combined with a large cathode
- Electrically insulating dissimilar metals
- Using coatings or protective layers

4.4.6 Pourbaix Diagrams

A Pourbaix diagram shows the stable phases of a metal as a function of potential and pH. It can be used to predict whether a metal will corrode, passivate, or remain immune under certain environmental conditions.

From a Pourbaix diagram, it is possible to estimate:

- Which reactions are likely at different pH and potentials
- Which corrosion products may form
- Conditions where the metal is protected by a passive layer
- Conditions where active corrosion is expected

4.4.7 Corrosion Kinetics and Overvoltage

Even if a corrosion reaction is thermodynamically possible, its rate depends on kinetics. In electrochemistry, overvoltage is the extra potential needed beyond the theoretical potential to make a reaction proceed at a measurable rate.

There are three main types of overvoltage:

- **Activation overvoltage:** energy barrier for the electrochemical reaction at the surface.
- **Concentration overvoltage:** caused by concentration differences between the electrode surface and the bulk electrolyte.
- **Resistance overvoltage:** caused by electrical resistance in the electrolyte, electrodes, or connections.

The total overvoltage may be written as:

$$\eta = \eta_a + \eta_c + \eta_R \quad (10)$$

where η_a is the activation overvoltage, η_c is the concentration overvoltage, and η_R is the resistance overvoltage.

4.4.8 Forms of Corrosion

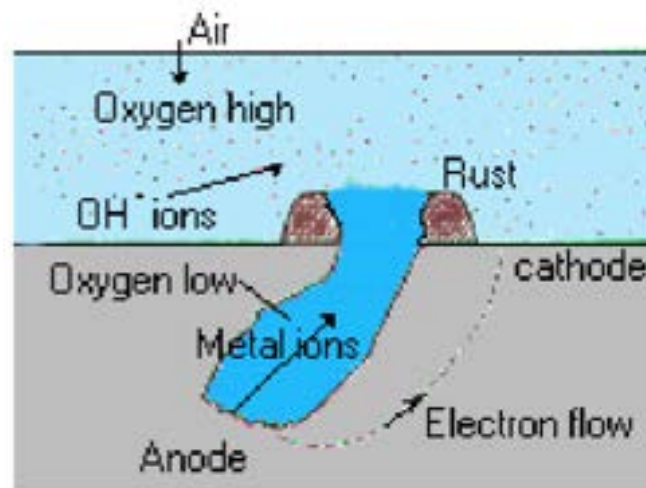
Uniform Corrosion Uniform corrosion occurs with approximately equal intensity over the entire exposed surface. It often produces a visible corrosion product or scale. This type of corrosion is usually easier to predict than localized corrosion.

Galvanic Corrosion Galvanic corrosion occurs when two dissimilar metals are electrically coupled in an electrolyte. The less noble metal becomes the anode and corrodes faster.

Crevice Corrosion Crevice corrosion occurs in narrow gaps where the electrolyte becomes trapped. These regions can develop different oxygen concentrations compared to the surrounding surface, creating a local corrosion cell.

Typical locations include:

- Under washers
- Under gaskets
- Inside joints
- Beneath deposits or dirt



Pitting Corrosion Pitting corrosion is a localized form of corrosion where small holes or pits form on the metal surface. It is dangerous because the surface damage may look small while the pit grows deeply into the material.

Stainless steels can suffer from pitting corrosion in halogen-rich environments, especially chloride-containing environments.

Intergranular Corrosion Intergranular corrosion occurs along grain boundaries. In stainless steels, this can happen when chromium carbides or nitrides form at the grain boundaries during inappropriate heat treatment.

When chromium is removed from solid solution near the grain boundary, the local corrosion resistance decreases. This weakens the passive protection in those regions.

Erosion Corrosion Erosion corrosion is caused by the combined action of corrosion and mechanical wear. The flowing fluid removes protective layers from the surface, exposing fresh metal to further corrosion.

Stress Corrosion Cracking Stress corrosion cracking occurs when a material is exposed to both tensile stress and a corrosive environment. The combination can cause cracks to form and grow even if the stress is below the normal yield strength of the material.

4.4.9 Protection Against Corrosion

Several methods can be used to reduce corrosion.

Cathodic Protection Cathodic protection makes the metal component act as a cathode, reducing or stopping its corrosion. This can be done by connecting the component to a more active sacrificial metal. For example, zinc is more anodic than iron. Therefore, zinc can corrode preferentially and protect steel.

Anodic Protection Anodic protection is used for active-passive metals. The metal is held at a potential where a stable passive film forms on the surface. This method can be used in aggressive environments, but it is only suitable for metals that show passivation behavior.

Inhibitors Corrosion inhibitors are chemicals added to the environment to reduce corrosion rate. They can work by:

- Adsorbing onto the metal surface
- Slowing down anodic or cathodic reactions
- Removing active corrosive species from the electrolyte

Inhibitors are mainly useful in confined or closed systems.

4.4.10 High-Temperature Corrosion

High-temperature corrosion occurs when metals react with dry gases at elevated temperatures. A common example is oxidation, where oxygen reacts with the metal surface to form a metal oxide.



The oxide layer may protect the metal if it is dense, adherent, and continuous. However, if the oxide layer cracks, flakes off, or becomes porous, fresh metal is exposed and oxidation continues.

4.4.11 Pilling-Bedworth Ratio

The Pilling-Bedworth ratio compares the volume of oxide formed to the volume of metal consumed:

$$\text{PBR} = \frac{V_{\text{oxide}}}{V_{\text{metal}}} \quad (12)$$

This ratio gives an indication of whether the oxide layer will be protective.

- If $\text{PBR} < 1$, the oxide layer is usually porous and does not fully cover the metal.
- If $\text{PBR} > 1$, compressive stresses develop in the oxide layer.
- If PBR is too high, often above about 2 to 3, the oxide layer may crack and spall off.

Therefore, a protective oxide layer should have a suitable volume ratio, good adhesion, and low tendency to crack.

4.4.12 Thermal Expansion Mismatch

During heating and cooling, the metal and oxide layer may expand or contract by different amounts. If their thermal expansion coefficients are different, stresses develop at the interface.

During cooling, these stresses can cause cracking or spallation of the oxide layer. Once the oxide layer flakes off, the metal surface is exposed again and high-temperature oxidation can continue.

4.5 Surface Modification of Metals

4.5.1 Objectives of Surface Modification

Metals are widely used in engineering due to their:

- High mechanical strength
- Good electrical and thermal conductivity
- Versatility in applications

However, their performance is often limited by surface-related degradation mechanisms:

- **Corrosion:** chemical or electrochemical deterioration
- **Wear:** material loss due to mechanical interaction
- **Fatigue:** failure under cyclic loading

Surface modification techniques are therefore used to improve:

- Surface hardness
- Wear resistance
- Corrosion resistance
- Fatigue life
- Aesthetic properties
- Adhesion for coatings or paints

Thermochemical Processes	Vapour Phase Processes	Conversion Coating Processes
Light elements: • Carburizing • Nitriding • Boriding Metallic elements: • Galvanizing • Aluminizing • Chromizing • Vanadizing • TPD Properties: • Surface hardness • Wear resistance • Corrosion resistance • Fatigue resistance	• Physical vapour deposition (PVD) • Chemical vapour deposition (CVD) • Ion implantation Properties: • Surface hardness • Wear resistance • Corrosion resistance • Decorative	Electrochemical: • Anodizing • Electroplating Chemical: • Phosphating • Chromating • Ti/Zr based • Silane Properties: • Surface hardness • Wear resistance • Corrosion resistance • Decorative • Base layer for painting

- Selection depends on the type of object, required properties, and process limitations
- Combination of processes is often used to achieve multiple properties

Table 11: Overview of surface modification techniques and their effects

4.5.2 Surface Engineering Approaches

The selection of a surface modification method depends on:

- Required property improvements
- Type of material
- Service environment
- Process limitations

Often, multiple processes are combined to achieve the desired performance.

Surface modification techniques can be broadly classified into:

- Mechanical methods
- Chemical methods
- Electrochemical methods
- Physical and thermal methods
- Coating and deposition techniques

4.5.3 Mechanical Surface Treatments

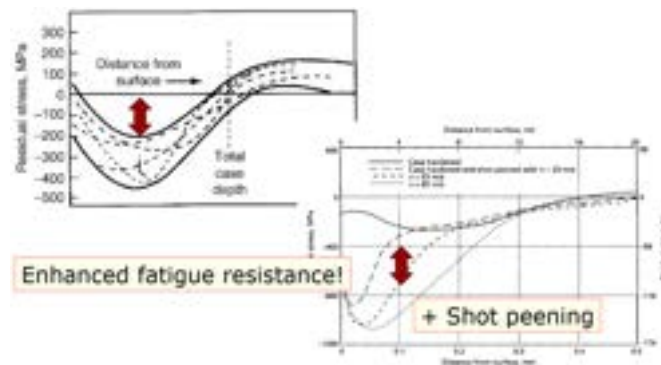
Mechanical treatments modify the surface without changing its chemistry.

Examples include:

- Shot peening
- Burnishing
- Surface grinding

These methods:

- Introduce compressive residual stresses
- Improve fatigue resistance
- Improve wear resistance



They are widely used in automotive and aerospace components.

4.5.4 Chemical Surface Treatments

Chemical methods involve reactions at the surface to improve properties.

Examples:

- Acid pickling
- Passivation
- Anodizing

These methods:

- Remove impurities
- Form protective oxide layers
- Improve corrosion resistance

4.5.5 Electrochemical Methods

Electrochemical techniques involve the use of electric current.

Examples:

- Electroplating
- Anodizing

These methods are used to:

- Deposit protective coatings
- Improve corrosion resistance
- Enhance surface appearance

4.5.6 Thermal and Thermochemical Treatments

These treatments modify the surface composition through diffusion processes.

Examples include:

- Carburizing
- Nitriding
- Nitrocarburising

These processes:

- Increase surface hardness
- Improve wear resistance
- Enhance fatigue strength

Nitriding Nitriding introduces nitrogen into the surface of steel, forming:

- A **compound layer** (hard, wear-resistant)
- A **diffusion zone** (improves fatigue strength)

It can also improve corrosion resistance when an oxidized layer forms.

4.5.7 Coatings and Deposition Techniques

Surface coatings are applied to protect or enhance surface properties.

Common techniques include:

- Physical Vapour Deposition (PVD)
- Chemical Vapour Deposition (CVD)
- Thermal spraying

- Electroplating

Coating materials include:

- Metals
- Ceramics
- Oxides

These coatings can provide:

- Corrosion resistance
- Wear resistance
- Biocompatibility

4.5.8 Residual Stresses and Fatigue

Mechanical treatments such as shot peening introduce compressive residual stresses at the surface.

These stresses:

- Delay crack initiation
- Improve fatigue life

4.6 Corrosion in Electronics

Corrosion plays a critical role in modern electronic systems, especially in high-power and high-voltage applications.

Applications include:

- Power electronics for renewable energy systems
- Electric vehicles (400–2000 V systems)
- Low-power electronics (12–48 V systems)

4.6.1 Failure Mechanisms

Corrosion in electronics often occurs through electrochemical processes.

Key mechanisms include:

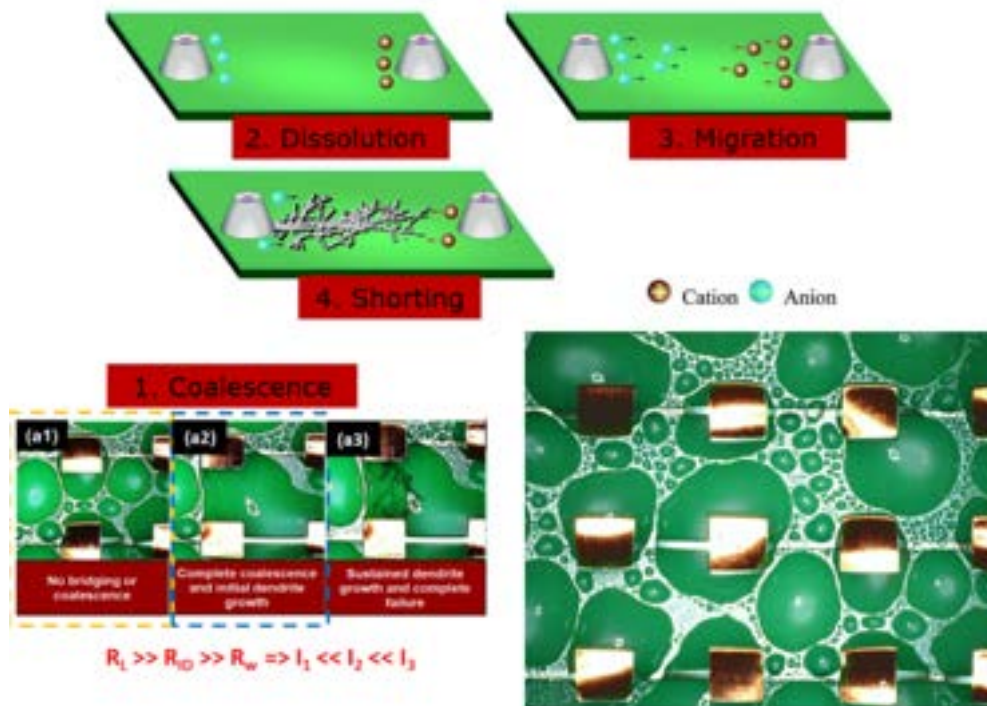
- Galvanic corrosion (e.g., Al–Au systems)
- Formation of electrolytic cells
- Electrochemical migration

4.6.2 Water-Induced Corrosion

A major issue is the presence of water droplets on printed circuit boards (PCBs).

The failure process typically involves:

1. Water droplet formation (coalescence)
2. Dissolution of metal ions
3. Ion migration under electric field
4. Short circuit formation



The probability of failure increases with higher voltage and power levels.

4.6.3 Protection of Electronic Components

To prevent corrosion, protective coatings are applied.

One example is:

- Plasma Enhanced Chemical Vapour Deposition (PECVD)

These coatings:

- Act as barriers against moisture
- Reduce corrosion risk
- Improve reliability of electronic systems